

# Level 4

## Watsonx Orchestrate

Hands-on lab guide – Integrate your AI Assistant in your web interface.



## LEGAL NOTICES

This information was developed for products and services offered in the USA. IBM may not offer the products, services, or features discussed in this document in other countries. Consult your local IBM representative for information on the products and services currently available in your area. Any reference to an IBM product, program, or service is not intended to state or imply that only that IBM product, program, or service may be used. Any functionally equivalent product, program, or service that does not infringe any IBM intellectual property right may be used instead. However, it is the user's responsibility to evaluate and verify the operation of any non-IBM product, program, or service. IBM may have patents or pending patent applications covering subject matter described in this document. The furnishing of this document does not grant you any license to these patents. You can send license inquiries, in writing, to:

IBM Director of Licensing IBM Corporation  
North Castle Drive, MD-NC119 Armonk, NY 10504-1785  
United States of America

The following paragraph does not apply to the United Kingdom or any other country where such provisions are inconsistent with local law: INTERNATIONAL BUSINESS MACHINES CORPORATION PROVIDES THIS PUBLICATION “AS IS” WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Some states do not allow disclaimer of express or implied warranties in certain transactions; therefore, this statement may not apply to you. This information could include technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in new editions of the publication. IBM may make improvements and/or changes in the product(s) and/or the program(s) described in this publication at any time without notice. Any references in this information to non-IBM websites are provided for convenience only and do not in any manner serve as an endorsement of those websites. The materials at those websites are not part of the materials for this IBM product and use of those websites is at your own risk. IBM may use or distribute any of the information you supply in any way it believes appropriate without incurring any obligation to you. Information concerning non-IBM products was obtained from the suppliers of those products, their published announcements or other publicly available sources. IBM has not tested those products and cannot confirm the accuracy of performance, compatibility or any other claims related to non-IBM products. Questions on the capabilities of non-IBM products should be addressed to the suppliers of those products.

This information contains examples of data and reports used in daily business operations. To illustrate them as completely as possible, the examples include the names of individuals, companies, brands, and products. All names and references for organizations and other business institutions used in this deliverable's scenarios are fictional. Any match with real organizations or institutions is coincidental. All names and associated information for people in this deliverable's scenarios are fictional. Any match with a real person is coincidental.

## TRADEMARKS

IBM, the IBM logo, and [ibm.com](https://www.ibm.com) are trademarks or registered trademarks of International Business Machines Corp., registered in many jurisdictions worldwide. Other product and service names might be trademarks of IBM or other companies. A current list of IBM trademarks is available on the web at “Copyright and trademark information” at <https://www.ibm.com/legal/copyright-trademark>.

Adobe, the Adobe logo, PostScript, and the PostScript logo are either registered trademarks or trademarks of Adobe Systems Incorporated in the United States, and/or other countries.

Java and all Java-based trademarks and logos are trademarks or registered trademarks of Oracle and/or its affiliates.

INFOBLOX is a trademark of Infoblox Inc. or its affiliated companies, registered in the United States and other countries.

Microsoft®, Azure, Windows, Windows NT, Windows Server, Microsoft® Excel®, Microsoft® Word®, Microsoft® Outlook®, Microsoft® Excel® logo, Microsoft® Word® logo, Microsoft® Outlook® logo, Microsoft® Windows App, Microsoft® Windows App logo and the Windows logo are trademarks of Microsoft® Corporation in the United States, other countries, or both.

Salesforce and Salesforce logo are registered trademarks of SALESFORCE, INC. (CORPORATION; CALIFORNIA, USA).

Slack is a registered trademark of Salesforce, Inc (CORPORATION; DELAWARE, USA).

Red Hat®, JBoss®, OpenShift®, Fedora®, Hibernate®, Ansible®, CloudForms®, RHCA®, RHCE®, RHCSA®, Ceph®, and Gluster® are trademarks or registered trademarks of Red Hat, Inc. or its subsidiaries in the United States and other countries.

Google, Google logo, Google Tasks, Google Cloud Platform, GMAIL, GMAIL logo are registered trademarks of GOOGLE LLC and this material is not endorsed by or affiliated with Google in any way.

Elasticsearch, Elastic search logo, Elastic, Elastic logo are registered trademarks of Elasticsearch BV (LIMITED LIABILITY COMPANY; NETHERLANDS) in United States, other countries or both.

Milvus, Milvus logo are registered trademarks of LF Projects, LLC (LIMITED LIABILITY COMPANY; DELAWARE, USA) in United States, other countries or both.

Meta, Meta logos, Llama, Llama logos are registered trademarks of META PLATFORMS, INC. (CORPORATION; California, USA) in United States, other countries or both.

Mistral, Mistral logo, Mistral AI, Mistral AI logo are registered trademarks of Mistral AI (SOCIÉTÉ PAR ACTIONS SIMPLIFIÉE (SAS); FRANCE) in United States, other countries or both.

PyCharm, PyCharm logo are registered trademarks of JetBrains s.r.o. (CORPORATION; CZECH REPUBLIC) in United States, other countries or both.

Bruno is a trademark held by Anoop M D. Bruno logo is sourced from OpenMoji. License: CC BY-SA 4.0

Apple, Apple logo, Apple store, App store, App store logo are registered trademarks of Apple Inc. (CORPORATION; CALIFORNIA, USA) in United States, other countries or both.

VMWARE, VMWARE logos are registered trademarks of VMware LLC (LIMITED LIABILITY COMPANY; Delaware, USA) in United States, other countries or both.

RANCHER, RANCHER logo, Rancher Desktop, Rancher Desktop logo are registered trademarks of SUSE LLC (LIMITED LIABILITY COMPANY; MASSACHUSETTS, USA) in United States, other countries or both.

Langfuse, Langfuse logo are registered trademarks of Langfuse GmbH (GESELLSCHAFT MIT BESCHRÄNKTER HAFTUNG (GMBH); Germany, Germany) in United States, other countries or both.

OpenMeteo is a registered trademark owned by Patrick Zippenfenig (Individual, Germany) in United States, other countries or both.

© Copyright International Business Machines Corporation 2025. This document may not be reproduced in whole or in part without the prior written permission of IBM.

# Table of Contents

|                                                                                                         |            |
|---------------------------------------------------------------------------------------------------------|------------|
| <b>Introduction .....</b>                                                                               | <b>6</b>   |
| Before you start .....                                                                                  | 6          |
| About this lab .....                                                                                    | 6          |
| Environment .....                                                                                       | 6          |
| Lab use-case .....                                                                                      | 7          |
| How to get support .....                                                                                | 7          |
| Prerequisites .....                                                                                     | 7          |
| <b>Enrich your assistant.....</b>                                                                       | <b>9</b>   |
| <b>Disambiguation and digression.....</b>                                                               | <b>14</b>  |
| Disambiguation: Core concepts.....                                                                      | 14         |
| Disambiguation in practice.....                                                                         | 14         |
| Digressions: Core concepts and in practice.....                                                         | 35         |
| <b>Advanced Action Configuration: Session Variables, Subactions, and Advanced Step Conditions .....</b> | <b>44</b>  |
| Session Variables: Core concepts .....                                                                  | 44         |
| Create Session Variables.....                                                                           | 45         |
| Session Variables in practice .....                                                                     | 48         |
| Subactions: Core concepts.....                                                                          | 73         |
| Subactions in practice.....                                                                             | 73         |
| Advanced Step Conditions .....                                                                          | 93         |
| Condition Groups.....                                                                                   | 103        |
| <b>Embed the Web Chat .....</b>                                                                         | <b>123</b> |
| Create a demo website .....                                                                             | 123        |
| Create or log in to your GitHub account .....                                                           | 123        |
| Get the required files .....                                                                            | 123        |
| Create the repository .....                                                                             | 124        |
| Add the files to the repository .....                                                                   | 127        |
| Configure your GitHub page.....                                                                         | 128        |
| Embed the assistant.....                                                                                | 132        |
| Publish to the Live environment .....                                                                   | 135        |
| Customize chat UI (Optional).....                                                                       | 139        |
| Customize the Home screen.....                                                                          | 141        |
| Get the Script code .....                                                                               | 145        |
| Embed the script .....                                                                                  | 146        |

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| <b>Annex - Export / re-import/ delete your assistant archive .....</b> | <b>151</b> |
| Export an assistant .....                                              | 151        |
| Create a version .....                                                 | 151        |
| Create the archive .....                                               | 154        |
| Import an assistant .....                                              | 157        |
| Delete your assistant.....                                             | 163        |

# Introduction

## Before you start

This lab is the second part of a series of 2 independent labs:

- Lab I: Create your AI Assistant
  - How to create an assistant and preview an assistant
  - How to define custom actions
  - How to define user interactions with the assistant
  - How to specify an end-to-end discussion
  - How to preview / publish an assistant
- Lab II: Integrate your AI Assistant in your web page
  - How to leverage the disambiguation and digression features
  - How to debug and trace an assistant before its deployment
  - How to work with sub-actions
  - How to configure and deploy your assistant an existing web site

**IMPORTANT NOTE:** The number of assistants is limited for each tenant; when you are finished with all the assistant labs, or should you need to suspend the learning for a moment, please DELETE YOUR ASSISTANT for the environment. Instructions to export your work and delete an assistant are provided at the end of this guide. **Assistants aged more than 3 weeks will be deleted by administrators.**

## About this lab

This Assistant lab 2 is designed to teach you how to Integrate an intelligent assistant in a web page to enrich it with the watsonx Orchestrate capabilities.

**Note:** It is strongly recommended to export your work at the end of each lab, should your assistant be deleted by the administrator for inactivity or delay expiration reasons. Instructions to export/import your assistant are given in this guide.

## Environment

This lab has been created and tested on watsonx Orchestrate Standard running on IBM Cloud. To avoid issues please use a watsonx Standard environment running on IBM Cloud as there are minor differences between AWS, IBM Cloud, and Cloud Pak for Data tenants. Instructions for requesting access to an enablement tenant to complete these labs are provided in the Preparation guide included in this learning plan.

## Lab use-case

During this lab, you will continue to implement and enrich the banking assistant you created during the Assistant lab 1 (Build an AI assistant). In Assistant lab 1, you created an AI assistant that is able to assist a customer with opening a new account. In this new lab, you will:

- Add additional actions from an imported assistant, expanding the assistant's banking capabilities
- Configure Disambiguation settings at the global / action level to improve the user experience
- Configure Digression settings at the global / action level to improve user experience
- Create Session Variables to track user information across actions
- Instantiate Subactions to reduce repetitive processes across actions, such as user authentication
- Configure advanced step conditions to implement complex business logic
- Implement an AI-guided action to answer questions in a specific topic
- Embed the AI assistant web chat on an external demo website

## How to get support

If you encounter an issue with this lab,

- IBM employees can open an IBM Support ticket at <http://ibm.biz/support>.
- Business Partners can request help at the [Partner Plus Support](#) website.

## Prerequisites

Please note the following prerequisites for this lab:

- You have Completed the wxO L4 Preparation guide
  - You have access to a wxO SaaS instance, as specified in wxO L4 Preparation guide.
- You have downloaded and extracted [LA Assistant Starter.zip](#) archive file on your computer



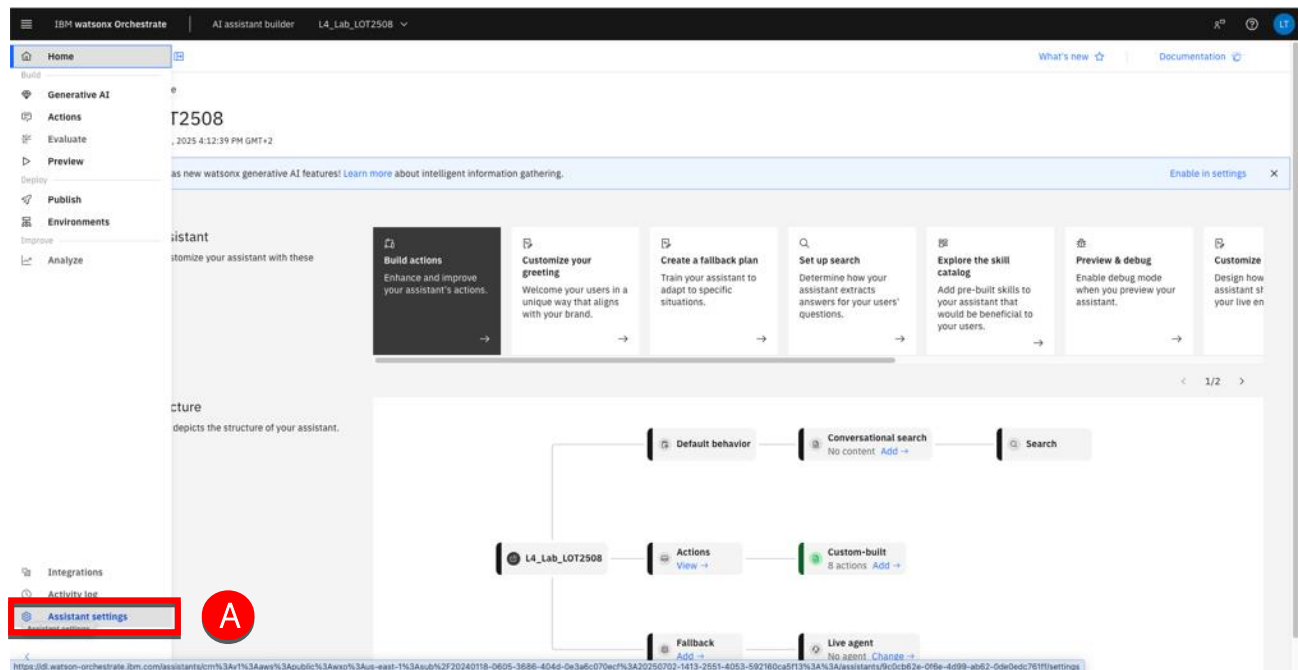
## Enrich your assistant

Before you continue with this lab, more actions must be added to your assistant. This chapter will explain you how to import a complete new pre-defined assistant that will replace your assistant developed in the Build an AI Assistant lab.

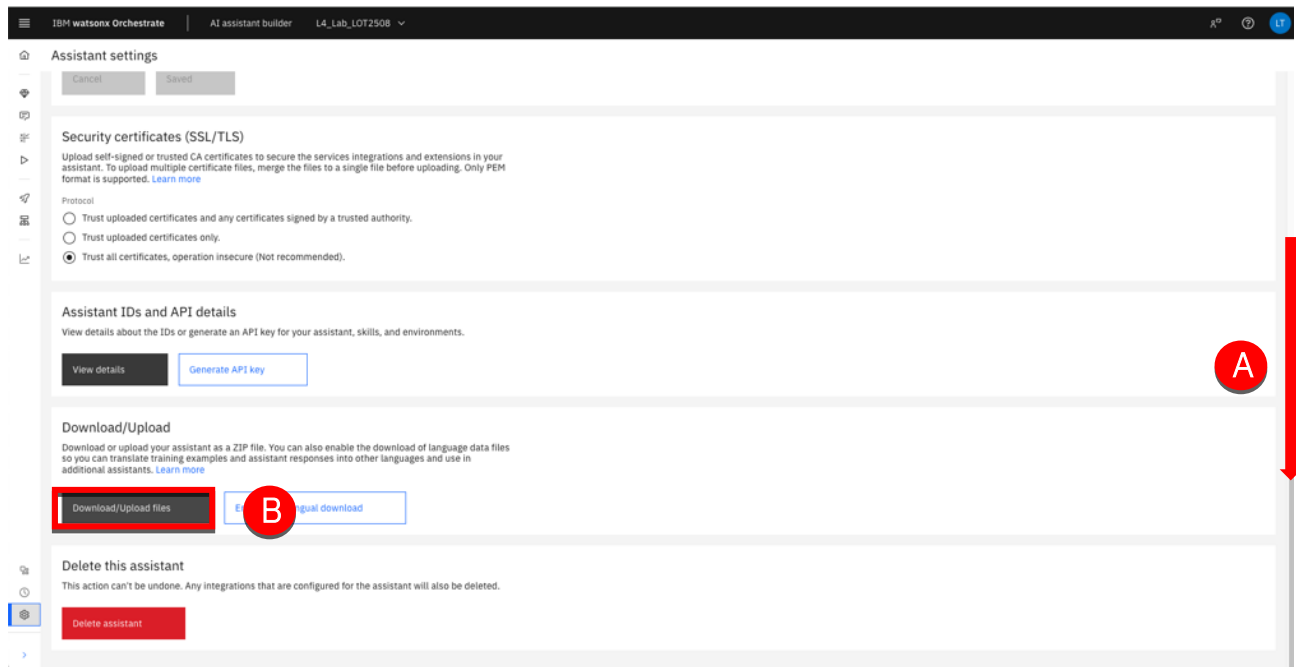
The assistant that you will import contains the Open an account action that you created in the previous lab. Additionally, the Assistant you will import contains more preconfigured banking related actions.

The import in AI Assistant builder will erase your current assistant content and replace it with the assistant you are about to import. Please export your current assistant should you want to keep a trace of your previous lab activities.

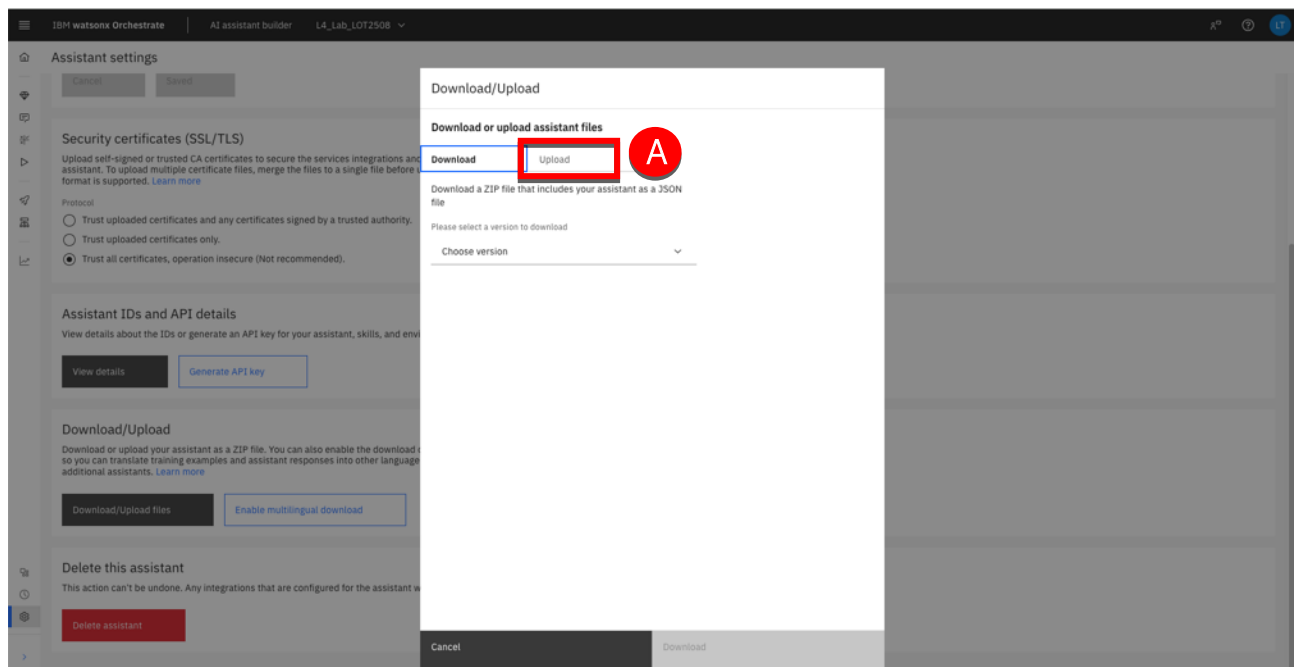
1. Open the **Assistant settings** menu (A).



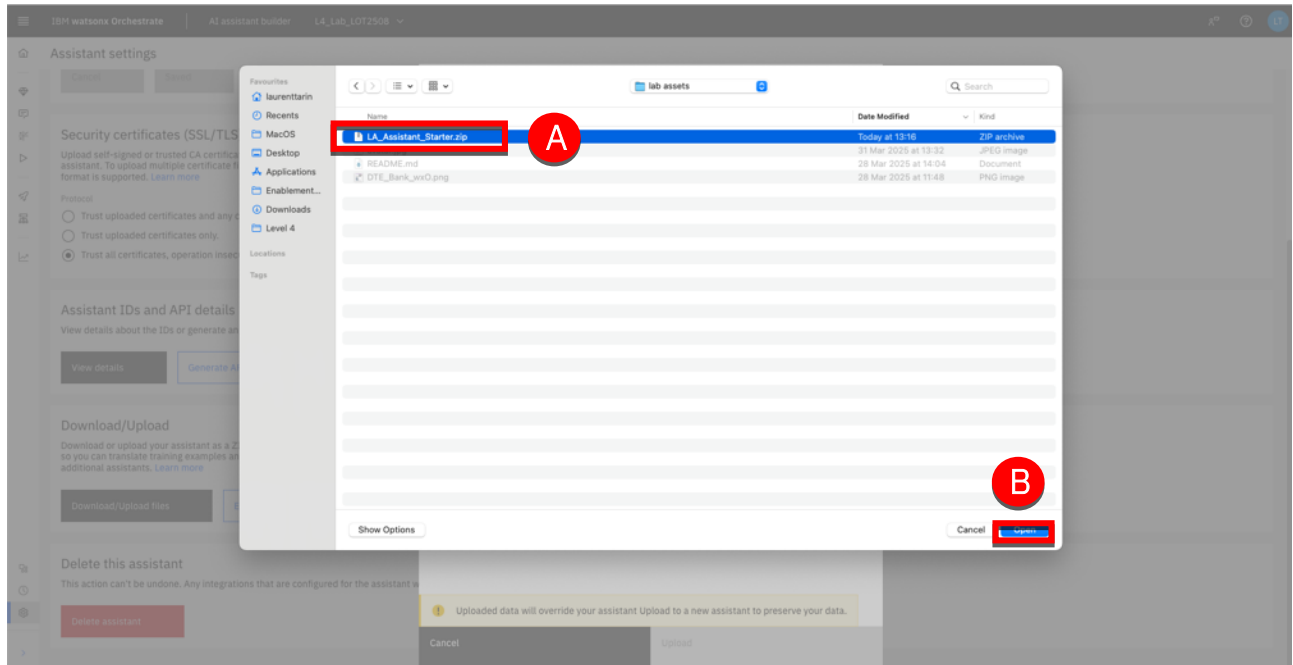
2. Scroll-down (A) and click **Download/Upload files** (B).



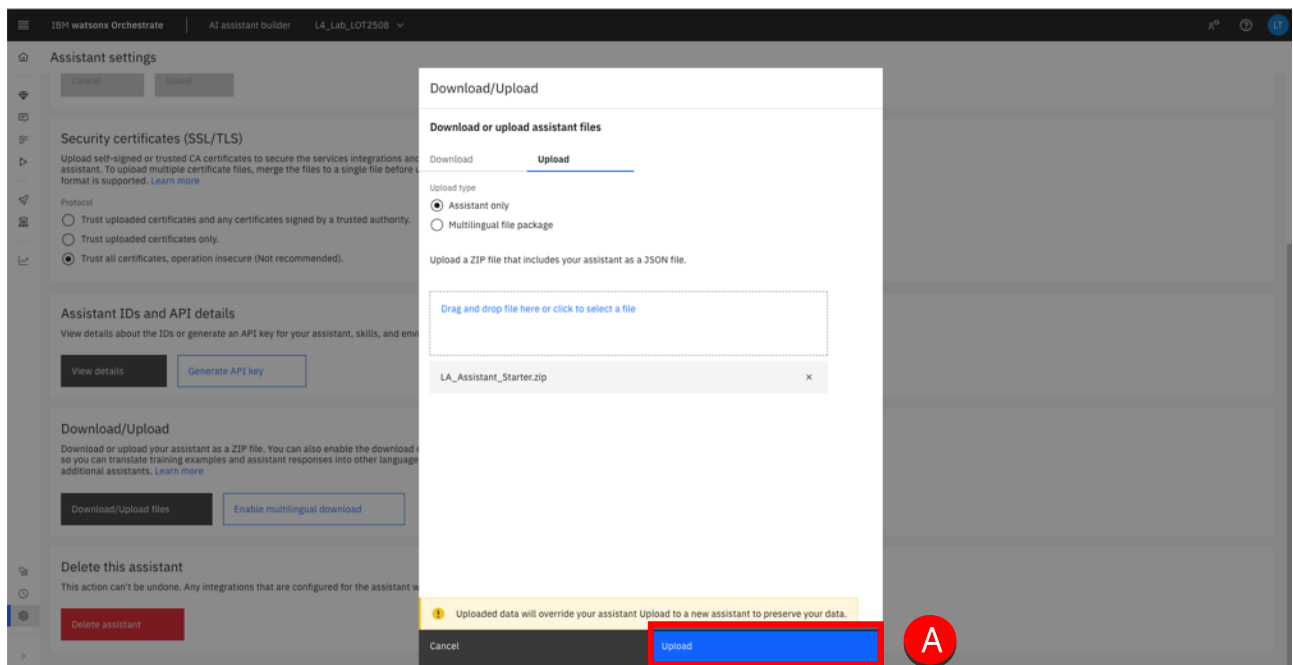
3. Select the **Upload** tab (A).



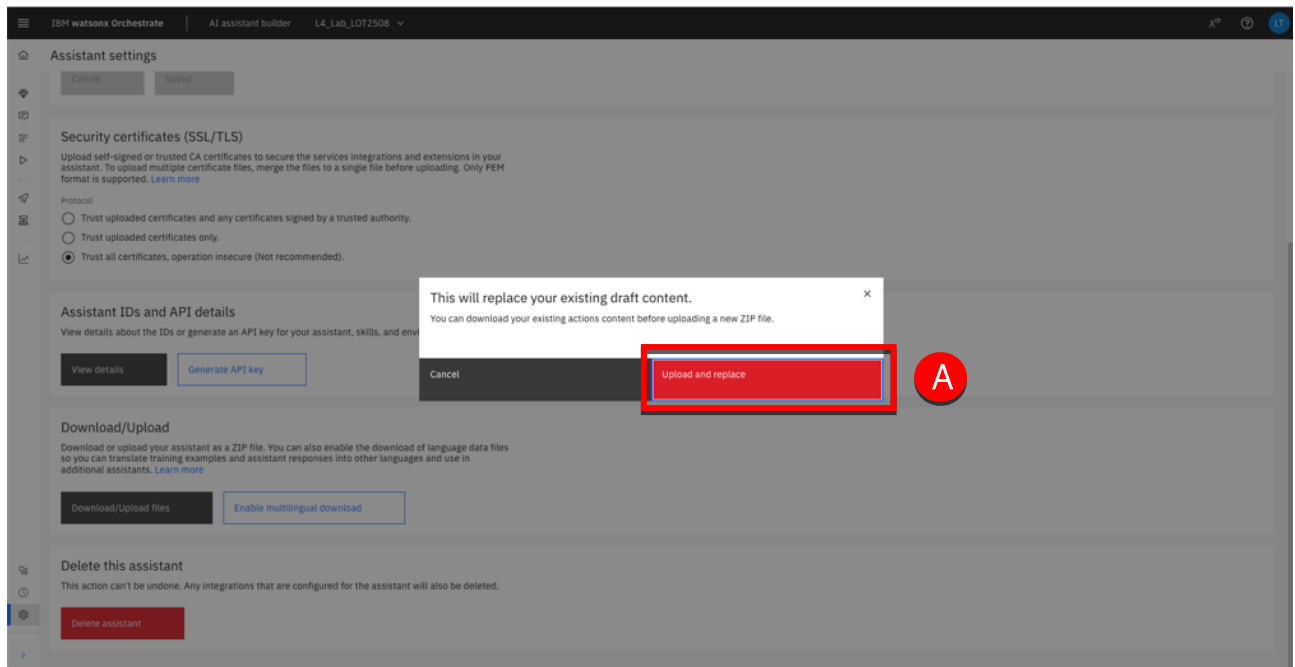
4. Select the **L4\_Assistant\_Starter.zip** file (A) you have extracted from the archive on your computer then click **Open** (B).



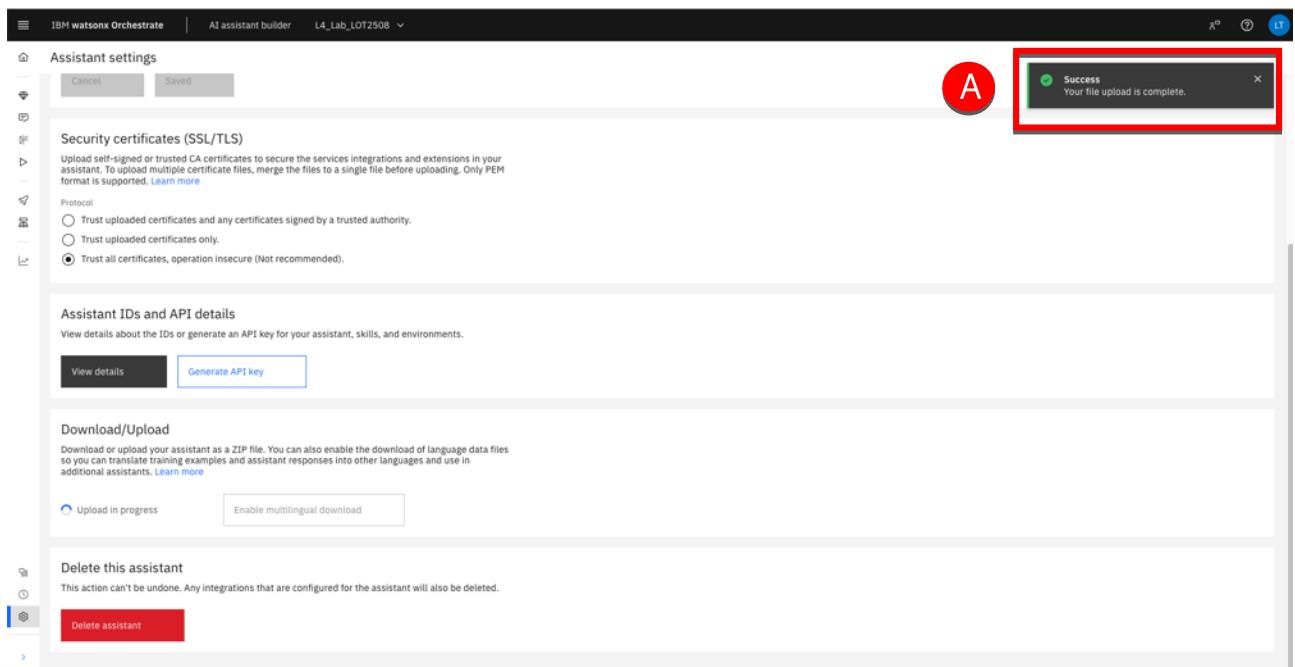
5. Click **Upload** (A).



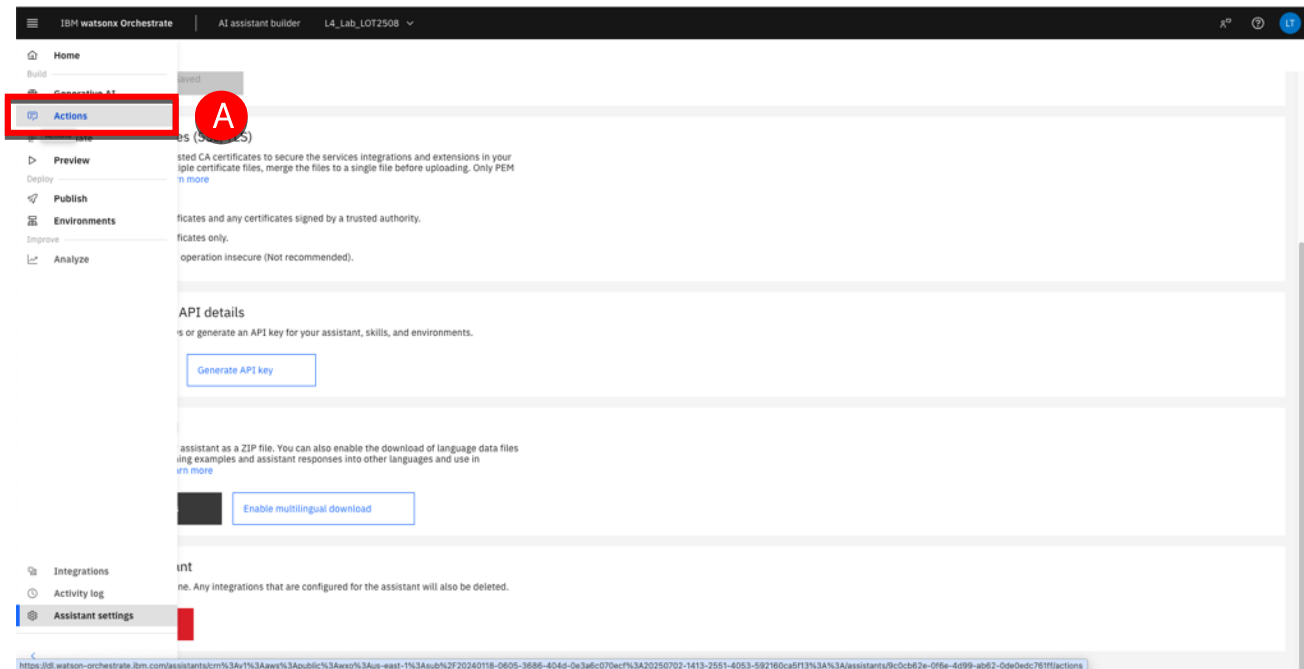
6. Click **Upload and replace** (A).



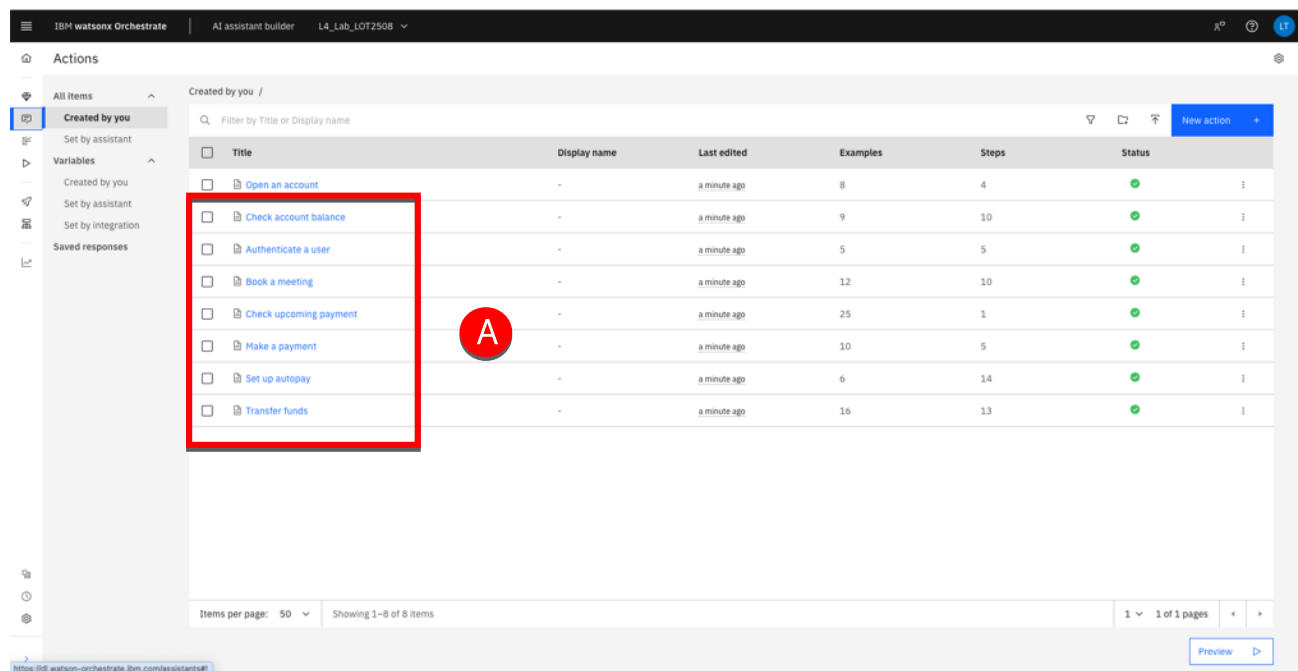
7. Wait for the upload Success message (A).



8. Click the **Actions** menu (A).



9. Note all the new actions added to your assistant (A). You are ready to continue this lab.



## Disambiguation and digression

### Disambiguation: Core concepts

In the context of watsonx Orchestrate's AI Assistant, disambiguation is a technical term for asking clarifying questions.

For example, imagine that a customer is using an AI assistant in a banking context and intend to take out a loan. So, they enter the term "business loan" in your assistant.

However, this hypothetical banking assistant may have several actions related to loans: take out a new loan, pay an existing loan, view loan estimates, etc. What action should it choose?

AI assistants determine the correct action to choose by assigning [confidence scores](#) to each action, based on the user input. The confidence scores represent the AI assistant's confidence that the sentence or phrase that you entered can be solved by the steps that are built into a specific action.

The action with the highest confidence score is typically selected by the assistant to address the user's request (assuming the action's confidence score is above 20%). However, multiple actions may have similar confidence scores, meanings that multiple actions may solve the user's request. If this is the situation, the AI assistant will ask a follow up question to the user to clarify this user's intent. This is Disambiguation.

In this lab exercise, you will observe disambiguation in action.

### Disambiguation in practice

Now that you have added more actions, you can discover the disambiguation capability of the AI assistant.

1. Click **Preview** (A).

The screenshot shows the IBM watsonx Orchestrate interface. The top bar includes the logo, 'AI assistant builder', and a dropdown menu. The left sidebar has a navigation menu with 'All items', 'Created by you', 'Set by assistant', 'Variables', 'Created by you', 'Set by assistant', 'Set by integration', and 'Saved responses'. The main area displays a table of actions created by the user. The table has columns for Name, Last edited, Examples count, Steps count, and Status. The actions listed are: Open an account, Check account balance, Authenticate a user, Book a meeting, Check upcoming payment, Make a payment, Set up autopay, and Transfer funds. At the bottom right, a red circle labeled 'A' highlights the 'Preview' button.

| Name                   | Last edited       | Examples count | Steps count | Status |
|------------------------|-------------------|----------------|-------------|--------|
| Open an account        | 2 hours ago       | 8              | 4           | ✓      |
| Check account balance  | 2 hours ago       | 9              | 10          | ✓      |
| Authenticate a user    | a few seconds ago | 5              | 5           | ✓      |
| Book a meeting         | a few seconds ago | 12             | 10          | ✓      |
| Check upcoming payment | a few seconds ago | 25             | 1           | ✓      |
| Make a payment         | a few seconds ago | 10             | 5           | ✓      |
| Set up autopay         | a few seconds ago | 6              | 14          | ✓      |
| Transfer funds         | a few seconds ago | 16             | 13          | ✓      |

2. Type account in the **Preview** window (A).

The screenshot shows the IBM watsonx Orchestrate interface with the 'Preview' window open. The window displays a chat conversation with a user. The user's message is 'Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?'. The assistant's response is 'Greet customer [default]'. At the bottom, there is an input field where the word 'account' has been typed. A red circle labeled 'A' highlights this input field.

3. Observe that this assistant did not immediately continue to an action. Instead, it recognized that the user input account could plausibly correspond to many different actions. So, it lists the potential plausible actions for the user to choose from – an example of [asking clarifying questions](#), also known as **Disambiguation**. Click the **Debug mode** icon (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, a sidebar contains navigation options: All Items, Created by you (selected), Set by assistant, Variables, Created by you, Set by assistant, Set by integration, and Saved responses. The main area displays a table of actions created by the user. The table has columns for Name, Last edited, Examples count, Steps count, and Status. The actions listed are: Open an account (5 hours ago, 8 examples, 4 steps), Check account balance (5 hours ago, 9 examples), Authenticate a user (4 hours ago, 4 examples), Book a meeting (4 hours ago, 4 examples), Check upcoming payment (4 hours ago, 4 examples), Make a payment (4 hours ago, 10 examples), Set up autopay (4 hours ago, 6 examples, 14 steps), and Transfer funds (4 hours ago, 16 examples, 13 steps). A 'Debug mode' tooltip is visible over the 'Authenticate a user' row, stating: 'You can see the trigger message's confidence score and navigate to each step from the preview panel when this option is turned on.' On the right, a 'Preview' chat window shows a conversation. The user input is 'account'. The assistant response is 'Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?'. Below the response, a 'Did you mean:' section lists four suggested actions: 'Open an account', 'Authenticate a user', 'Check account balance', and 'None of the above'. A red circle with the letter 'A' is placed over the 'Debug mode' icon in the chat window.

| Name                   | Last edited | Examples count | Steps count | Status |
|------------------------|-------------|----------------|-------------|--------|
| Open an account        | 5 hours ago | 8              | 4           | ✓      |
| Check account balance  | 5 hours ago | 9              |             | ✓      |
| Authenticate a user    | 4 hours ago | 4              |             |        |
| Book a meeting         | 4 hours ago | 4              |             |        |
| Check upcoming payment | 4 hours ago | 4              |             |        |
| Make a payment         | 4 hours ago | 10             |             |        |
| Set up autopay         | 4 hours ago | 6              | 14          |        |
| Transfer funds         | 4 hours ago | 16             | 13          |        |

4. Hover over the [Confidence Score](#) icon (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface, similar to the previous one. The table of actions is the same. The 'Preview' chat window shows the same conversation. However, a 'Confidence Score' tooltip is now visible over the 'Open an account' suggestion in the 'Did you mean:' section. The tooltip text is: 'Confidence Score: The top three Actions that may be a match for your customer's query. Open an account: 48%, Authenticate a user: 42%, Check account balance: 32%'. A red circle with the letter 'A' is placed over the 'Confidence Score' icon in the chat window.

| Name                   | Last edited | Examples count | Steps count | Status |
|------------------------|-------------|----------------|-------------|--------|
| Open an account        | 5 hours ago | 8              | 4           | ✓      |
| Check account balance  | 5 hours ago | 9              |             | ✓      |
| Authenticate a user    | 4 hours ago | 4              |             |        |
| Book a meeting         | 4 hours ago | 4              |             |        |
| Check upcoming payment | 4 hours ago | 4              |             |        |
| Make a payment         | 4 hours ago | 10             |             |        |
| Set up autopay         | 4 hours ago | 6              | 14          |        |
| Transfer funds         | 4 hours ago | 16             | 13          |        |

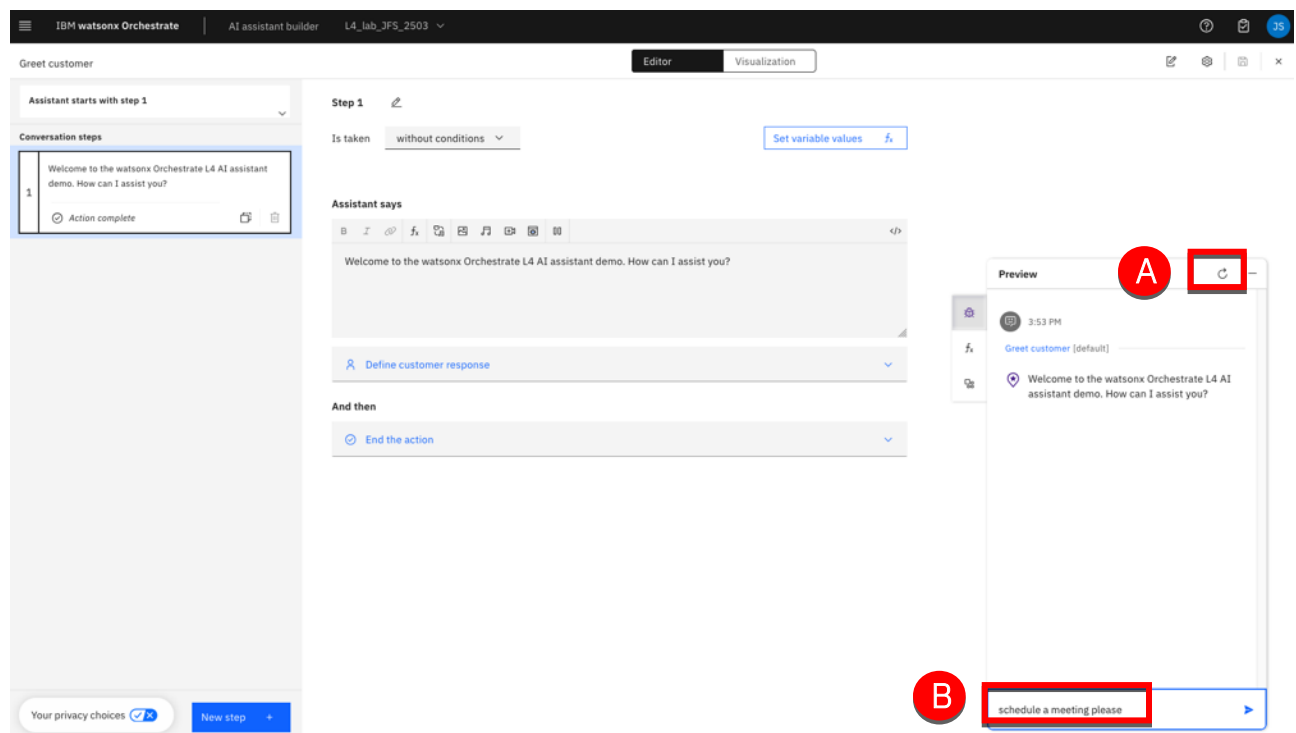
The confidence scores represent the AI assistant’s confidence that the sentence or phrase that you entered can be solved by the steps that are built into a specific action.

Your exact confidence score values may differ from the values shown in the screenshot.

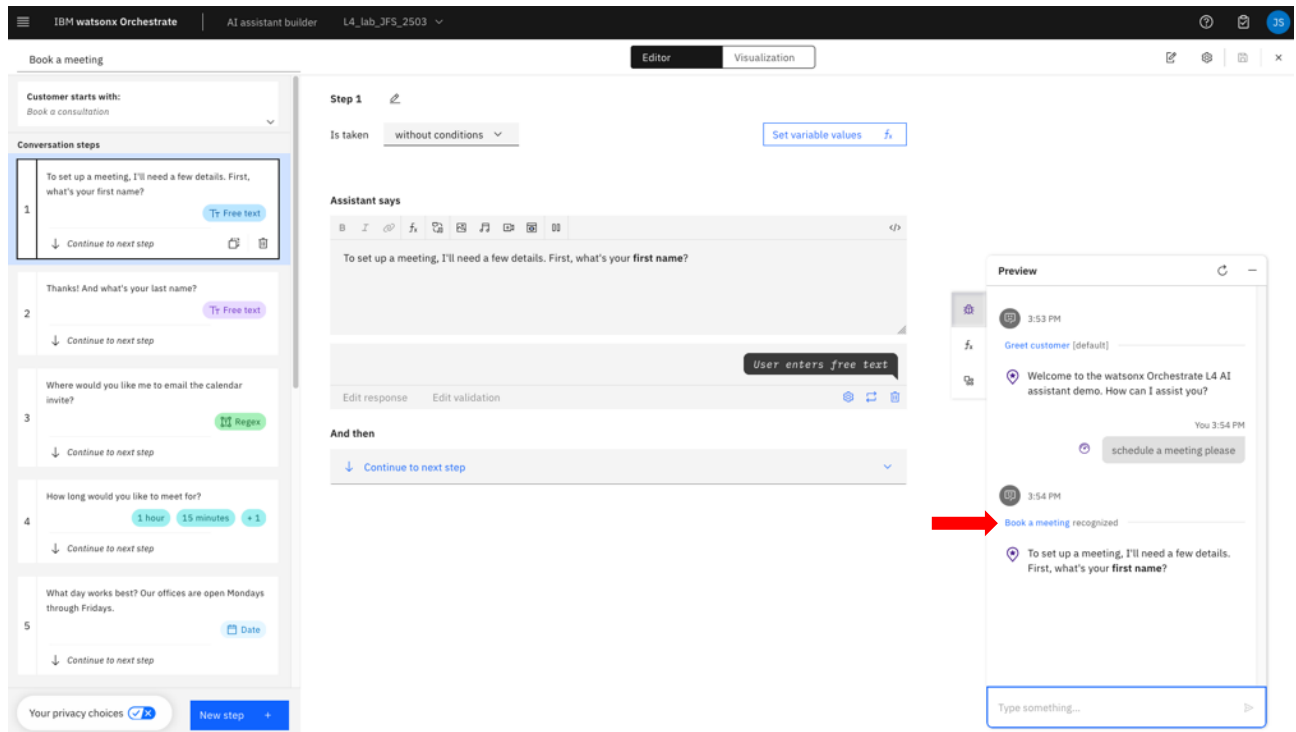
Notice the 3 plausible actions that your assistant selected. In this case, the assistant selected **Open an account**, **Authenticate a user**, and **Check account balance** as options. This is a reasonable selection because the standalone term “account” could be related to any of these actions. So, it is helpful to have the assistant disambiguate.

Now, let’s contrast this with a scenario where disambiguation is not needed.

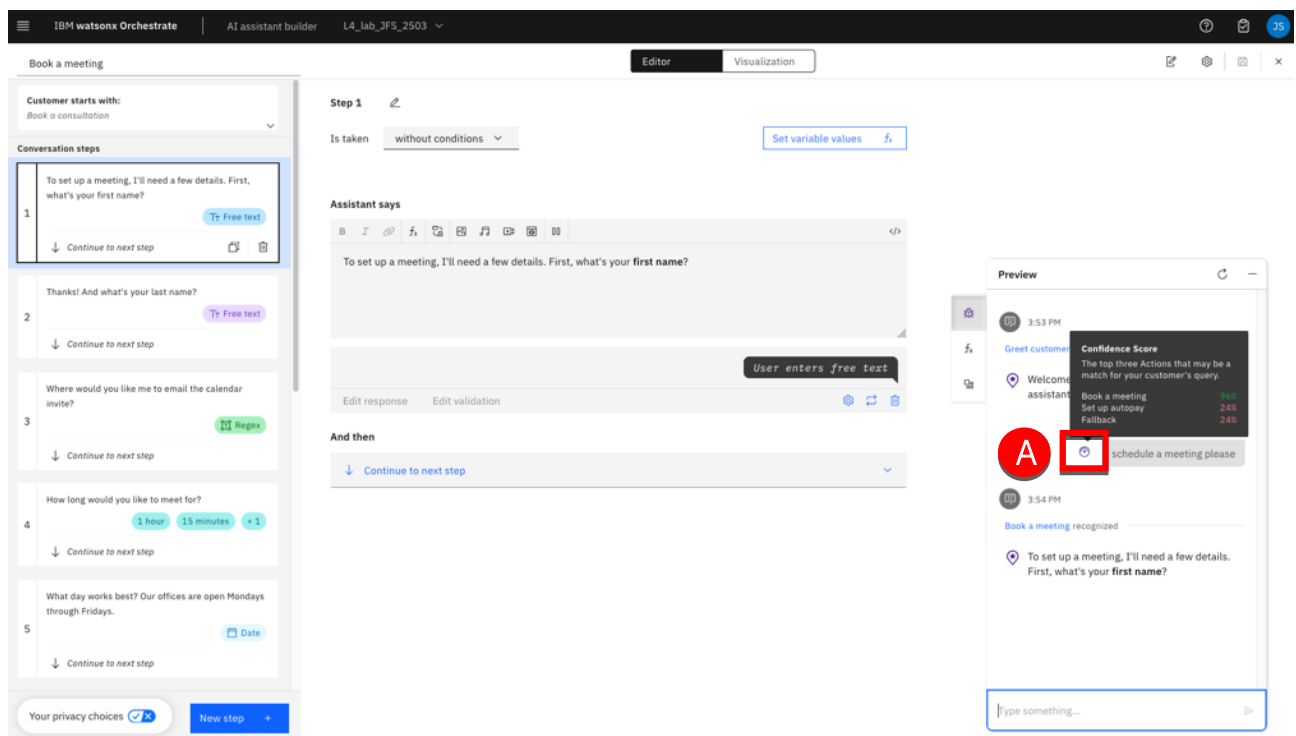
5. Refresh the **Preview** window (A), then type schedule a meeting please (B).



6. Notice that disambiguation was not invoked. Instead, the assistant recognized that this user request corresponded to the **Book a meeting** action, so the **Book a meeting** action was started.



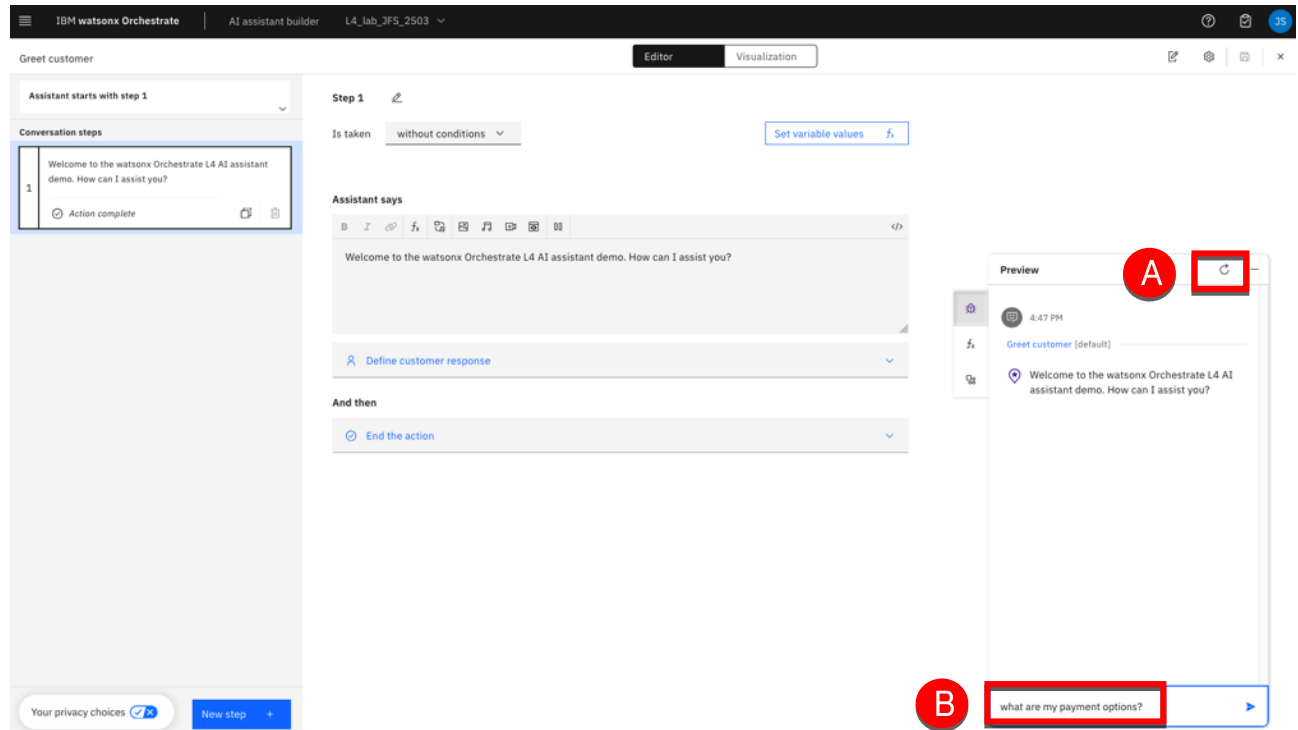
7. Hover over the Confidence Score icon (A).



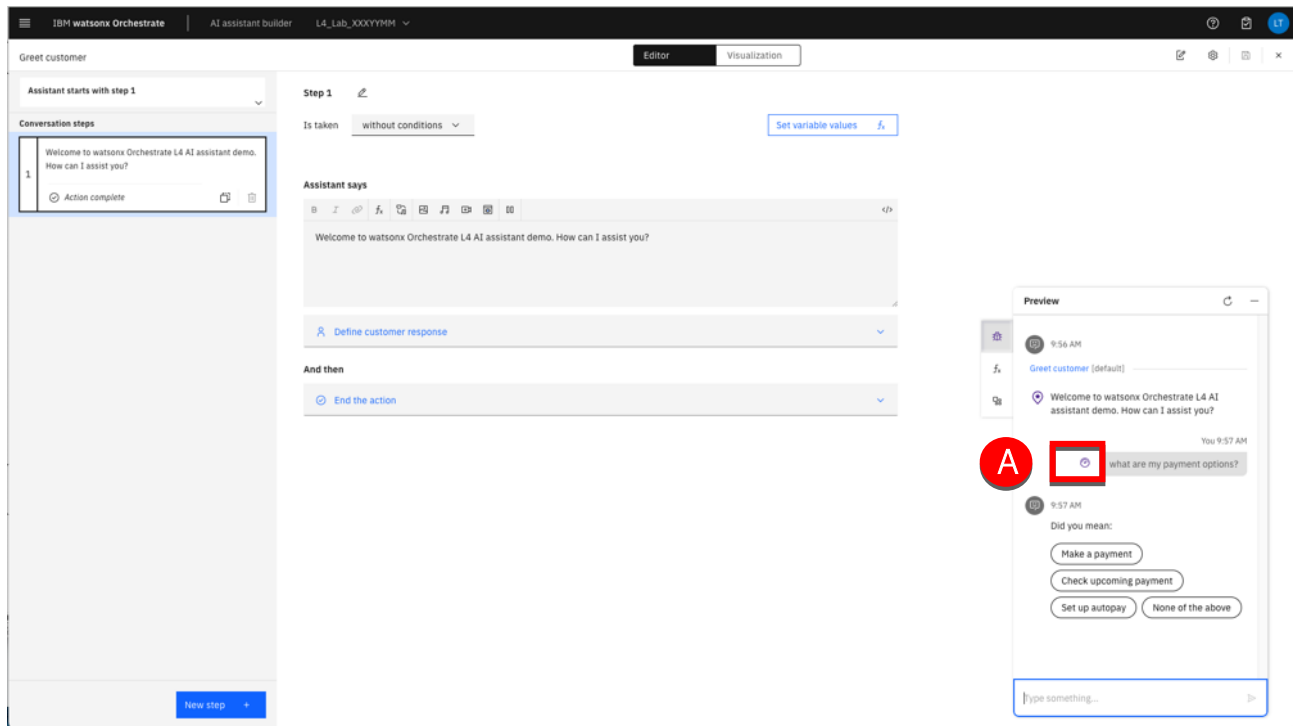
Notice that the action **Book a meeting** has a high confidence score, which is colored green. The two next actions listed have low confidence scores, which are both colored red. Because the **Book a meeting** action has a high confidence score, the assistant does not disambiguate.

Next, you will discover another example of disambiguation.

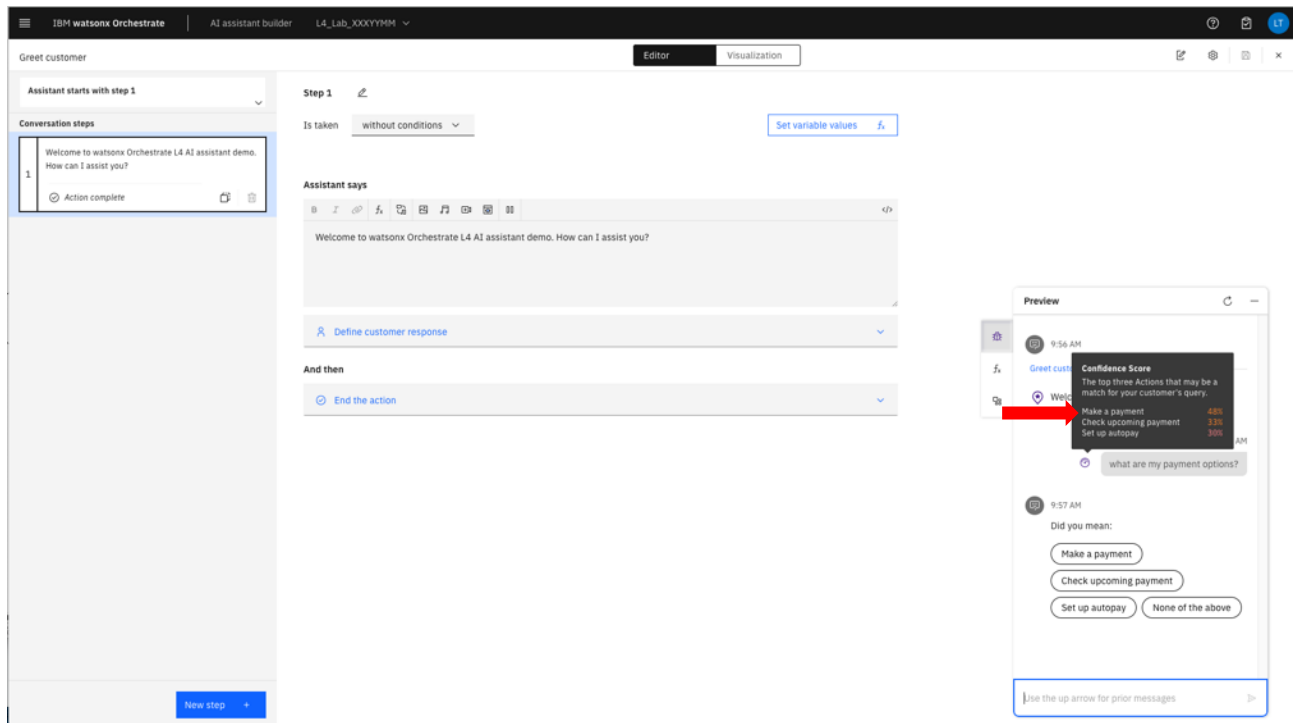
8. Refresh your **Preview** window (A), then type what are my payment options? (B).



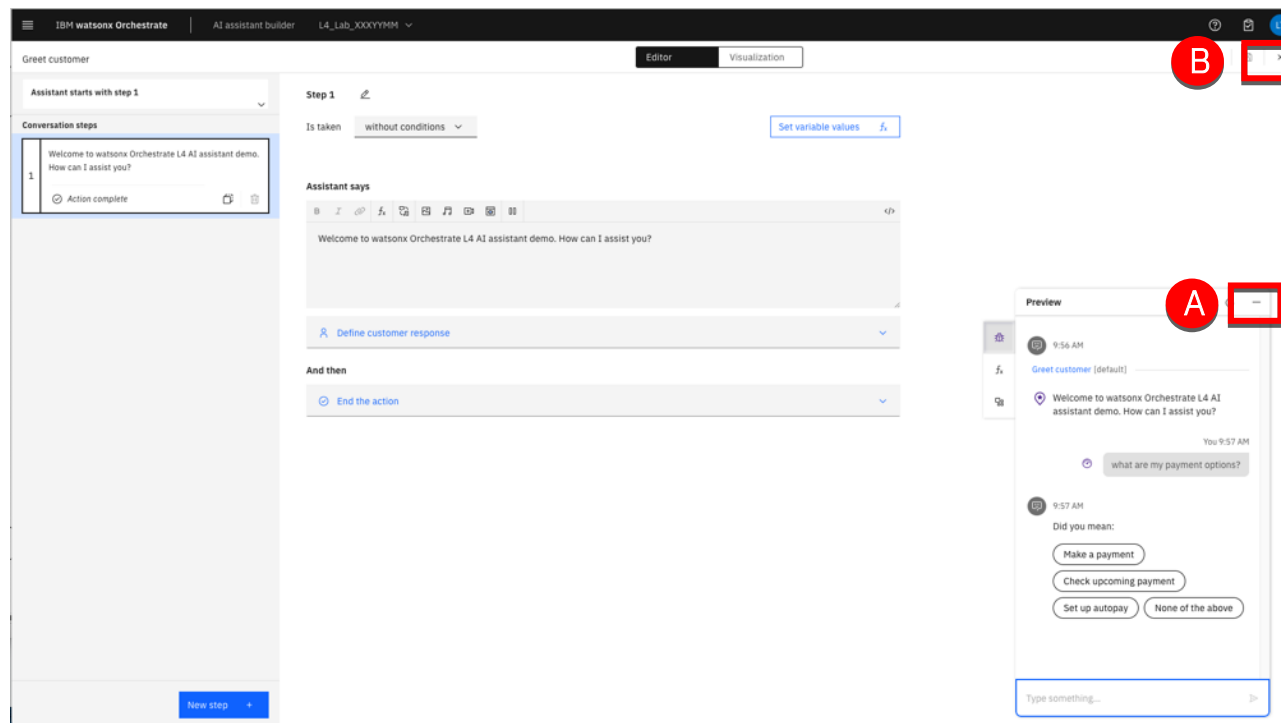
9. Hover over the confidence score icon (A).



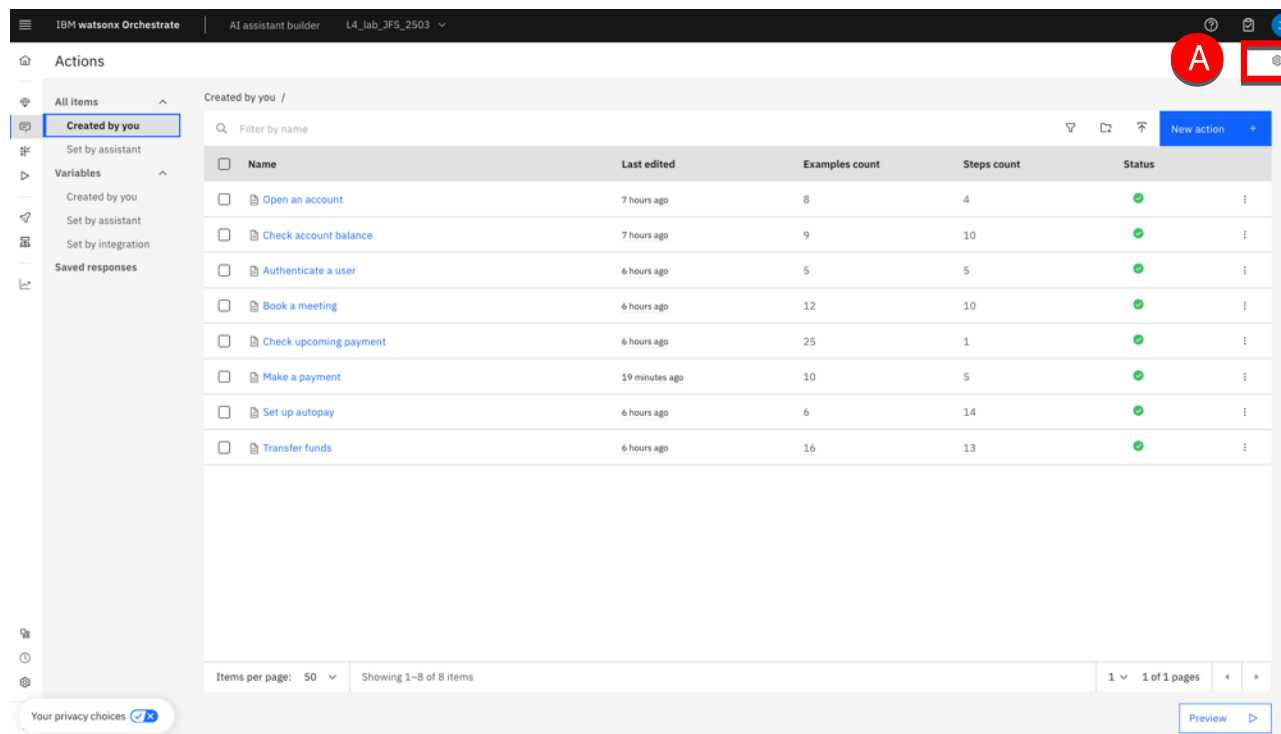
10. Notice that **Make a payment** is the action with the highest confidence score. However, the assistant is still invoking disambiguation.



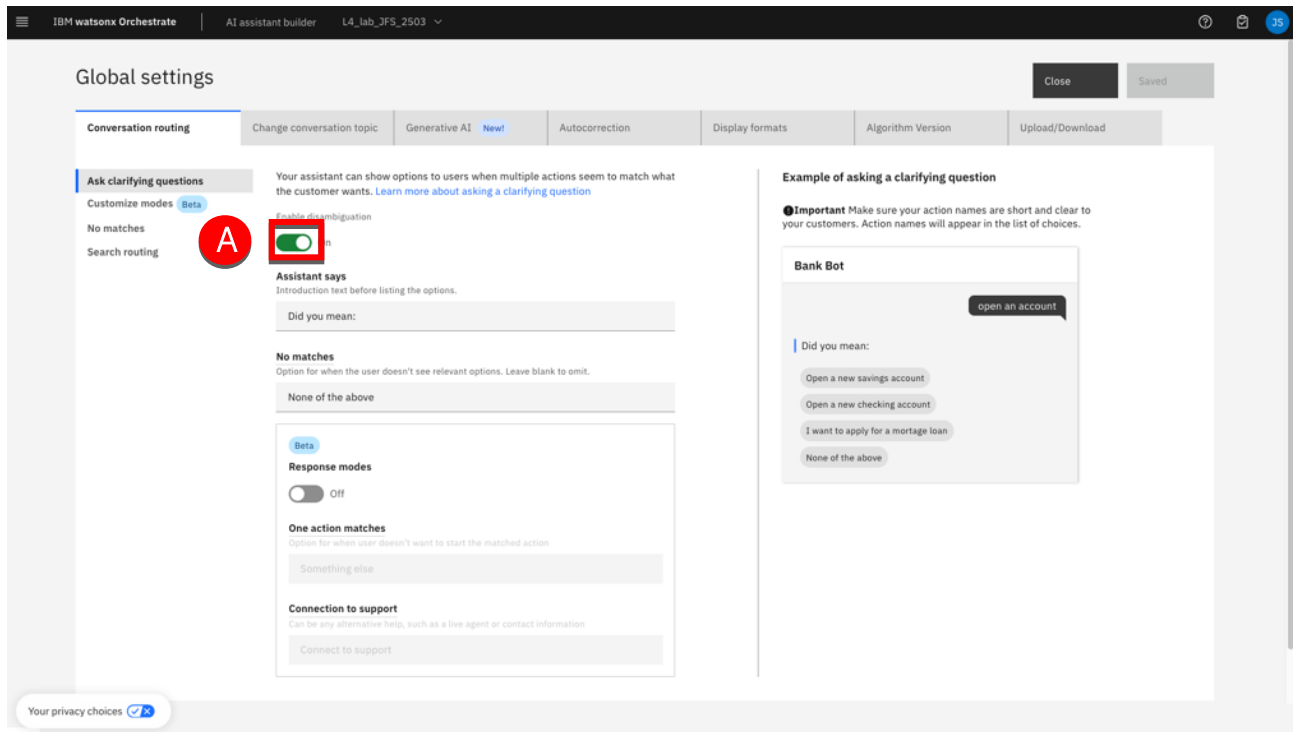
11. Now, let's demonstrate how the assistant would behave without disambiguation enabled. Close the **Preview** window (A) then close the action editor (B).



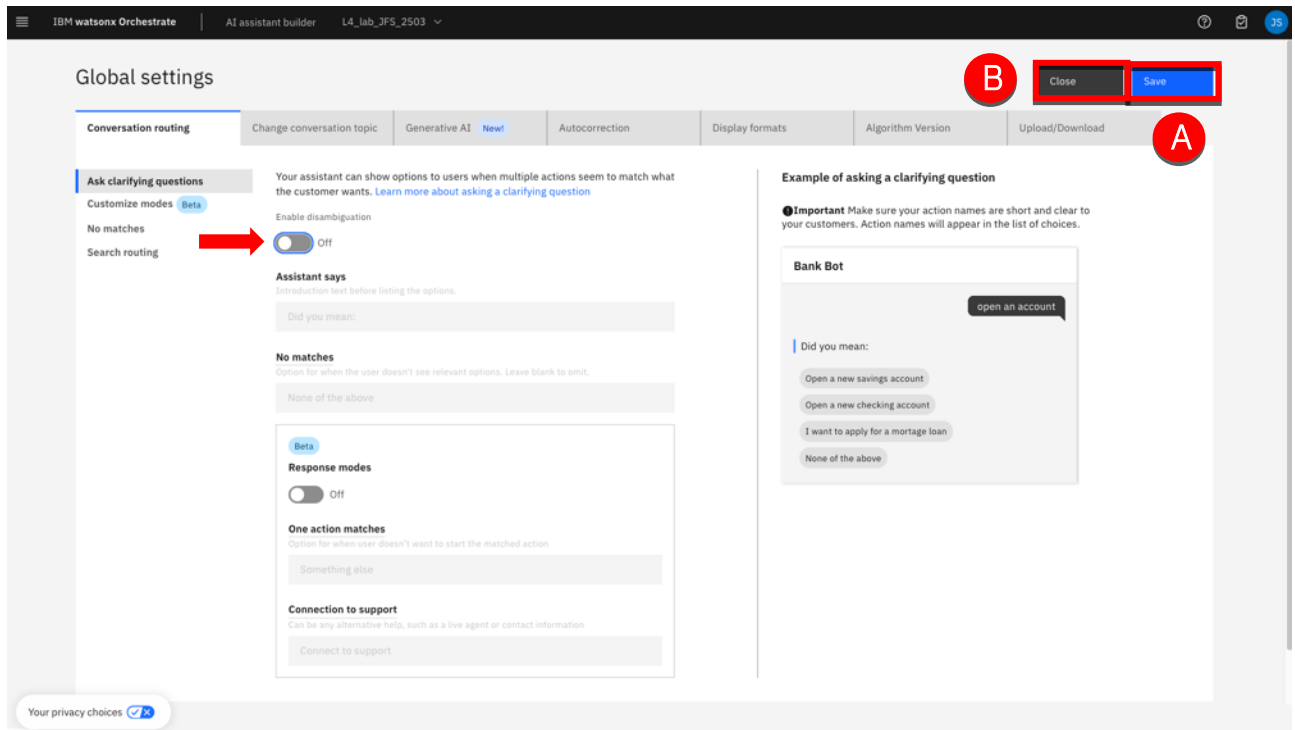
12. In the upper right corner, click the global settings icon (A).



13. Click the **Enable disambiguation** toggle switch (A) to turn off disambiguation.



14. Verify disambiguation is turned off, then click **Save** (A), then click **Close** (B).



15. Click **Preview** (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Actions

Global settings

Created by you

Filter by name

| Name                   | Last edited    | Examples count | Steps count | Status |
|------------------------|----------------|----------------|-------------|--------|
| Open an account        | 7 hours ago    | 8              | 4           | ✓      |
| Check account balance  | 7 hours ago    | 9              | 10          | ✓      |
| Authenticate a user    | 6 hours ago    | 5              | 5           | ✓      |
| Book a meeting         | 6 hours ago    | 12             | 10          | ✓      |
| Check upcoming payment | 6 hours ago    | 25             | 1           | ✓      |
| Make a payment         | 23 minutes ago | 10             | 5           | ✓      |
| Set up autopay         | 6 hours ago    | 6              | 14          | ✓      |
| Transfer funds         | 6 hours ago    | 16             | 13          | ✓      |

Items per page: 50 | Showing 1–8 of 8 items | 1 of 1 pages

Your privacy choices

**A** Preview

16. Refresh the **Preview** window (A), then type again what are my payment options? (B).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Greet customer

Editor | Visualization

Assistant starts with step 1

Conversation steps

1 Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?

Action complete

Step 1

Is taken without conditions

Set variable values

Assistant says

Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?

Define customer response

And then

End the action

Preview

5:05 PM

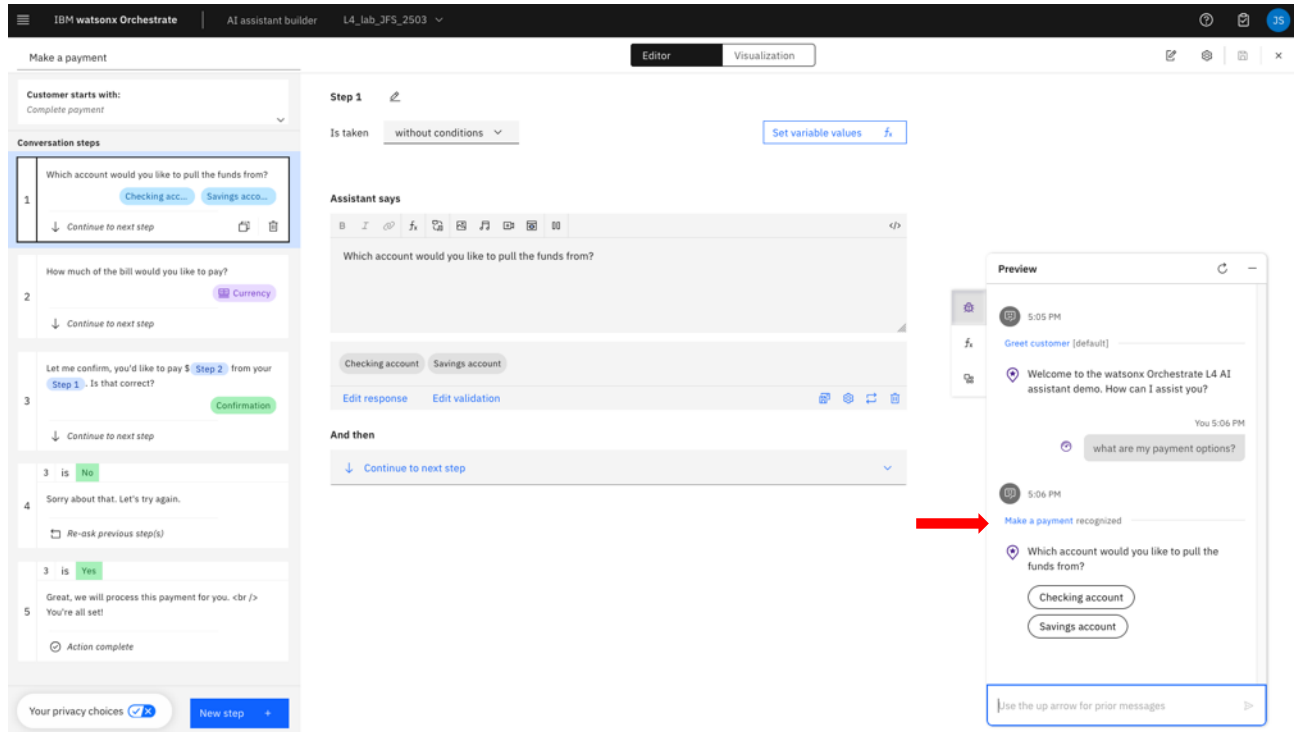
Greet customer [default]

Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?

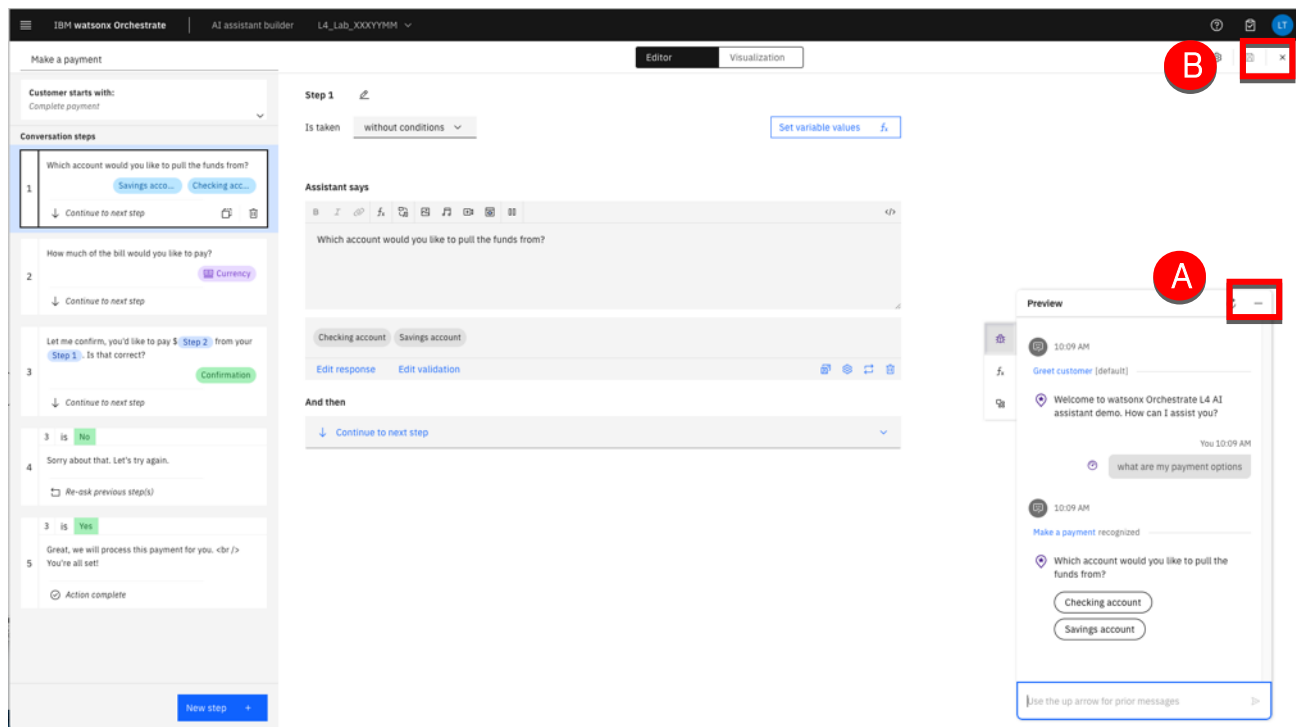
**A**

**B** what are my payment options?

17. Notice that the action **Make a Payment** was automatically selected, and no disambiguation occurred.

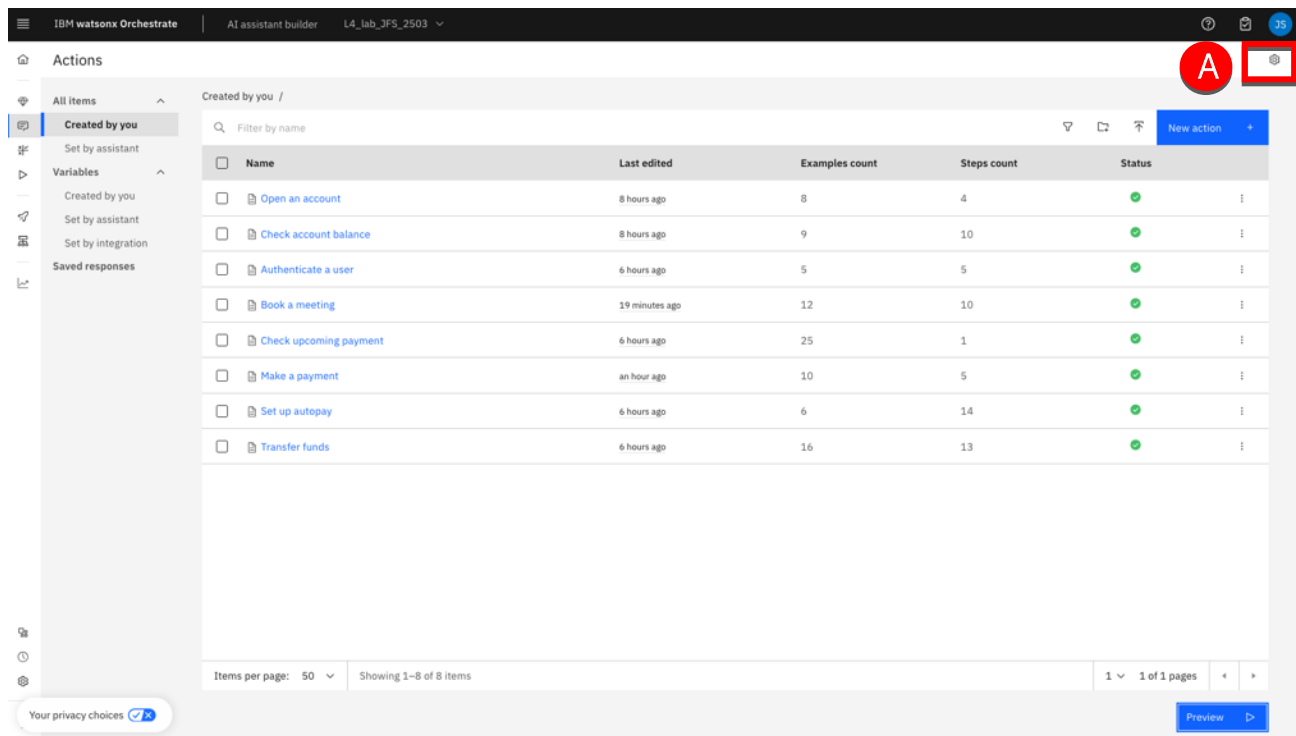


18. Minimize the **Preview** window (A) then close the action window (B).



Let's now turn on the disambiguation back.

19. Click the global settings icon (A).



IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Actions

All Items

Created by you

Filter by name

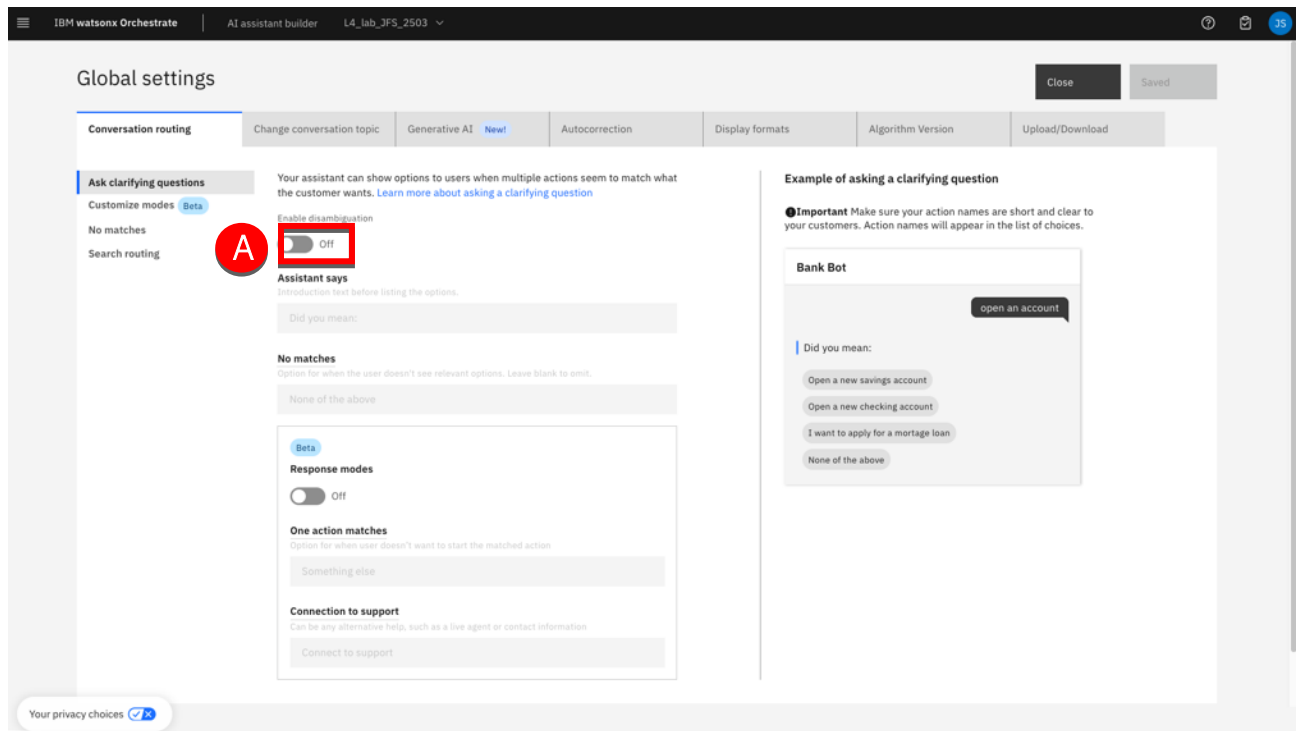
| Name                   | Last edited    | Examples count | Steps count | Status |
|------------------------|----------------|----------------|-------------|--------|
| Open an account        | 8 hours ago    | 8              | 4           | ✓      |
| Check account balance  | 8 hours ago    | 9              | 10          | ✓      |
| Authenticate a user    | 6 hours ago    | 5              | 5           | ✓      |
| Book a meeting         | 19 minutes ago | 12             | 10          | ✓      |
| Check upcoming payment | 6 hours ago    | 25             | 1           | ✓      |
| Make a payment         | an hour ago    | 10             | 5           | ✓      |
| Set up autopay         | 6 hours ago    | 6              | 14          | ✓      |
| Transfer funds         | 6 hours ago    | 16             | 13          | ✓      |

Items per page: 50 | Showing 1-8 of 8 items | 1 of 1 pages

Your privacy choices

Preview

20. Click the **Enable disambiguation** toggle (A) to ensure disambiguation is enabled.



IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Global settings

Close | Saved

Conversation routing | Change conversation topic | Generative AI | Autocorrection | Display formats | Algorithm Version | Upload/Download

Ask clarifying questions

Customize modes

No matches

Search routing

Enable disambiguation

Assistant says

Did you mean:

No matches

Response modes

One action matches

Connection to support

Example of asking a clarifying question

Bank Bot

open an account

Did you mean:

Open a new savings account

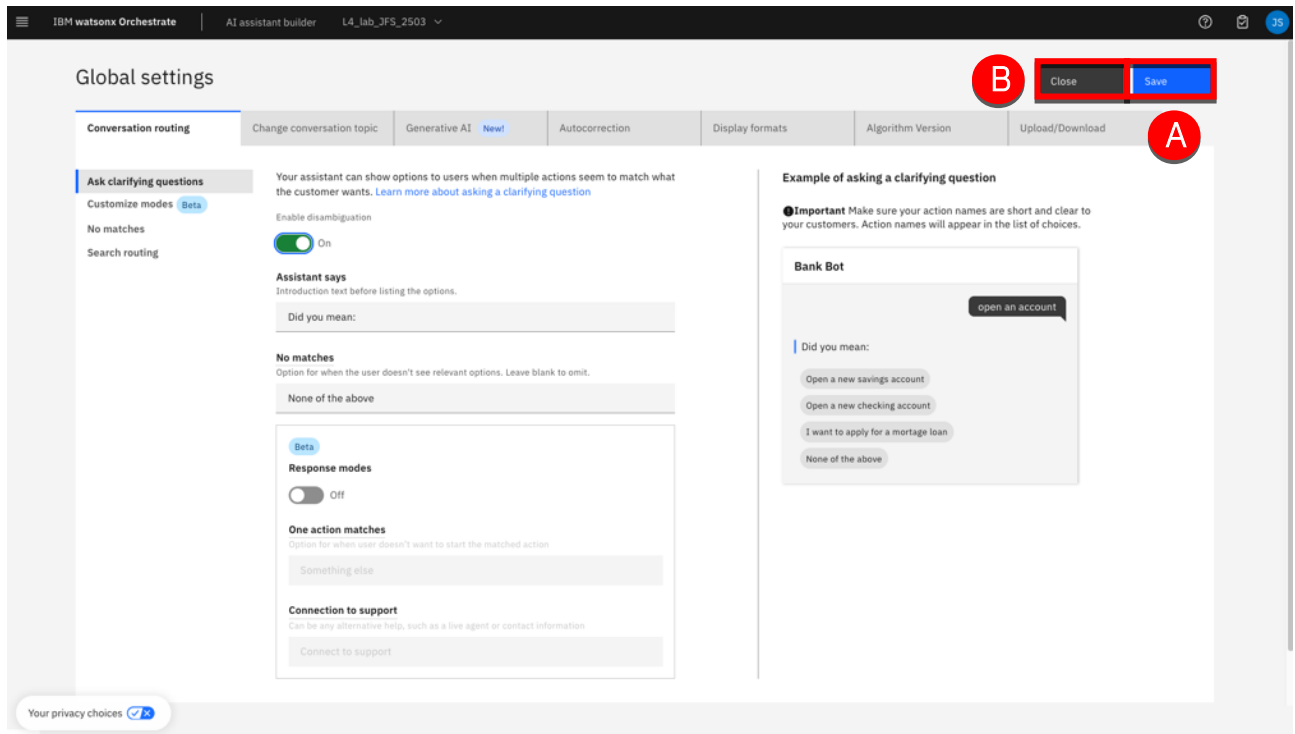
Open a new checking account

I want to apply for a mortgage loan

None of the above

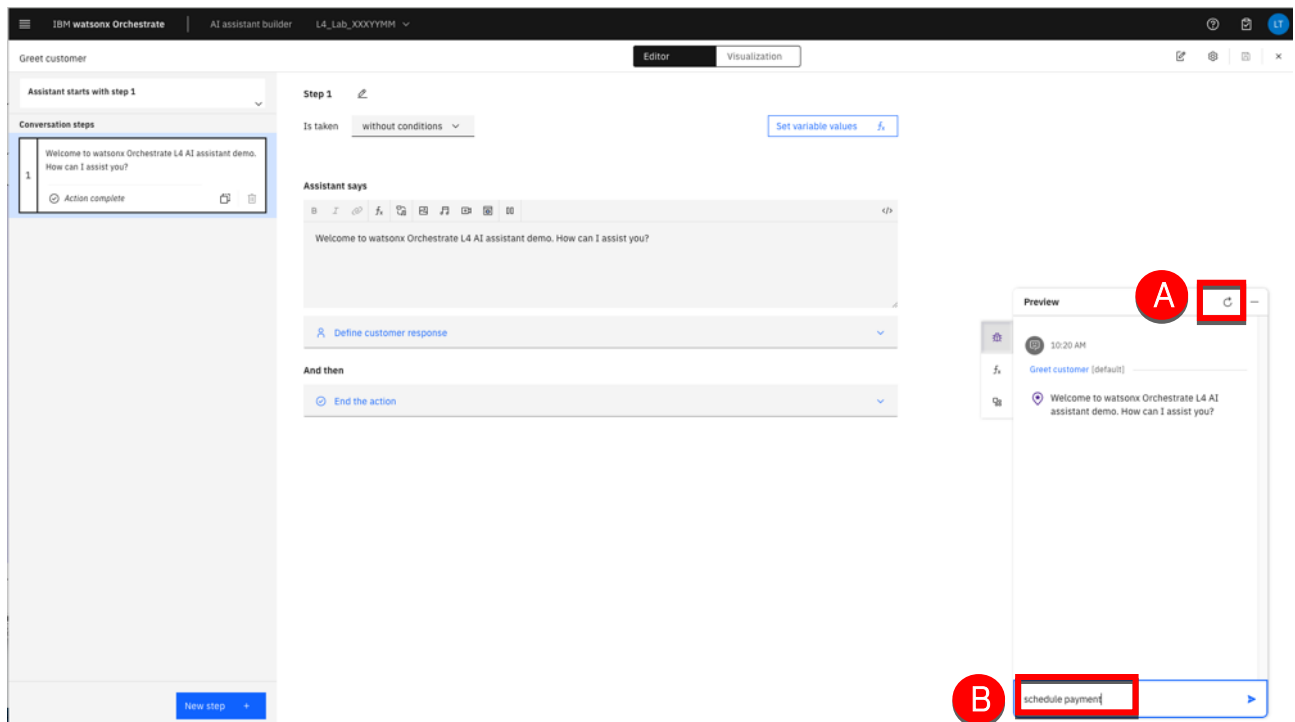
Your privacy choices

21. Click **Save** (A) then click **Close** (B).

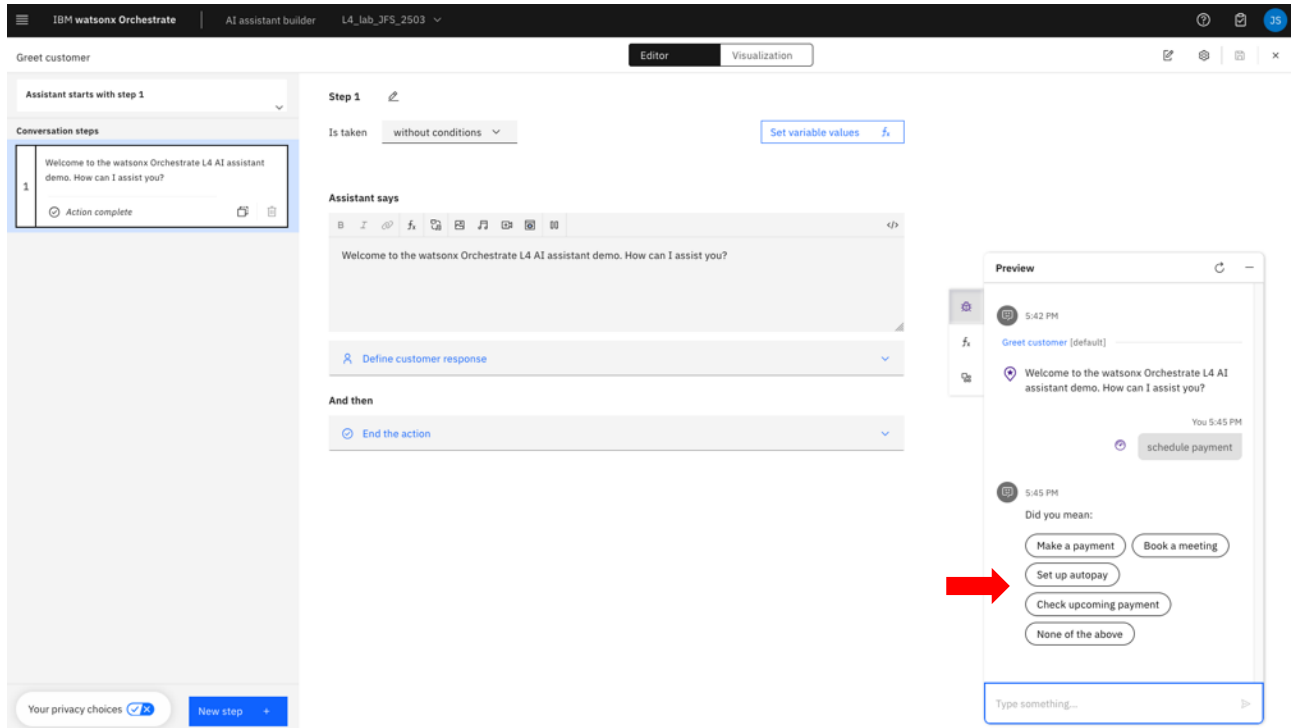


You have seen the effects of enabling and disabling Disambiguation at a **global assistant level**. Now, let's discover the impact of enabling and disabling Disambiguation at an **action level**.

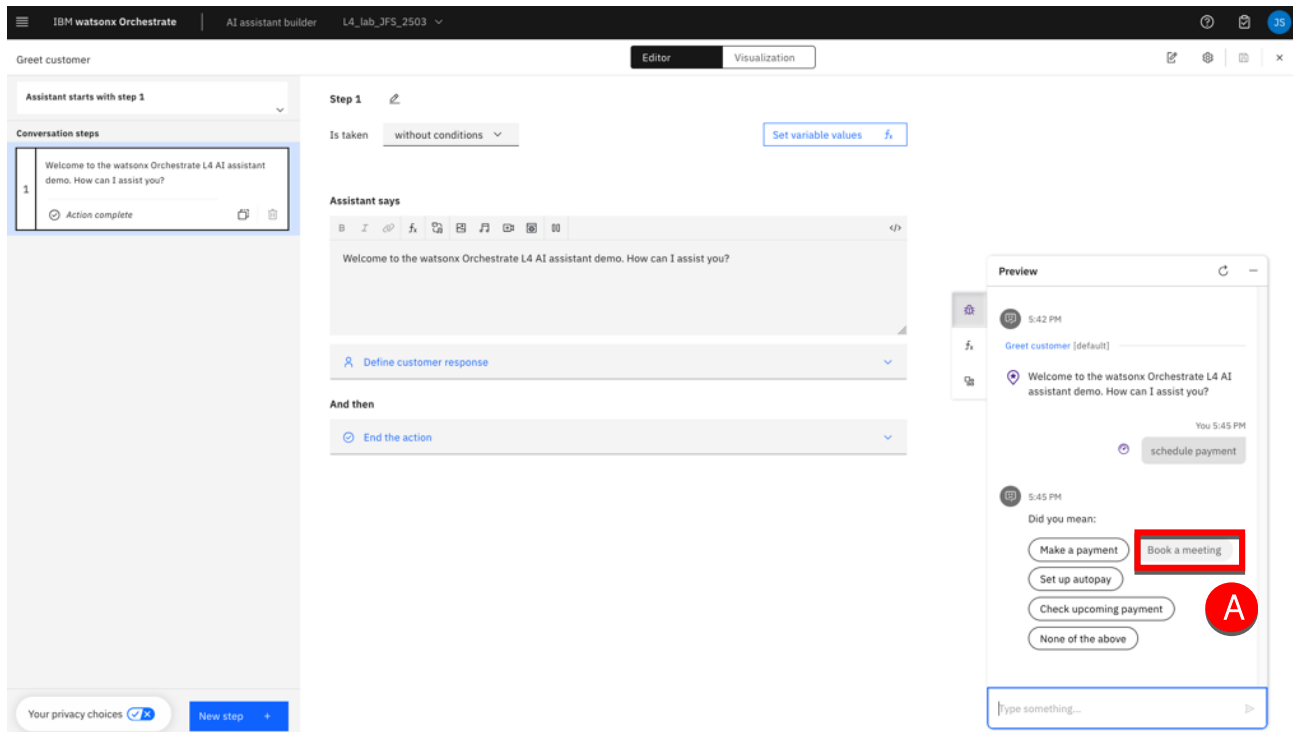
22. Open and refresh the preview window (A), then type `schedule payment` (B).



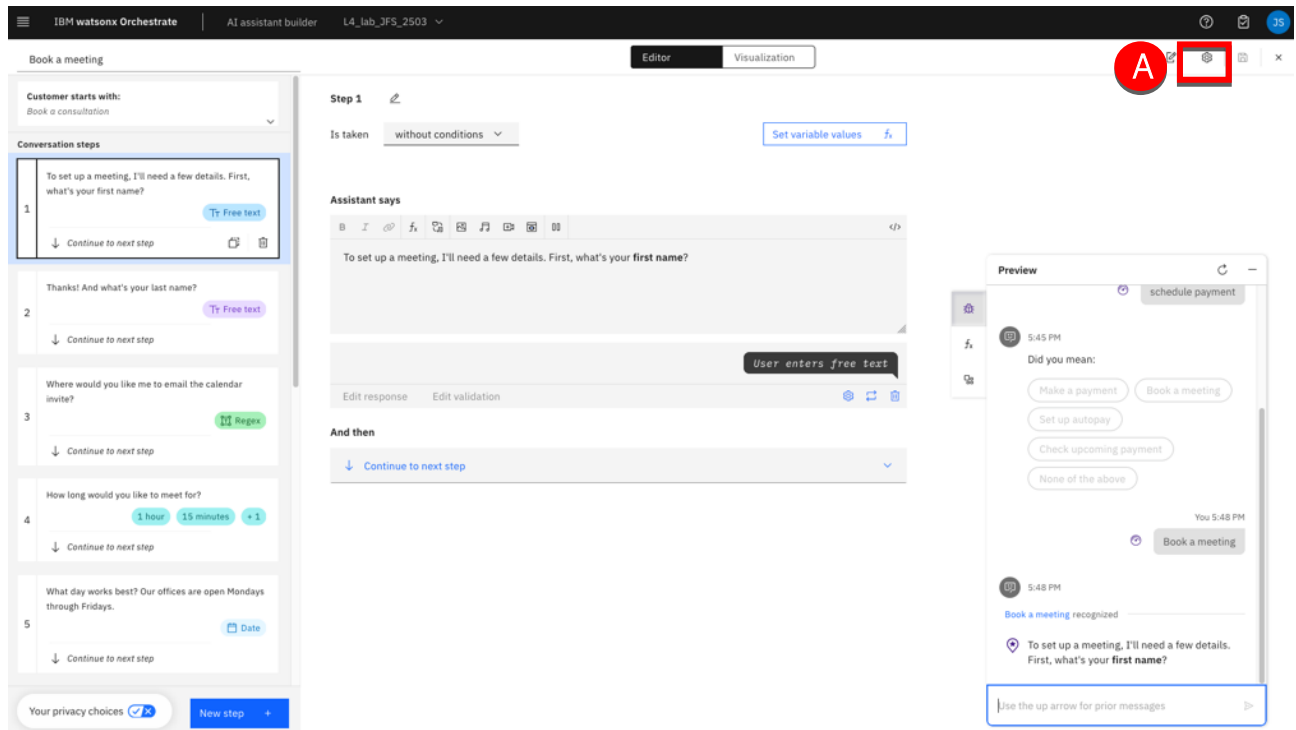
23. Observe the options listed. They include many actions, such as **Book a meeting**. Optionally, you may examine the confidence scores of the schedule payment user input.



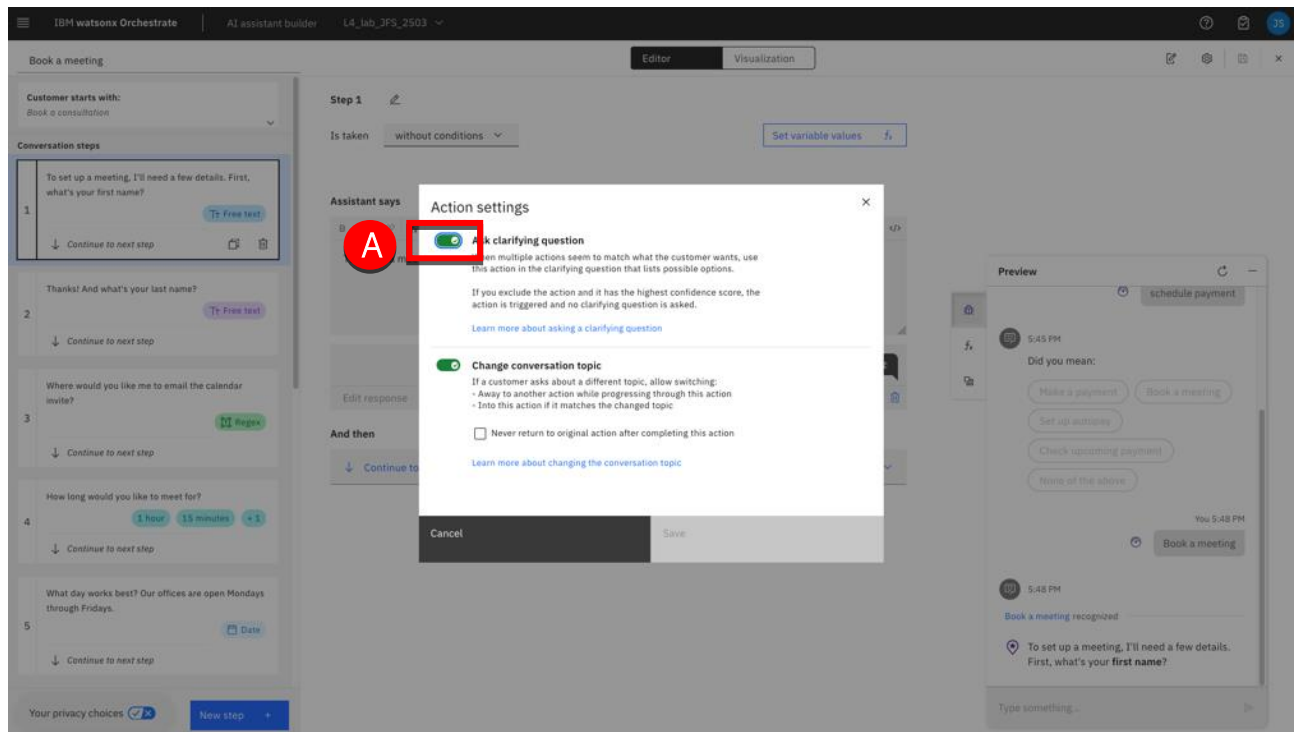
24. Click the **Book a meeting** option (A). You are then redirected to the Book a meeting action editor page.



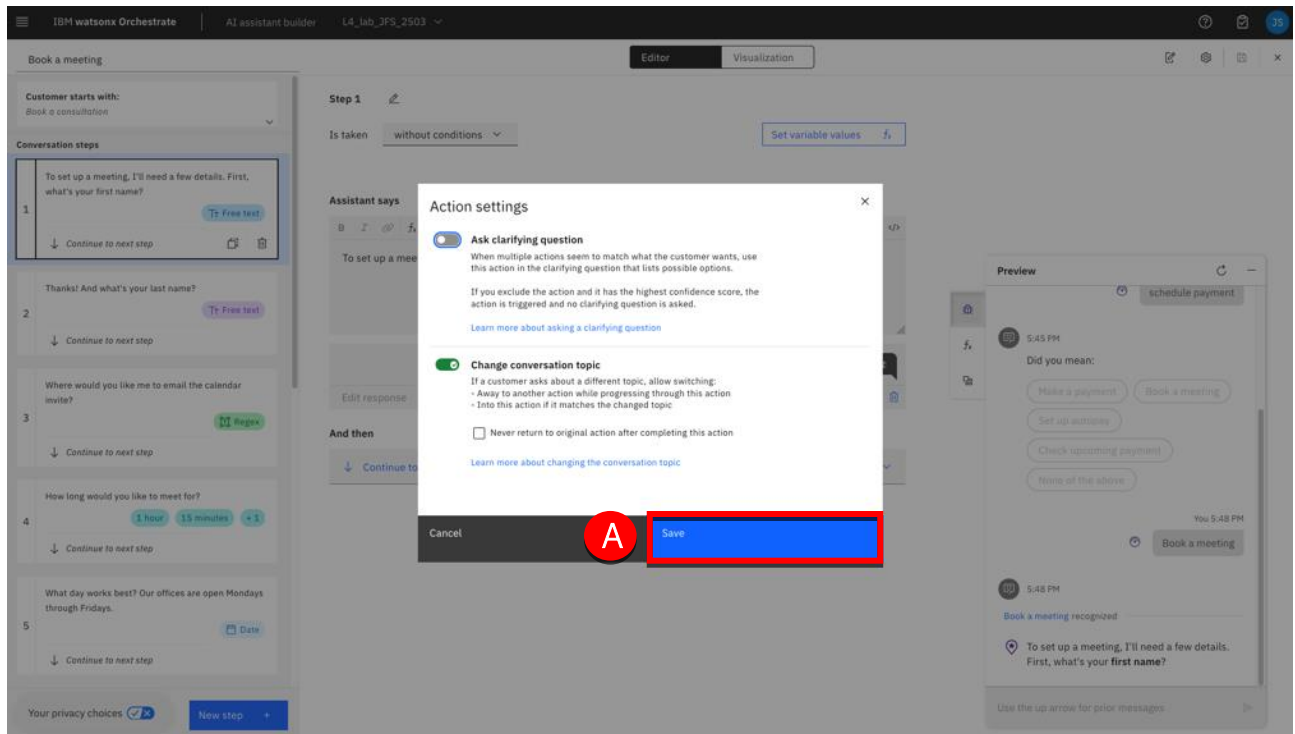
25. Click the Action settings icon at the top right corner.



26. Click the Ask clarifying questions toggle (A) to off.

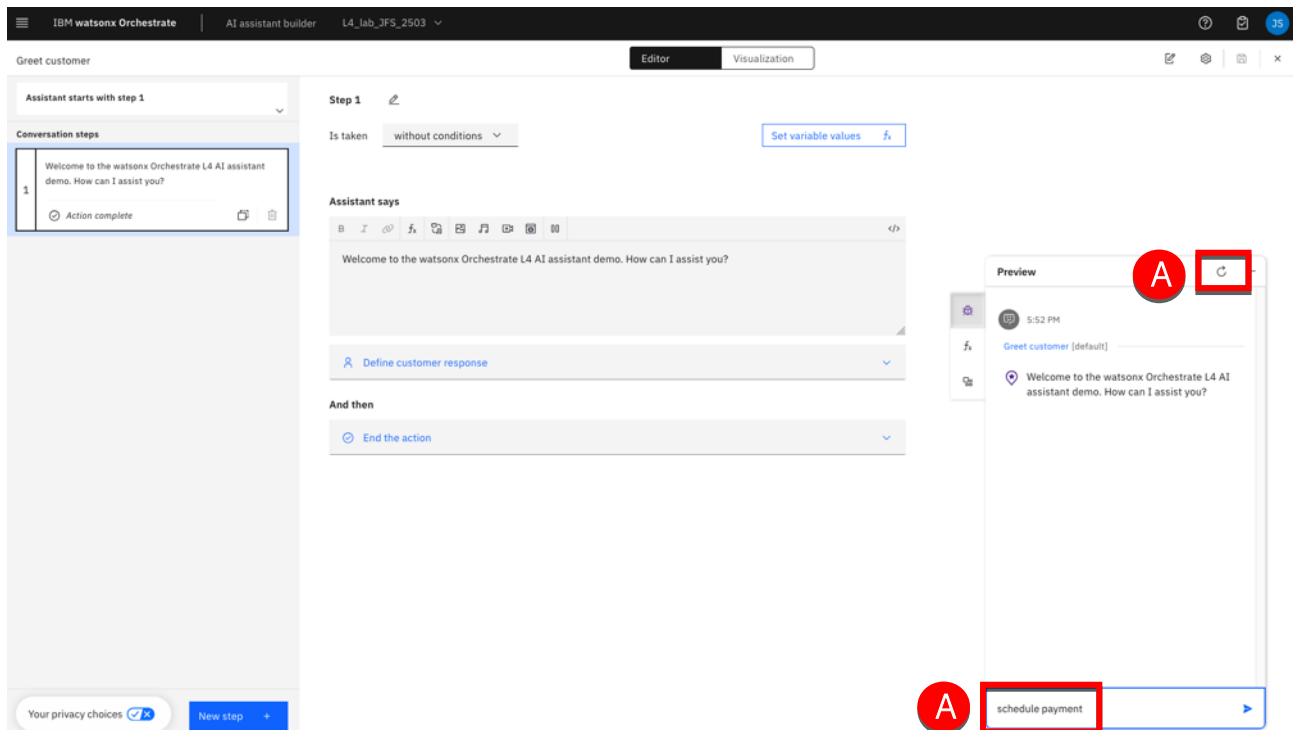


27. Click **Save** (A).



You have disabled the Ask clarying question (Disambiguation) feature for the Book a Meeting action.

28. Refresh the **Preview** window (A), then type again `schedule payment` (B).



29. Observe the listed options. Notice that the Book a meeting action is no longer listed (A). Hover over the confidence scores (B).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The top navigation bar includes the IBM watsonx Orchestrate logo, the AI assistant builder title, and the current project name 'L4\_lab\_3FS\_2503'. The interface is divided into two main sections: 'Editor' and 'Visualization'.

**Editor Section:**

- Conversation steps:** A list of steps is shown on the left. Step 1 is selected and contains the text 'Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?'. The status 'Action complete' is indicated.
- Step 1 configuration:** The right side shows the configuration for Step 1. It is set to 'Is taken' without conditions. The 'Assistant says' section contains the same welcome message. Below this, there are options to 'Define customer response' and 'And then' (currently set to 'End the action').

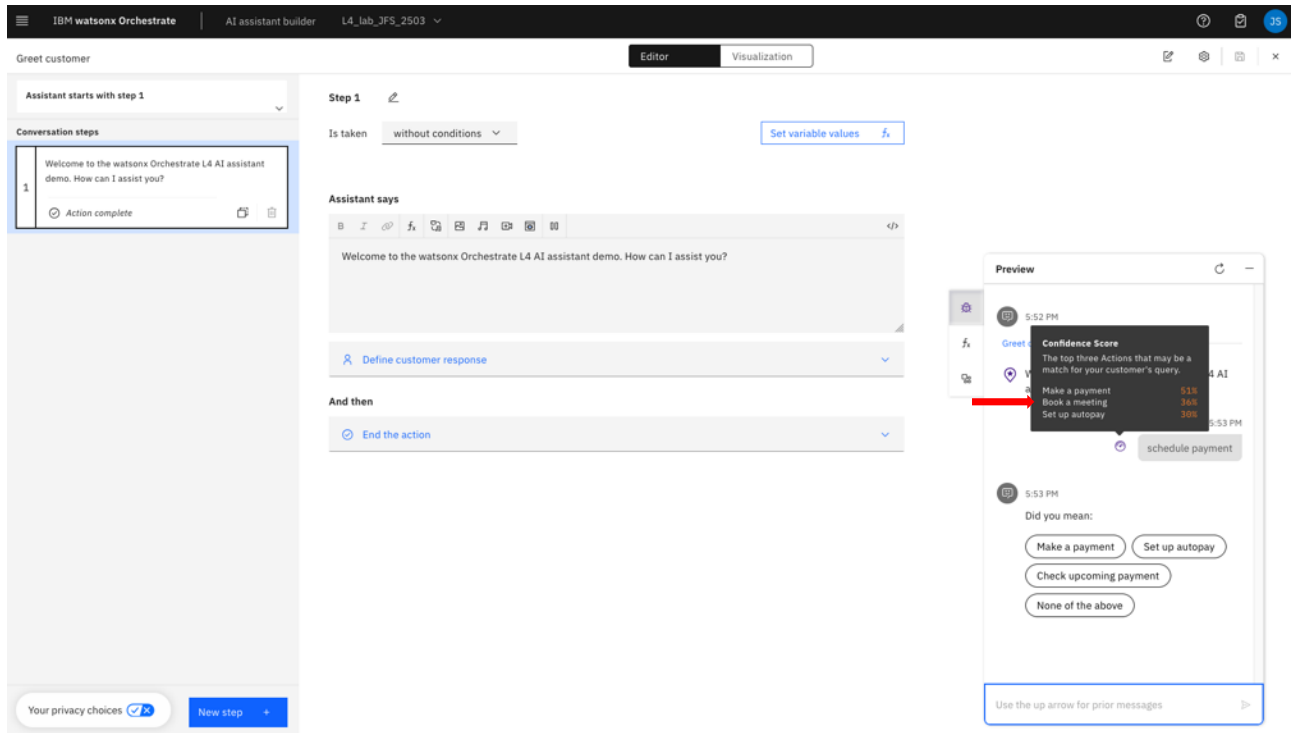
**Visualization Section:**

The 'Preview' window on the right shows a simulated chat conversation:

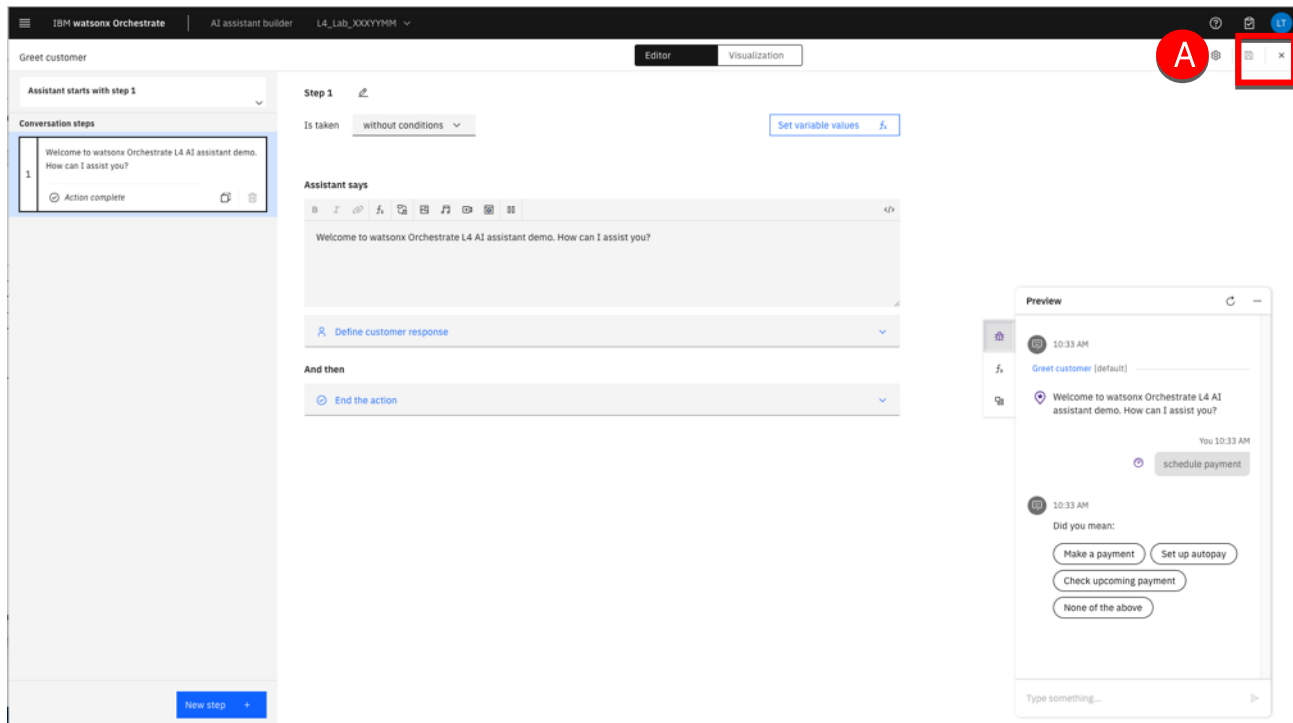
- Customer:** 'Greet customer [default]'
- Assistant:** 'Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?'
- Customer:** 'schedule payment' (This message is highlighted with a red box and labeled 'B').
- Assistant:** 'Did you mean:' followed by four buttons: 'Make a payment', 'Set up autopay', 'Check upcoming payment', and 'None of the above'. These buttons are grouped in a red box and labeled 'A'.

At the bottom of the preview, there is a prompt: 'Use the up arrow for prior messages'.

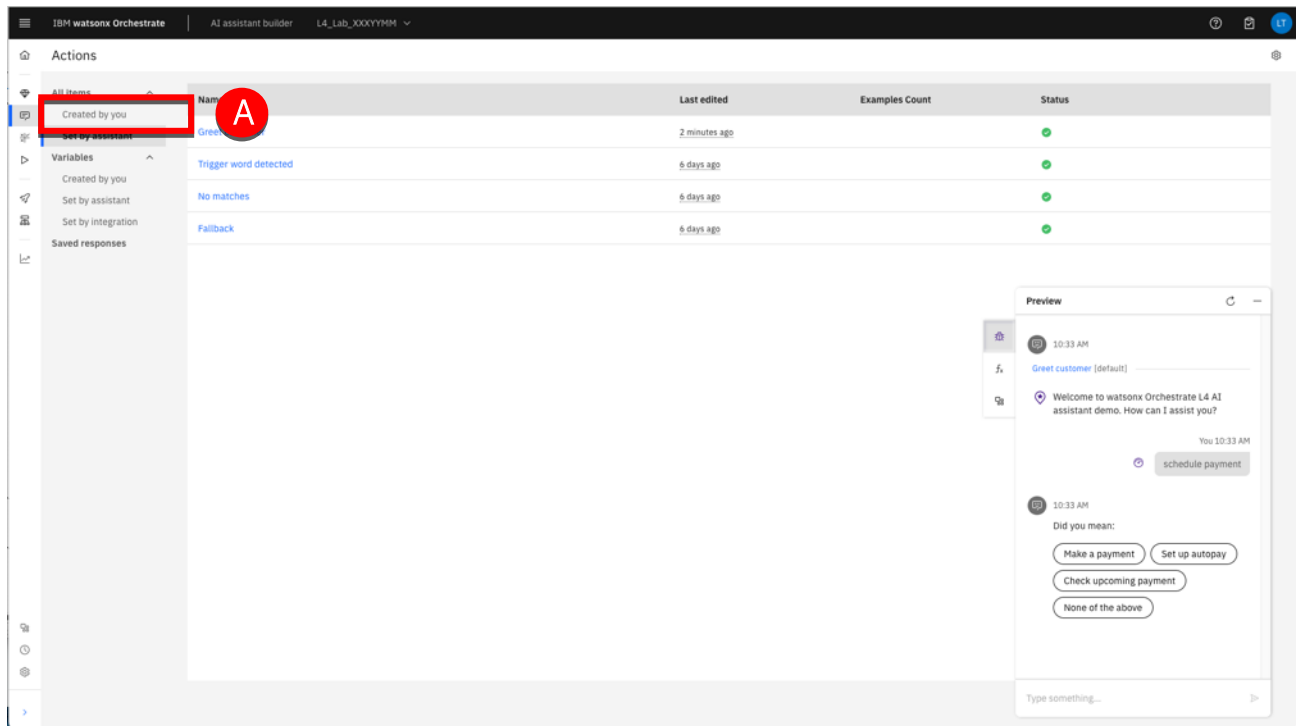
30. Observe that **Book a meeting** still has a confidence score listed. But **Book a meeting** does not appear as a possible option. This is because **Book a meeting** is not visible to the user when the assistant is disambiguating potential actions.



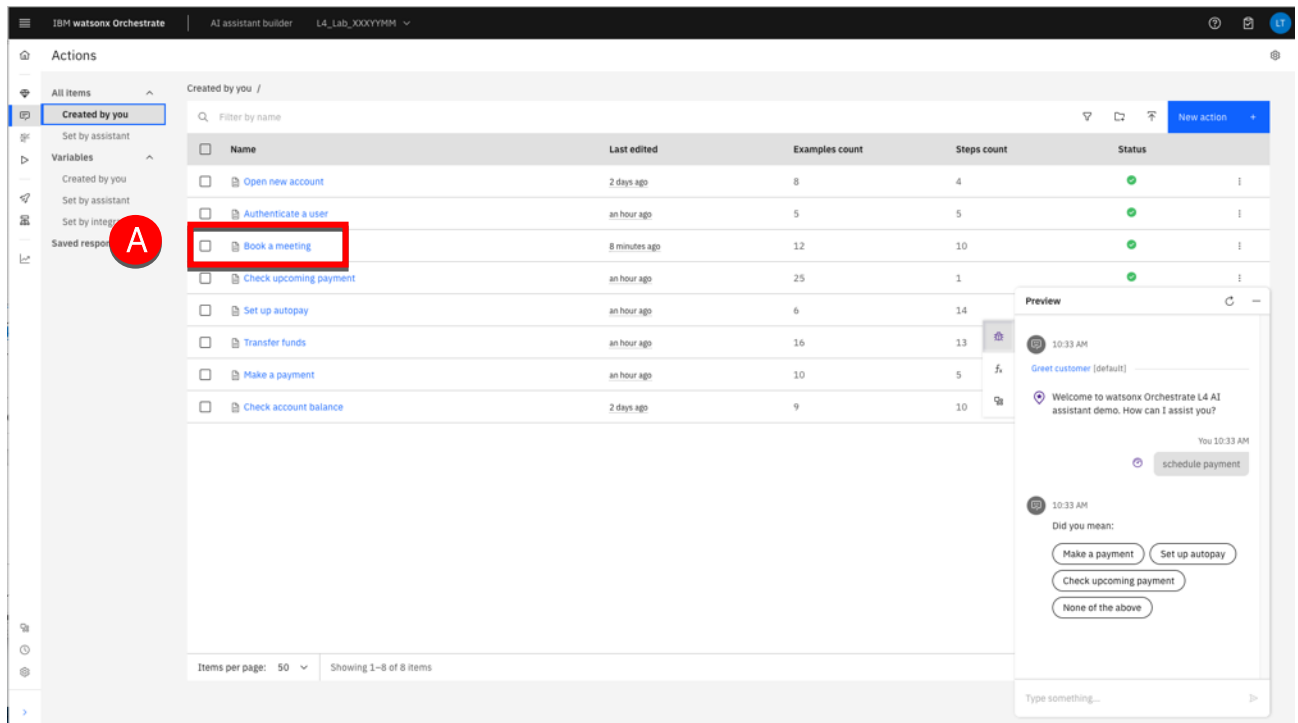
31. Close the **Greet customers** action editor (A).



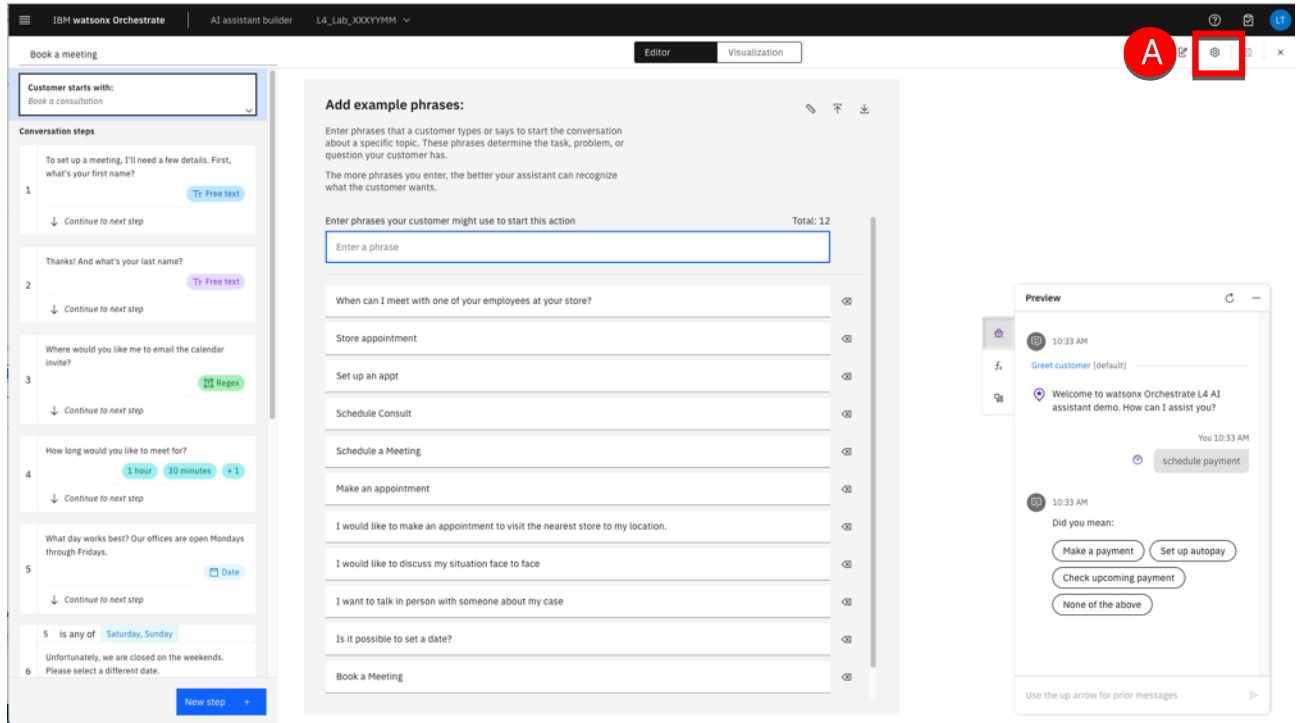
32. In the left-hand side menu, select **Created by you** (A).



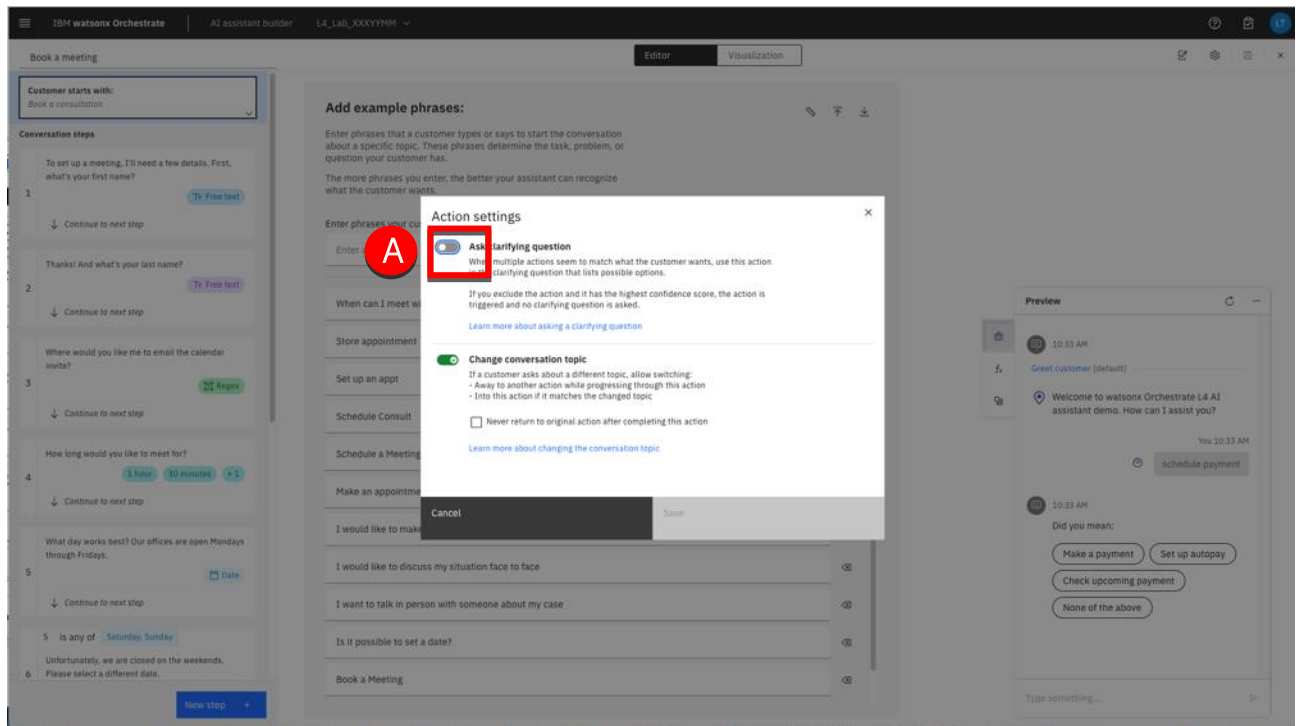
33. Click **Book a meeting** (A).



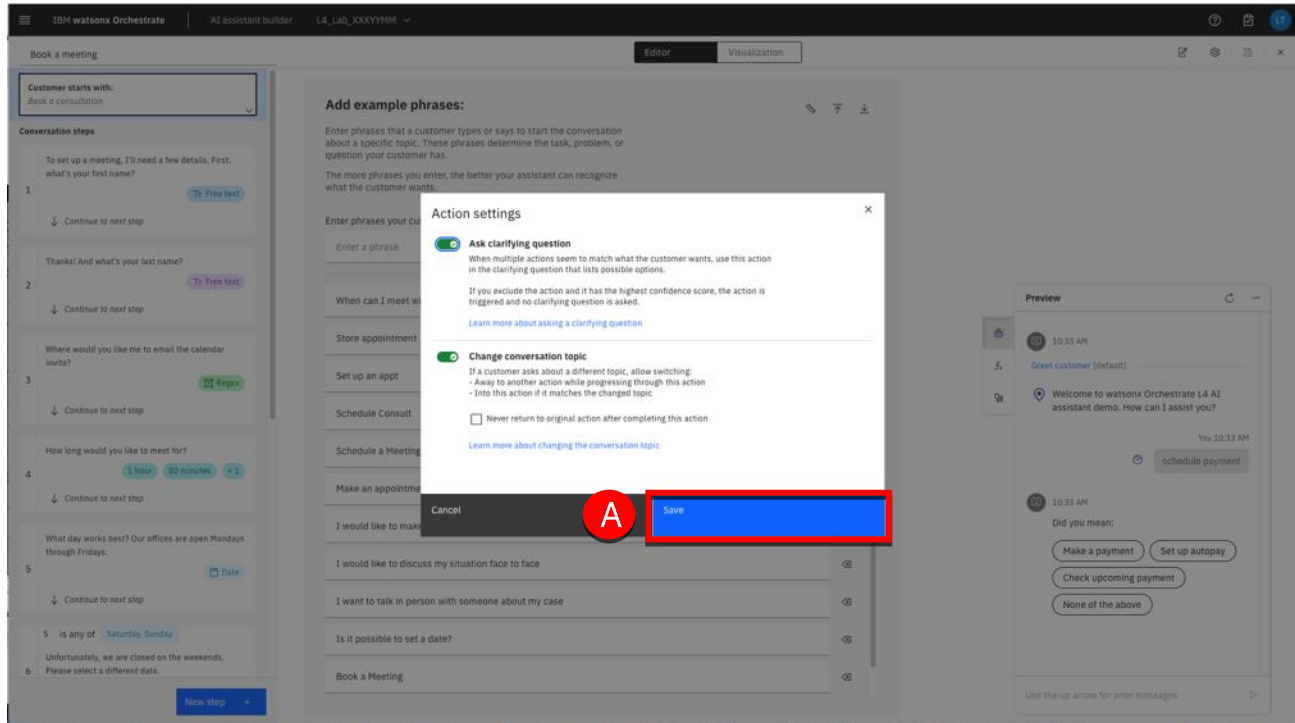
34. Click the action settings icon (A).



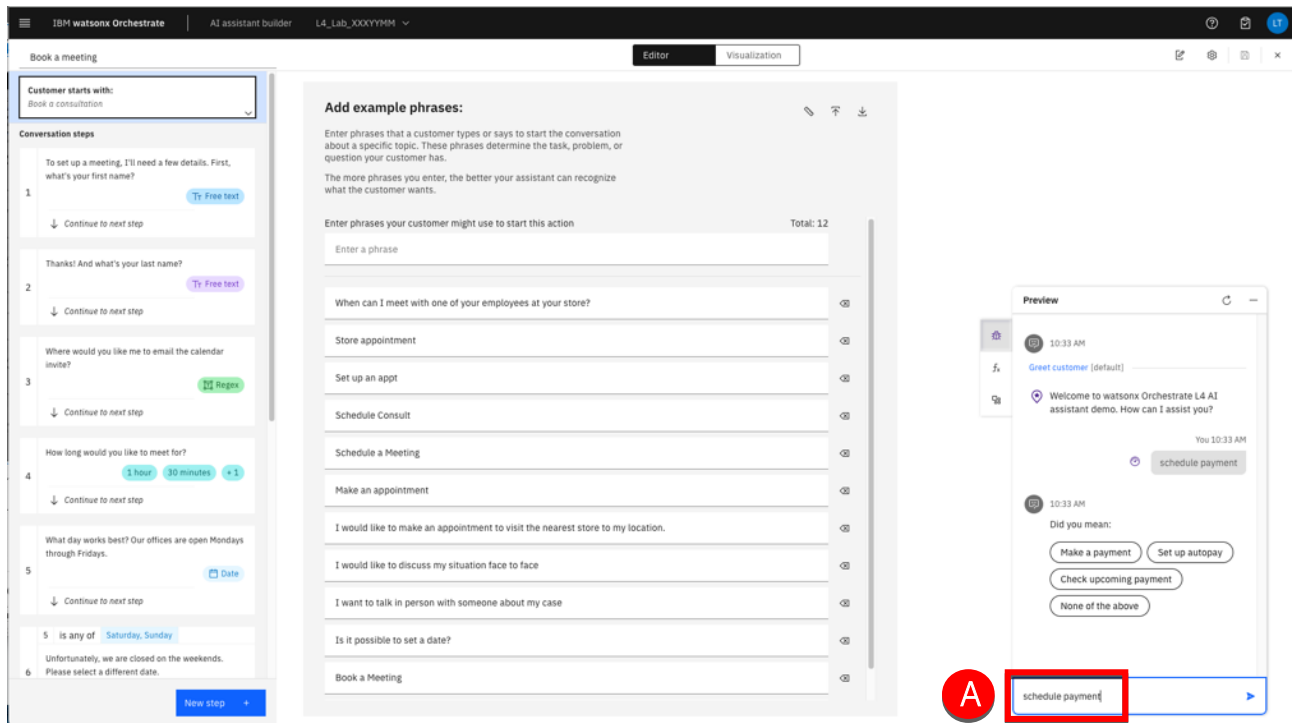
35. Click Ask clarifying questions toggle to turn it on (A).



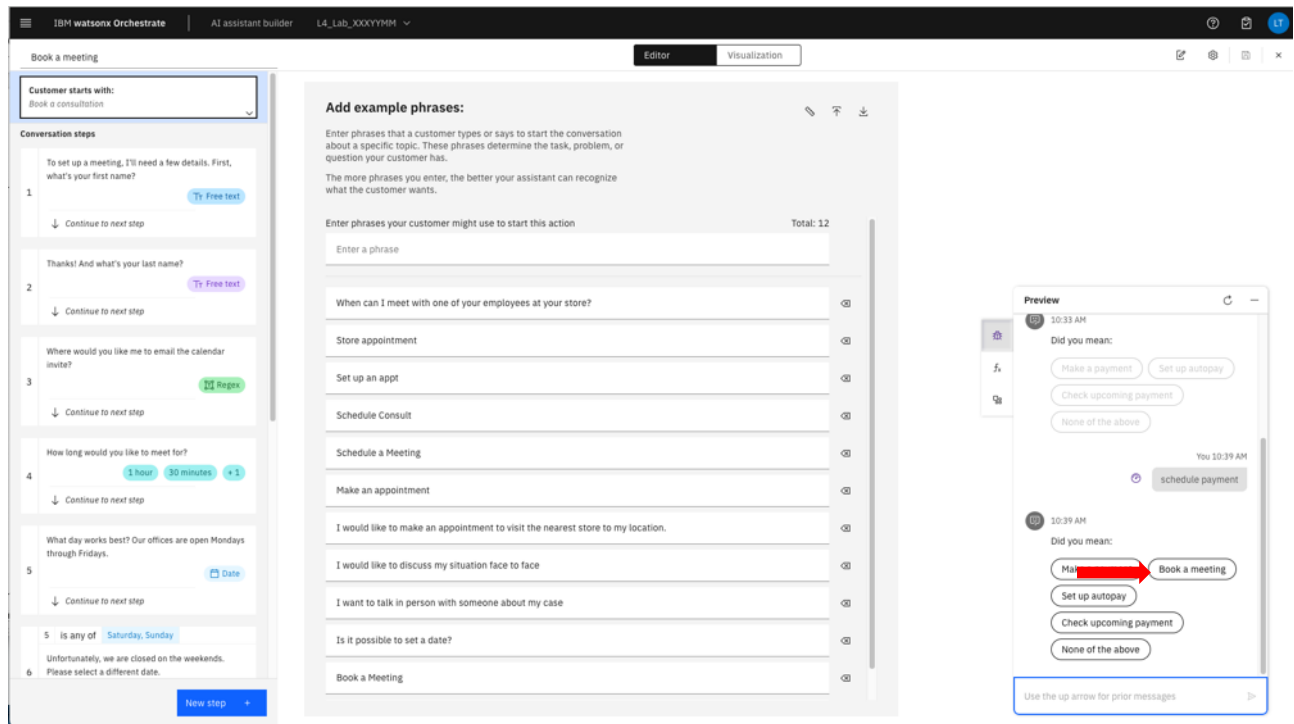
36. Click **Save** (A).



37. Type again **schedule payment** (A).



38. Note that **Book a meeting** is back in the options.



You have demonstrated disambiguation at a global and action level. Once you finish this lab, you can optionally experiment with using [Response modes](#) to manage disambiguation at a more advanced level.

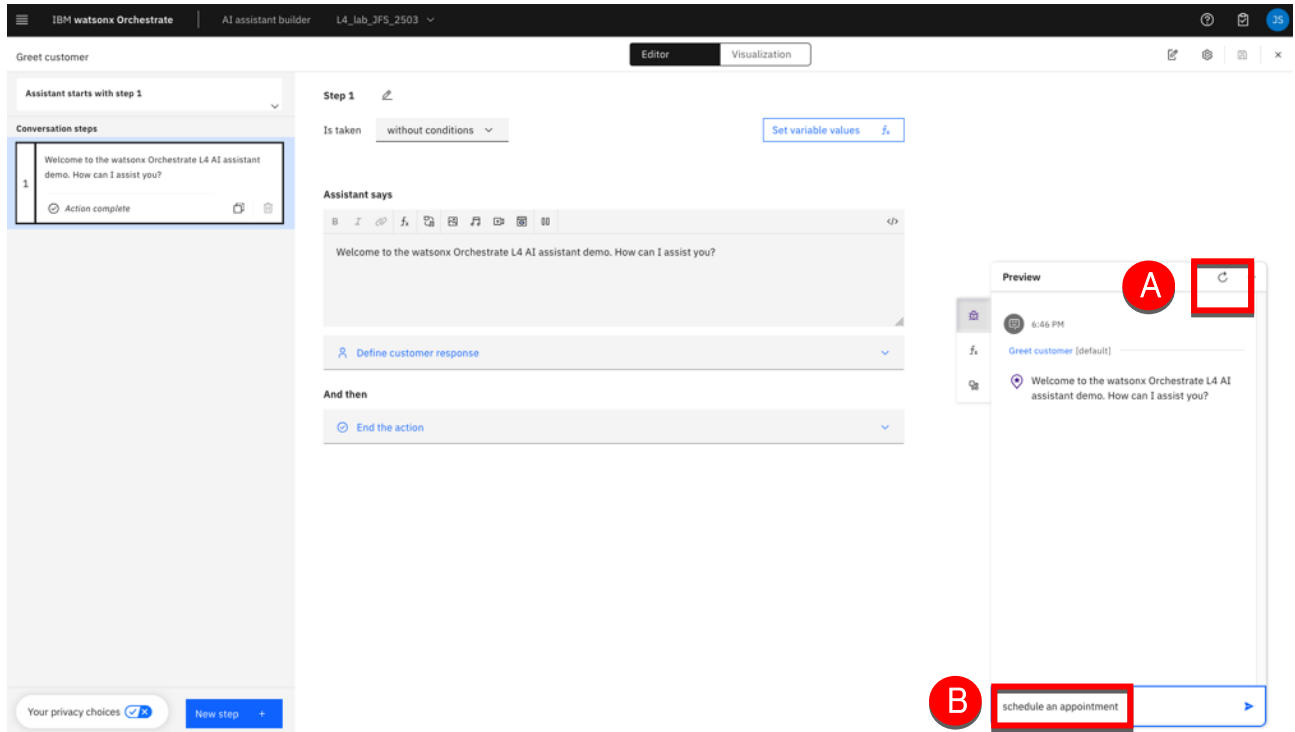
## Digressions: Core concepts and in practice

In general, an action is designed to lead a customer through a particular process without any interruptions. However, real conversations almost never follow such a simple flow. In the middle of a conversation, customers might get distracted, ask questions about related issues, misunderstand something, or change their minds about what they want to do.

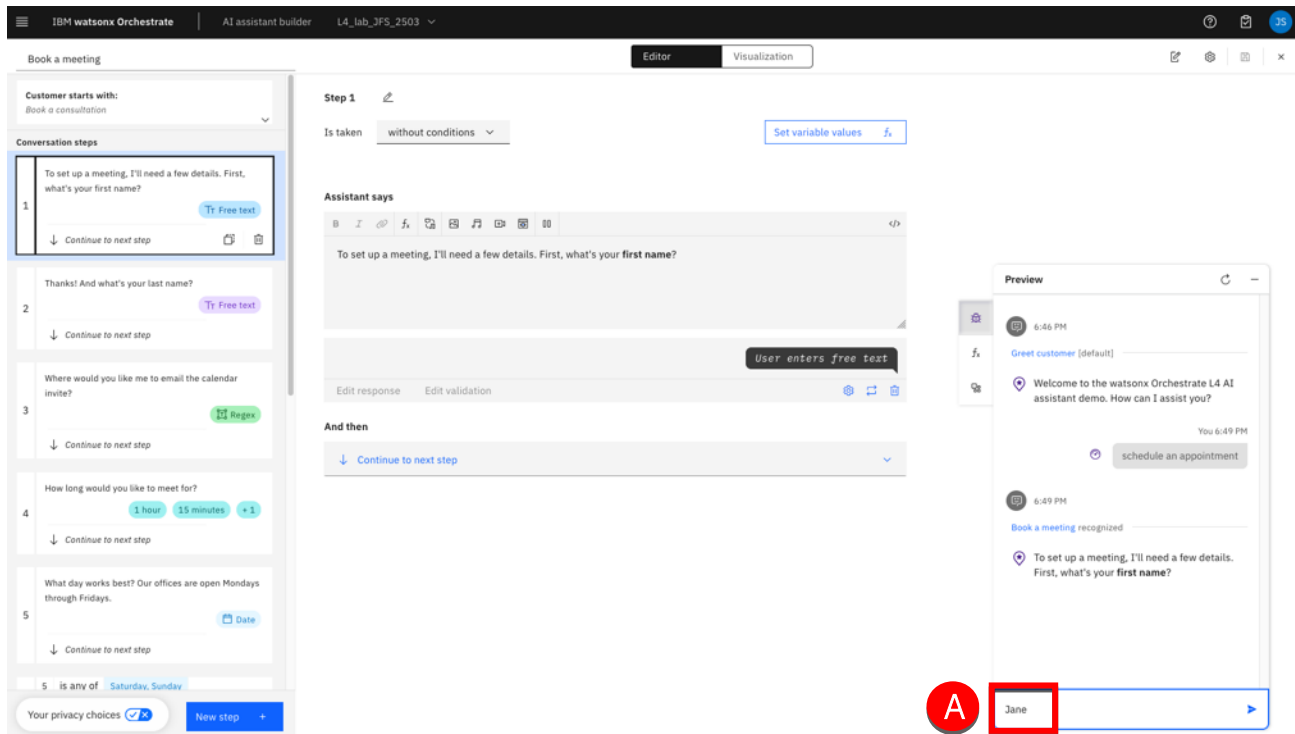
The [Change conversation topic](#) feature enables your AI assistant to handle these digressions, dynamically responding to the user by changing the conversation topic as needed.

Now, you will see an example of the Digression feature in action.

1. In the actions menu, restart the conversation in the Preview window (A), then type **schedule an appointment** (B).



2. The assistant will recognize this user input and route you to the **Book a meeting** action. Type your first name (A).



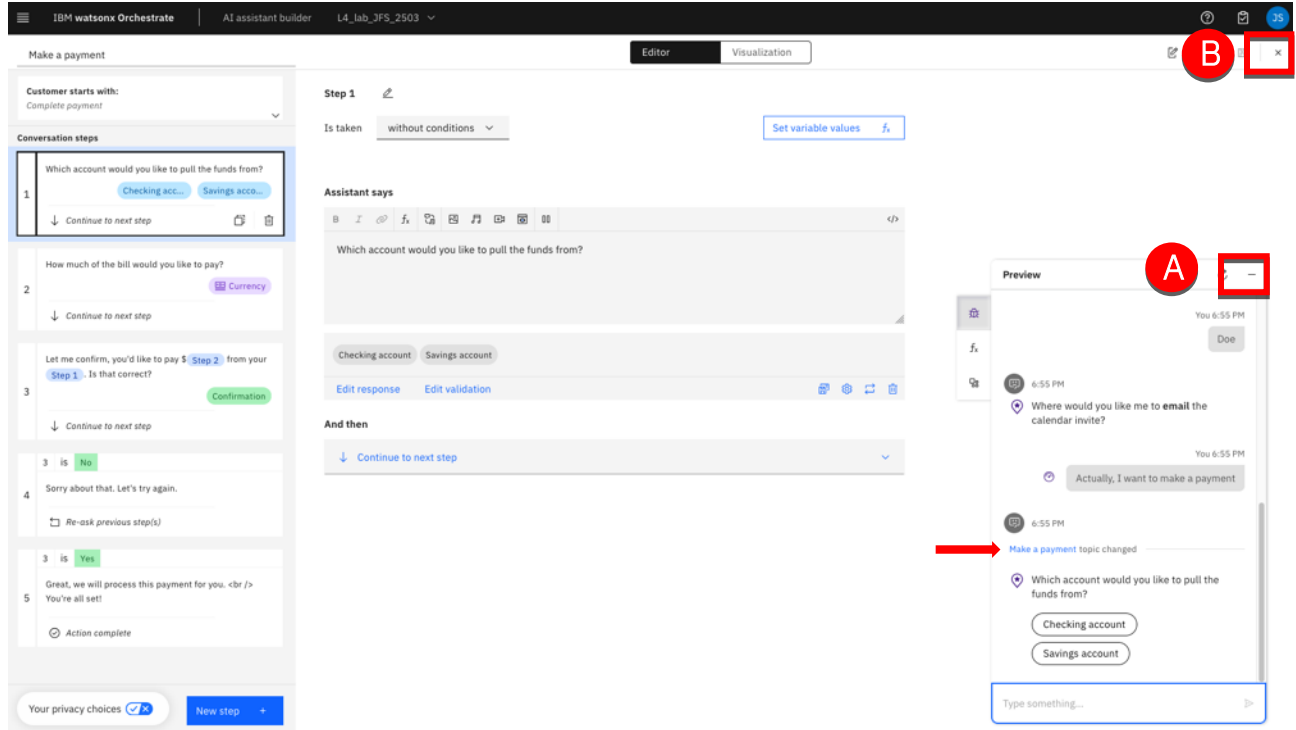
3. Then type your last name (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main editor is in 'Step 2' configuration. The 'Is taken' dropdown is set to 'without conditions'. The 'Assistant says' section contains the text 'Thanks! And what's your last name?'. The 'And then' section has a 'Continue to next step' action. On the left, the 'Conversation steps' list shows five steps, with Step 2 highlighted. On the right, the 'Preview' pane shows a chat history: 'Greet customer [default]', 'Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?', 'schedule an appointment', 'Book a meeting recognized', 'To set up a meeting, I'll need a few details. First, what's your first name?' (with input 'Jane'), and 'Thanks! And what's your last name?' (with input 'Doe' highlighted by a red box and labeled 'B').

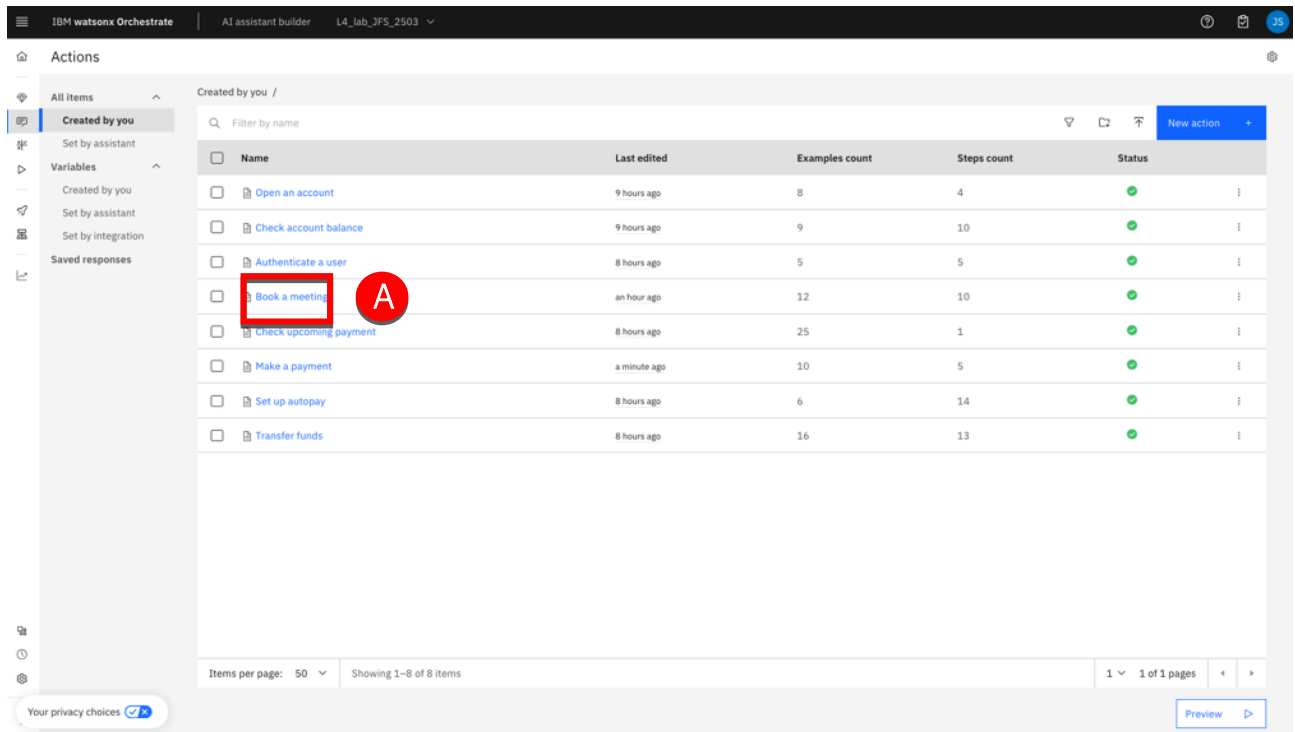
4. You are now in Step 3 of the action. Instead of entering your email, type Actually, I want to make a payment (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main editor is in 'Step 3' configuration. The 'Is taken' dropdown is set to 'without conditions'. The 'Assistant says' section contains the text 'Where would you like me to email the calendar invite?'. The 'And then' section has a 'Continue to next step' action. On the left, the 'Conversation steps' list shows five steps, with Step 3 highlighted by a red arrow. On the right, the 'Preview' pane shows a chat history: 'Greet customer [default]', 'Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?', 'schedule an appointment', 'Book a meeting recognized', 'To set up a meeting, I'll need a few details. First, what's your first name?' (with input 'Jane'), 'Thanks! And what's your last name?' (with input 'Doe'), and 'Where would you like me to email the calendar invite?' (with input 'Actually, I want to make a payment' highlighted by a red box and labeled 'A').

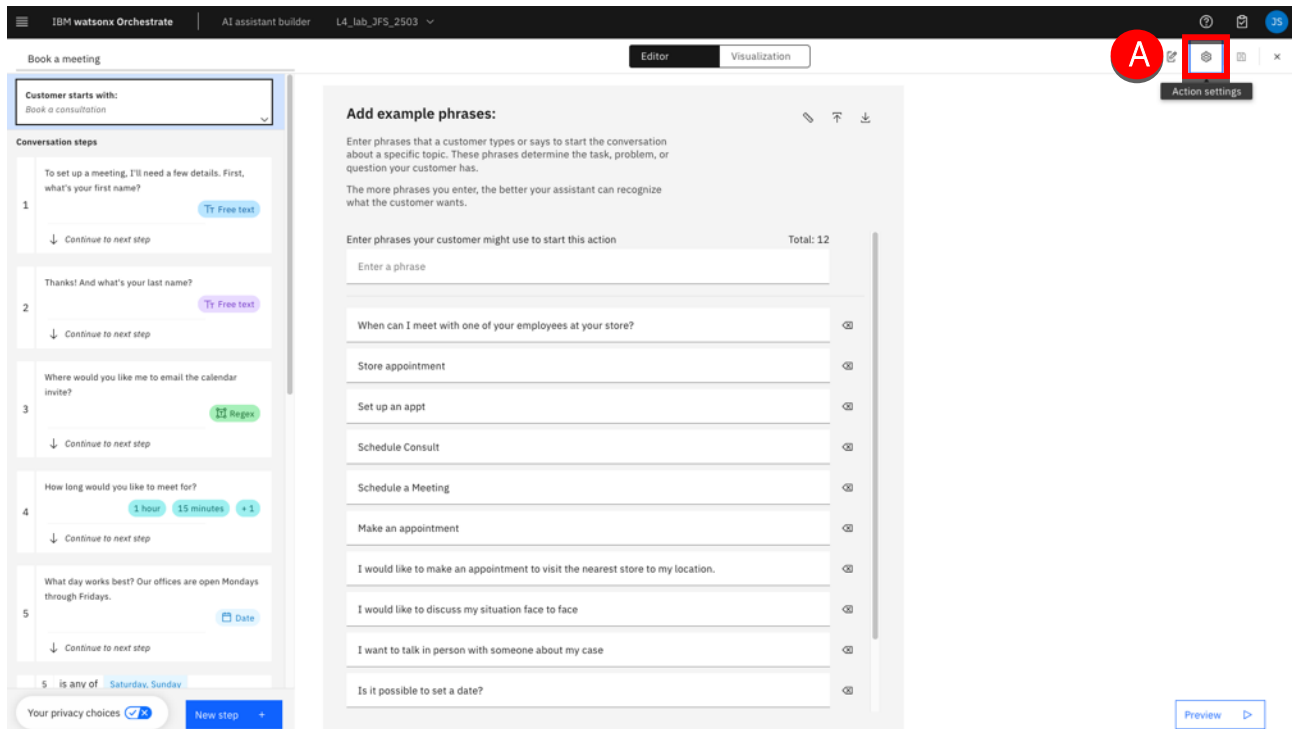
5. Notice that the assistant changed the topic **Make a payment**. Instead of continuing the conversation flow in the Book a meeting action, the assistant recognized the user intent to abandon the current action and switch to the **Make a payment** action, so it performed a topic change also called **Digression**. Minimize the Preview window (A), then close the action editor (B).



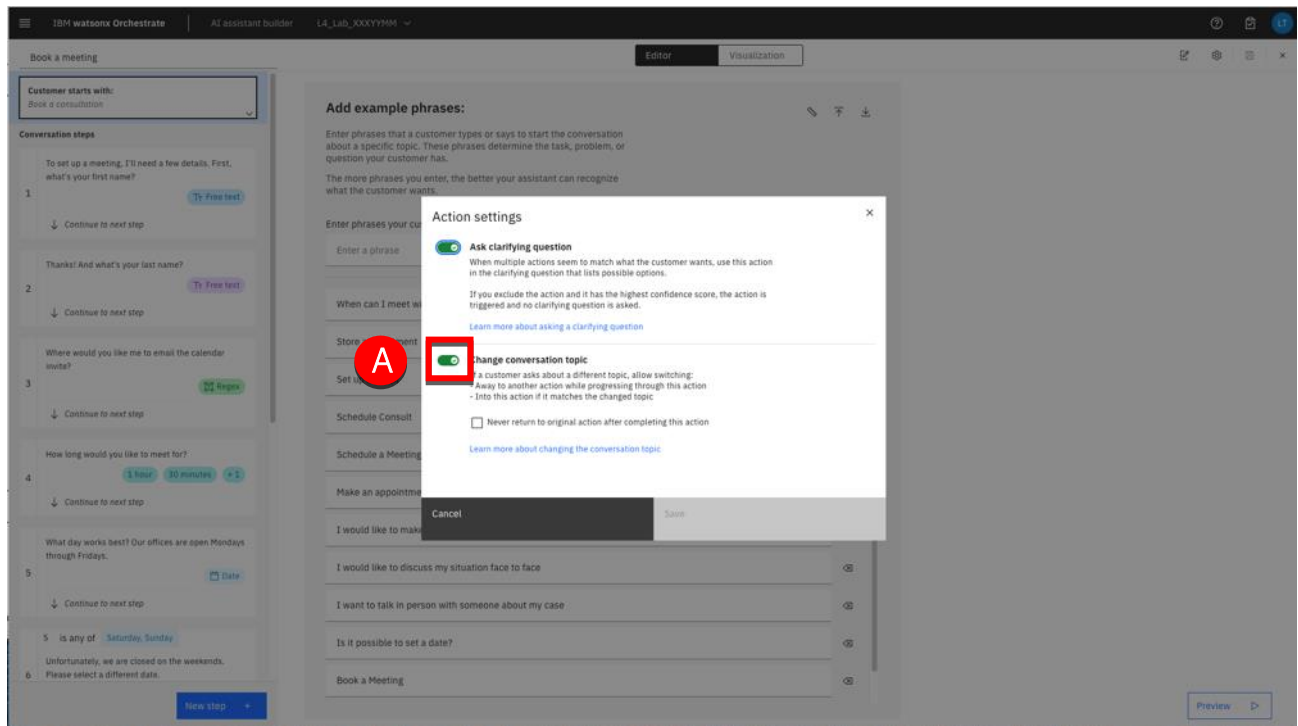
6. In the Actions window, click **Book a meeting** (A).



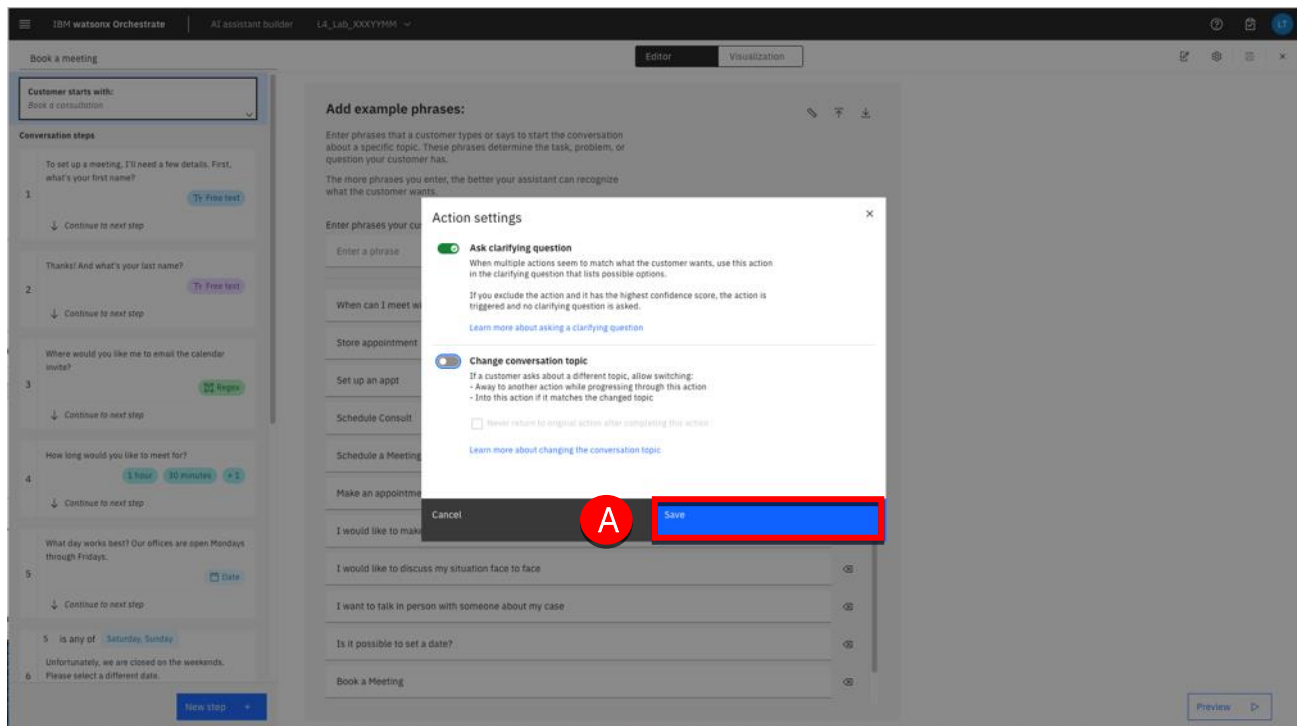
7. Click the action settings icon (A).



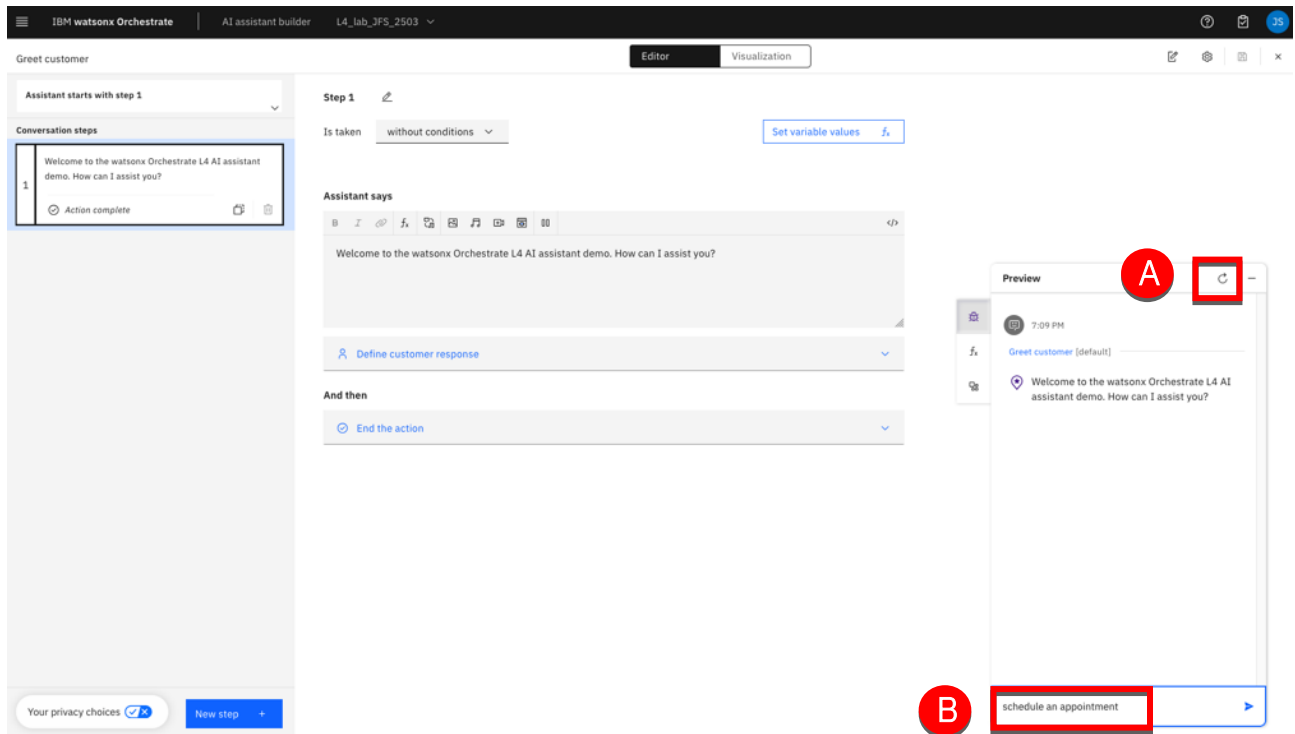
8. Click the **Change Conversation topic** toggle (A).



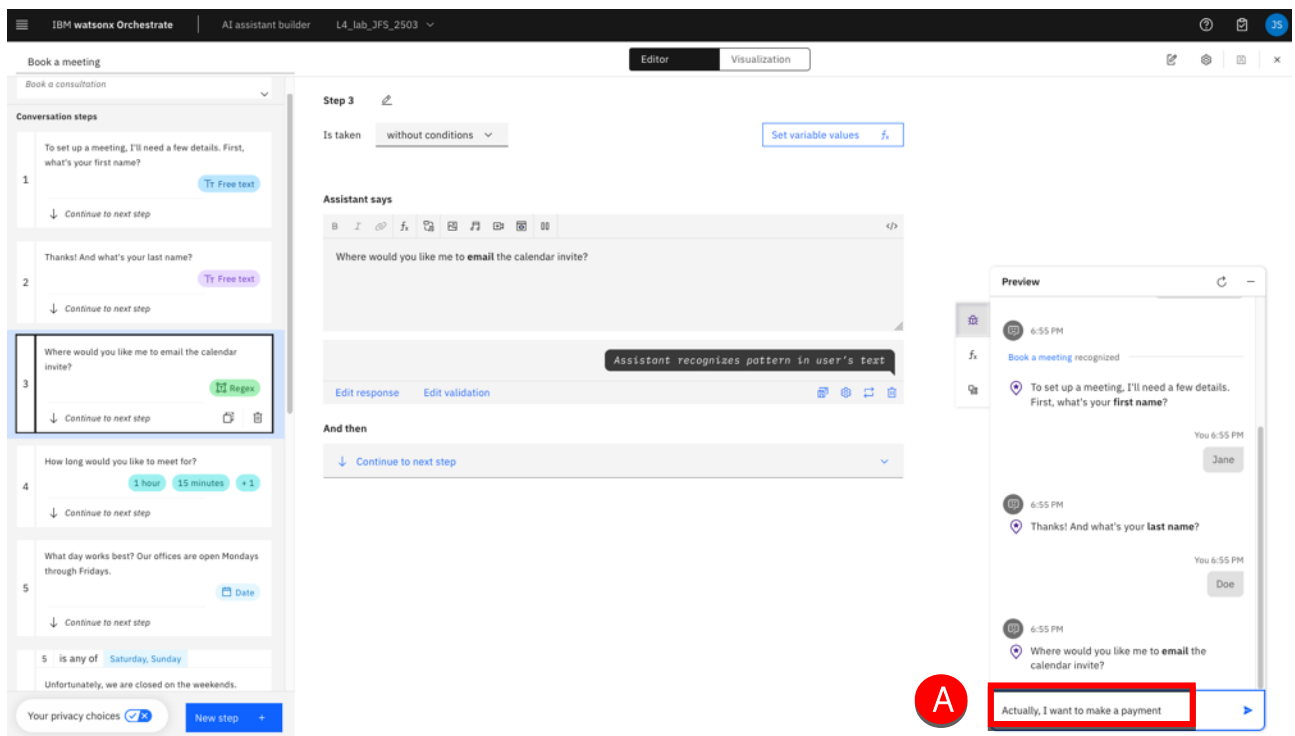
9. Click **Save** (B).



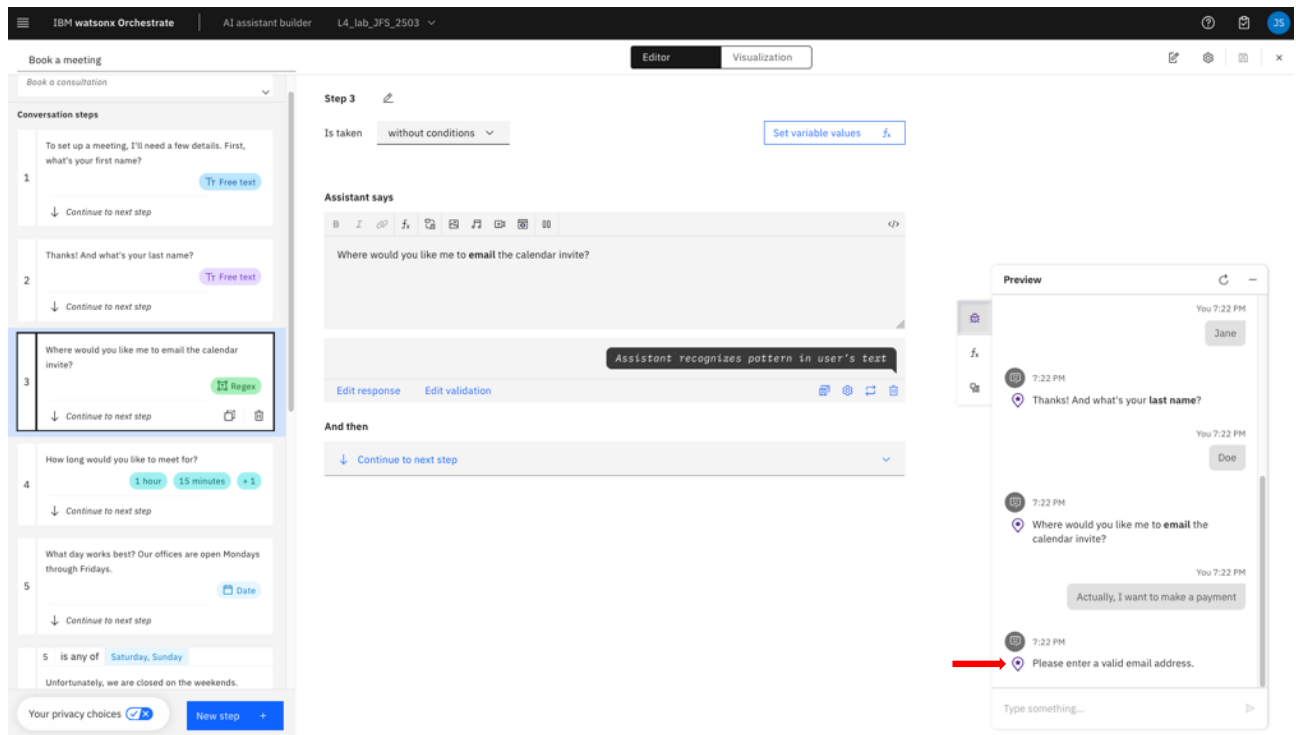
10. Open and refresh the preview window (A), then type `schedule an appointment` (B).



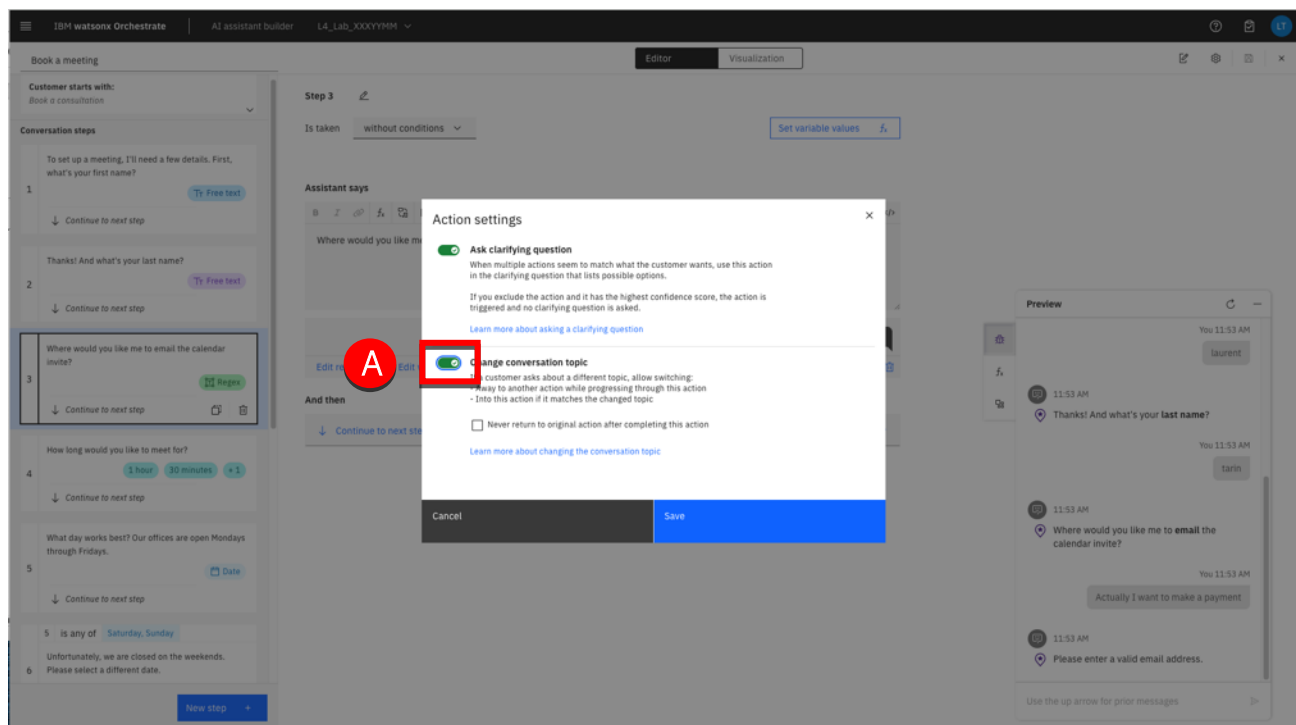
11. Repeat steps 2 – 4 of this section providing your first name, last name, and type `Actually I want to make a payment` (A).



12. Notice that you cannot switch topics from the Book a meeting action. Instead, the Assistant remains in the Book a meeting action. The user input text does not match the email formatting specified by a Regular Expression, so the assistant displays a validation message.



13. Repeat steps 7-9 of this section, but instead setting the **Change conversation topic** (Digression) to enabled (A).



**Note:** Disabling Change Conversation topic (Digression) for an action also prevents users from entering an action through a digression.

For example, if you disable digression for the Make a payment action and repeat the previous flow, you will not be able to digress from the Book a meeting action into the Make a payment action (Even if Digression is enabled in Book a meeting).

With that, you have demonstrated Digression in action. Like the Disambiguation feature, you can enable and disable the Digression feature at both the action level and the global assistant level.

# Advanced Action Configuration: Session Variables, Subactions, and Advanced Step Conditions

## Session Variables: Core concepts

In the Assistant I lab, you successfully built an end-to-end action (Open an account). Moreover, you used action variables and conditions to instantiate a logical flow into your Open an account action.

[Action variables](#) store a customer response to a step throughout an action – you can think of them as the assistant's short-term memory: they persist only during the current action.

Unlike action variables, you can think of [session variables](#) as the assistant's long-term memory: they persist throughout the user's interaction with the AI assistant, and your AI assistant can reference them from any action.

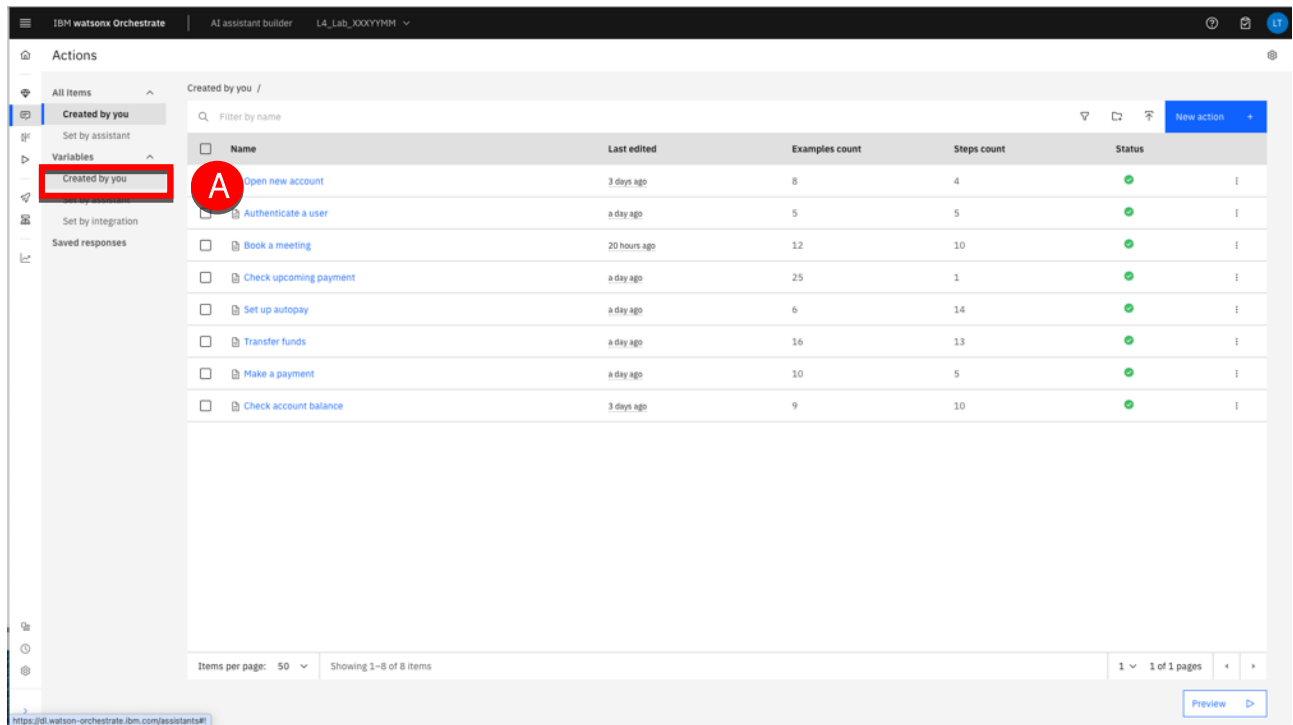
You can create a session variable to store the value from an action variable if you want to keep the value available for other actions. You can also define a session variable based on another session variable or by using a value defined in an expression. In addition to the variables you create, the AI assistant provides a set of built-in session variables for global values like the current time and date.

For this exercise, you will use session variables to collect user information that is useful across multiple actions. Specifically, you will create session variables to collect this information:

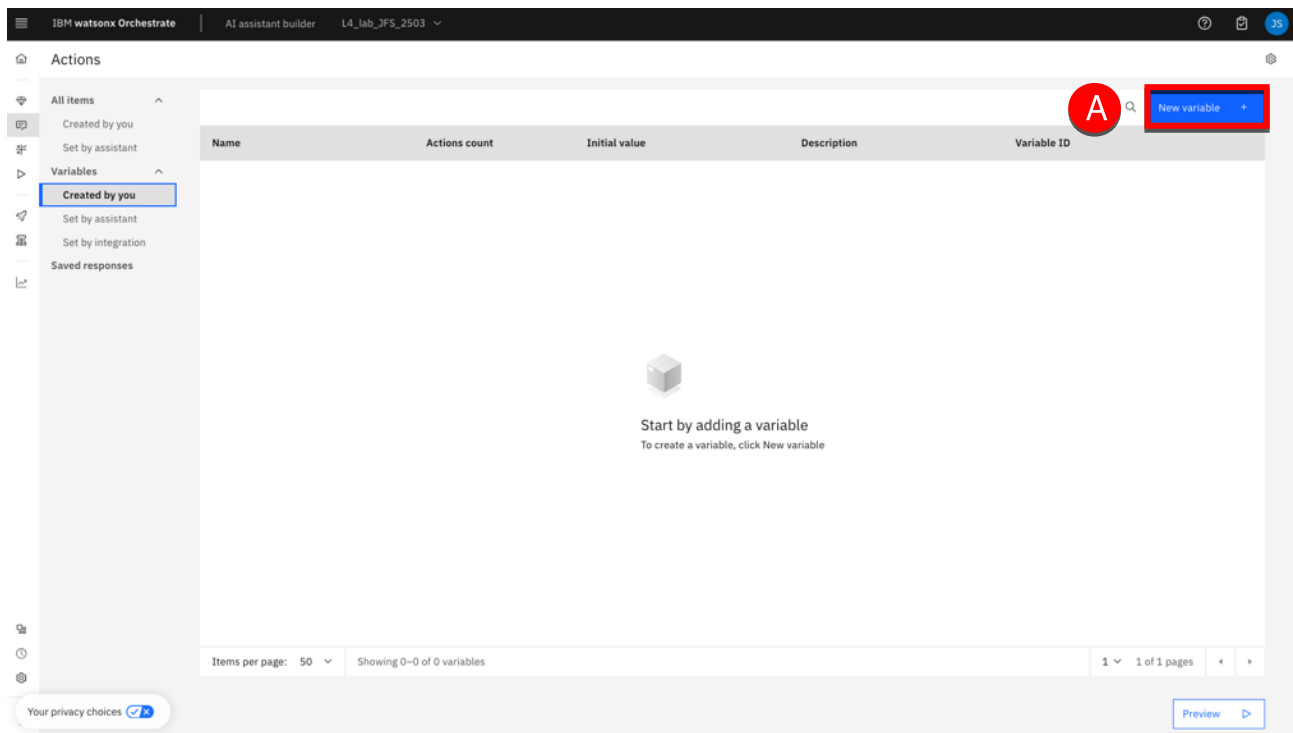
- User's first name
- User's last name
- User's email address
- Is the user authenticated?

## Create Session Variables

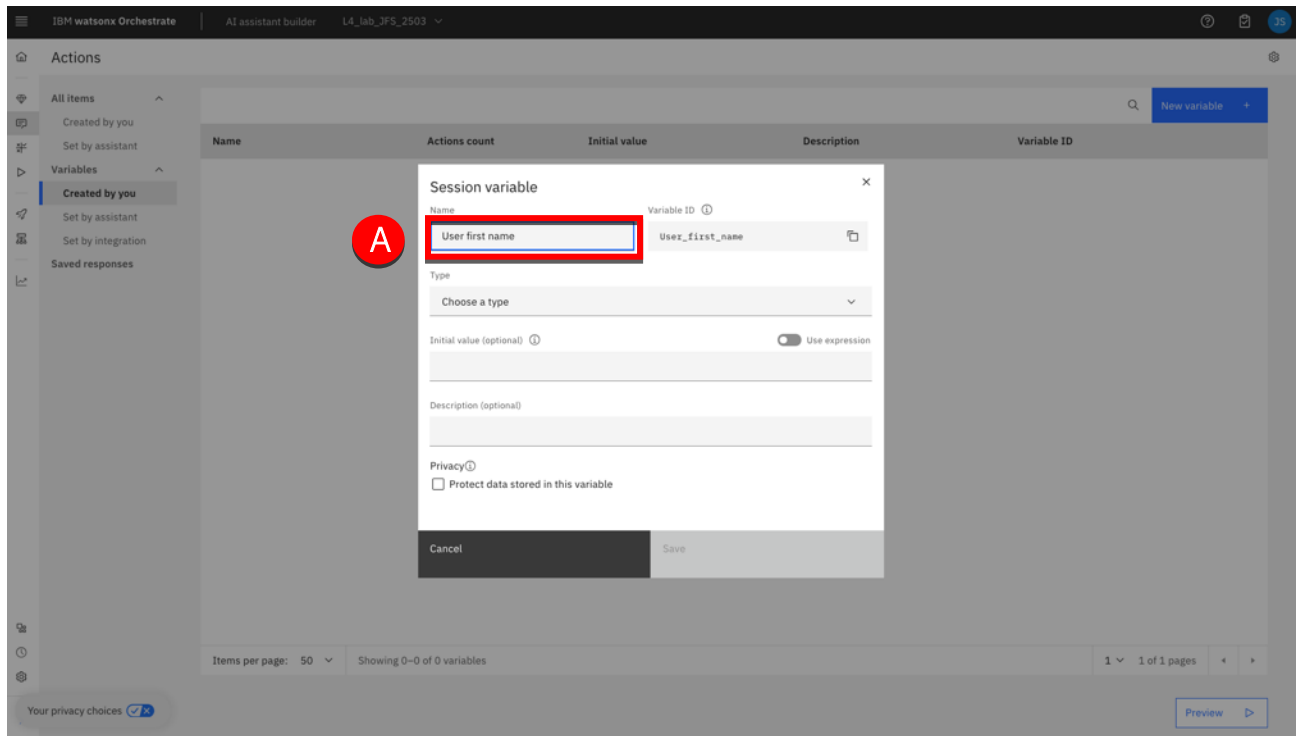
1. Go the actions window, then click **Created by you** under the **Variables** menu (A).



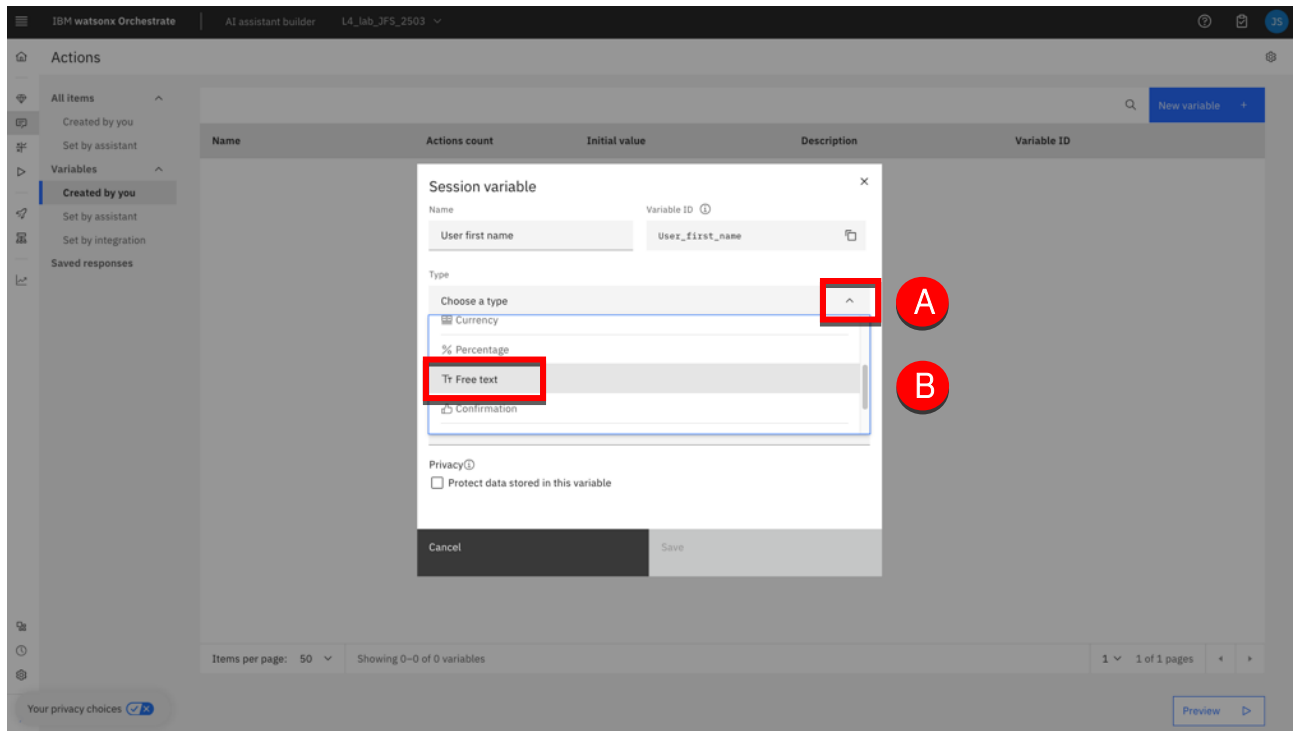
2. Click **New variable** (A).



3. Type `User first name (A)` in the **Name** field.



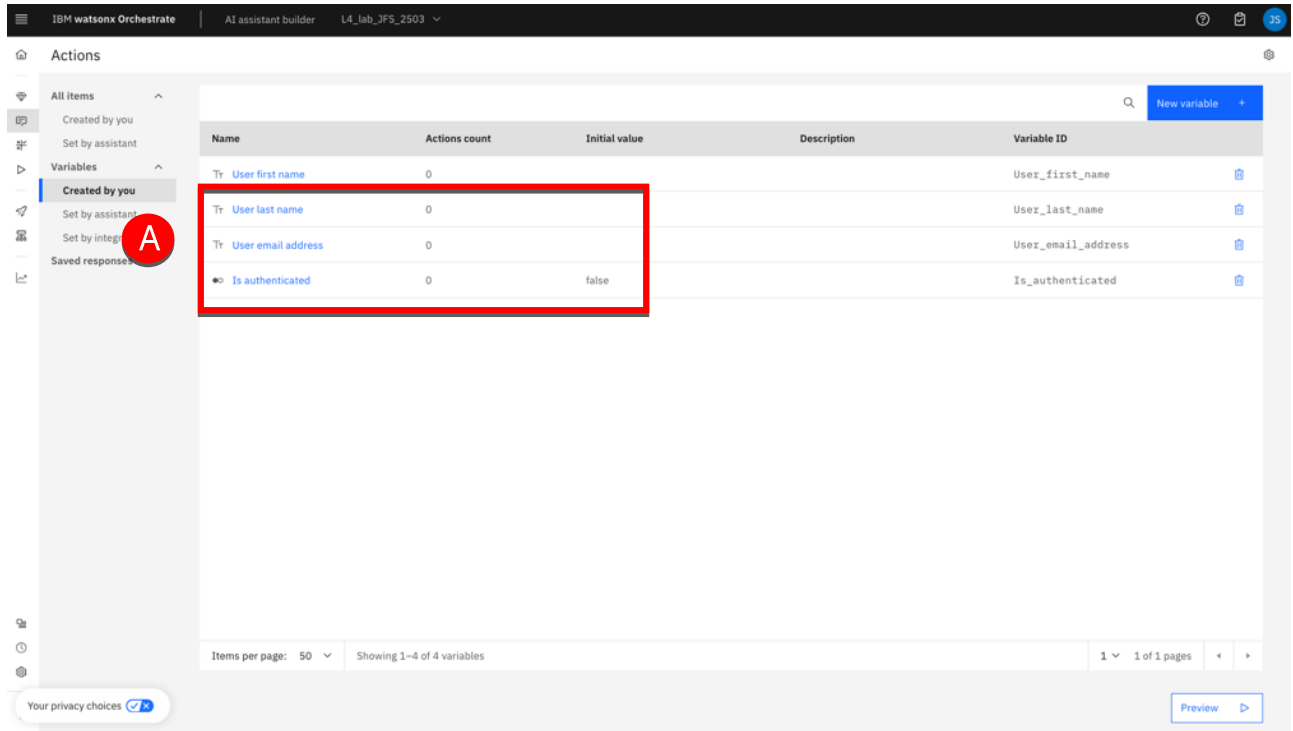
4. Click the dropdown menu for **Type** (A), then scroll down to select **Free text** (B).



5. Click **Save** (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a sidebar lists navigation options: All items, Created by you, Set by assistant, Variables, Created by you, Set by assistant, Set by integration, and Saved responses. The main area displays a table with columns: Name, Actions count, Initial value, Description, and Variable ID. A modal dialog titled 'Session variable' is open, allowing configuration of a new variable. The 'Name' field contains 'User first name' and the 'Variable ID' field contains 'User\_first\_name'. The 'Type' is set to 'Tr Free text'. There are optional fields for 'Initial value' and 'Description', and a 'Privacy' section with a checkbox for 'Protect data stored in this variable'. At the bottom of the dialog, there are 'Cancel' and 'Save' buttons. The 'Save' button is highlighted with a red rectangular box, and a red circle with the letter 'A' is placed next to it. The bottom of the interface shows pagination information: 'Items per page: 50', 'Showing 0-0 of 0 variables', and '1 of 1 pages'. A 'Preview' button is also visible in the bottom right corner.

6. Repeat Steps 2 – 5 of this Section, creating 3 additional variables (A) with the information:
- Variable Name: User last name; Variable Type: **Free text**
  - Variable Name: User email address; Variable Type: **Free text**
  - Variable Name: Is authenticated; Variable Type: **Boolean**; Initial Value: **False**



The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar has a filter 'Created by you' highlighted with a red circle and a white 'A'. The main area displays a table of variables:

| Name                  | Actions count | Initial value | Description | Variable ID        |
|-----------------------|---------------|---------------|-------------|--------------------|
| Tr User first name    | 0             |               |             | User_first_name    |
| Tr User last name     | 0             |               |             | User_last_name     |
| Tr User email address | 0             |               |             | User_email_address |
| Is authenticated      | 0             | false         |             | Is_authenticated   |

At the bottom, it says 'Items per page: 50' and 'Showing 1-4 of 4 variables'. A 'Preview' button is at the bottom right.

## Session Variables in practice

In this section, your goal is to edit the Authenticate a user action so that the user information is stored in Session variables.

1. Under **All items**, click **Created by you** (A).

The screenshot shows the IBM watsonx Orchestrate interface. The top navigation bar includes 'IBM watsonx Orchestrate', 'AI assistant builder', and a dropdown menu 'L4\_lab\_JFS\_2503'. The left sidebar has a 'Variables' section with a 'Created by you' sub-section highlighted by a red box and labeled with a red circle 'A'. The main content area displays a table of variables created by the user.

| Name                  | Actions count | Initial value | Description | Variable ID        |
|-----------------------|---------------|---------------|-------------|--------------------|
| Tr User first name    | 0             |               |             | User_first_name    |
| Tr User last name     | 0             |               |             | User_last_name     |
| Tr User email address | 0             |               |             | User_email_address |
| Is authenticated      | 0             | false         |             | Is_authenticated   |

At the bottom of the table, it says 'Items per page: 50' and 'Showing 1-4 of 4 variables'. There is a 'New variable' button in the top right and a 'Preview' button in the bottom right.

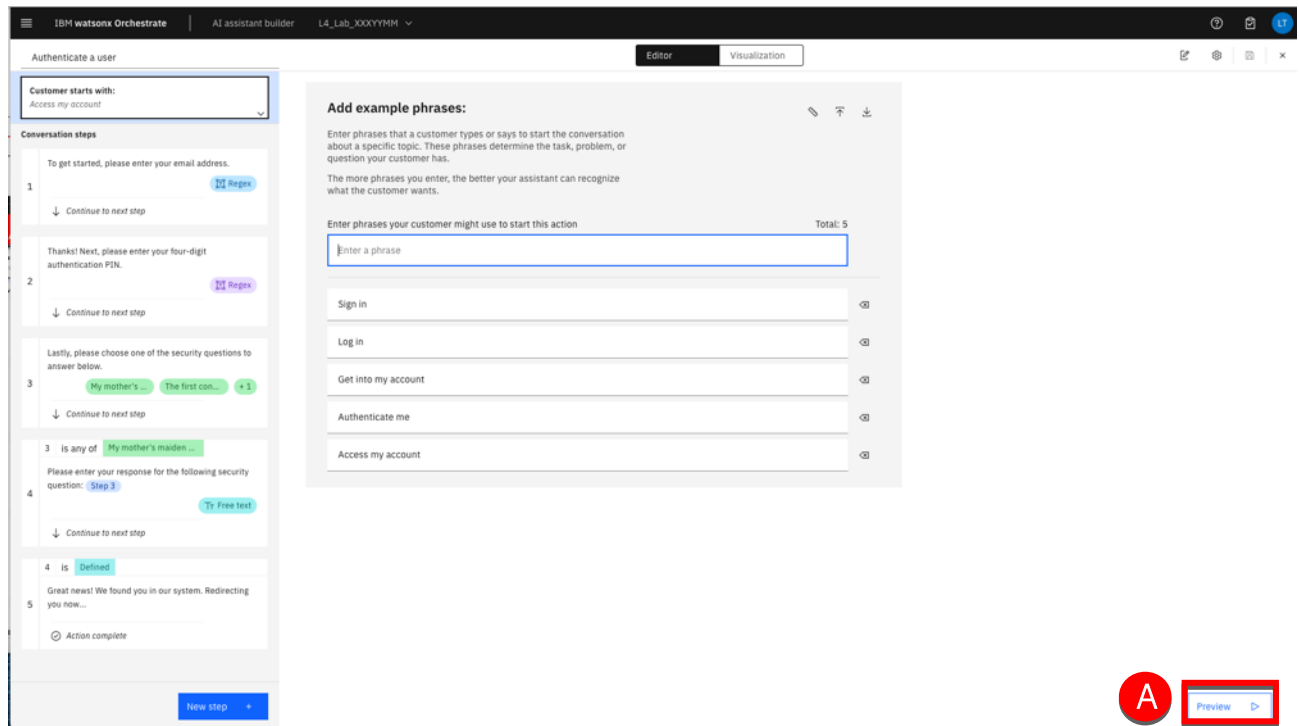
2. Click **Authenticate a user** (A).

The screenshot shows the IBM watsonx Orchestrate interface. The top navigation bar includes 'IBM watsonx Orchestrate', 'AI assistant builder', and a dropdown menu 'L4\_lab\_JFS\_2503'. The left sidebar has an 'Actions' section with a 'Created by you' sub-section highlighted by a blue box. The main content area displays a table of actions created by the user.

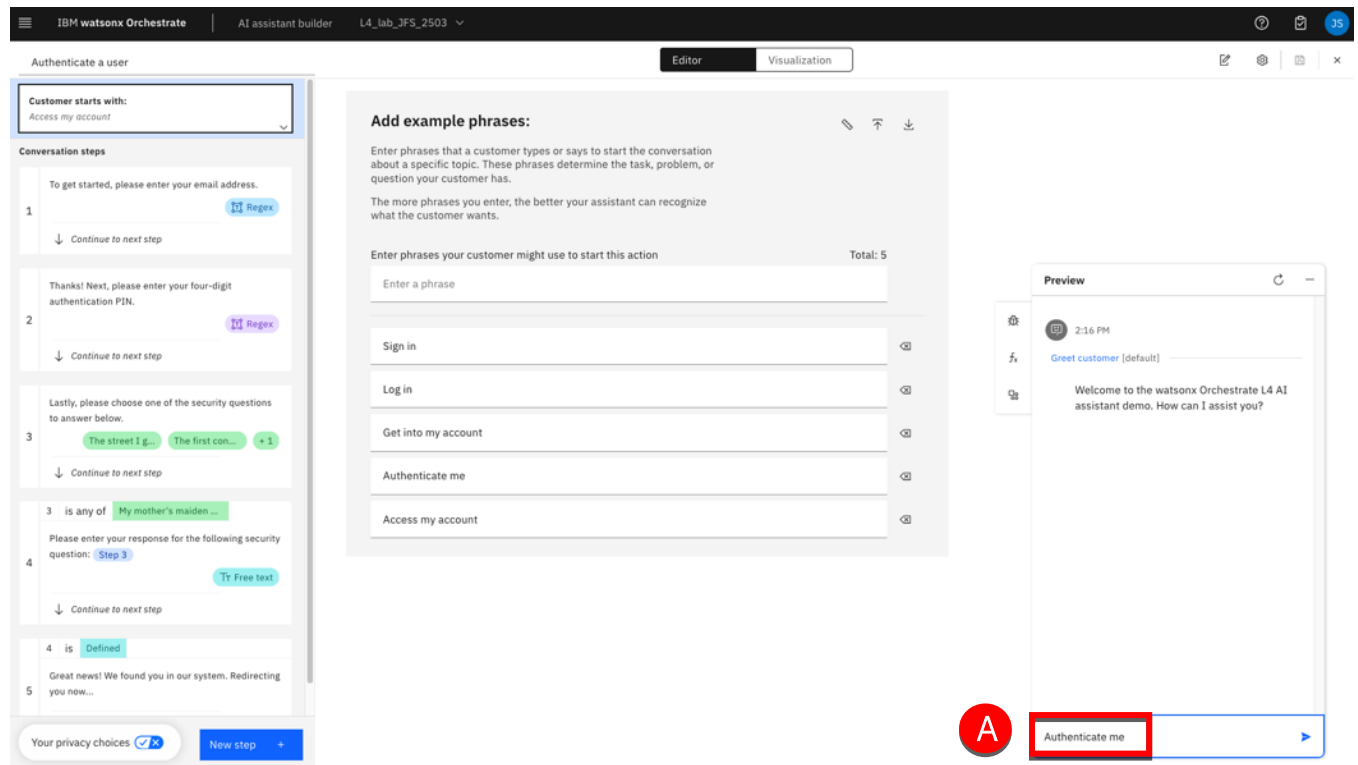
| Name                   | Last edited  | Examples count | Steps count | Status |
|------------------------|--------------|----------------|-------------|--------|
| Open an account        | a day ago    | 8              | 4           | ✓      |
| Check account balance  | a day ago    | 9              | 10          | ✓      |
| Authenticate a user    | a day ago    | 5              | 5           | ✓      |
| Book a meeting         | 19 hours ago | 12             | 10          | ✓      |
| Check upcoming payment | a day ago    | 25             | 1           | ✓      |
| Make a payment         | 19 hours ago | 10             | 5           | ✓      |
| Set up autopay         | a day ago    | 6              | 14          | ✓      |
| Transfer funds         | a day ago    | 16             | 13          | ✓      |

At the bottom of the table, it says 'Items per page: 50' and 'Showing 1-8 of 8 items'. There is a 'New action' button in the top right and a 'Preview' button in the bottom right.

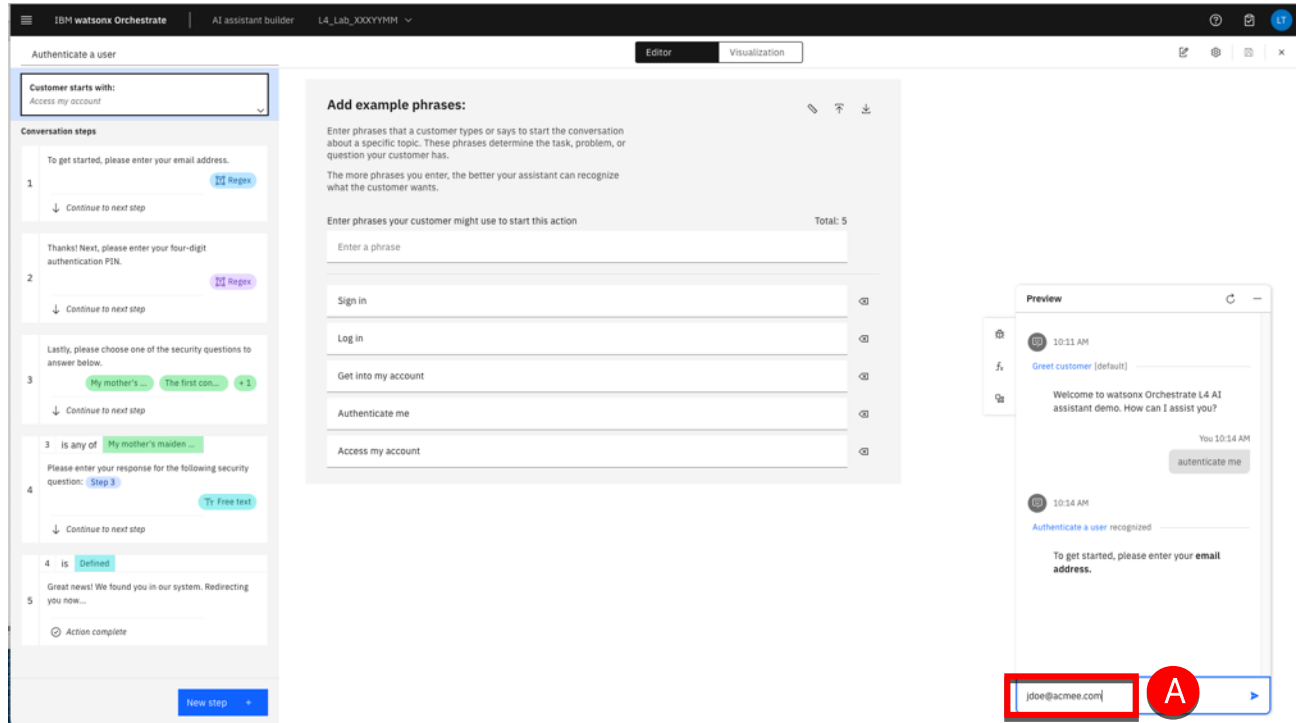
3. Click **Preview** (A).



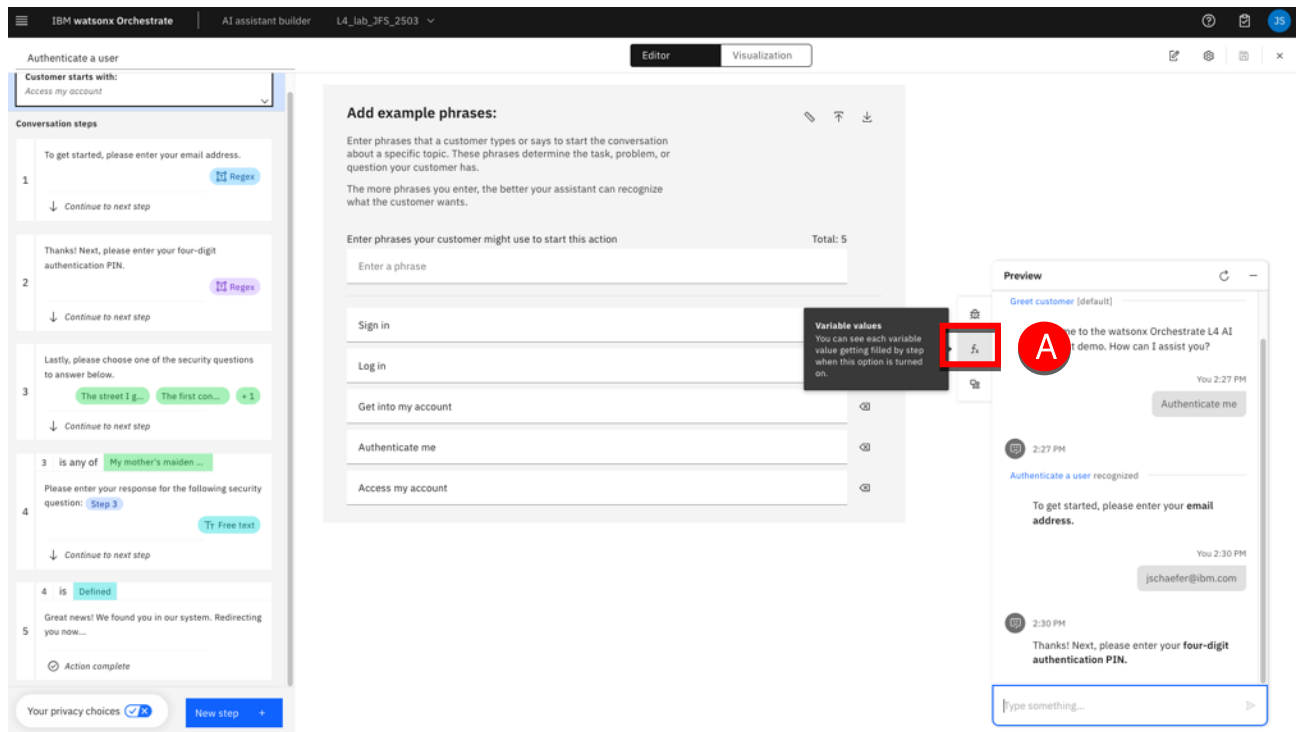
4. Type **authenticate me** (A).



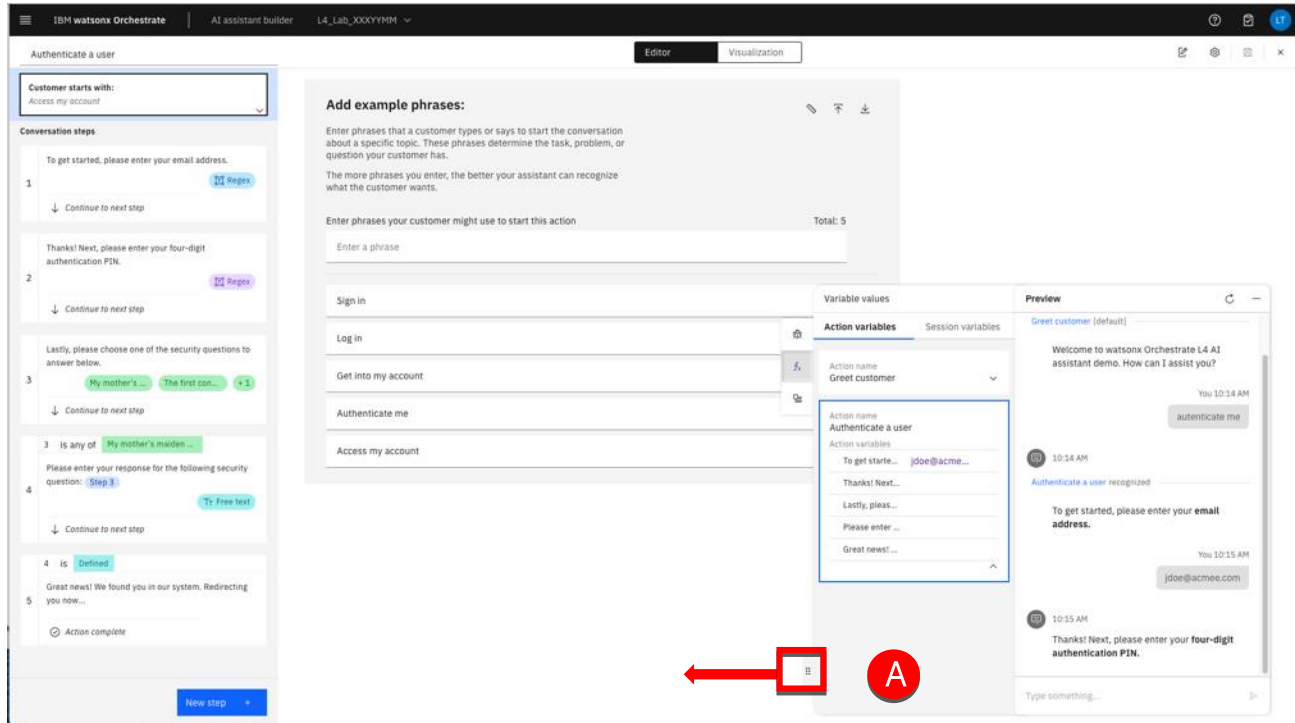
5. Type a valid email address (A).



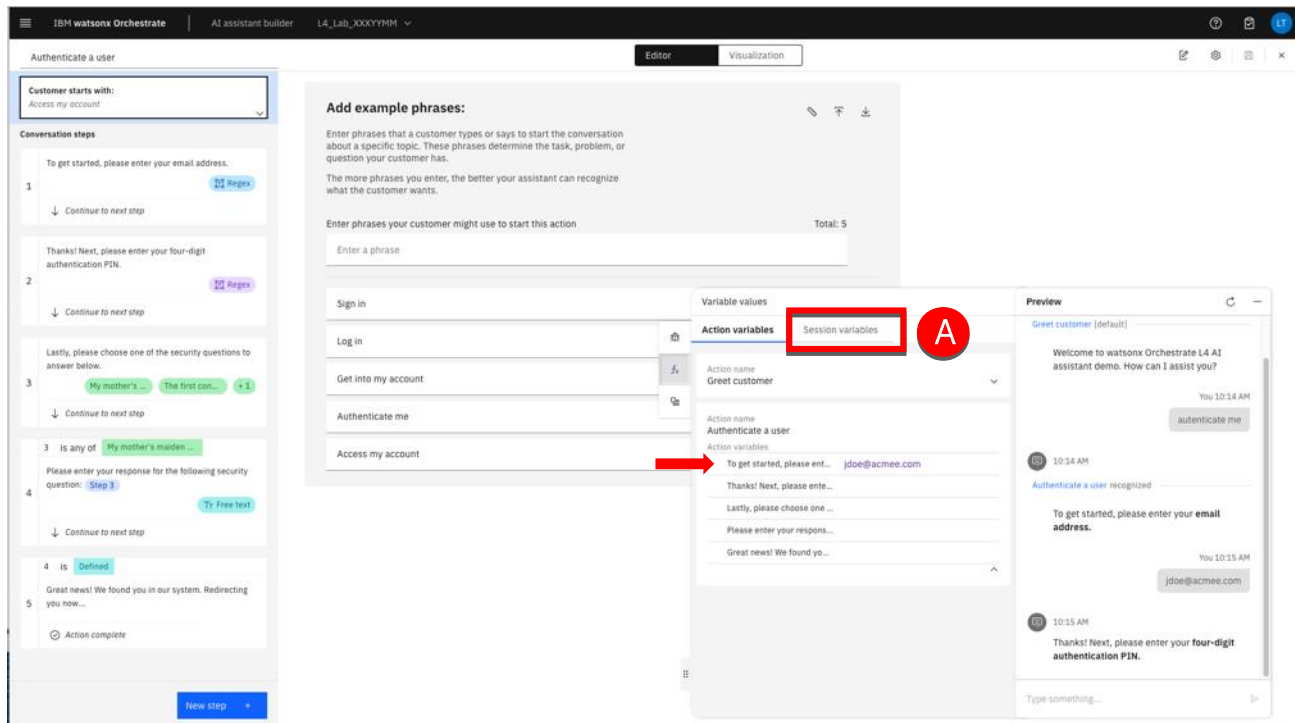
6. In the preview window, click [Variable values](#)



7. Use the lower left side handle (A) to expand the Variable values view



8. Notice that the Step 1 (To get started, please enter your email address) action variable is populated with the email value you entered in the assistant chat earlier. In the Variable values view, select **Session variables** (A).



9. Notice that the only session variable currently defined is the **Is\_authenticated** variable. This is because the **Is\_authenticated** variable is configured to initialize with a **false** Boolean value. Minimize the **Preview** window (A).

The screenshot displays the IBM Watsonx Orchestrate AI Assistant Builder interface. The main workspace is titled 'Authenticate a user' and shows a conversation flow with five steps. A 'Variable values' panel is open, showing 'Session variables' with 'Is\_authenticated' set to 'false'. A red arrow points to this variable. A 'Preview' window on the right shows a chat simulation, with a red box and letter 'A' highlighting its minimize button.

Let's now initialize the `User_email_address` variable with the value from the Step 1 response.

10. Click on step 2 (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_JFS\_2503

Authenticate a user

Editor Visualization

Customer starts with: Access my account

Conversation steps

1 To get started, please enter your email address. [Regex]

2 Thanks! Next, please enter your four-digit authentication PIN. [Regex] (A)

3 Lastly, please choose one of the security questions to answer below. [The street I g... The first con... +1]

3 is any of My mother's maiden ...

4 Please enter your response for the following security question: Step 3 [Tr Free text]

4 is Defined

5 Great news! We found you in our system. Redirecting you now...

Action complete

Your privacy choices [Checkmark] [New step +]

Add example phrases:

Enter phrases that a customer types or says to start the conversation about a specific topic. These phrases determine the task, problem, or question your customer has.

The more phrases you enter, the better your assistant can recognize what the customer wants.

Enter phrases your customer might use to start this action Total: 5

Enter a phrase

Sign in

Log in

Get into my account

Authenticate me

Access my account

Preview

11. Click Set variable values (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_JFS\_2503

Authenticate a user

Editor Visualization

Customer starts with: Access my account

Conversation steps

1 To get started, please enter your email address. [Regex]

2 Thanks! Next, please enter your four-digit authentication PIN. [Regex] (A)

3 Lastly, please choose one of the security questions to answer below. [The street I g... The first con... +1]

3 is any of My mother's maiden ...

4 Please enter your response for the following security question: Step 3 [Tr Free text]

4 is Defined

5 Great news! We found you in our system. Redirecting you now...

Action complete

Your privacy choices [Checkmark] [New step +]

Step 2

Is taken without conditions

Set variable values (A)

Assistant says

Thanks! Next, please enter your four-digit authentication PIN.

Assistant recognizes pattern in user's test

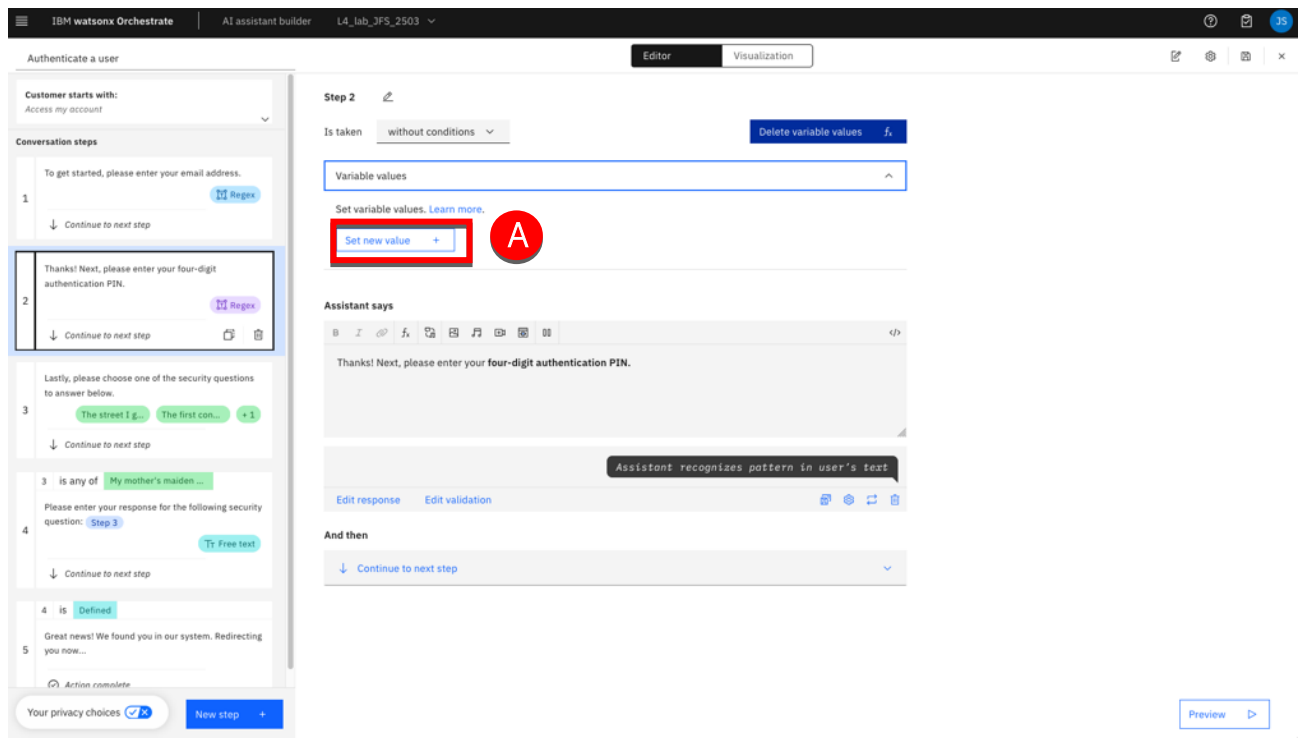
Edit response Edit validation

And then

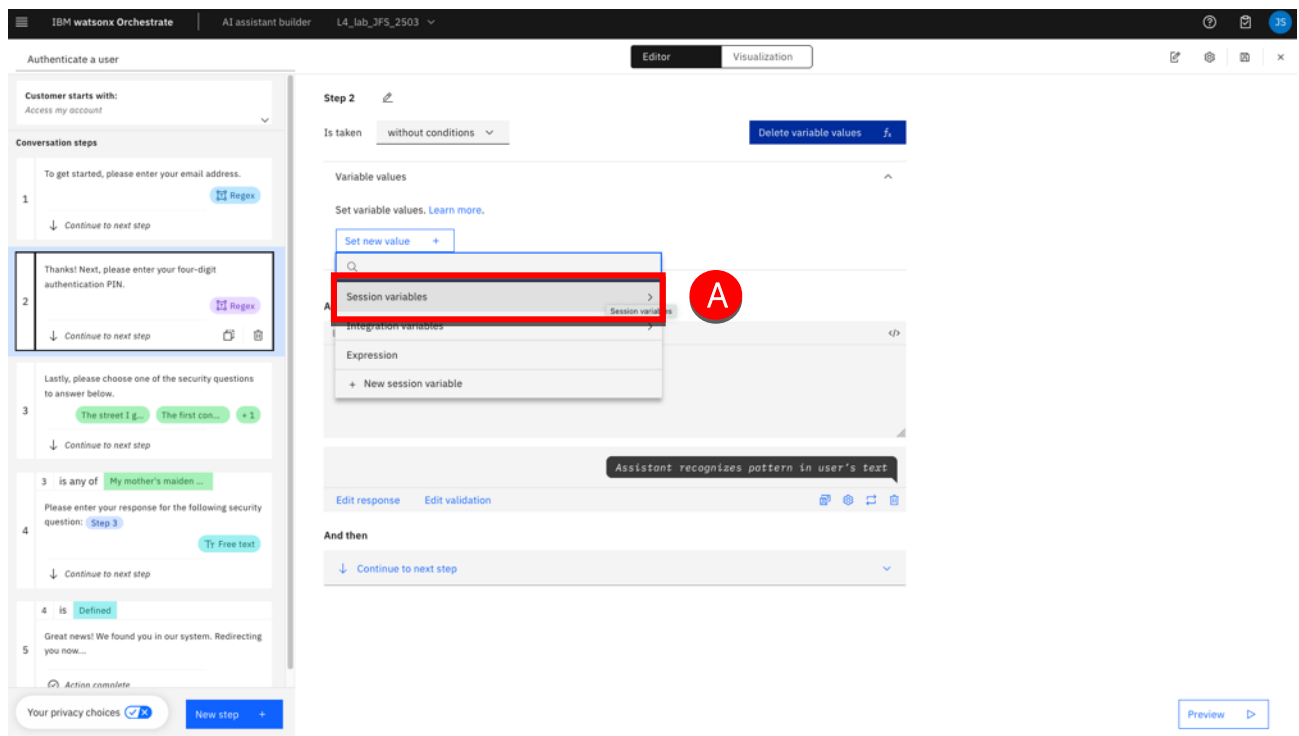
Continue to next step

Preview

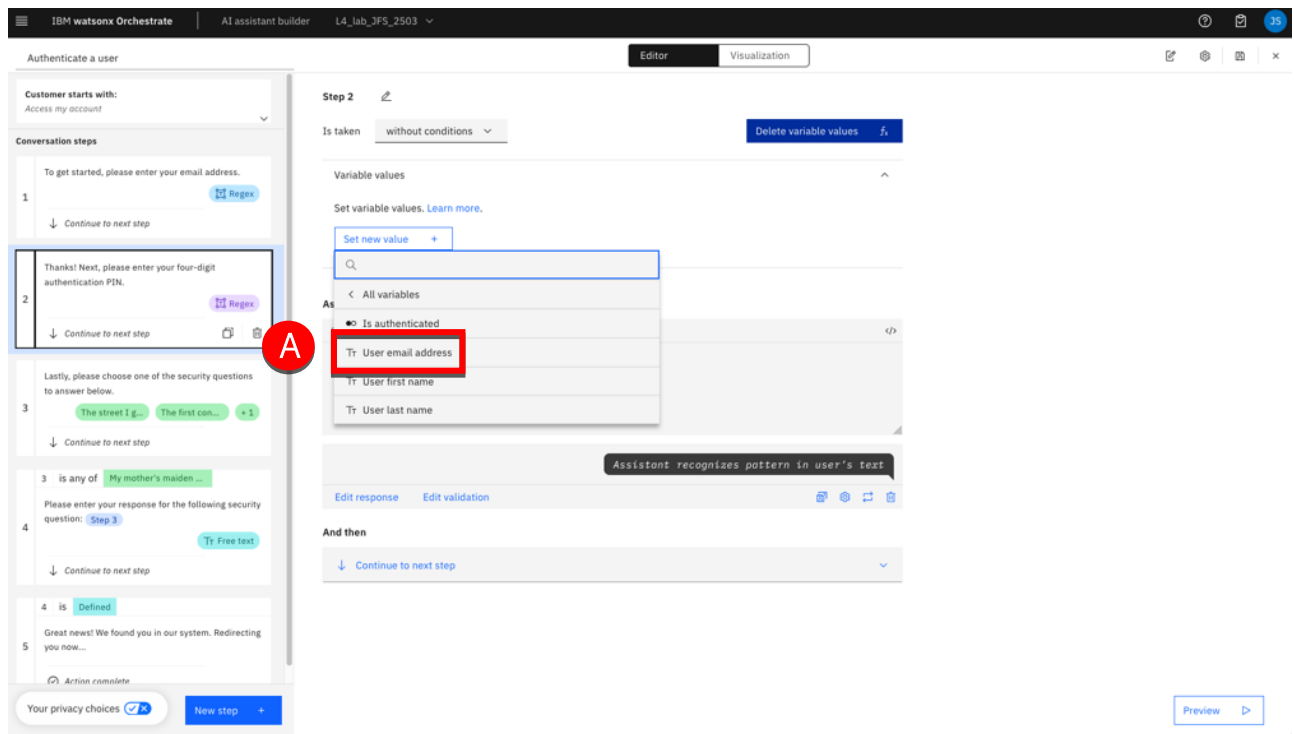
12. Click **Set new value** (A).



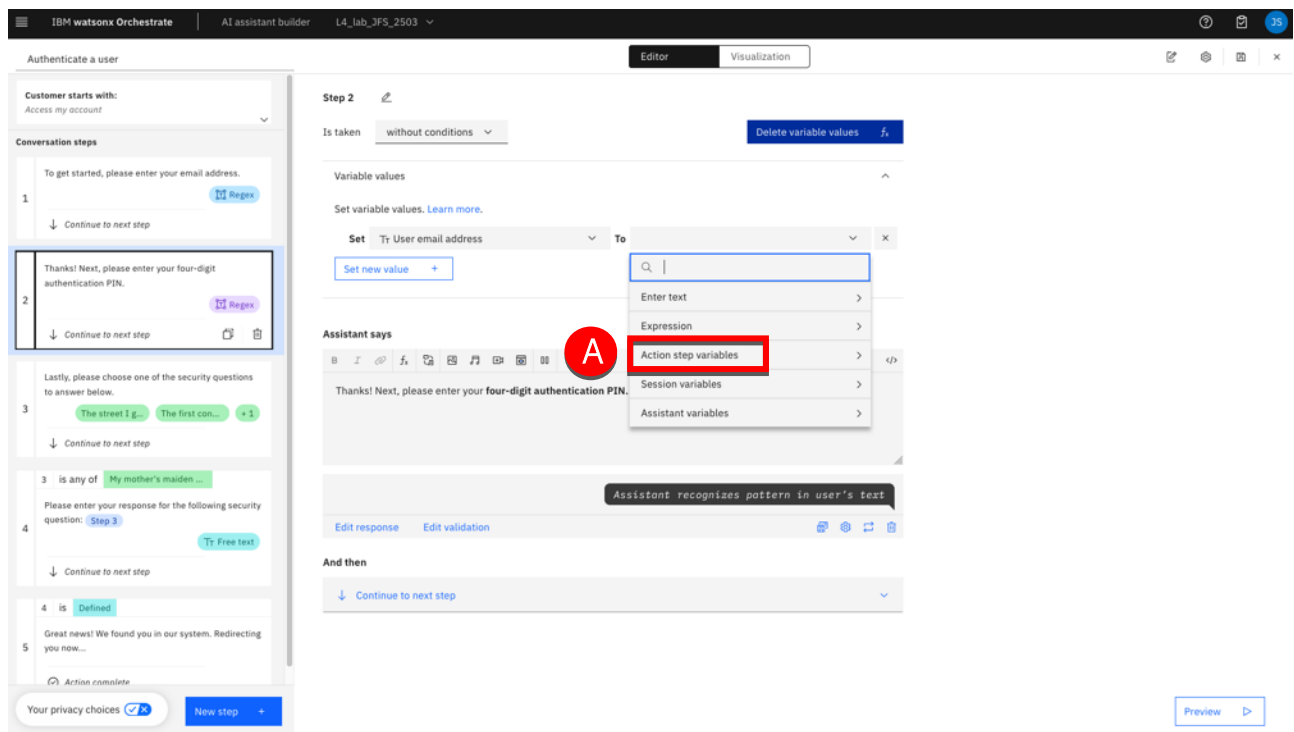
13. Select **Session variables** (A).



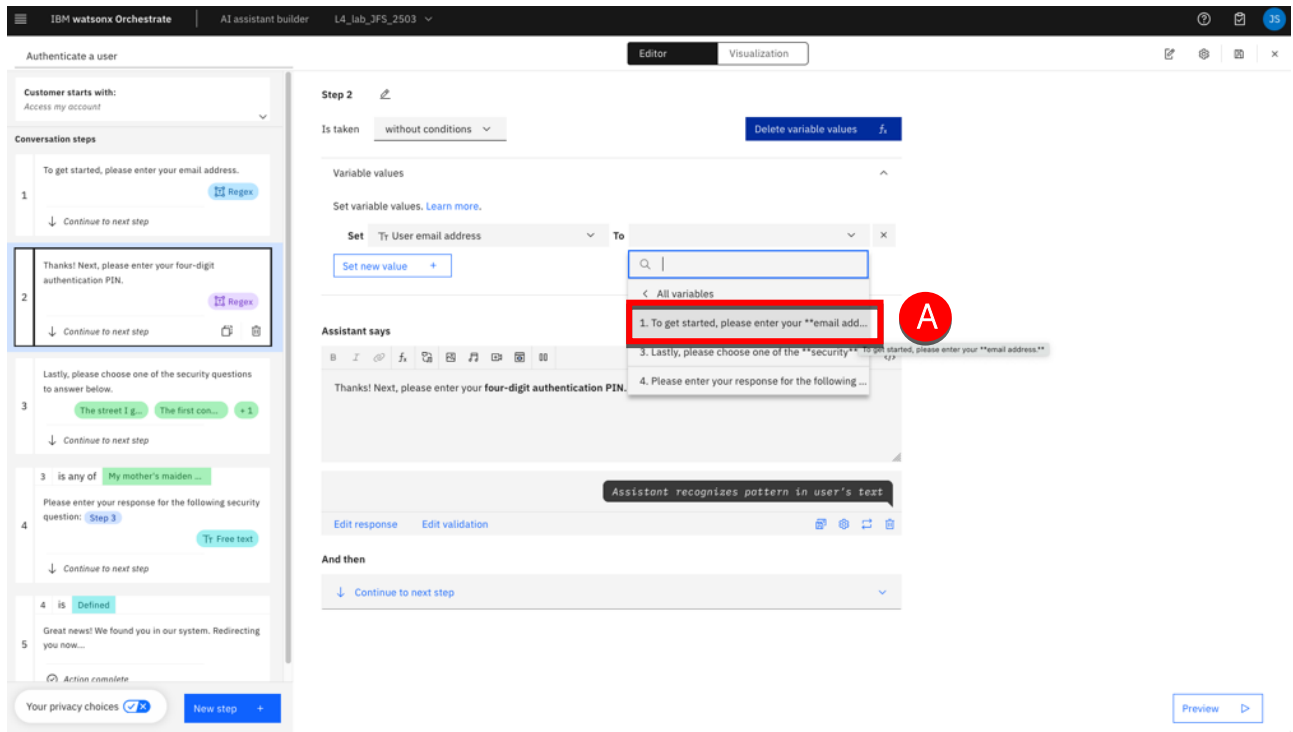
14. Select User email address (A).



15. Click Action step variable (A).

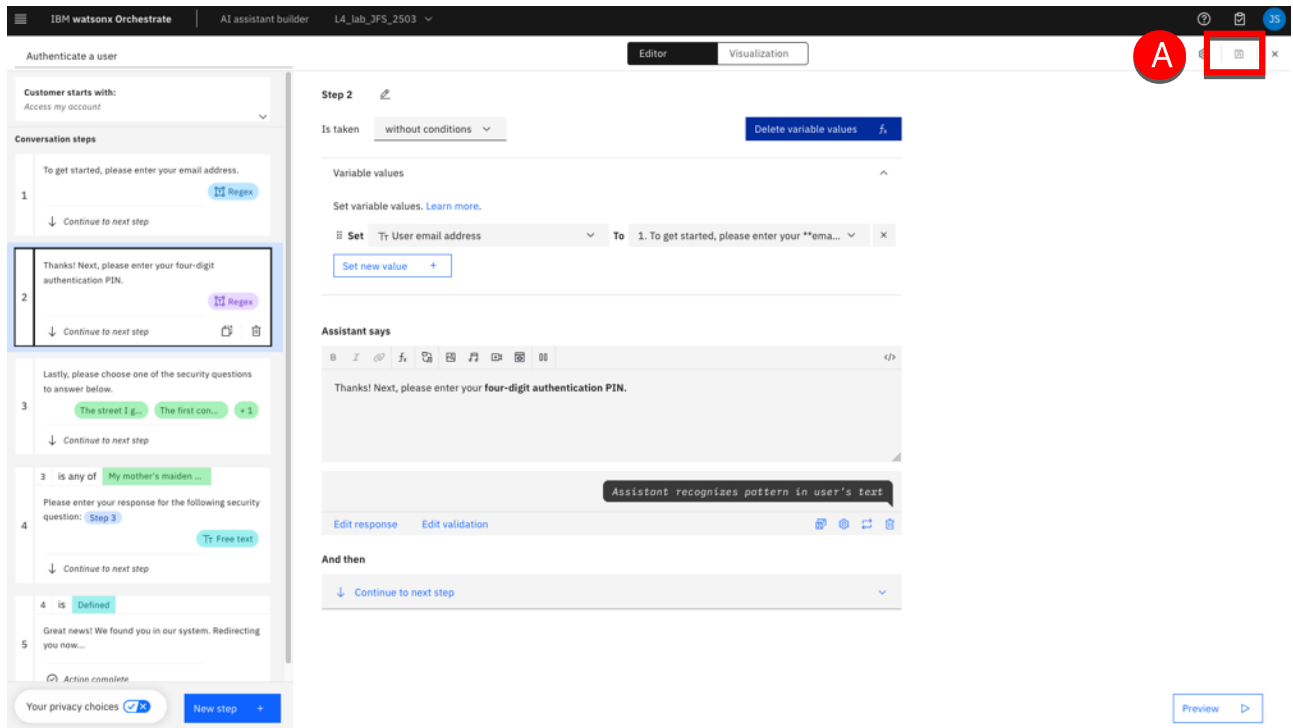


16. Select **1. To get started, please enter your email address (A).**

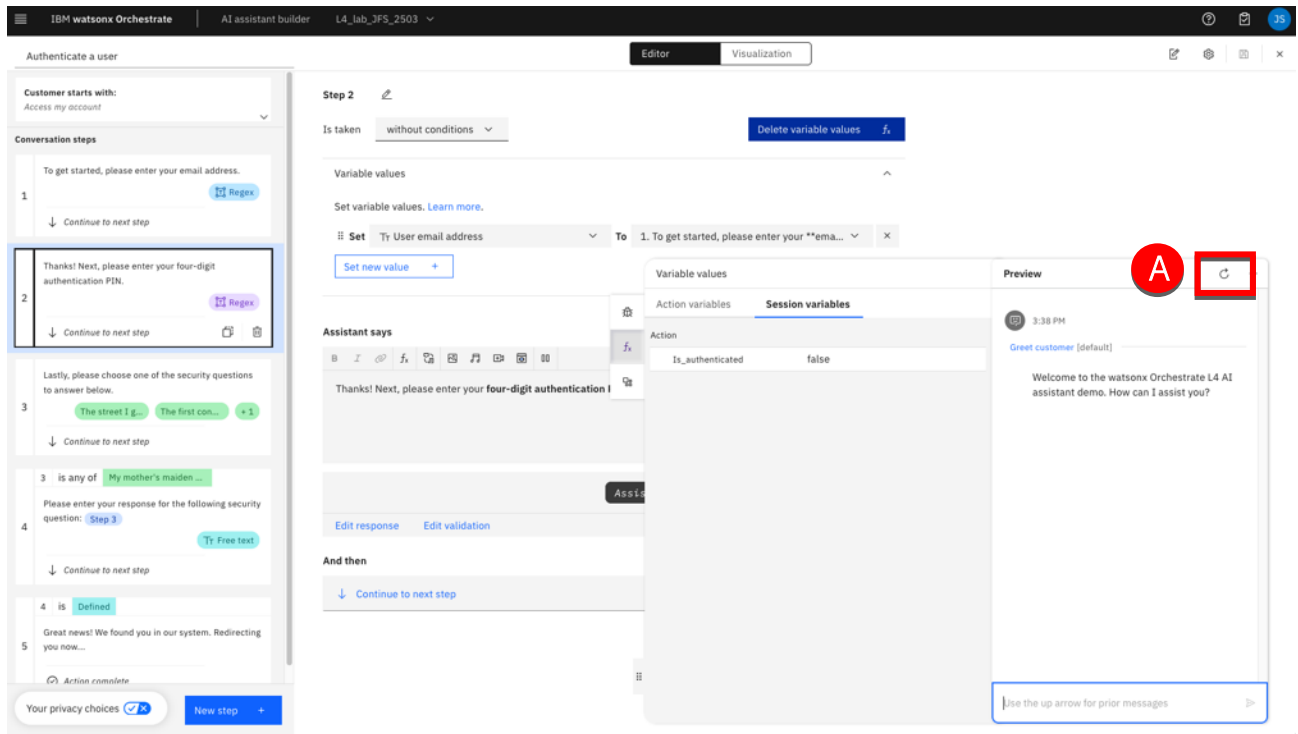


At this point in the action, the **User email address** session variable is initialized with the user answer from step 1.

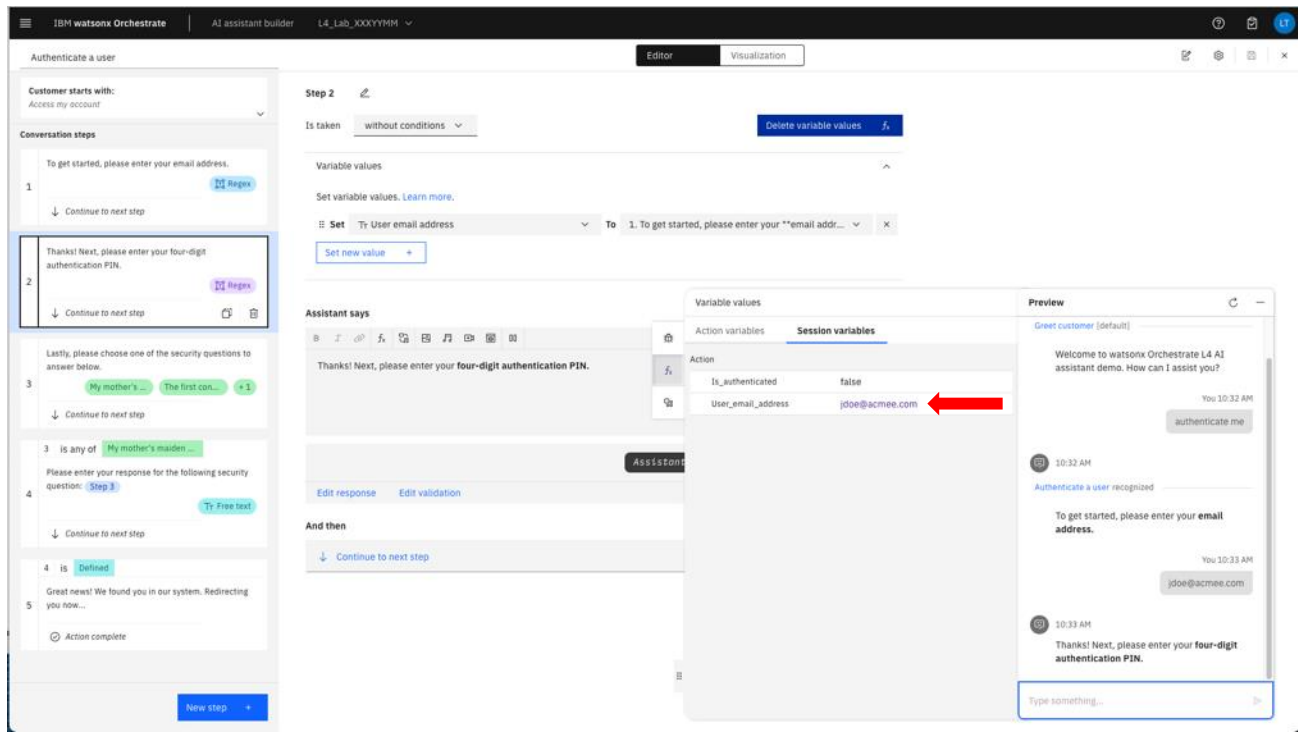
17. Click **Save (A).**



18. Open and refresh the **Preview** window (A).



19. Repeat steps 3 and 4 from this Section (Type Authenticate me, then type your email address). Note that the **User\_email\_address** Session Variable is defined with the value you provided.



20. Type a four digit number to act as your PIN (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main workspace is titled "Authenticate a user" and is in "Step 2" of the flow. The "Conversation steps" panel on the left shows a sequence of steps: 1. "To get started, please enter your email address." (Regex), 2. "Thanks! Next, please enter your four-digit authentication PIN." (Regex), 3. "Lastly, please choose one of the security questions to answer below." (Free text), 4. "Please enter your response for the following security question: (Step 3)." (Free text), and 5. "Great news! We found you in our system. Redirecting you now..." (Defined). The "Assistant says" panel in the center shows the current step's response: "Thanks! Next, please enter your four-digit authentication PIN." The "Variable values" panel on the right shows session variables: "Is\_authenticated" (false) and "User\_email\_address" (jdoe@acmee.com). The "Preview" panel on the far right shows a chat conversation. In the preview, a red box highlights the input field containing "1234", and a red circle with the letter "A" is placed next to it.

21. Select a Security question (A).

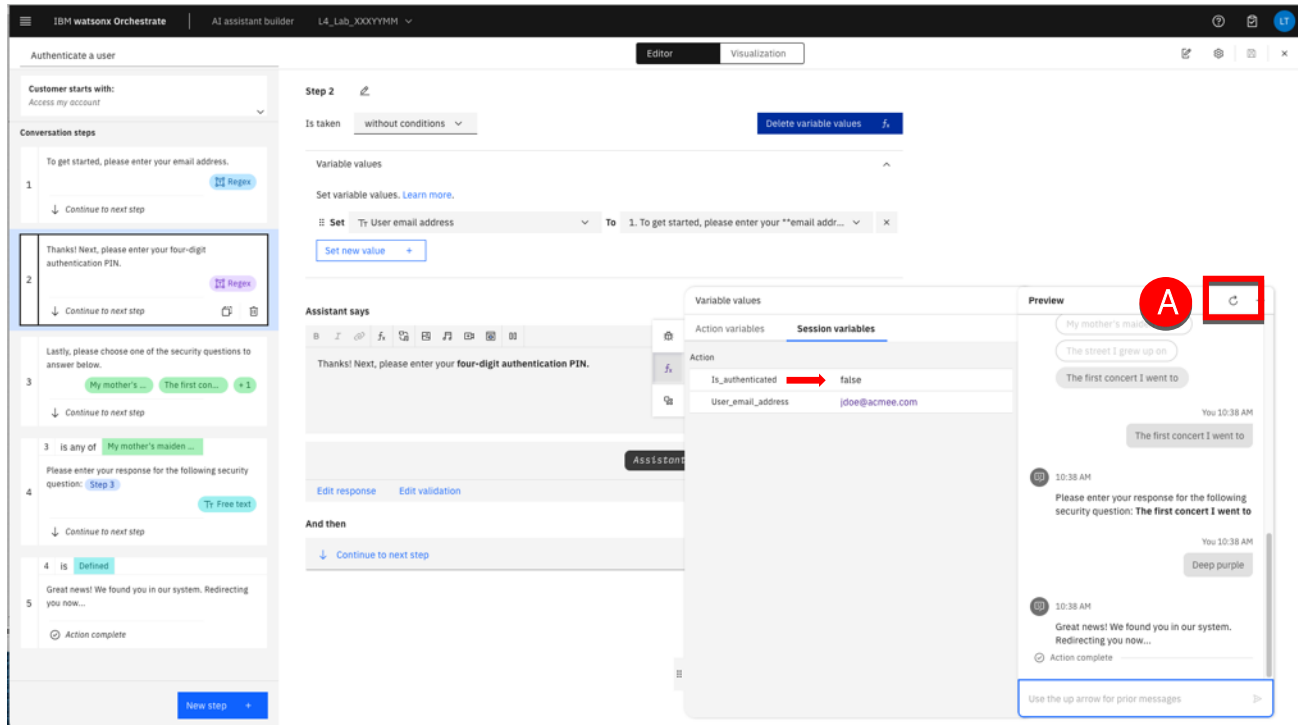
The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main workspace is titled "Authenticate a user" and is in "Step 2" of the flow. The "Conversation steps" panel on the left shows a sequence of steps: 1. "To get started, please enter your email address." (Regex), 2. "Thanks! Next, please enter your four-digit authentication PIN." (Regex), 3. "Lastly, please choose one of the security questions to answer below." (Free text), 4. "Please enter your response for the following security question: (Step 3)." (Free text), and 5. "Great news! We found you in our system. Redirecting you now..." (Defined). The "Assistant says" panel in the center shows the current step's response: "Thanks! Next, please enter your four-digit authentication PIN." The "Variable values" panel on the right shows session variables: "Is\_authenticated" (false) and "User\_email\_address" (jdoe@acmee.com). The "Preview" panel on the far right shows a chat conversation. In the preview, a red box highlights the selected security question "The first concert I went to", and a red circle with the letter "A" is placed next to it.

22. Type a free text answer to this question (A) (You do not have to input sensitive information – any free text value will work for this demonstration)

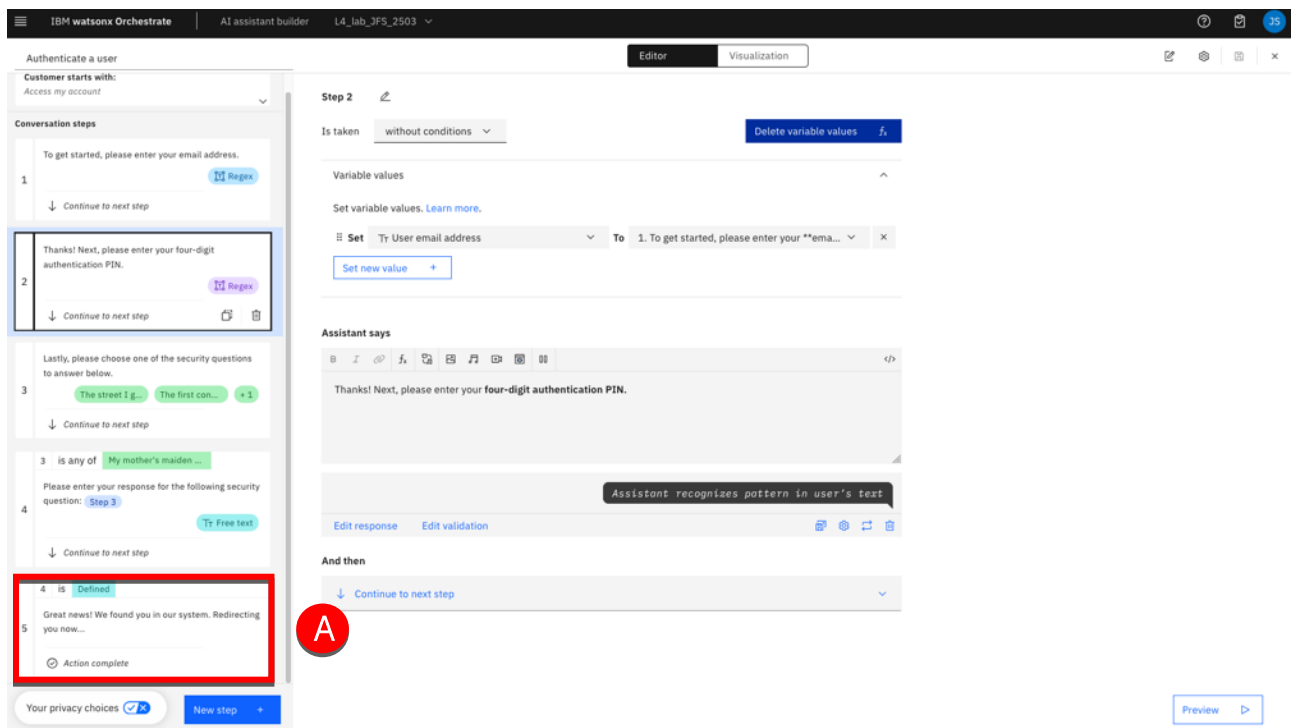
The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The main workspace is divided into several panels:

- Conversation steps:** A list of steps for the authentication process. Step 2, "Thanks! Next, please enter your four-digit authentication PIN.", is highlighted with a blue border.
- Step 2 configuration:** The "Is taken" dropdown is set to "without conditions". The "Variable values" section shows a set of variables: "Set" to "User email address" and "To" to "1. To get started, please enter your \*\*email addr...".
- Assistant says:** The assistant's response is "Thanks! Next, please enter your four-digit authentication PIN.".
- Variable values:** A panel showing session variables: "Is\_authenticated" is false, and "User\_email\_address" is "jdoe@acmee.com".
- Preview:** A chat window showing the conversation history. The assistant's message "Thanks! Next, please enter your four-digit authentication PIN." is visible. Below it, a user input field contains the text "Deep purple" (highlighted with a red box) and a red circle with the letter "A" next to it.

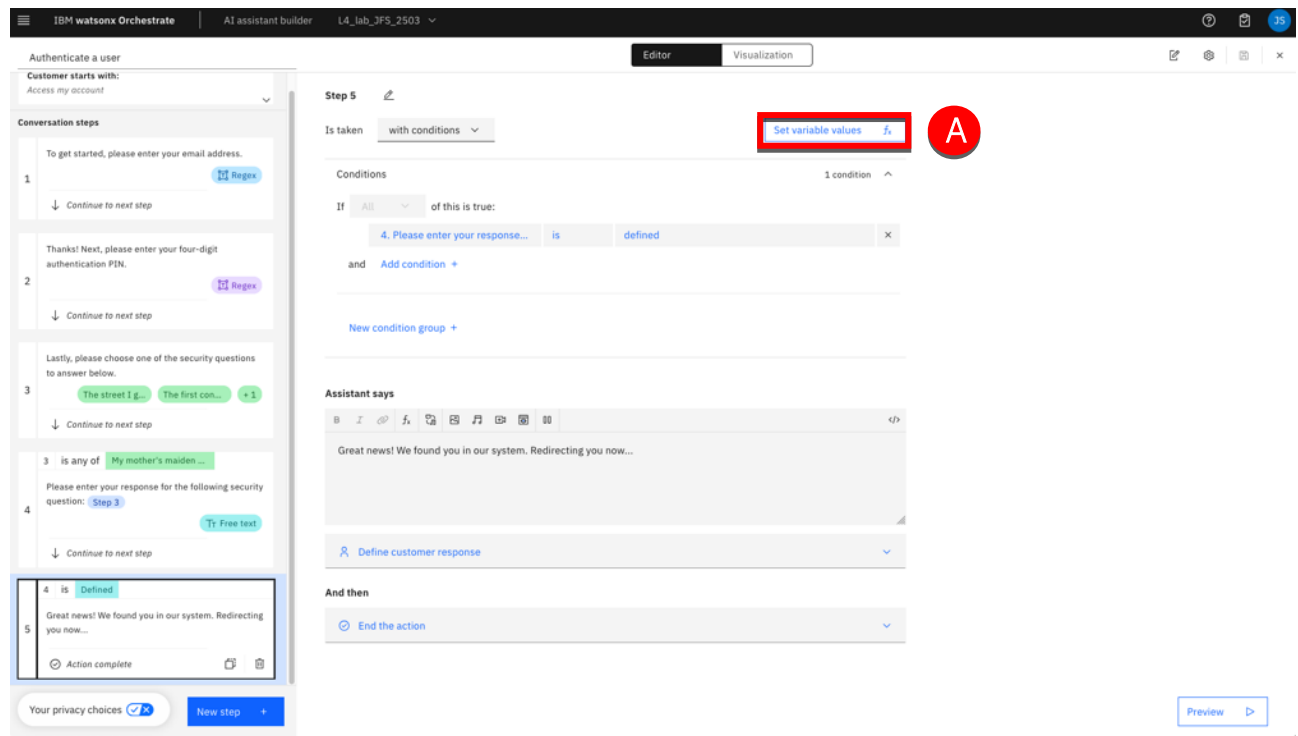
23. You have now completed the Authenticate a user action. However, the **Is\_authenticated** session variable value is still set to **false**, even though you have successfully authenticated. Additionally, you have not defined the first or last name values. You will address these requirements in your next edits to this action. Minimize the Preview window.



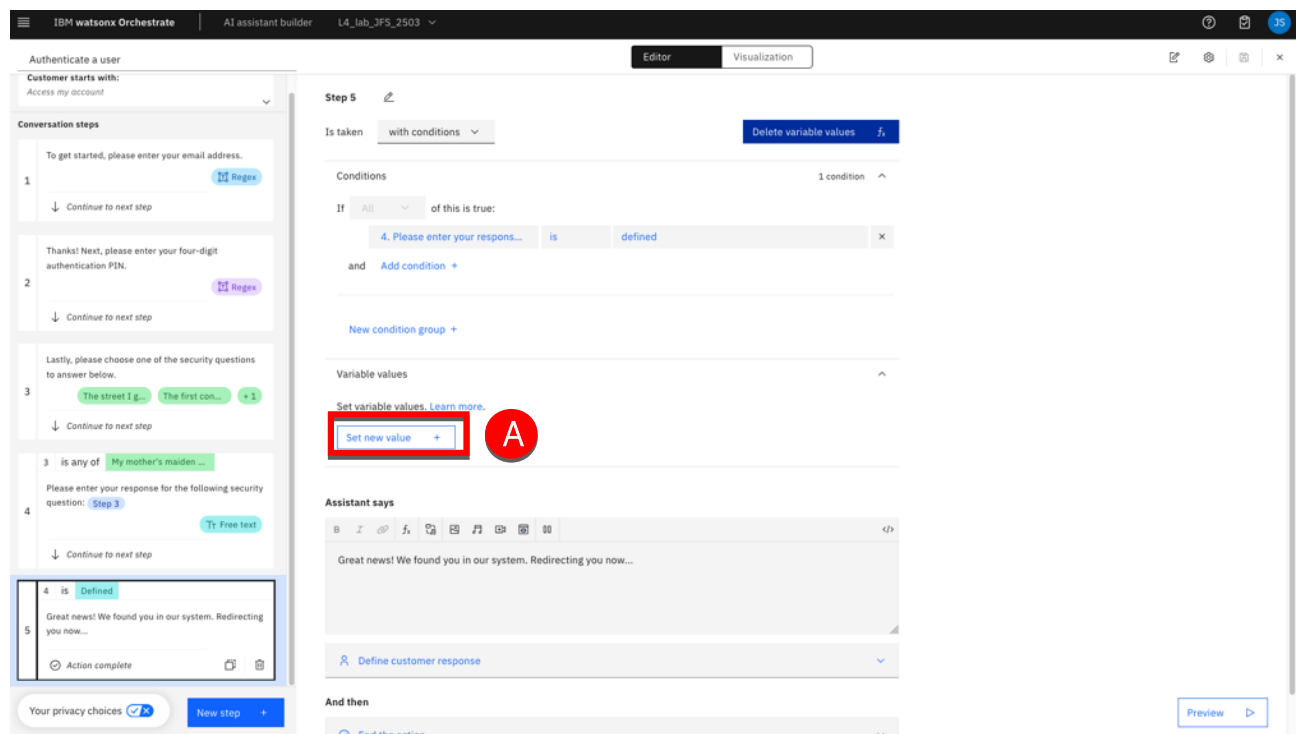
24. Click Step 5 (A).



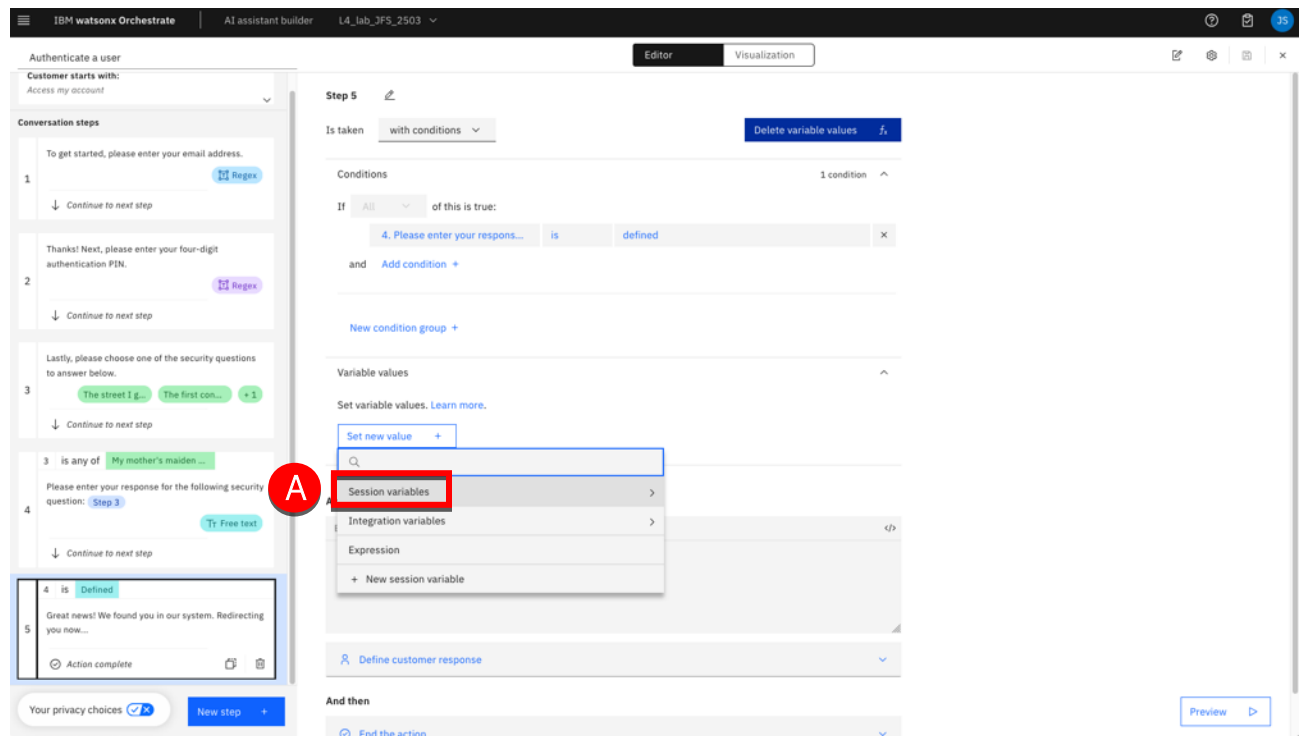
25. Click Set variable values (A).



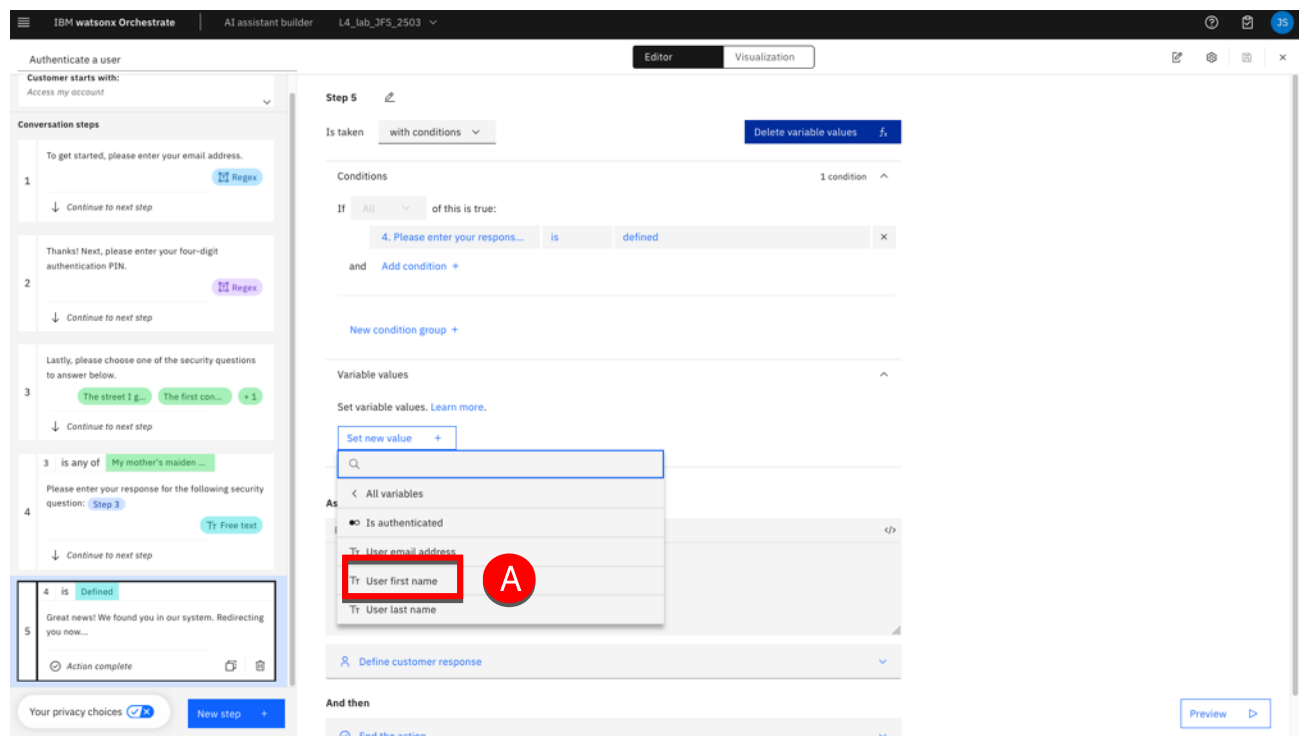
26. Click Set new value (A).



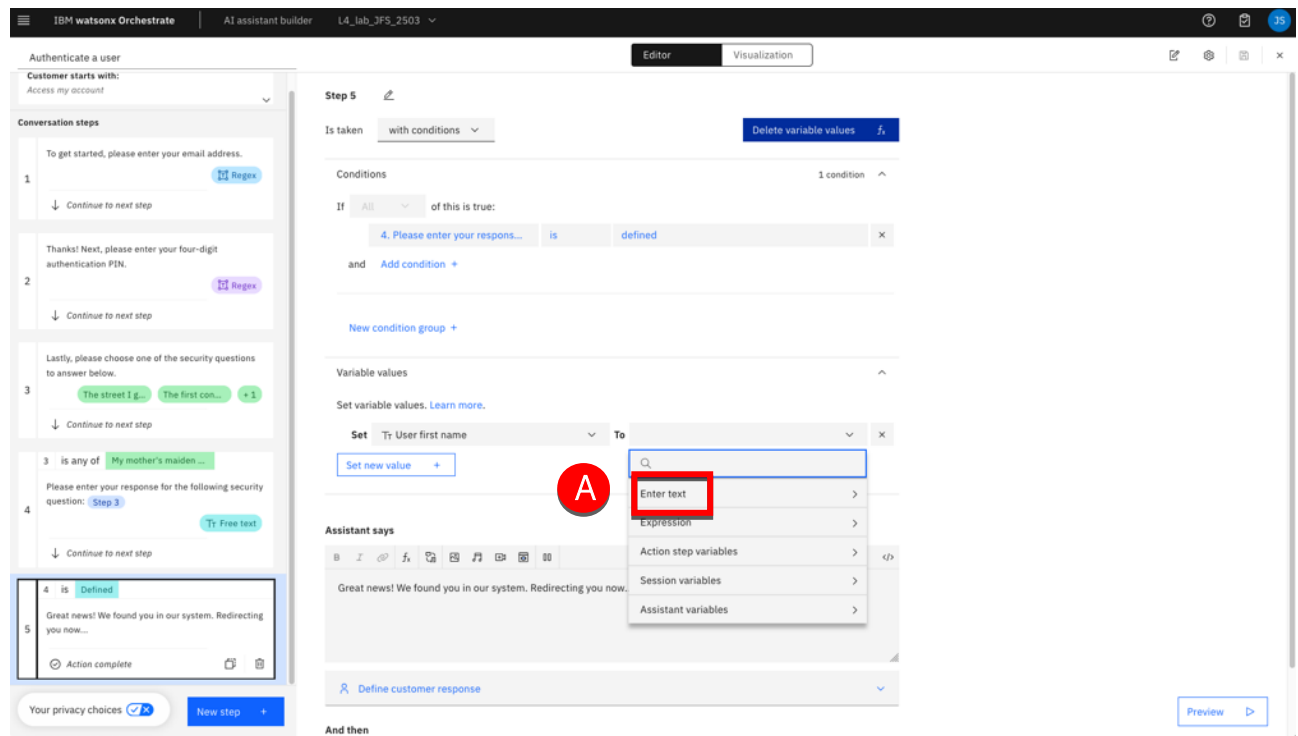
## 27. Select Session variables (A).



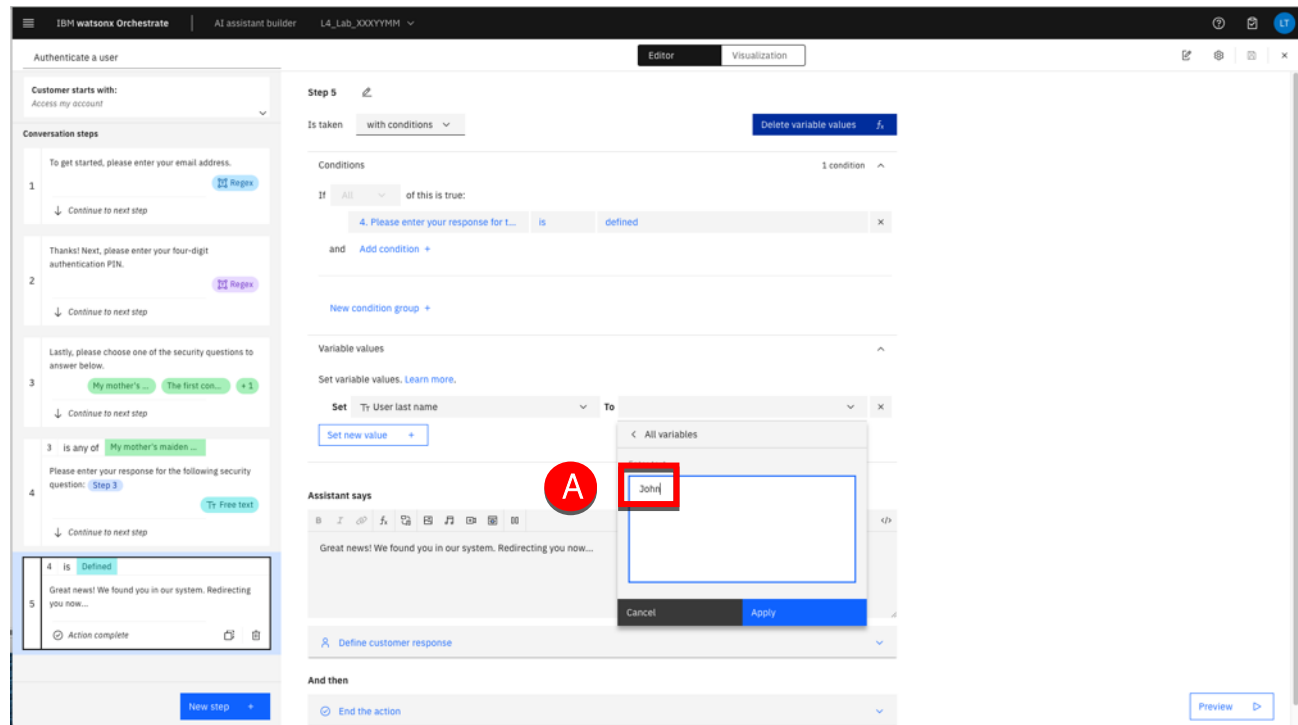
## 28. Select User first name (A).



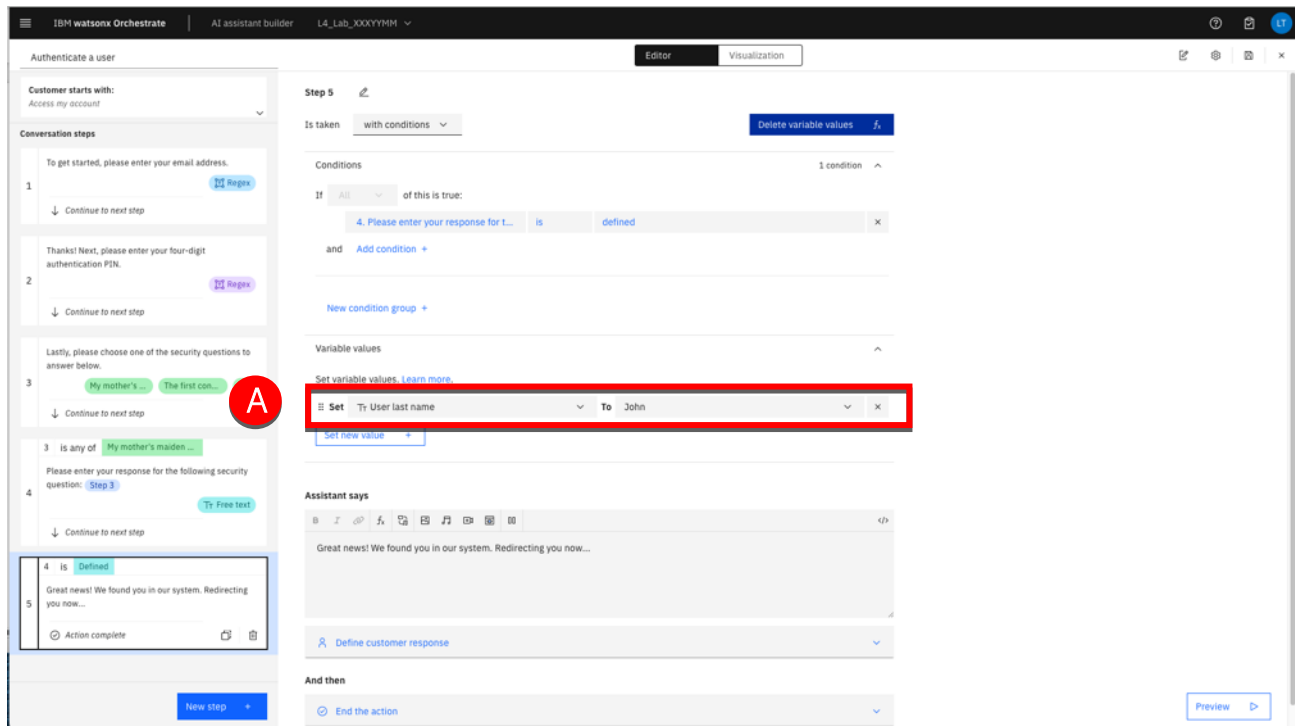
29. Select Enter text (A).



30. Type your first name (A), then click Apply (B).



31. Verify that you have set the **User first name** to a text value (A).



**Note:** In a production setting, an external Identify Authentication Management system would verify that a user inputs a valid email address, PIN, and security question.

If the user input were valid, the external Authentication system would respond with a confirmation and return information such as the user's first and last name.

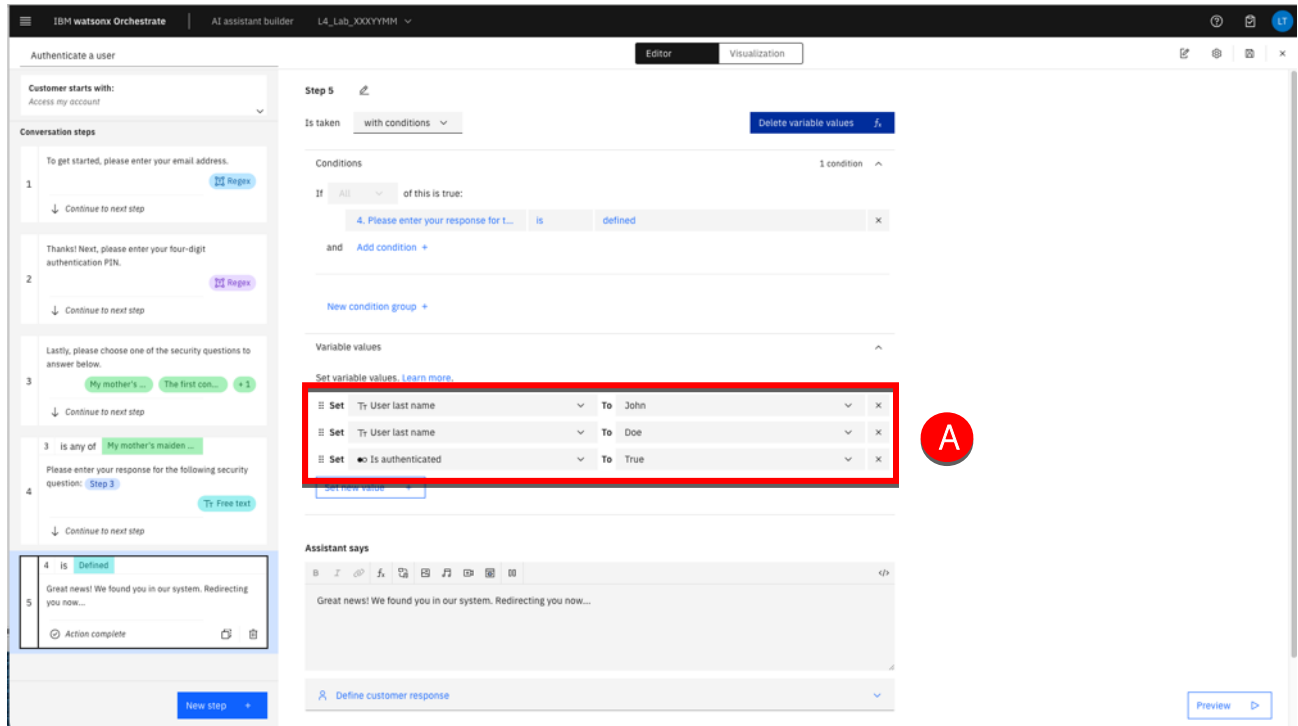
Step 5 simulates this process (no external Authentication system is used in this example). Because you are simulating this process, you provide a first and last name value that an external Authentication system would provide.

32. Repeat steps 26 – 29 in this section, but this time configuring:

- The session variable **User last name** to a **text** value of your choice
- The session variable **Is authenticated** to **True**

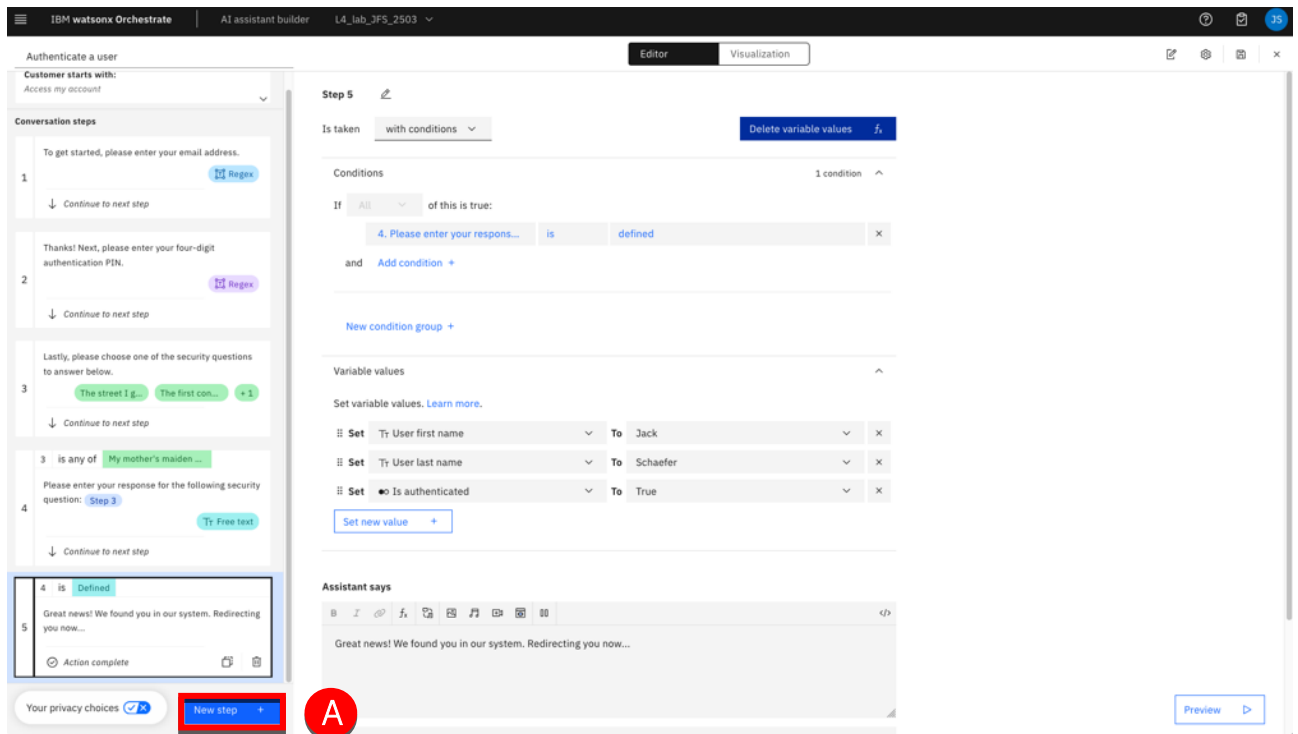
**Note:** When setting the Session Variable Is\_authenticated value, click Select an option, then click True (Do not set to an Expression).

Your **Step 5** should then be configured as shown (A):

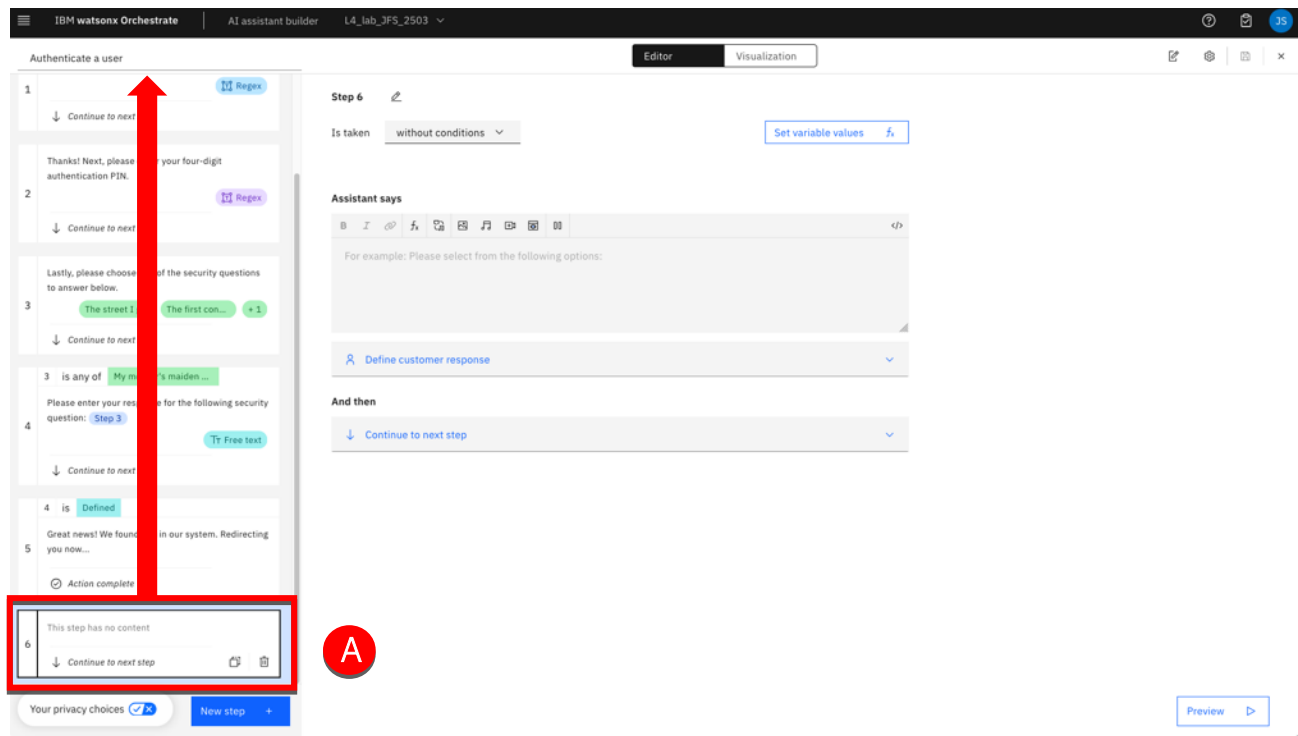


Now, you will configure the **Authenticate a user** action to skip through the action if the user is already authenticated.

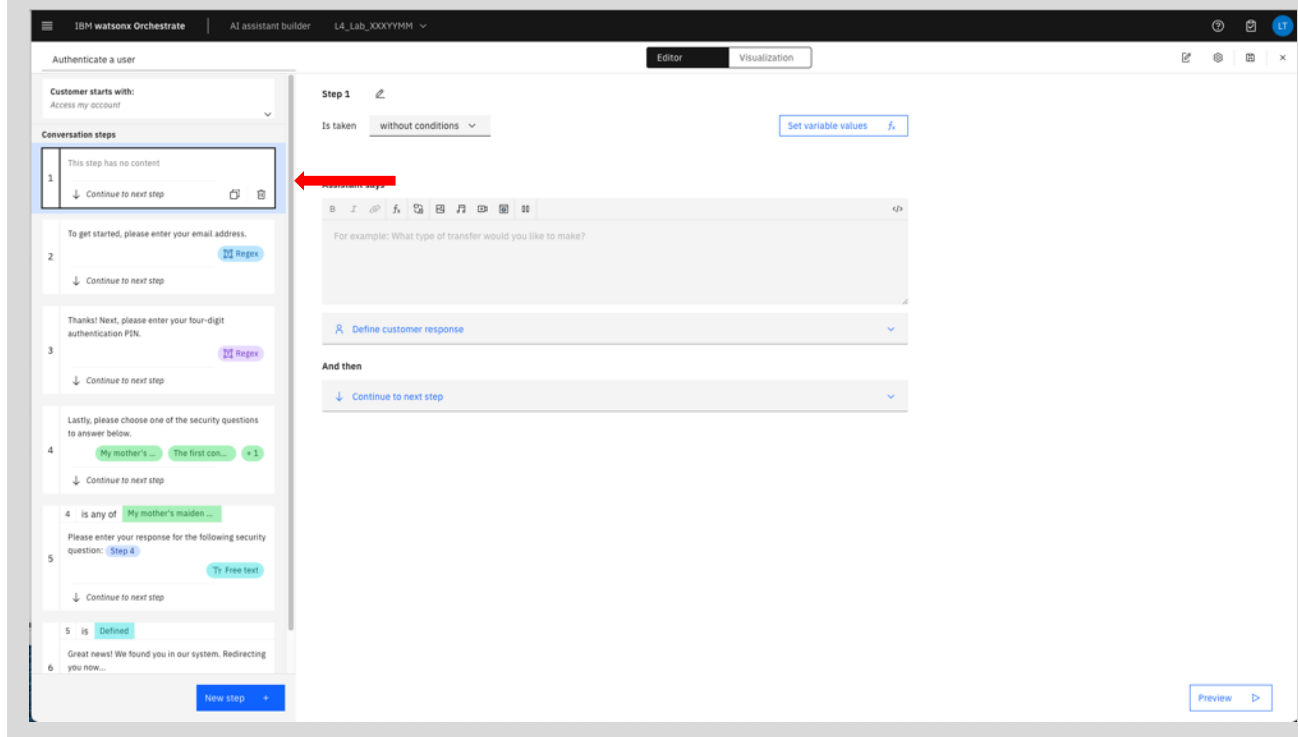
33. Click **New step** (A).



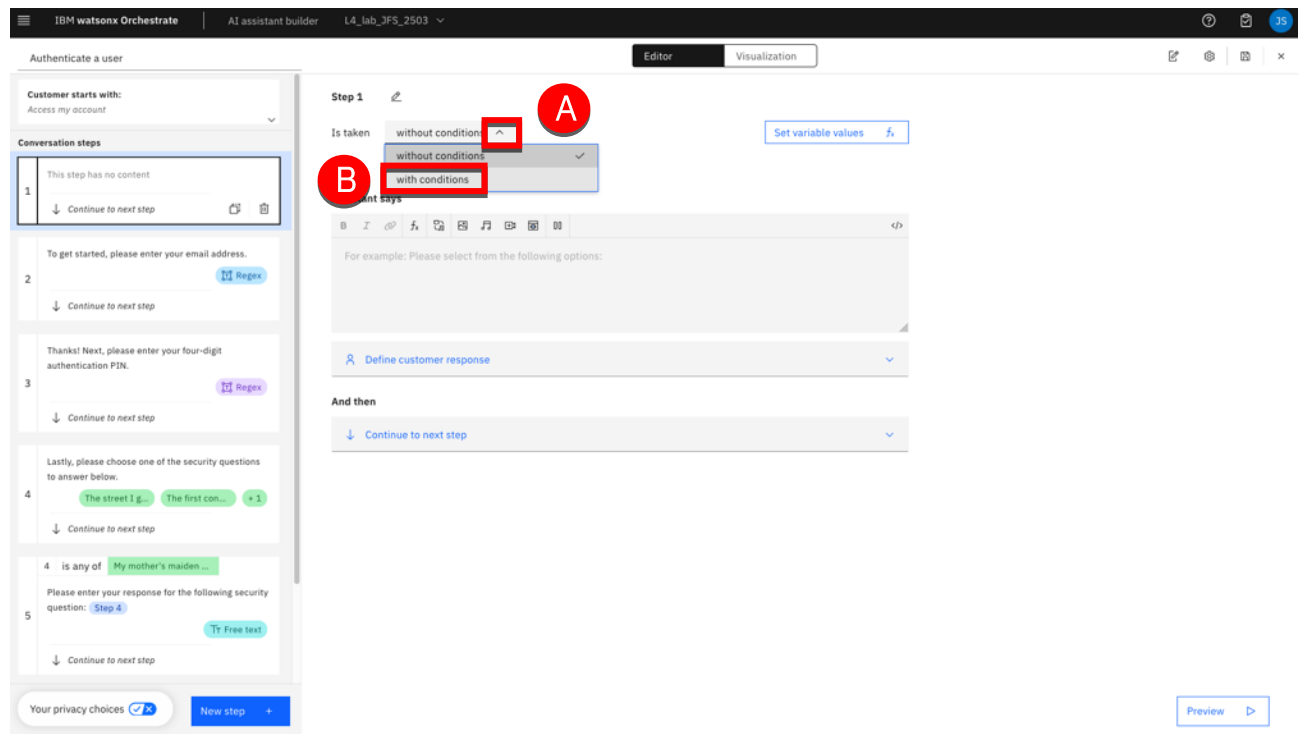
34. Drag the new step to the top position of the action (A).



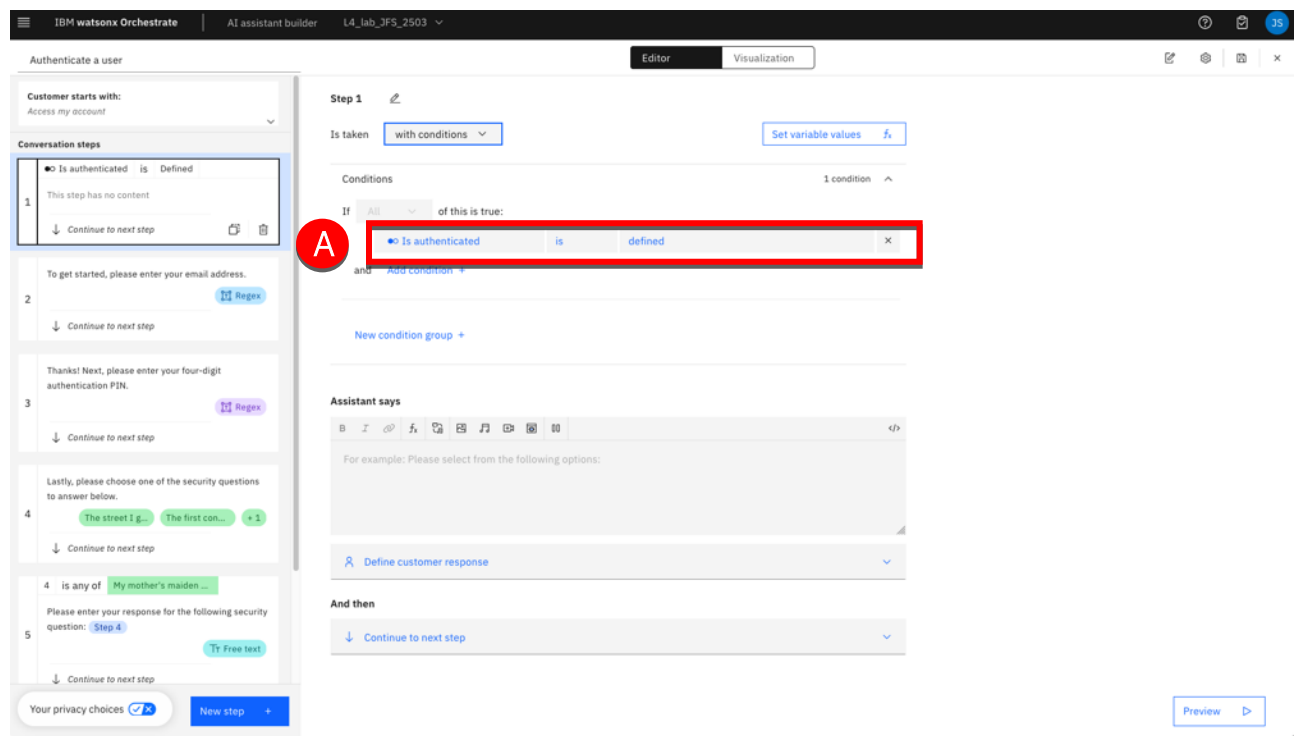
**Note:** The new step is now step 1



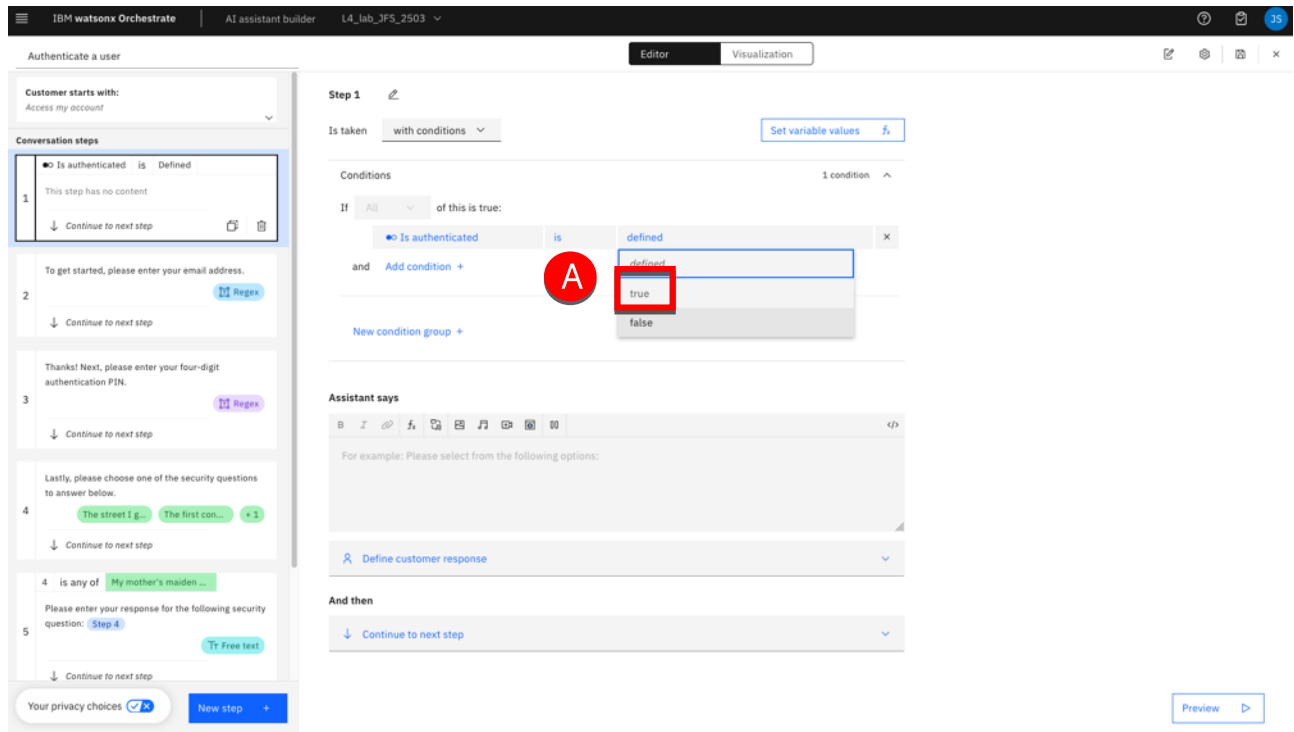
35. Click **Is taken** on step 1 (A) then click set to **with conditions** (B).



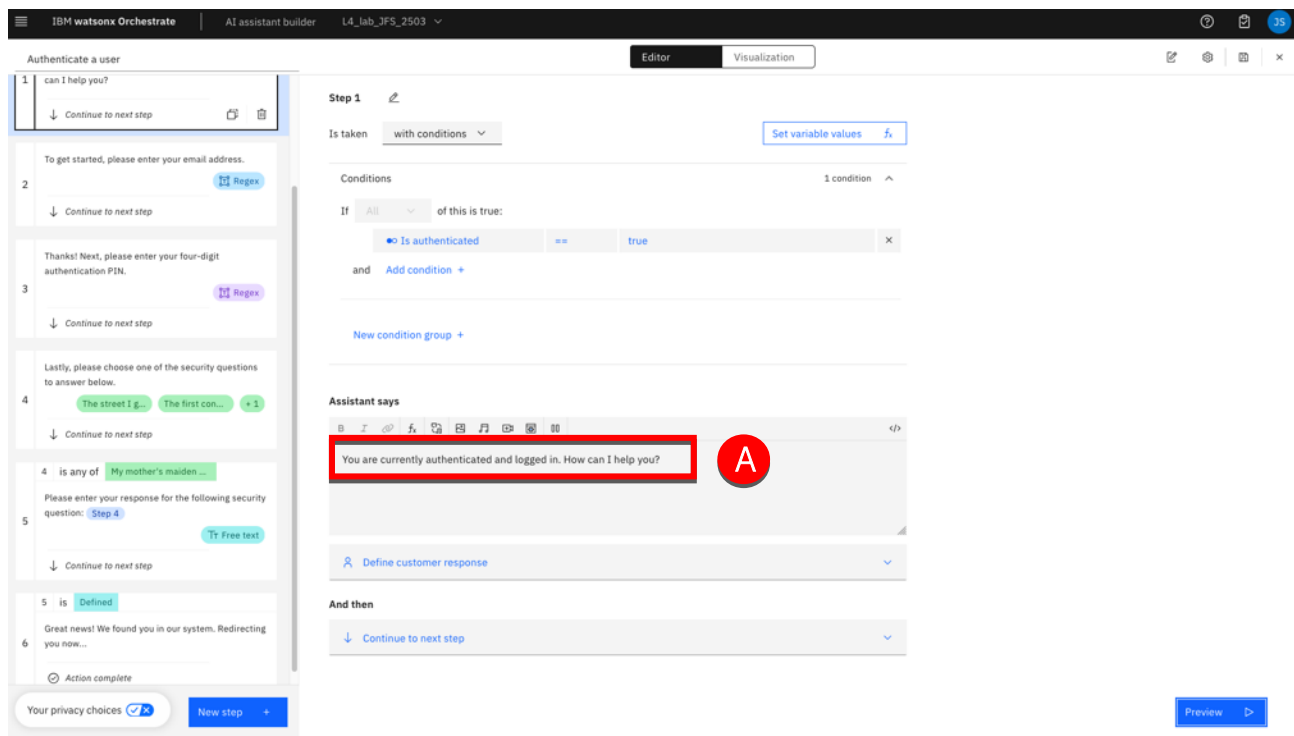
36. Your assistant will automatically generate a condition. Verify that the session variable **Is Authenticated** is listed (if it is not, select the Session Variable Is Authenticated).



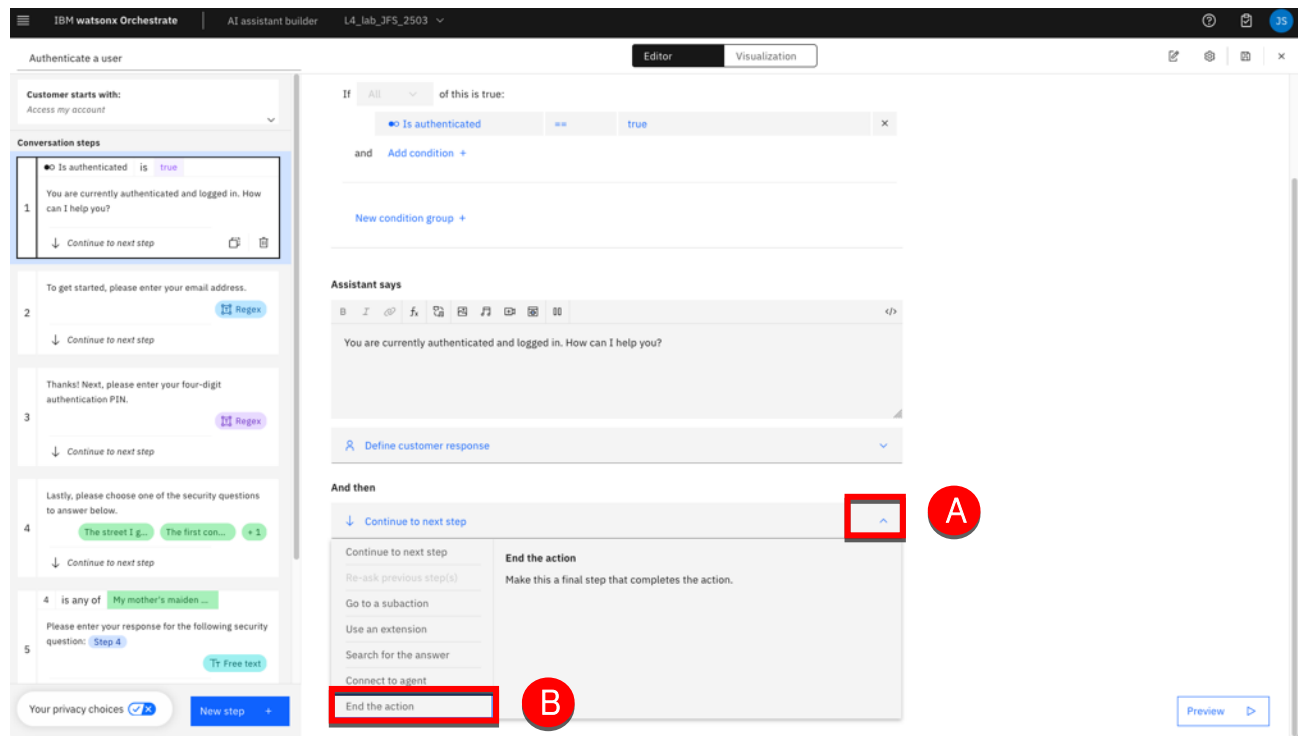
37. Set the value of the condition to **true** (A).



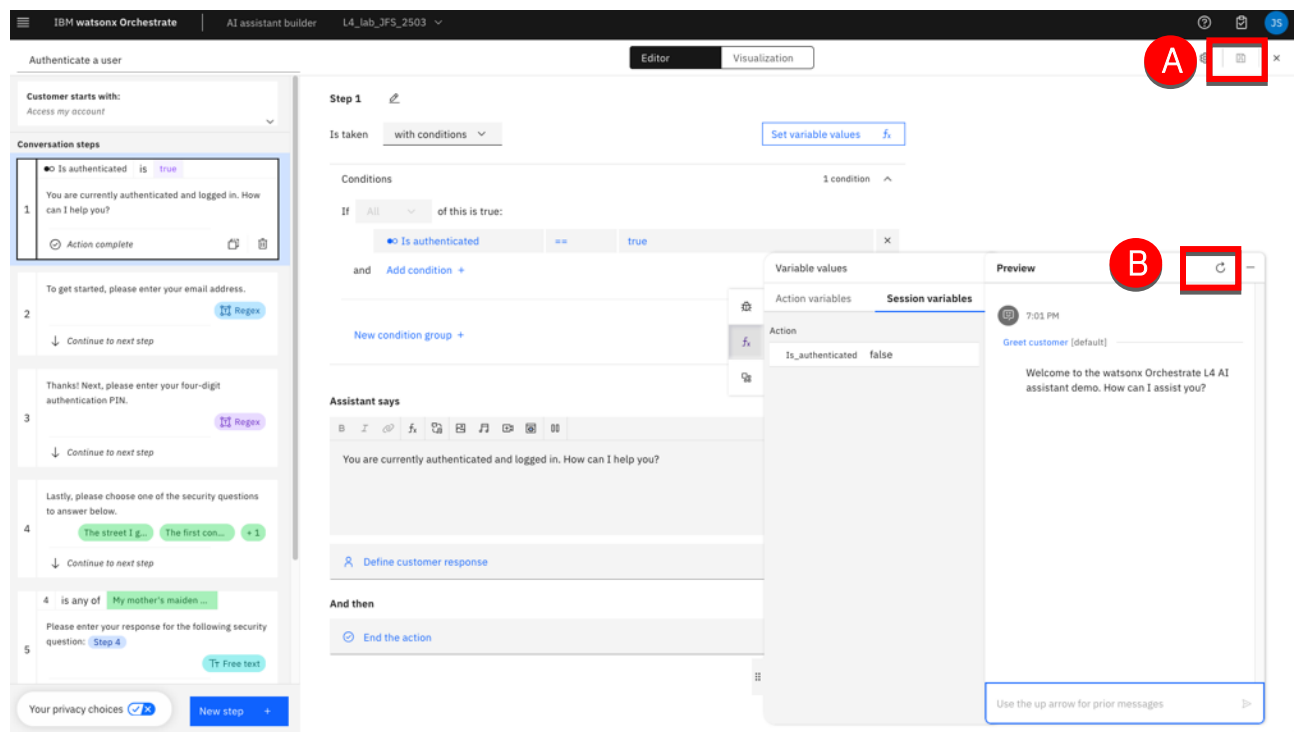
38. In **Assistant says**, type You are currently authenticated and logged in. How can I help you? (A).



39. Click the **And then** dropdown menu (A). Select End the action (B).



40. **Save** the action (A), then open and refresh the **Preview** window (B).



41. Repeat steps 18 – 20 of this section (Type Authenticate me, then provide valid values for the remainder of the Authenticate a user flow). Once finished, observe the Session variables listed (A).

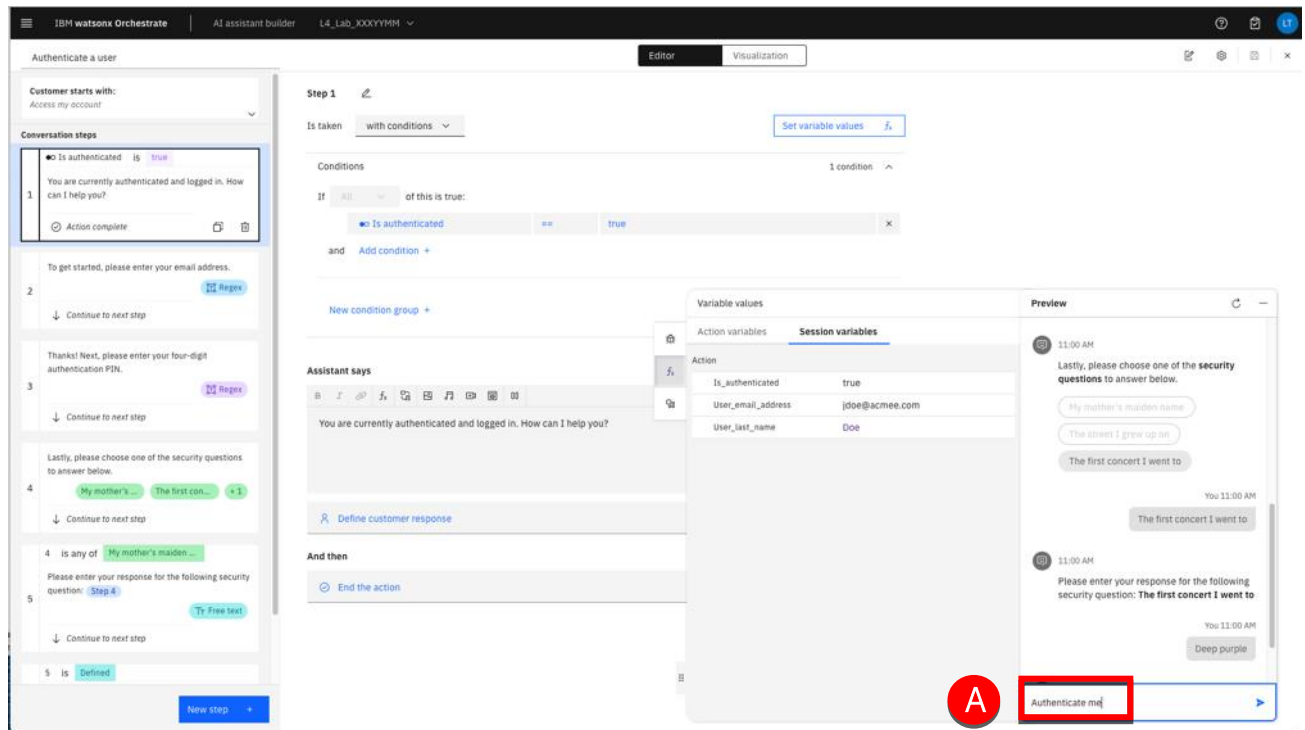
The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface for a flow titled "Authenticate a user".

- Conversation steps (Left Panel):** Shows a sequence of steps. Step 1 is "Is authenticated" with a value of "true". Step 2 is "To get started, please enter your email address." Step 3 is "Thank! Next, please enter your four-digit authentication PIN." Step 4 is "Lastly, please choose one of the security questions to answer below." Step 5 is "Please enter your response for the following security question: The first concert I went to".
- Step 1 Editor (Main Panel):** Shows the configuration for Step 1. The condition is "Is authenticated" set to "true". The action is "End the action".
- Session variables (Right Panel):** A red circle with the letter "A" points to this panel. It lists the following variables:

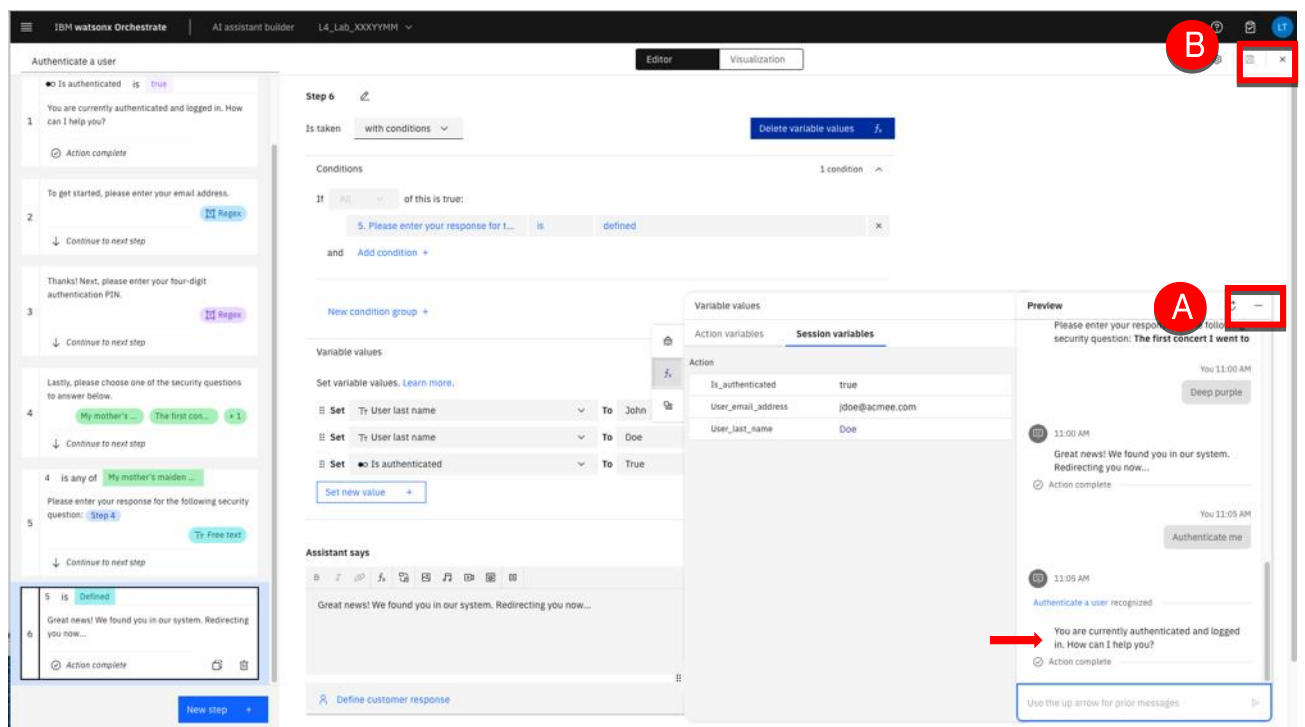
| Variable           | Value          |
|--------------------|----------------|
| Is_authenticated   | true           |
| User_email_address | jdoe@acmee.com |
| User_last_name     | Doe            |
- Preview (Far Right Panel):** Shows a chat conversation. The assistant asks for the user's email address, PIN, and a security question. The user provides "The first concert I went to". The assistant responds with "Great news! We found you in our system. Redirecting you now..." and "Action complete".

Note that **Is authenticated** is now set to true, and that **User first name** and **User last name** are now defined thanks to the direct initialization in Step 6, even though the user did not input their first and last name directly. Remember, you are simulating an external authentication system.

42. Type **Authenticate me** (A).



43. Note that the Authentication action ended without asking the user for duplicate information. Minimize the **Preview** window (A) and click **x** (B) to exit the action editor window.



## Subactions: Core concepts

An action called from another action is referred to as a [subaction](#). Selecting this option switches the conversation flow to another action, then back to the original action once the subaction has been completed.

In other words, you can build an action once and reuse it in the flow of other actions. Leveraging both subactions and session variables is a great way to build reusable actions, encouraging a modular design approach.

In our use case, you will configure the assistant to call the **Authenticate a user action** as a subaction. This serves two purposes

- You only have to build one Authenticate a user action, reducing the building effort
- You refrain from requesting personal information from the user (such as email and name) repeatedly, enhancing the overall user experience.

## Subactions in practice

1. In the Actions window, click **Transfer funds** (A).

The screenshot shows the IBM watsonx Orchestrate interface. The top bar indicates the current project is 'AI assistant builder' with ID 'L4\_lab\_JFS\_2503'. The left sidebar shows the 'Actions' section with a list of items: 'All items', 'Created by you', 'Set by assistant', 'Variables', 'Created by you', 'Set by assistant', 'Set by integration', and 'Saved responses'. The main area displays a table of actions created by the user. The table has columns for 'Name', 'Last edited', 'Examples count', 'Steps count', and 'Status'. The 'Transfer funds' action is highlighted with a red box and a red circle labeled 'A'.

| Name                   | Last edited    | Examples count | Steps count | Status |
|------------------------|----------------|----------------|-------------|--------|
| Open an account        | a day ago      | 8              | 4           | ✓      |
| Check account balance  | a day ago      | 9              | 10          | ✓      |
| Authenticate a user    | 30 minutes ago | 5              | 6           | ✓      |
| Book a meeting         | a day ago      | 12             | 10          | ✓      |
| Check upcoming payment | a day ago      | 25             | 1           | ✓      |
| Make a payment         | a day ago      | 10             | 5           | ✓      |
| Set up autopay         | a day ago      | 6              | 14          | ✓      |
| Transfer funds         | a day ago      | 16             | 13          | ✓      |

2. Navigate through all the action steps to understand the business logic (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Editor Visualization

Customer starts with:  
can you help me send some money

Conversation steps

Before we can transfer any funds, please confirm where you'd like to transfer funds from.

1 is Savings account Checking acco...

Continue to next step

Transfers from savings accounts must be made over the phone. <br /> Please give us a call at +1 (800)...

Action complete

1 is Checking a

Okay! How much would you like to transfer?

3 >= 10000

Transfers over \$10,000 must be made with a live agent.

Connect to agent

3 < 10000

Alright. When would you like to make the transfer?

Date

Your privacy choices New step

Add example phrases:

Enter phrases that a customer types or says to start the conversation about a specific topic. These phrases determine the task, problem, or question your customer has. The more phrases you enter, the better your assistant can recognize what the customer wants.

Enter phrases your customer might use to start this action Total: 16

Enter a phrase

Transfer to checking

transfer to another account

transfer money from one card to another

transfer money from my savings account

transfer money

Transfer funds between my accounts

Transfer funds

set up money transfer

send money into my savings account from checking

send money from checking into my savings account

Preview

While this is a great start for this action, it can be improved. Note that Steps 6 and 7 are a short form of Authentication. It would be better to go to the Authenticate a user action instead.

Moreover, this action will make the user authenticate each time they want to Transfer < \$1000 from their checking account (transfers > \$1000 or from savings must be done with an agent), even if they just authenticated moments before.

You will improve this action by instantiating a subaction.

3. Scroll down, then click Step 6 (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Editor Visualization

Transfers from savings accounts must be made over the phone. <br /> Please give us a call at +1 (800)...

Action complete

1 is Checking account

Okay! How much would you like to transfer?

Continue to next step

3 ≥ 10000

Transfers over \$10,000 must be made with a live agent.

Connect to agent

3 < 10000

Alright. When would you like to make the transfer?

Continue to next step

6 Okay. I'll need to authenticate you before moving forward. <br /> What's the email address associated...

Continue to next step

Thanks. Please enter your four-digit authentication or PIN code.

7

Your privacy choices

New step

Add example phrases:

Enter phrases that a customer types or says to start the conversation about a specific topic. These phrases determine the task, problem, or question your customer has.

The more phrases you enter, the better your assistant can recognize what the customer wants.

Enter phrases your customer might use to start this action Total: 16

Enter a phrase

Transfer to checking

transfer to another account

transfer money from one card to another

transfer money from my savings account

transfer money

Transfer funds between my accounts

Transfer funds

set up money transfer

send money into my savings account from checking

send money from checking into my savings account

Preview

4. Change Step 6 to begin with conditions (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Editor Visualization

3 ≥ 10000

Transfers over \$10,000 must be made with a live agent.

Connect to agent

3 < 10000

Alright. When would you like to make the transfer?

Continue to next step

6 Okay. I'll need to authenticate you before moving forward. <br /> What's the email address associated...

Continue to next step

Thanks. Please enter your four-digit authentication or PIN code.

7

Great. I found you in our system! <br /> Finally, to which account would you like to transfer to?

8 A different ac... Account Cx0... +1

Continue to next step

8 is Account Bx01974917 Account Cx0197...

Your privacy choices

New step

Step 6

Is taken without conditions

without conditions

with conditions

Set variable values

Okay. I'll need to authenticate you before moving forward.

What's the email address associated with your account?

Assistant recognizes pattern in user's text

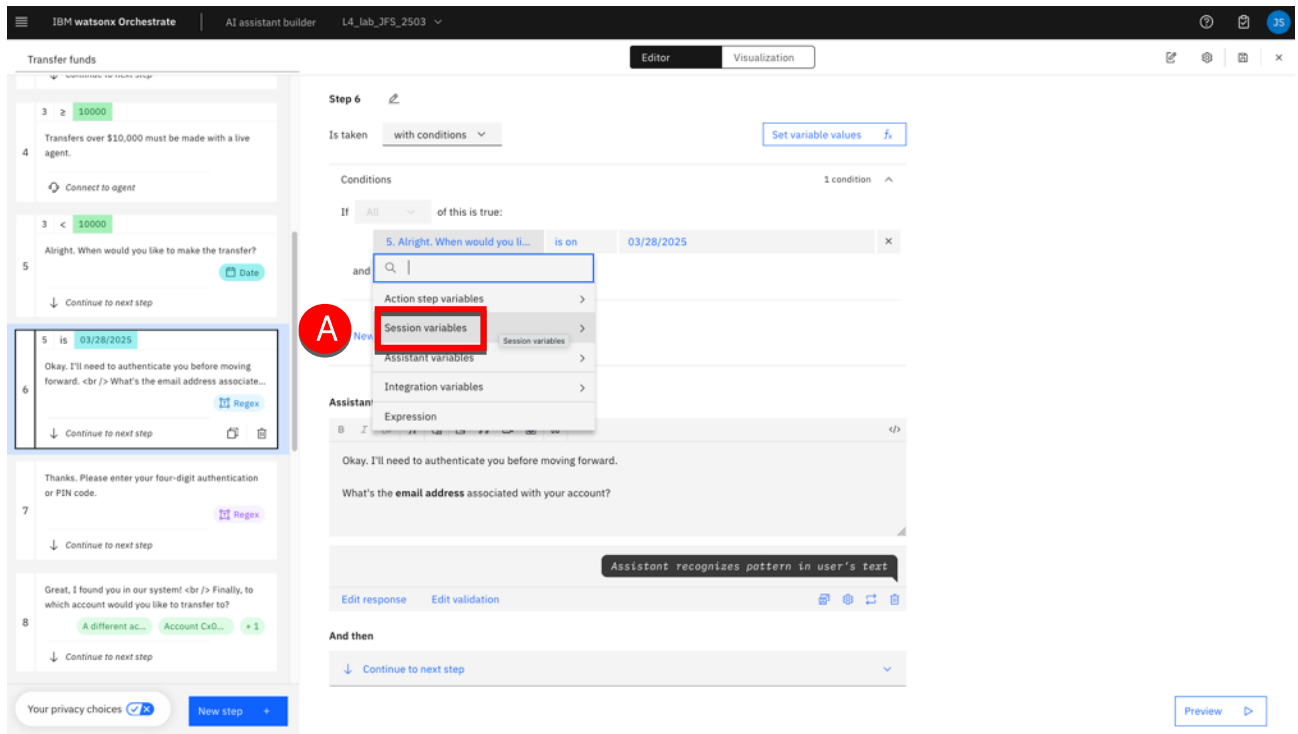
Edit response Edit validation

And then

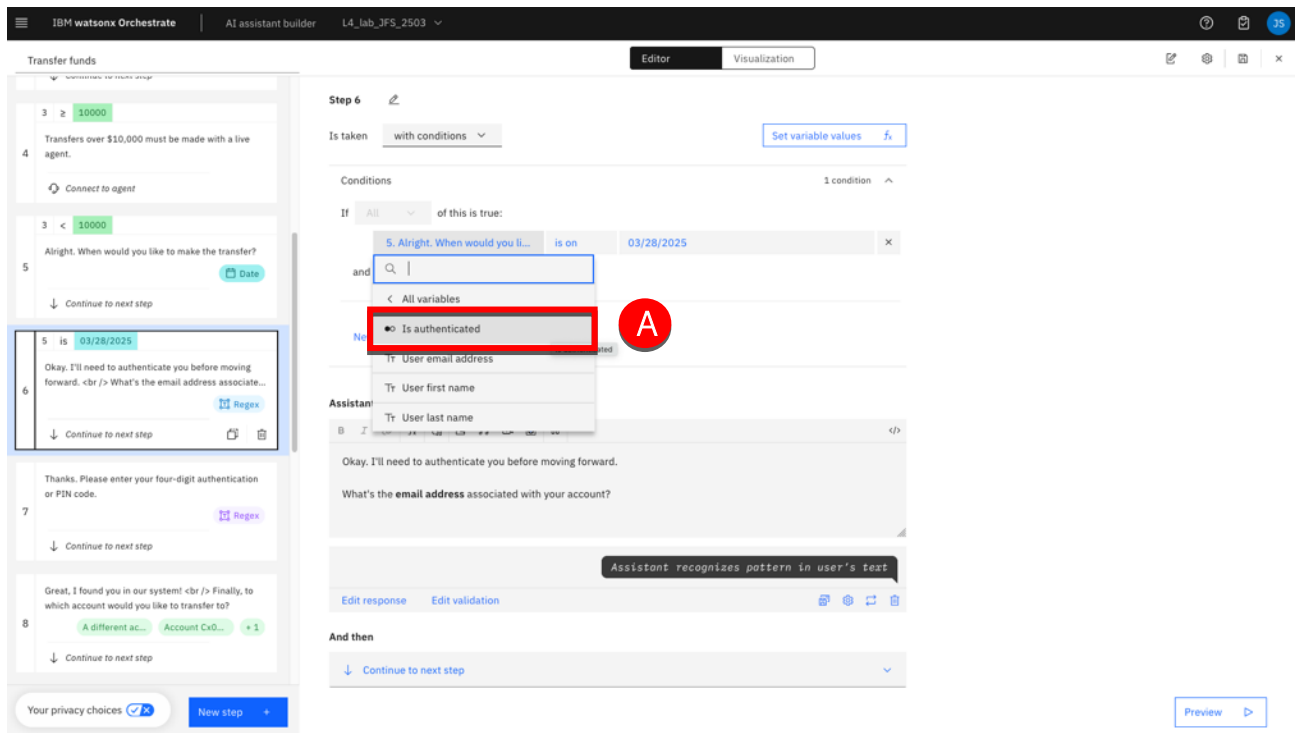
Continue to next step

Preview

5. Set the variable to **Session variables** (A) in the condition editor.



6. Select the variable **Is authenticated** (A).



7. Set the condition's value to **false** (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Step 6

Is taken with conditions

Set variable values

Conditions 1 condition

If All of this is true:

- Is authenticated == true

and Add condition +

defined

true

false

New condition group +

Assistant says

Okay, I'll need to authenticate you before moving forward.

What's the email address associated with your account?

Assistant recognizes pattern in user's text

Edit response Edit validation

And then

Continue to next step

Preview

8. In the Assistant says section, delete the text **What's the email address associated with your account?** (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Step 6

Is taken with conditions

Set variable values

Conditions 1 condition

If All of this is true:

- Is authenticated == false

and Add condition +

New condition group +

Assistant says

Okay, I'll need to authenticate you before moving forward.

What's the email address associated with your account?

Assistant recognizes pattern in user's text

Edit response Edit validation

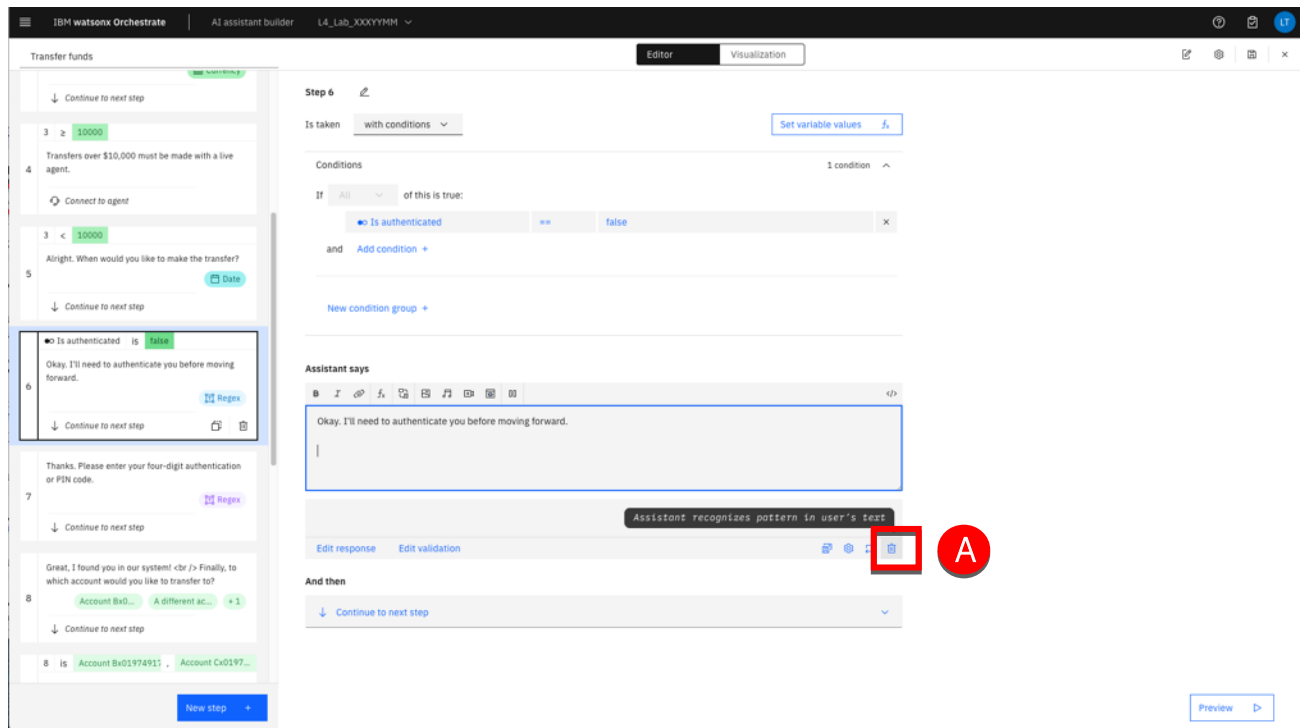
And then

Continue to next step

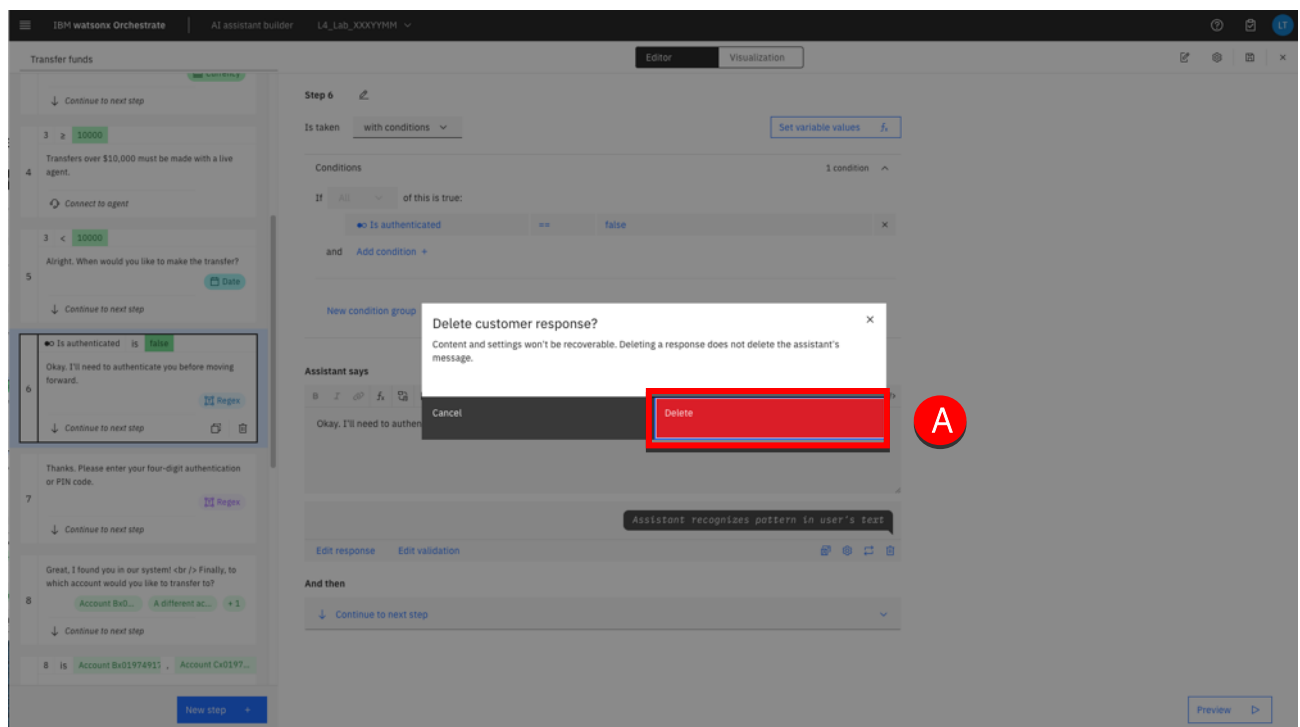
Preview

Let's now invoke the Authentication subaction instead of asking for the credentials from the action itself.

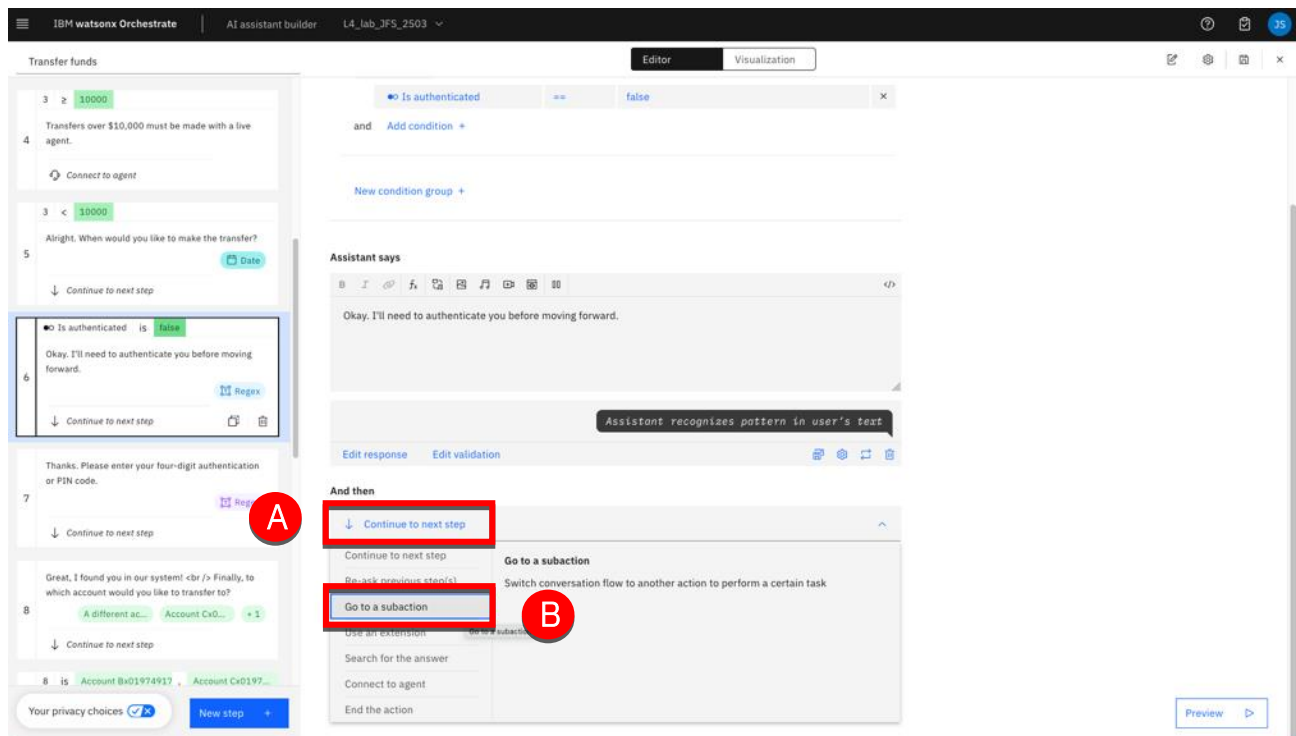
9. In the define customer response, click the delete button (A).



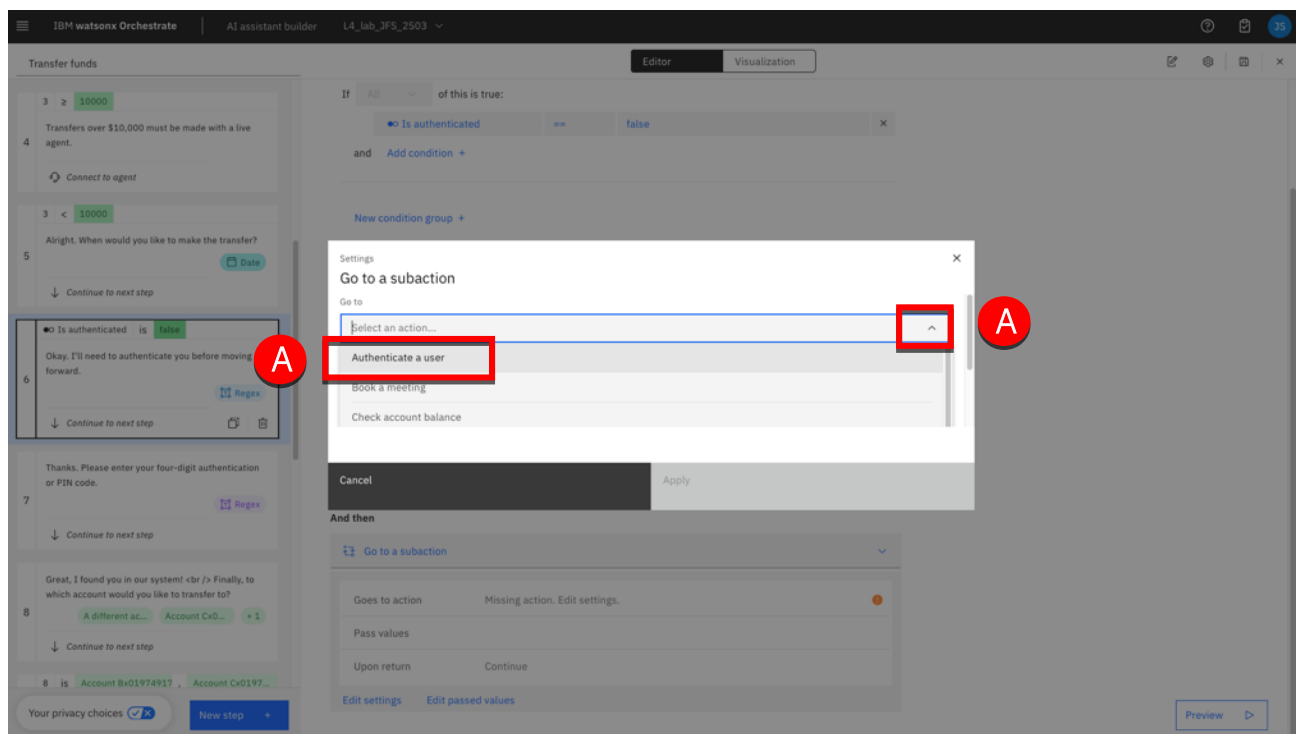
10. Click Delete (A).



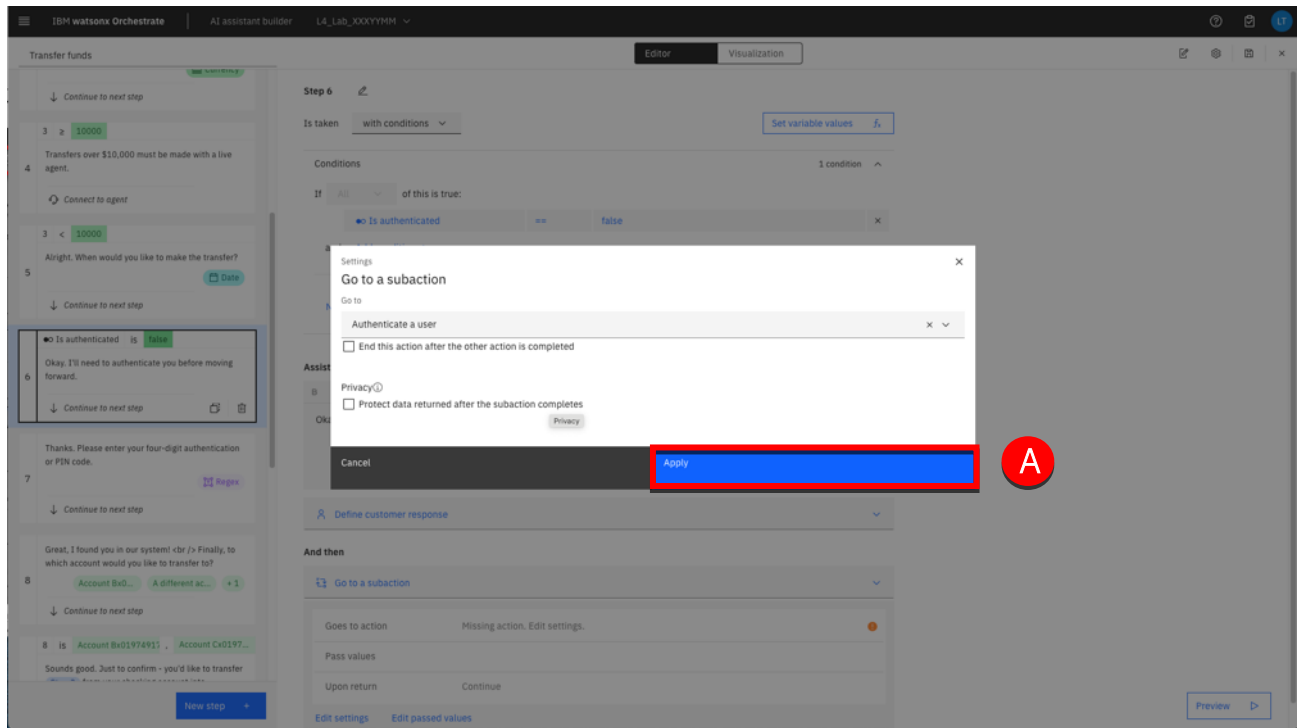
11. Click the **And then** field (A) then select **Go to a subaction** (B).



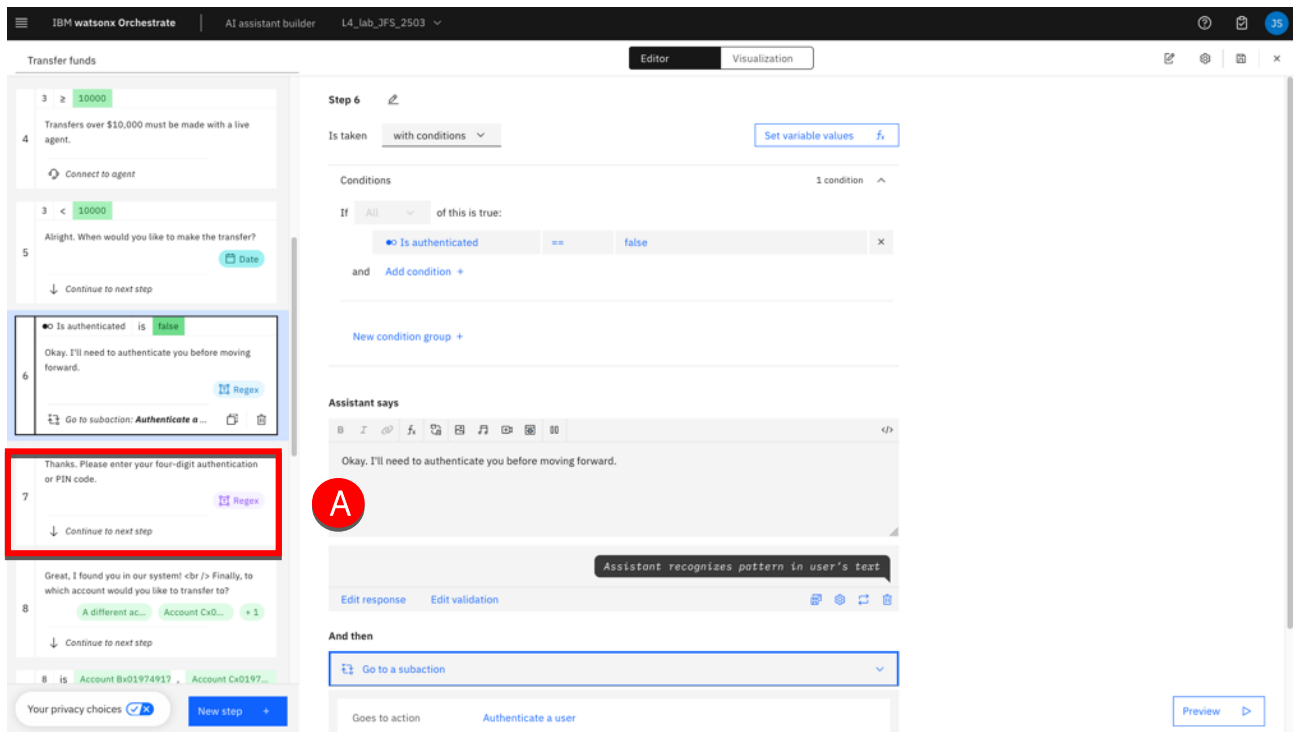
12. Expand the **Select an action...** dropdown menu (A), then select **Authenticate a user** (B).



13. Click **Apply** (A).

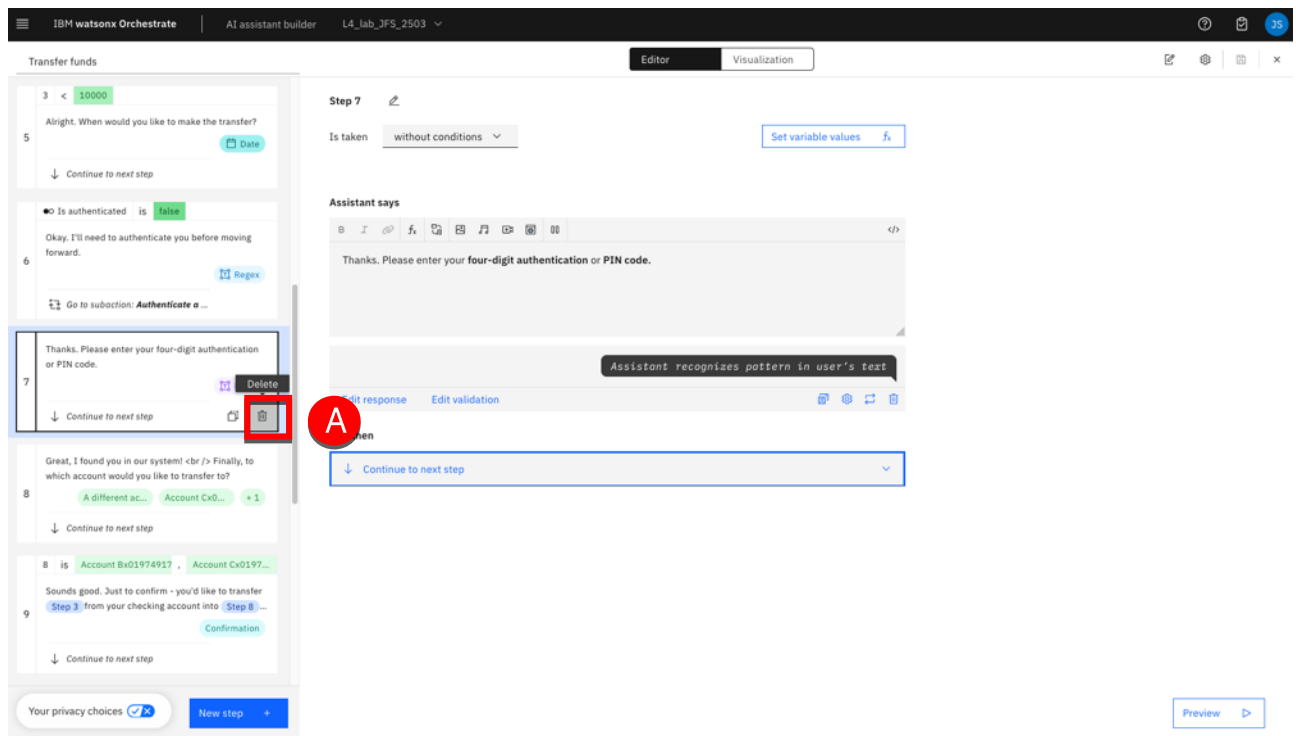


14. Select step 7 (A).

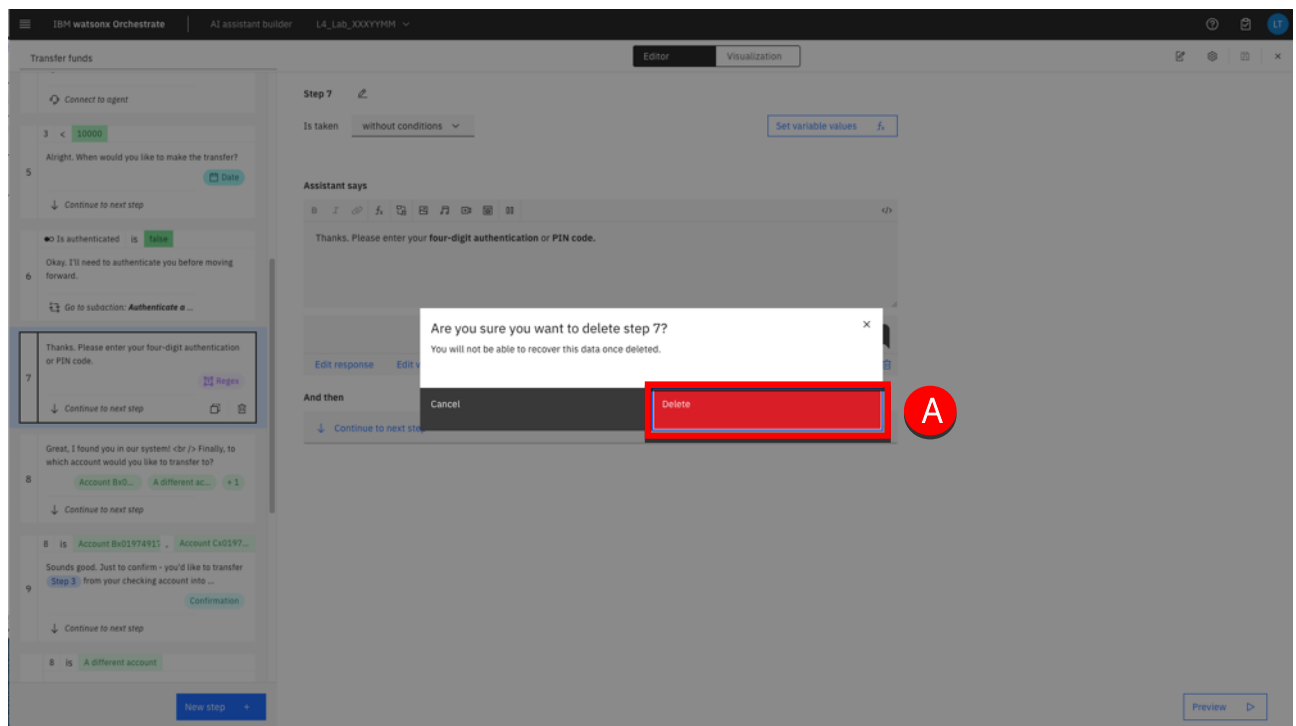


Step 7 is no longer required since the authentication is now performed from the subaction invoked from step 6.

15. Click Delete (A).



16. Click Delete (A).



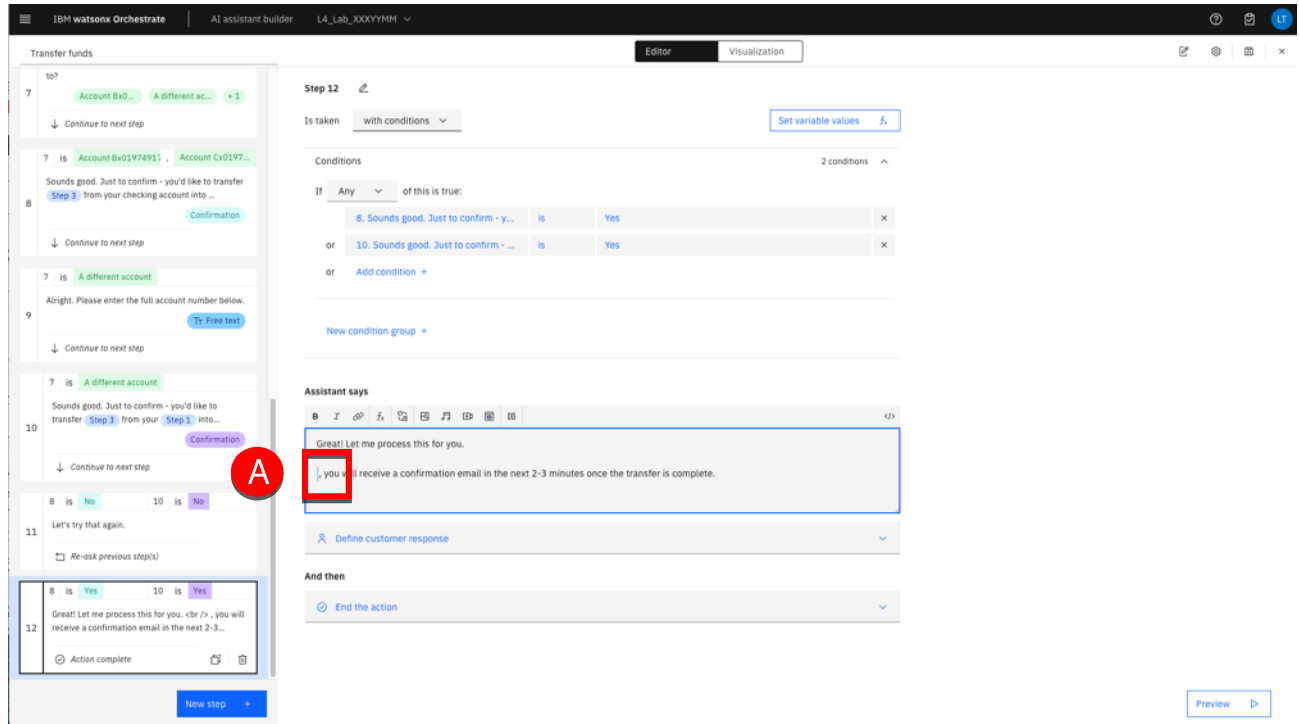
17. Now, you should be viewing step 7 (previously, step 8 before you performed the deletion). In assistant says, delete the text: **Great, I found you in our system!** and the following empty lines (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar displays a flowchart with Step 7 highlighted. The main editor shows the 'Assistant says' section with a red box around the text 'Great, I found you in our system!' and a red circle labeled 'A' indicating the area to be deleted. The 'Is taken' section is set to 'without conditions'. The 'And then' section has a 'Continue to next step' action.

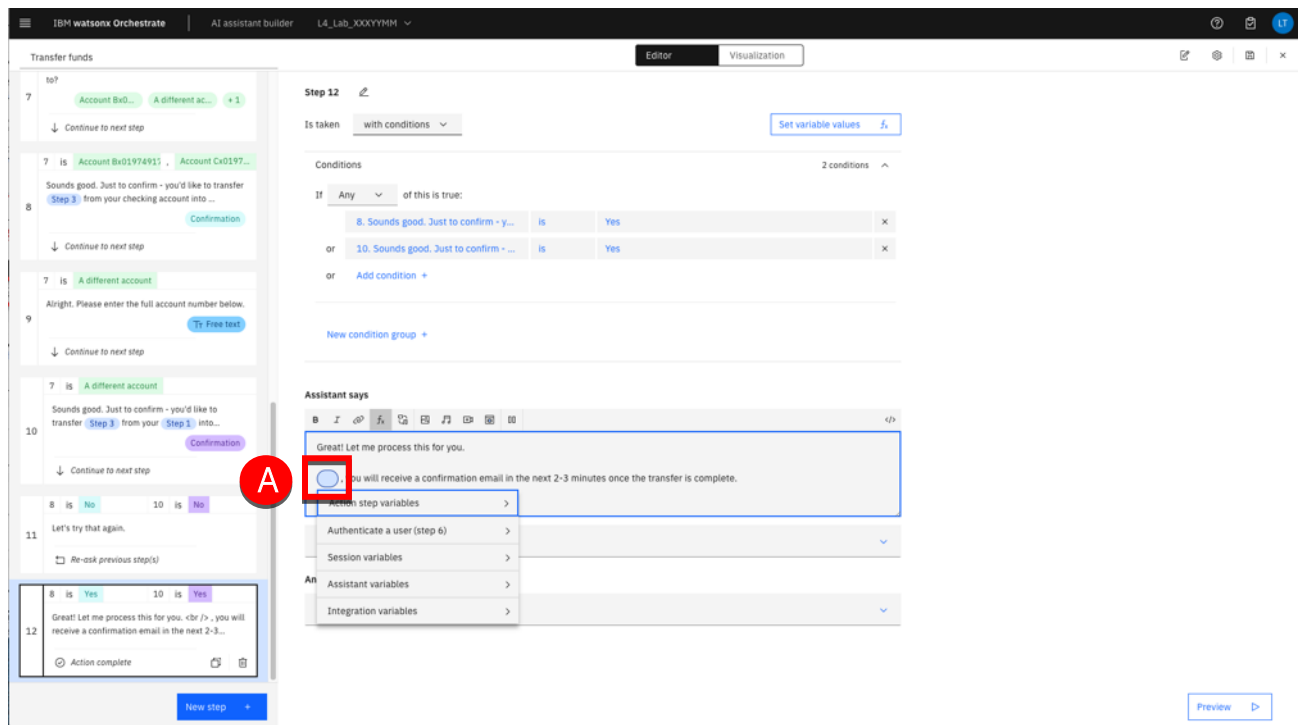
18. Scroll down, then click step 12 (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar displays a flowchart with Step 12 highlighted. The main editor shows the 'Assistant says' section with a red box around the text 'Great! Let me process this for you. <br /> You will receive a confirmation email in the next 2-3...' and a red circle labeled 'A' indicating the area to be deleted. The 'Is taken' section is set to 'without conditions'. The 'And then' section has a 'Continue to next step' action.

19. Type a space character followed by ‘, ’ in front the **You** (A) replace **You** with you (B) then move the cursor at the beginning of the line.



20. type \$. This should trigger the variable selector menu (A).



## 21. Select Session variables (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, a flowchart titled "Transfer funds" shows steps 7 through 12. Step 12 is highlighted. On the right, the configuration for Step 12 is shown. The "Is taken" dropdown is set to "with conditions". Below this, there are two conditions: "8. Sounds good. Just to confirm - y..." and "10. Sounds good. Just to confirm - ...". The "Assistant says" section contains a text block: "Great! Let me process this for you. <br> <br> ,you will receive a confirmation email in the next 2-3 minutes once the transfer is complete." Below this, there is a dropdown menu for "Action step variables". The "Session variables" option is highlighted with a red box and a red circle labeled "A".

## 22. Select User first name (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, a flowchart titled "Transfer funds" shows steps 7 through 12. Step 12 is highlighted. On the right, the configuration for Step 12 is shown. The "Is taken" dropdown is set to "with conditions". Below this, there are two conditions: "8. Sounds good. Just to conf..." and "10. Sounds good. Just to co...". The "Assistant says" section contains a text block: "Great! Let me process this for you. <br> <br> ,will receive a confirmation email in the next 2-3 minutes once the transfer is complete." Below this, there is a dropdown menu for "Action step variables". The "User first name" option is highlighted with a red box and a red circle labeled "A".

23. In the **Assistant says** text, insert a \$ character so that the text becomes: ... will receive a confirmation email at \$ in the next 2-3 minutes once the transfer is complete. (A). Click **Session variables** (B).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The top navigation bar shows 'IBM watsonx Orchestrate', 'AI assistant builder', and 'L4\_lab\_JFS\_2503'. The main workspace is titled 'Transfer funds' and has tabs for 'Editor' and 'Visualization'. On the left, a flowchart shows steps 7 through 12. Step 12 is highlighted, showing a 'Great! Let me process this for you.' message and a confirmation email prompt. The right panel shows the configuration for Step 12. Under 'Is taken', it is set to 'with conditions'. Below this, two conditions are listed: '8. Sounds good. Just to conf...' and '10. Sounds good. Just to co...'. The 'Assistant says' section contains the text 'Great! Let me process this for you.' followed by a prompt: 'Tr User first name', you will receive a confirmation email at \$ in the next 2-3 minutes once the transfer is complete. A red box labeled 'A' highlights the '\$' character. Below the text, a dropdown menu is open, showing options: 'Action step variables', 'Authenticate a user (step 6)', 'Session variables', 'Assistant variables', and 'Integration variables'. A red box labeled 'B' highlights the 'Session variables' option. At the bottom right, there is a 'Preview' button.

**Note:** Make sure the \$ is followed by a space character to have the insert variable menu displayed.

24. Select User email address (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_JFS\_2503

Transfer funds

Editor Visualization

Step 12

Is taken with conditions

Set variable values

Conditions 2 conditions

If Any of this is true:

|                                 |    |     |   |
|---------------------------------|----|-----|---|
| 8. Sounds good. Just to conf... | is | Yes | x |
| 10. Sounds good. Just to co...  | is | Yes | x |

or Add condition +

New condition group +

Assistant says

Great! Let me process this for you.

Tr User first name, you will receive a confirmation email at in the next 2-3 minutes once the transfer is complete.

Define customer response

And then

End the action

Tr User first name

Tr User last name

Tr User email address

Preview

25. Click Save (A), then open and refresh the Preview window (B).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_JFS\_2503

Transfer funds

Editor Visualization

Step 12

Is taken with conditions

Set variable values

Conditions 2 conditions

If Any of this is true:

|                                 |    |     |   |
|---------------------------------|----|-----|---|
| 8. Sounds good. Just to conf... | is | Yes | x |
| 10. Sounds good. Just to co...  | is | Yes | x |

or Add condition +

New condition group +

Assistant says

Great! Let me process this for you.

Tr User first name, you will receive a confirmation email at Tr User email address in the next 2-3 minutes once the transfer is complete.

Define customer response

And then

End the action

Tr User first name

Tr User last name

Tr User email address

Preview

## 26. Type Transfer money (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The workflow is titled "Transfer funds". On the left, a list of steps is visible, with Step 7 selected. The main editor area shows Step 7 with the following configuration:

- Is taken:** without conditions
- Assistant says:** "Finally, to which account would you like to transfer to?"
- Action:** A dropdown menu is open, showing "Is\_authenticated" with a value of "false".
- And then:** "Continue to next step"

On the right, a "Preview" window shows a chat conversation. The user's message is "Transfer money" (highlighted with a red box and a red circle with 'A'). The assistant's response is "Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?"

## 27. Select Checking account (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The workflow is titled "Transfer funds". On the left, a list of steps is visible, with Step 12 selected. The main editor area shows Step 12 with the following configuration:

- Is taken:** with conditions
- Conditions:** 2 conditions are listed:
  - 8. Sounds good. Just to conf... is Yes
  - 10. Sounds good. Just to co... is Yes
- Assistant says:** "Great! Let me process this for you. You will receive a confirmation email at [Tr user email address] in t complete."
- And then:** "End the action"

On the right, a "Preview" window shows a chat conversation. The user's message is "Checking account" (highlighted with a red box and a red circle with 'A'). The assistant's response is "Before we can transfer any funds, please confirm where you'd like to transfer funds from." Below this, a dropdown menu is open, showing "Checking account" (highlighted with a red box) and "Savings account".

28. Type 100 (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The top bar indicates the project is 'L4\_lab\_3FS\_2503'. The left sidebar shows a flowchart with steps 7 through 12. Step 12 is highlighted, showing a 'Confirmation' action. The main editor area displays the configuration for Step 12, which is 'Is taken with conditions'. The conditions are: 'If Any of this is true: 8. Sounds good. Just to conf... is Yes, or 10. Sounds good. Just to co... is Yes'. The 'Assistant says' section shows a message: 'Great! Let me process this for you. You will receive a confirmation email at [Tr User email address] in t complete.' The 'And then' section shows 'End the action'. The 'Preview' panel on the right shows a chat conversation. The user says 'Transfer funds' at 3:41 PM. The assistant responds: 'Before we can transfer any funds, please confirm where you'd like to transfer funds from.' The user selects 'Checking account' at 3:42 PM. The assistant asks: 'Okay! How much would you like to transfer?'. The user types '100' in the input field, which is highlighted with a red circle and labeled 'A'.

29. Select the next available business day (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The top bar indicates the project is 'L4\_lab\_3FS\_2503'. The left sidebar shows a flowchart with steps 7 through 12. Step 12 is highlighted, showing a 'Confirmation' action. The main editor area displays the configuration for Step 12, which is 'Is taken with conditions'. The conditions are: 'If Any of this is true: 8. Sounds good. Just to conf... is Yes, or 10. Sounds good. Just to co... is Yes'. The 'Assistant says' section shows a message: 'Great! Let me process this for you. You will receive a confirmation email at [Tr User email address] in t complete.' The 'And then' section shows 'End the action'. The 'Preview' panel on the right shows a chat conversation. The user says 'Transfer funds' at 3:41 PM. The assistant responds: 'Before we can transfer any funds, please confirm where you'd like to transfer funds from.' The user selects 'Checking account' at 3:42 PM. The assistant asks: 'Okay! How much would you like to transfer?'. The user types '100' in the input field, which is highlighted with a red circle and labeled 'A'. The assistant then asks: 'Alright. When would you like to make the transfer?'. The user selects a date from a calendar picker. The calendar shows 'March 2025'. The date '31' is selected, which is highlighted with a red circle and labeled 'A'.

30. Click Confirm date (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Editor Visualization

Step 12

Is taken with conditions

Set variable values

Conditions

2 conditions

If Any of this is true:

8. Sounds good. Just to confirm - you'd like to transfer Step 3 from your checking account into Step 7

10. Sounds good. Just to confirm - you'd like to transfer Step 3 from your Step 1 into...

Assistant says

Great! Let me process this for you. You will receive a confirmation email at Tr User email address in t complete.

Preview

You 3:42 PM

Checking account

3:42 PM

Okay! How much would you like to transfer?

You 3:43 PM

100

3:43 PM

Alright. When would you like to make the transfer?

Choose a date (mm/dd/yyyy)

03/31/2025

Confirm date

31. Type a correctly formatted email address (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Transfer funds

Editor Visualization

Step 6

Is taken with conditions

Set variable values

Conditions

1 condition

If All of this is true:

Is authenticated is false

Assistant says

Okay. I'll need to authenticate you before moving forward.

Preview

You 4:08 PM

Alright. When would you like to make the transfer?

Choose a date (mm/dd/yyyy)

2025-03-31

4:08 PM

Okay. I'll need to authenticate you before moving forward.

Authenticate a user

To get started, please enter your email address.

jd@acme.com

Notice that the subaction occurred – the AI assistant switched to the Authenticate a user action.

32. Type a four digit PIN (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The workflow is titled "Transfer funds". The left sidebar shows steps 4 through 9. Step 6 is selected, showing a condition "Is authenticated" set to "false". The central editor for Step 6 shows the condition and a "Go to a subaction" button labeled "Authenticate a user". The right preview pane shows a chat conversation. A red circle labeled "A" highlights a text input field containing "0451".

33. Choose a security question (A), then provide a valid text input (B).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The workflow is titled "Transfer funds". The left sidebar shows steps 4 through 9. Step 6 is selected, showing a condition "Is authenticated" set to "false". The central editor for Step 6 shows the condition and a "Go to a subaction" button labeled "Authenticate a user". The right preview pane shows a chat conversation. A red circle labeled "A" highlights a security question selection "My mother's maiden name". A red circle labeled "B" highlights a text input field containing "smith".

34. Select the first listed account (A).

The screenshot displays the IBM watsonx Orchestrate interface for a workflow titled "Transfer funds". The interface is divided into three main sections: a left-hand pane showing a sequence of steps, a central editor for the current step, and a right-hand preview pane.

- Left Pane:** Shows a list of steps. Step 6 is highlighted, indicating the current step in the workflow.
- Central Editor (Step 6):**
  - Is taken:** Set to "with conditions".
  - Conditions:** A single condition "Is authenticated" is set to "false".
  - Assistant says:** The assistant's response is "Okay, I'll need to authenticate you before moving forward."
  - And then:** The action is "Go to a subaction", which leads to "Authenticate a user".
- Variable values:** A table showing session variables:

| Variable           | Value         |
|--------------------|---------------|
| Is_authenticated   | true          |
| User_email_address | jdoe@acme.com |
| User_first_name    | Jack          |
| User_last_name     | Schaefer      |
- Preview:** A chat conversation showing the user's input and the assistant's response. The user asks to transfer funds. The assistant responds with a security question. The user provides the answer. The assistant then asks for confirmation of the transfer to "Account Bx01974917-01", which is highlighted with a red box and labeled "A".

Note that after completing the Authenticate a user subaction, you returned to the Transfer funds action.

35. Click **Yes** to confirm (A).

The screenshot displays the IBM watsonx Orchestrate interface for a workflow titled "Transfer funds". The interface is divided into three main sections: a left-hand pane showing a sequence of steps, a central editor for the current step, and a right-hand preview pane.

- Left Pane:** Shows a list of steps. Step 12 is highlighted, indicating the current step in the workflow.
- Central Editor (Step 12):**
  - Is taken:** Set to "with conditions".
  - Conditions:** Two conditions are listed: "8. Sounds good. Just to confirm..." and "10. Sounds good. Just to confirm...".
  - Assistant says:** The assistant's response is "Great! Let me process this for you."
  - And then:** The action is "End the action".
- Variable values:** A table showing session variables:

| Variable           | Value         |
|--------------------|---------------|
| Is_authenticated   | true          |
| User_email_address | jdoe@acme.com |
| User_first_name    | John          |
| User_last_name     | Doe           |
- Preview:** A chat conversation showing the user's input and the assistant's response. The user asks to transfer funds. The assistant responds with a security question. The user provides the answer. The assistant then asks for confirmation of the transfer to "Account Bx01974917-01", which is highlighted with a red box and labeled "A".

36. You have completed the Transfer funds action. Now type I'd like to transfer funds again please (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar displays a flowchart with steps 7 through 12. Step 12 is highlighted. The central editor shows the configuration for Step 12, including conditions and actions. The right sidebar shows a preview of the assistant's conversation. A red circle with the letter 'A' is placed over the input field in the preview, which contains the text "I'd like to transfer funds again please".

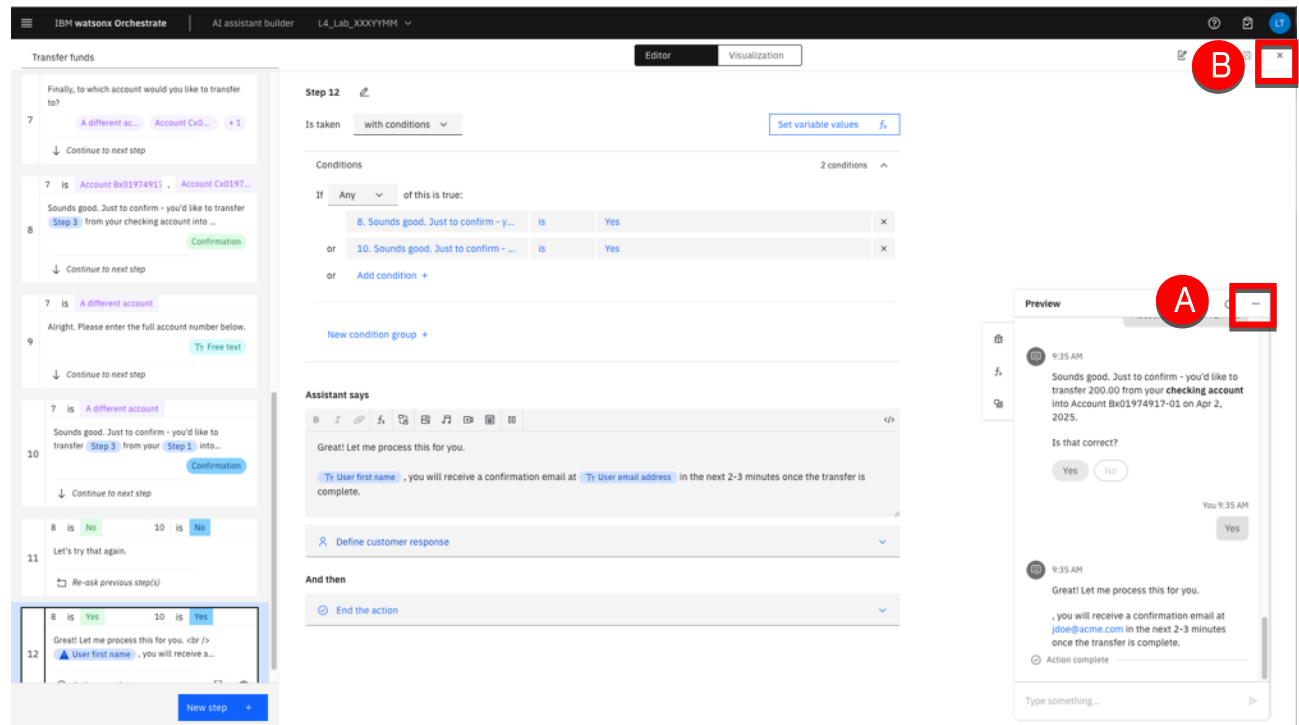
37. Repeat the flow you just performed, this time transferring 200 dollars (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface, similar to the previous one. The left sidebar displays a flowchart with steps 7 through 12. Step 12 is highlighted. The central editor shows the configuration for Step 12, including conditions and actions. The right sidebar shows a preview of the assistant's conversation. A red box highlights the preview area, which shows the assistant's response to the user's input, including a confirmation message and a red circle with the letter 'A'.

Notice that you do not need to authorize yourself again. The session variable 'Is authenticated' tracks your authentication status, ensuring that the authentication process is not repeated unnecessarily.

You can optionally implement calling Authenticate a user as a subaction for other actions in the assistant. The actions Set up autopay, Make a payment, and Check account balance could benefit from this design approach.

38. Minimize the **Preview** (A) then close the action editor (B).



## Advanced Step Conditions

Throughout this lab, you have used conditions to control the business logic of an action.

As a reminder, a [step condition](#) is a Boolean test based on a runtime value; the step executes only if the test evaluates to true. This test can be applied to any variable, such as an action variable that contains the customer response from a previous step. By defining step conditions, you can create multiple pathways through an action based on different possible runtime values.

A basic step condition is expressed in the following form:

If {variable} {operator} {value}

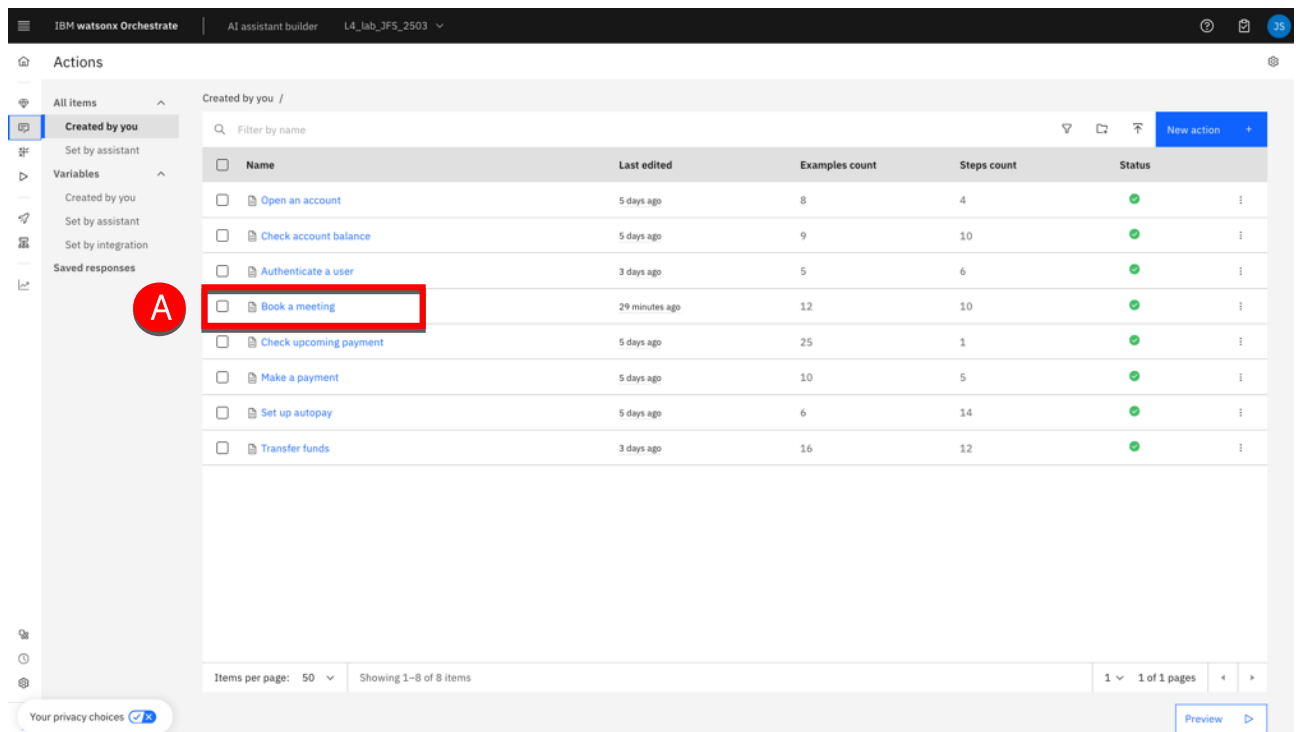
where:

- {variable} is the name of a variable or an expression.
- {operator} is the type of evaluation to apply to the variable value (for example, is or is not).
- {value} is the value to compare to the variable.

Thus far, you have used relatively simple conditions to control the business logic of an action. For this exercise, you will apply multiple conditions to the steps of an action.

Specifically, you will modify the **Book a meeting** action to incorporate holidays and revised business hours into its logic.

1. In the actions window, click **Book a meeting** (A).



2. Take a moment to understand the flow of the action. Optionally, you may open Preview, type “I’d like to book a meeting”, then enter valid values. Once you understand the business logic, click on Step 5 (A).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The top navigation bar includes the IBM watsonx Orchestrate logo, the AI assistant builder title, and a dropdown menu showing 'L4\_lab\_JFS\_2503'. The interface is divided into two main sections: 'Editor' and 'Visualization'. The 'Editor' section on the left shows a workflow titled 'Book a meeting' with a 'Customer starts with: Book a consultation' trigger. The workflow consists of five steps:

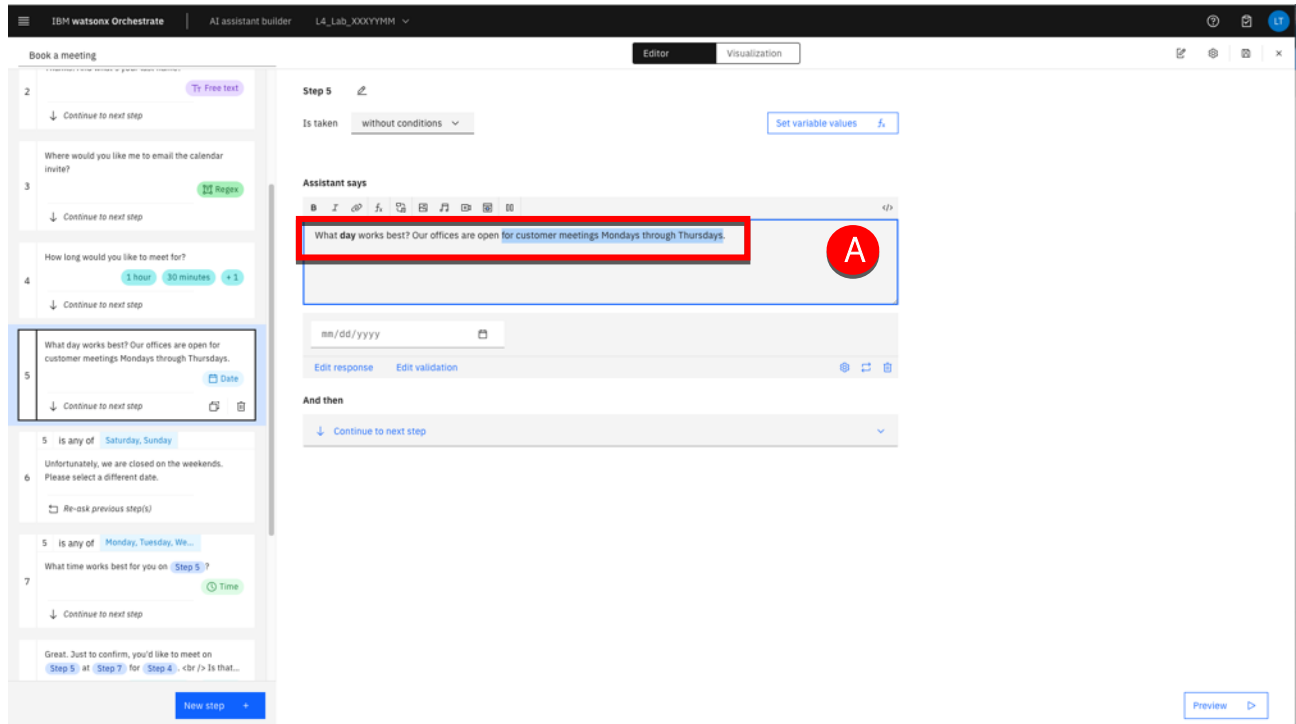
- Step 1: 'To set up a meeting, I'll need a few details. First, what's your first name?' with a 'Tr Free text' input field and a 'Continue to next step' button.
- Step 2: 'Thanks! And what's your last name?' with a 'Tr Free text' input field and a 'Continue to next step' button.
- Step 3: 'Where would you like me to email the calendar invite?' with a 'Regex' input field and a 'Continue to next step' button.
- Step 4: 'How long would you like to meet for?' with a '1 hour', '15 minutes', and '+ 1' input field and a 'Continue to next step' button.
- Step 5: 'What day works best? Our offices are open Mondays through Fridays.' with a 'Date' input field and a 'Continue to next step' button. This step is highlighted with a red box and a red circle with the letter 'A'.

The 'Visualization' section on the right shows the 'Add example phrases' section. It includes a text area for entering phrases, a 'Total: 12' counter, and a list of example phrases with expand/collapse icons:

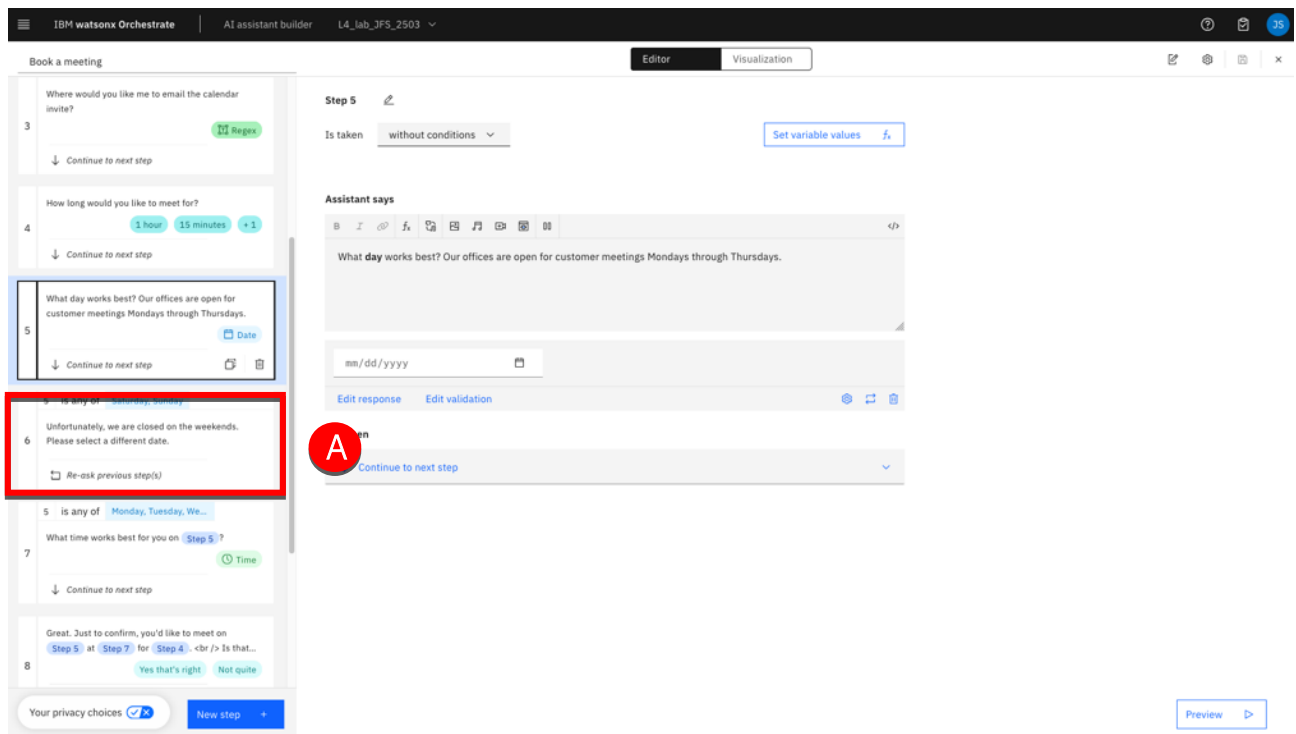
- When can I meet with one of your employees at your store?
- Store appointment
- Set up an appt
- Schedule Consult
- Schedule a Meeting
- Make an appointment
- I would like to make an appointment to visit the nearest store to my location.
- I would like to discuss my situation face to face
- I want to talk in person with someone about my case
- Is it possible to set a date?

At the bottom right, there is a 'Preview' button with a play icon.

3. Modify **Assistant says** to say: What **day** works best? Our offices are open for customer meetings Mondays through Thursdays.(A).



4. Click Step 6 (A).



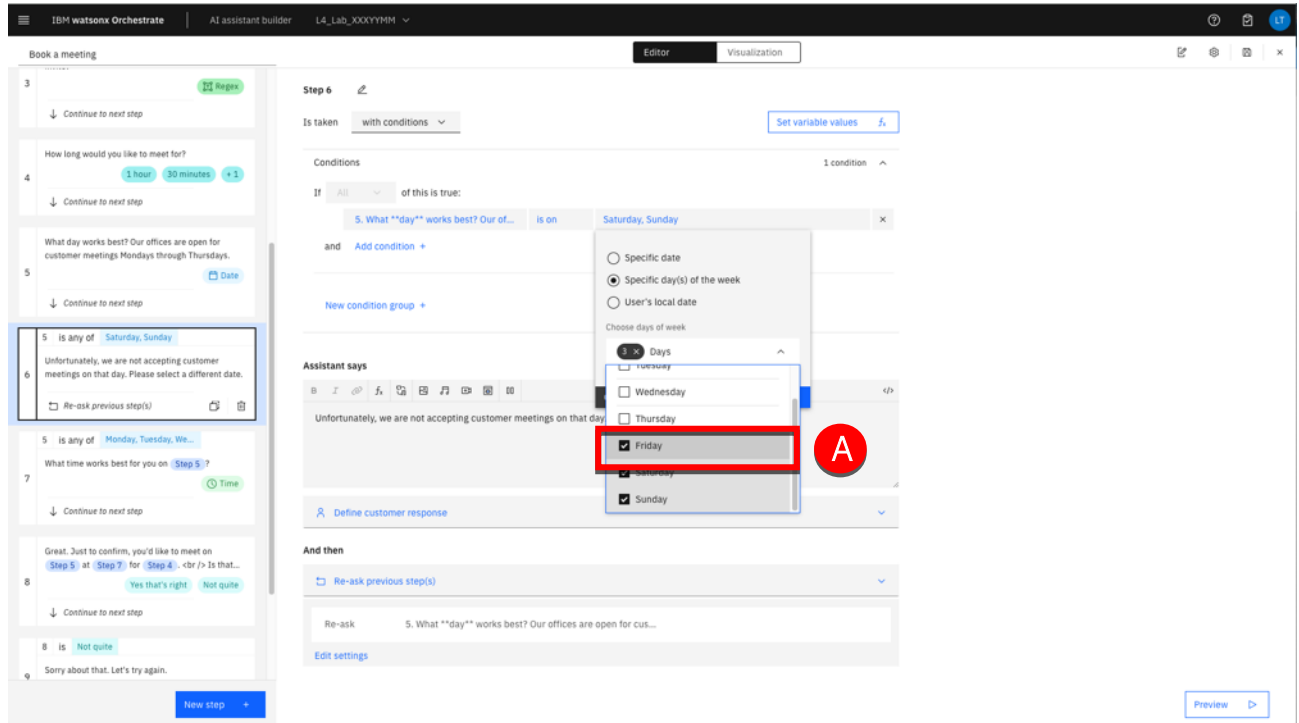
5. In **Assistant says** type Unfortunately, we are not accepting customer meetings on that day. Please select a different date. (A).

The screenshot shows the IBM Watsonx Orchestrate AI Assistant Builder interface. On the left, a flowchart titled "Book a meeting" shows steps 3 through 9. Step 6 is highlighted with a blue box. The main editor area shows the configuration for Step 6. Under "Is taken", the dropdown is set to "with conditions". Under "Conditions", there is one condition: "If All of this is true: 5. What \*\*day\*\* works best? Our of... is on Saturday, Sunday". Below the conditions, the "Assistant says" section contains the text "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." which is highlighted with a red box and labeled with a red circle 'A'. Below this, the "And then" section shows a "Re-ask previous step(s)" action.

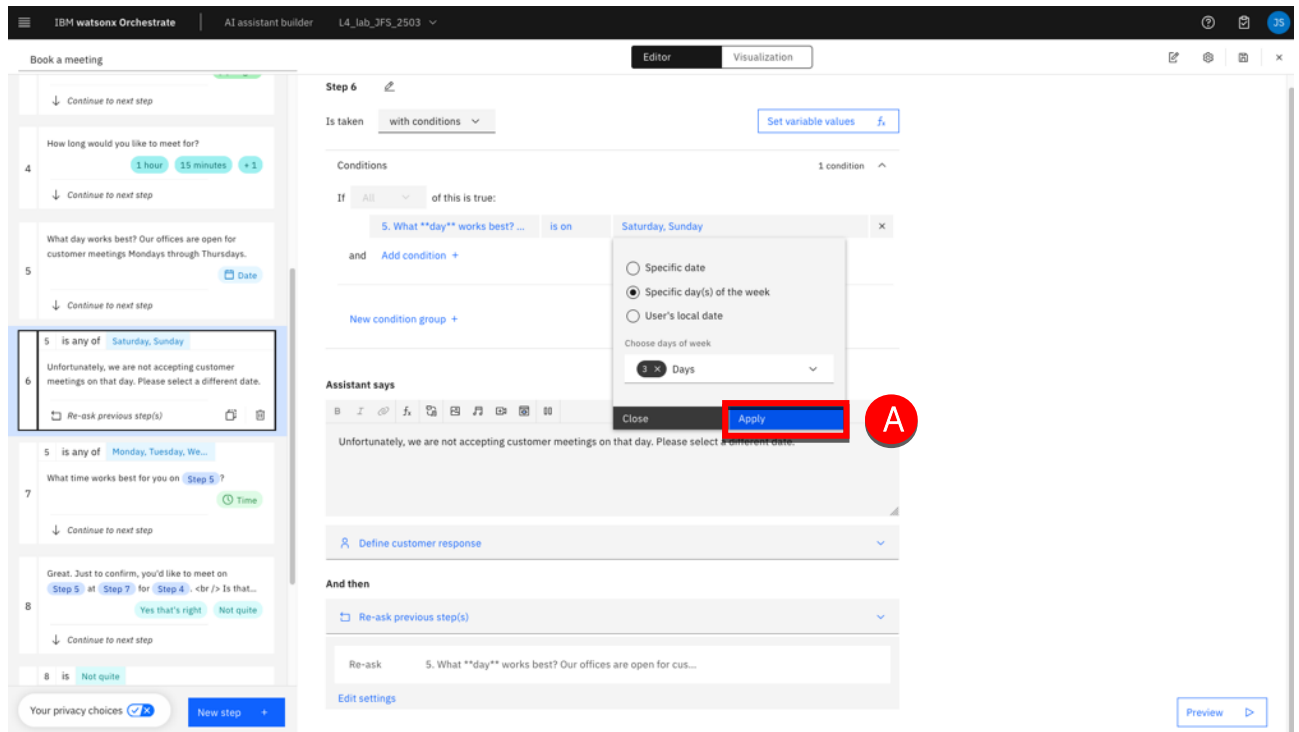
6. In the conditions section of Step 6, click the value of the current condition

This screenshot is similar to the previous one, showing the same Step 6 configuration. However, in the "Conditions" section, the value "Saturday, Sunday" in the condition "5. What \*\*day\*\* works best? ... is on Saturday, Sunday" is highlighted with a red box. A red circle 'A' is placed next to this box, indicating where to click.

7. Click the **Choose days of the week** dropdown menu, then add **Friday** to the selected days (A).



8. Click **Apply** (A).



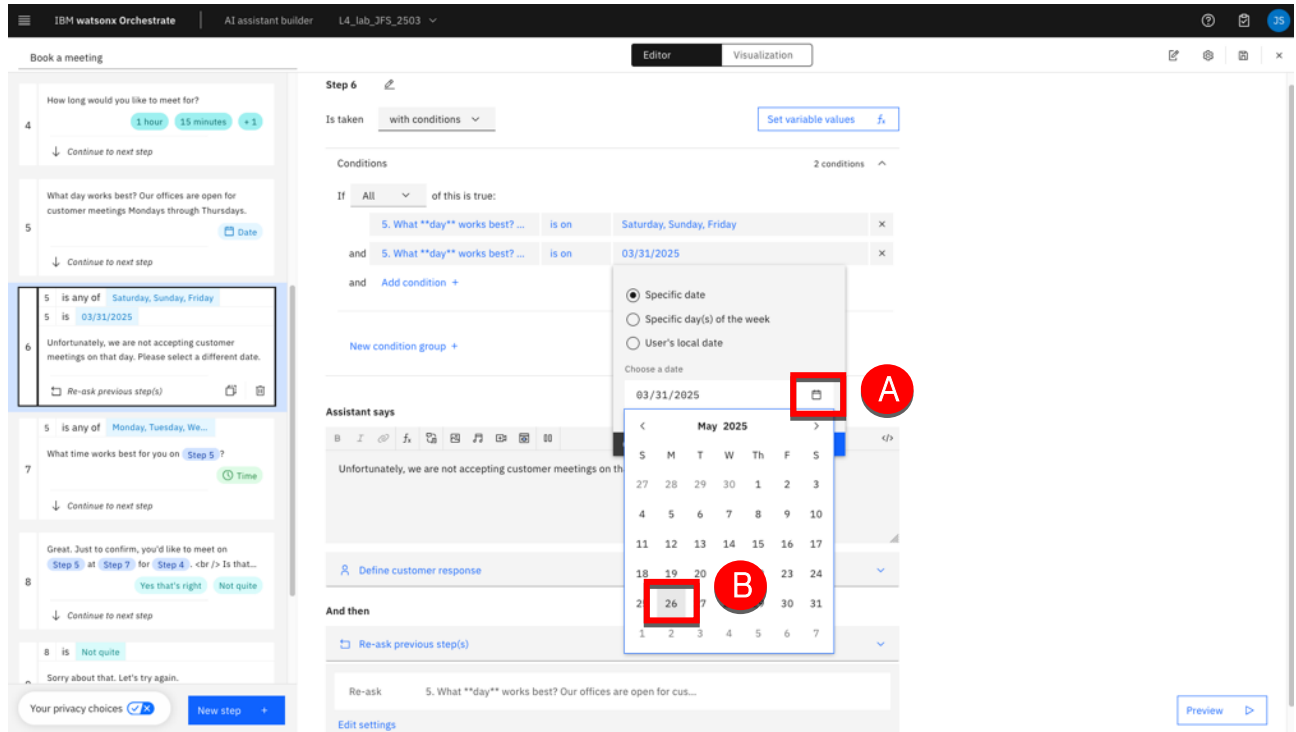
9. Click Add condition (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, a flowchart titled "Book a meeting" shows steps 4 through 8. Step 6 is highlighted with a blue box. On the right, the "Editor" tab for Step 6 is active. The "Conditions" section shows a single condition: "If All of this is true: 5. What \*\*day\*\* works best? ... is on Saturday, Sunday, Friday". A red box highlights the "Add condition +" button, and a red circle with the letter "A" is placed next to it. Below the conditions, the "Assistant says" section contains the text: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." The "And then" section shows a "Re-ask previous step(s)" action.

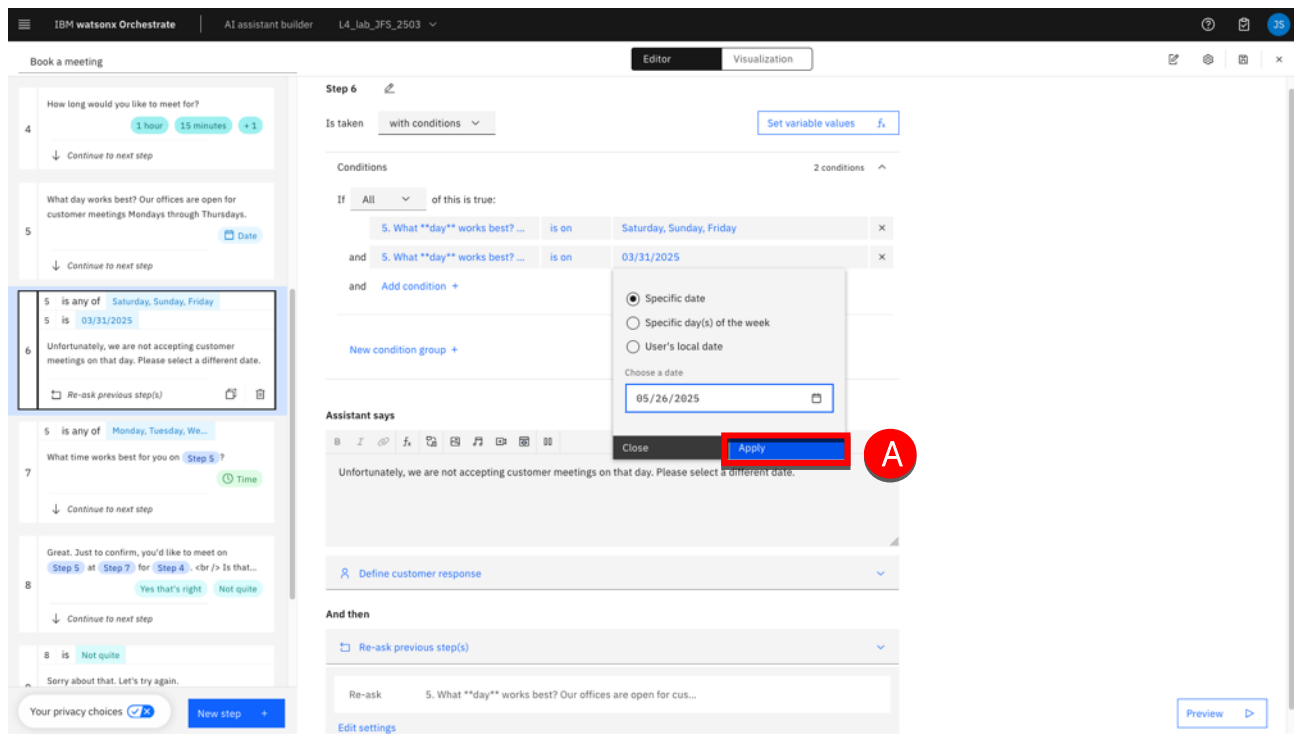
10. Click the value of the newly created condition (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, the flowchart shows step 6 highlighted. On the right, the "Editor" tab for Step 6 is active. The "Conditions" section now shows two conditions: "If All of this is true: 5. What \*\*day\*\* works best? ... is on Saturday, Sunday, Friday" and "5. What \*\*day\*\* works best? ... is on 03/31/2025". A red box highlights the value "03/31/2025", and a red circle with the letter "A" is placed next to it. The "Assistant says" section contains the text: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." The "And then" section shows a "Re-ask previous step(s)" action.

11. Click the calendar widget icon under **Choose a date** (A), then select the next upcoming company holiday (B).



12. Click **Apply** (A).



13. Repeat steps 9 -12 of this section, this time for another upcoming company holiday (A).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar shows a flowchart with steps 4 through 8. Step 5 is highlighted, showing a condition: "5 is any of Saturday, Sunday, Friday" and "5 is 05/26/2025 - 07/04/2025". The main editor area shows Step 6 configuration. Under "Conditions", there is a table with three conditions, all set to "is on". The third condition, "5. What \*\*day\*\* works best? ... is on 07/04/2025", is highlighted with a red box and a red circle labeled 'A'. The "Assistant says" section contains the text: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." The "And then" section shows a "Re-ask previous step(s)" action.

| Condition                       | Operator | Value                    |
|---------------------------------|----------|--------------------------|
| 5. What **day** works best? ... | is on    | Saturday, Sunday, Friday |
| 5. What **day** works best? ... | is on    | 05/26/2025               |
| 5. What **day** works best? ... | is on    | 07/04/2025               |

**Note:** This is a quick way to input company holidays and is useful to demonstrate the skills you learn in this lab. However, in a production environment, a better solution would involve integrating an external HR database with the AI assistant. The AI assistant would provide the user's entered date, and the HR system would return a Boolean value indicating if it was a company holiday. This way, the AI assistant could guide the conversation flow based on the most current information in the HR system.

14. Under conditions, click **All** (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a conversation flow is visible with steps 4 through 8. Step 6 is highlighted, showing a message: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." On the right, the "Step 6" configuration panel is open. Under the "Conditions" section, there are three conditions listed. The first condition is "If All of this is true:", where "All" is highlighted with a red box and a red circle labeled "A". The conditions are:

- 5. What \*\*day\*\* works best? ... is on Saturday, Sunday, Friday
- and 5. What \*\*day\*\* works best? ... is on 05/26/2025
- and 5. What \*\*day\*\* works best? ... is on 07/04/2025

Below the conditions, the "Assistant says" section shows the message: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." The "And then" section shows a "Re-ask previous step(s)" action.

15. Set the value to **Any** (A).

The screenshot shows the IBM watsonx Orchestrate interface, similar to the previous one, but with the "Any" option selected in the "Conditions" section. The "All" option is now greyed out, and "Any" is highlighted with a red box and a red circle labeled "A". The conditions remain the same:

- 5. What \*\*day\*\* works best? ... is on Saturday, Sunday, Friday
- and 5. What \*\*day\*\* works best? ... is on 05/26/2025
- and 5. What \*\*day\*\* works best? ... is on 07/04/2025

The "Assistant says" and "And then" sections are also visible, showing the same message and action as before.

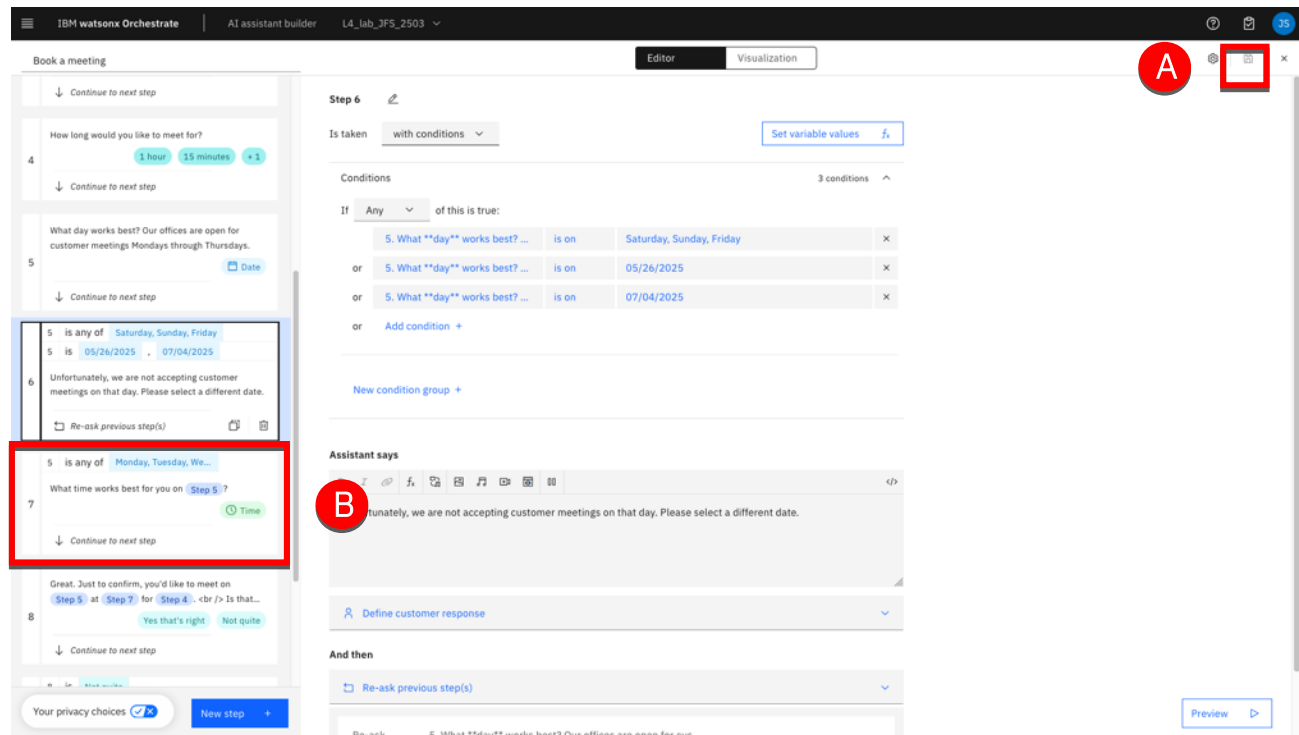
Notice that this change is altering the conditional logic to go from:

IF (the user inputted meeting date is on a Saturday, Sunday or Friday) **AND** Company Holiday 1 **AND** Company Holiday 2, THEN prompt the user for a new date

To:

IF (the user inputted meeting date is on a Saturday, Sunday or Friday) **OR** Company Holiday 1 **OR** Company Holiday 2, THEN prompt the user for a new date

16. Click **Save** (A), then click step **7** (B).



You have successfully used conditions to ensure users book meetings on valid dates.

## Condition Groups

You can use condition groups to build complex step conditions. Each condition group is individually evaluated as true or false, and then each group's results are evaluated together.

You will use condition groups to ensure customers book their meetings at valid times of the day.

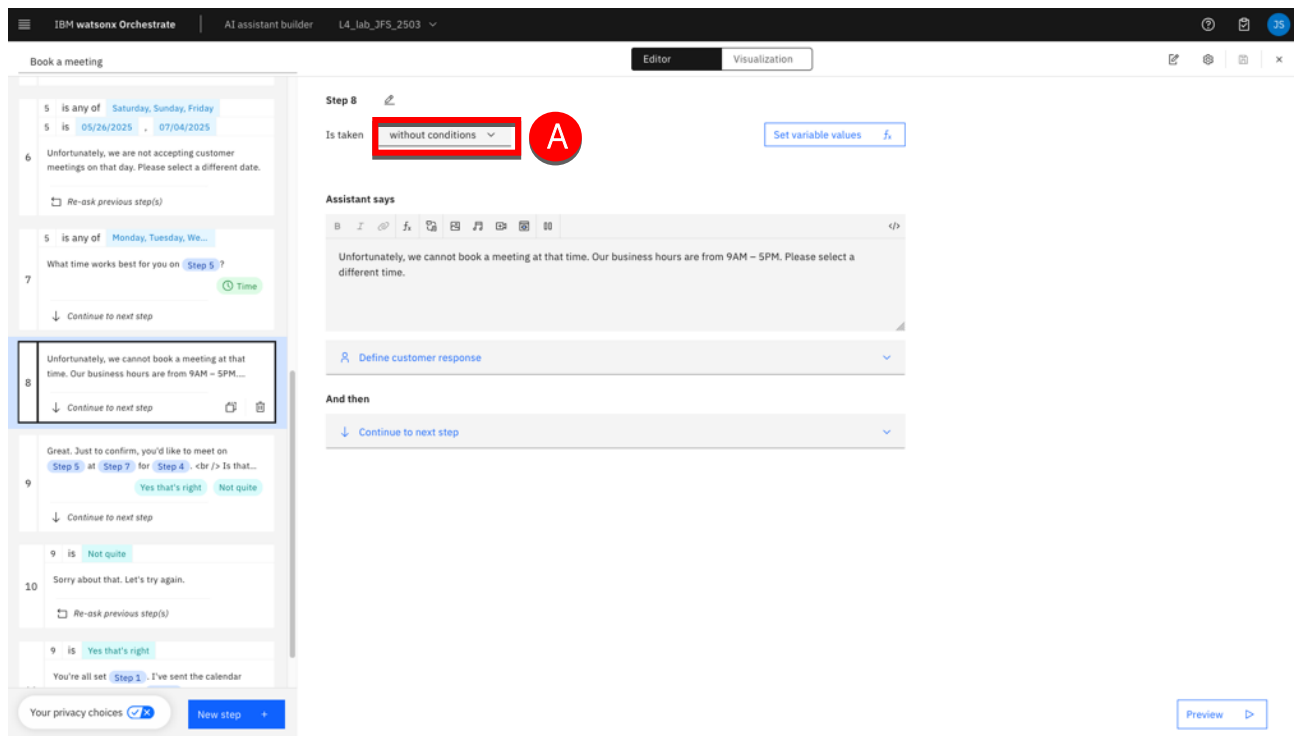
1. Click **New step** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, a list of steps is visible, with Step 7 highlighted. Step 7 is titled "What time works best for you on [Step 5]?" and includes a "Time" input field. A red box highlights the "New step" button at the bottom of the step list, and a red circle with the letter "A" is placed next to it. The main editor area shows the configuration for Step 7. It is set to "Is taken with conditions" and has a single condition: "If [All] of this is true: [5. What \*\*day\*\* works best? ...] is on [Monday, Tuesday, Wednesday, Friday, Thurs...]. Below the conditions, the "Assistant says" section contains a text input field with the placeholder "What time works best for you on [5. What \*\*day\*\* works best? Our offices ?]". The "And then" section has a dropdown menu set to "Continue to next step".

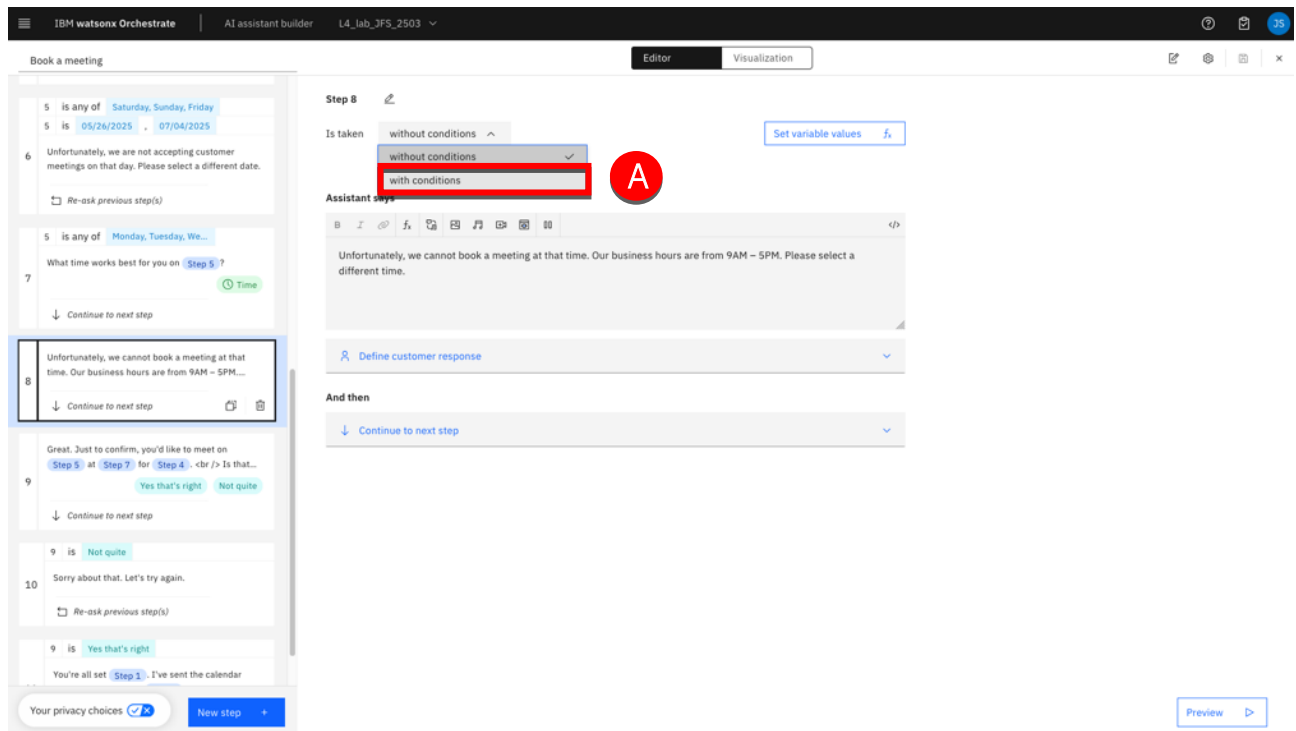
2. In the newly created step 8, within the assistant says section, type Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM – 5PM. Please select a different time. (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface, now with Step 8 highlighted. Step 8 is titled "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM – 5PM..." and includes a "Time" input field. A red box highlights the text input field in the "Assistant says" section, and a red circle with the letter "A" is placed next to it. The main editor area shows the configuration for Step 8. It is set to "Is taken without conditions". The "Assistant says" section contains a text input field with the placeholder "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM – 5PM. Please select a different time.". The "And then" section has a dropdown menu set to "Continue to next step".

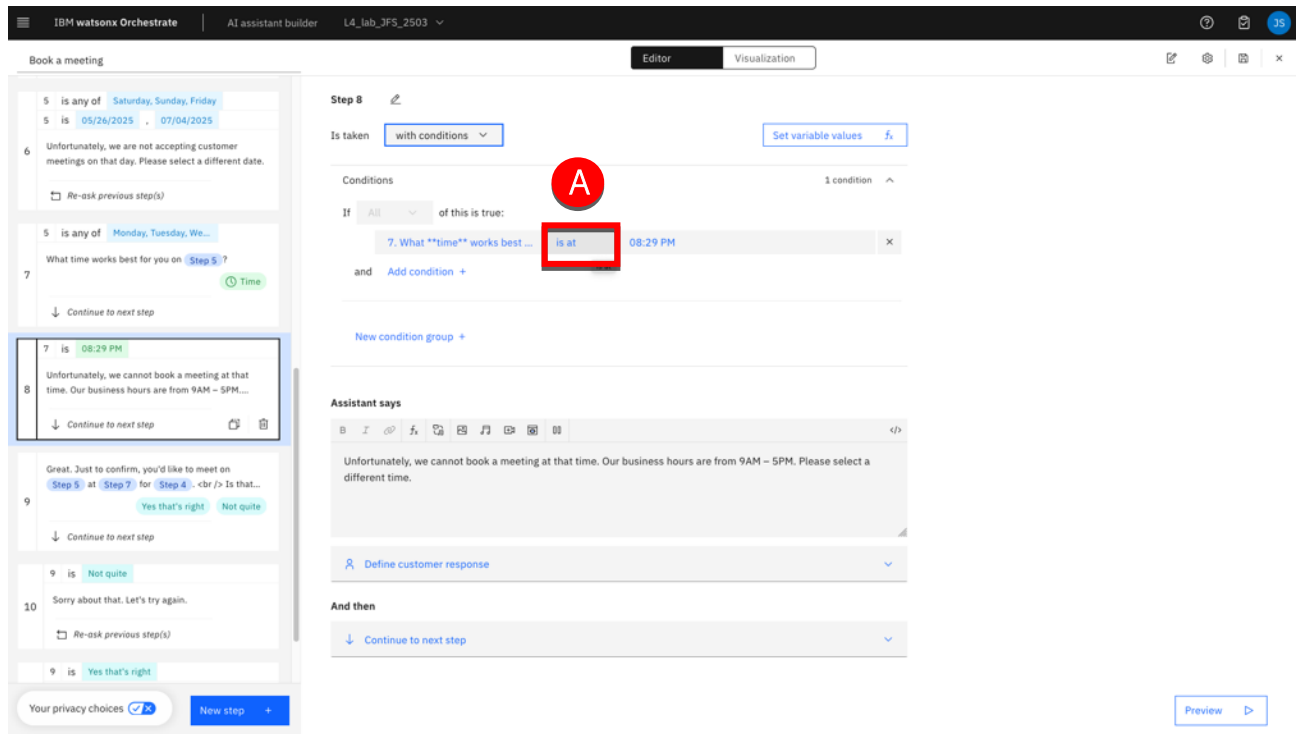
- Click **without conditions** to open the dropdown menu (A).



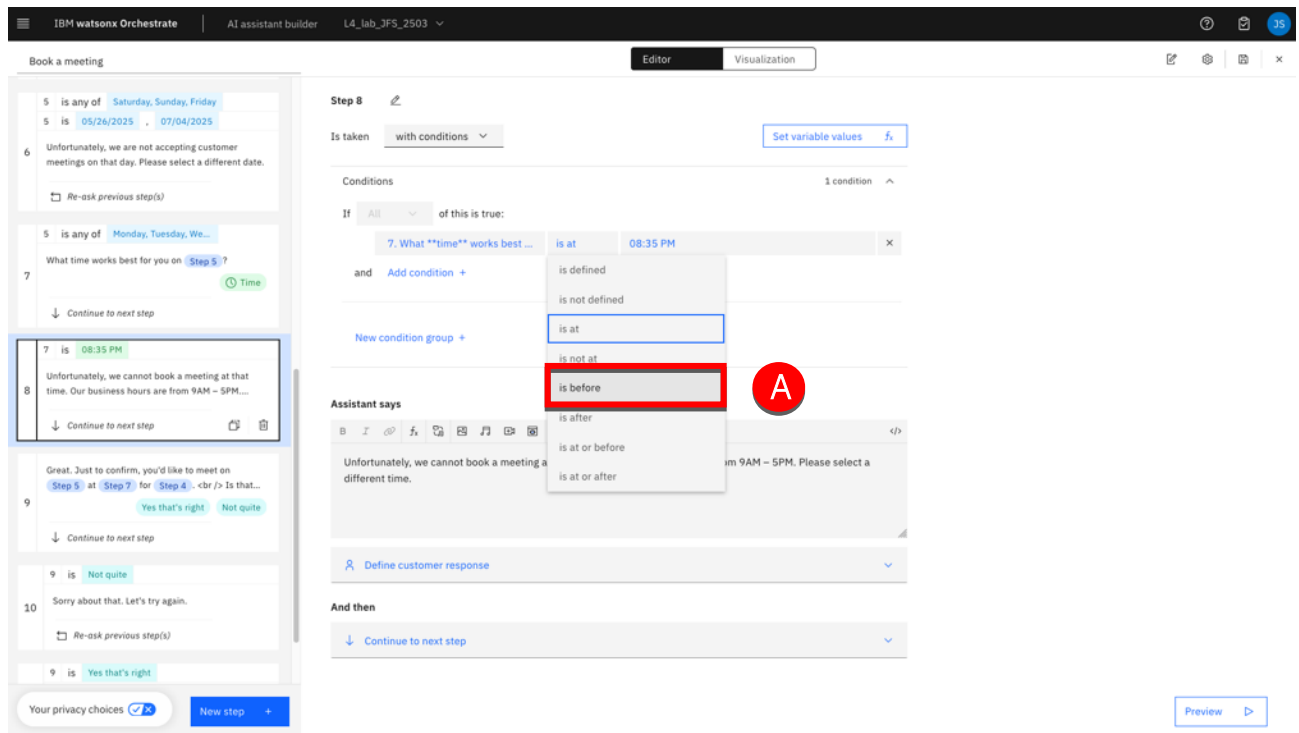
- In the dropdown menu, select **with conditions** (A).



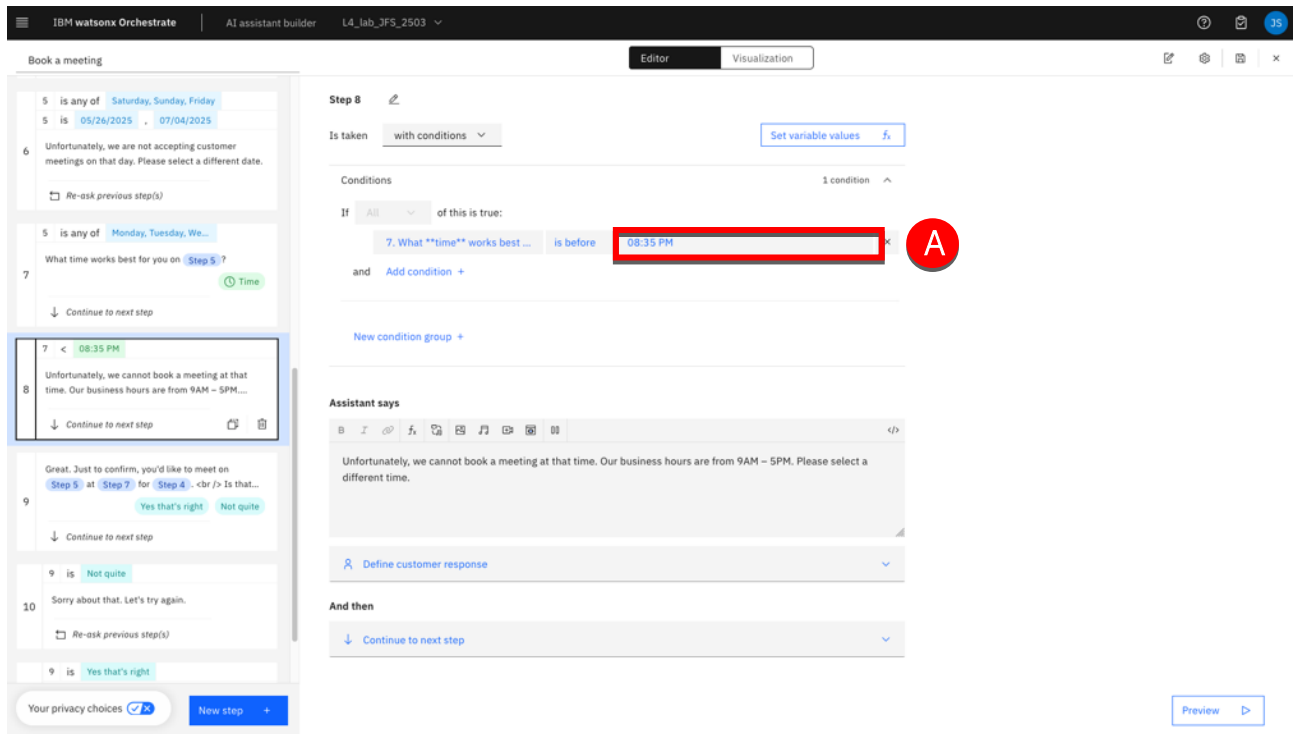
5. Click the operator of the condition (A).



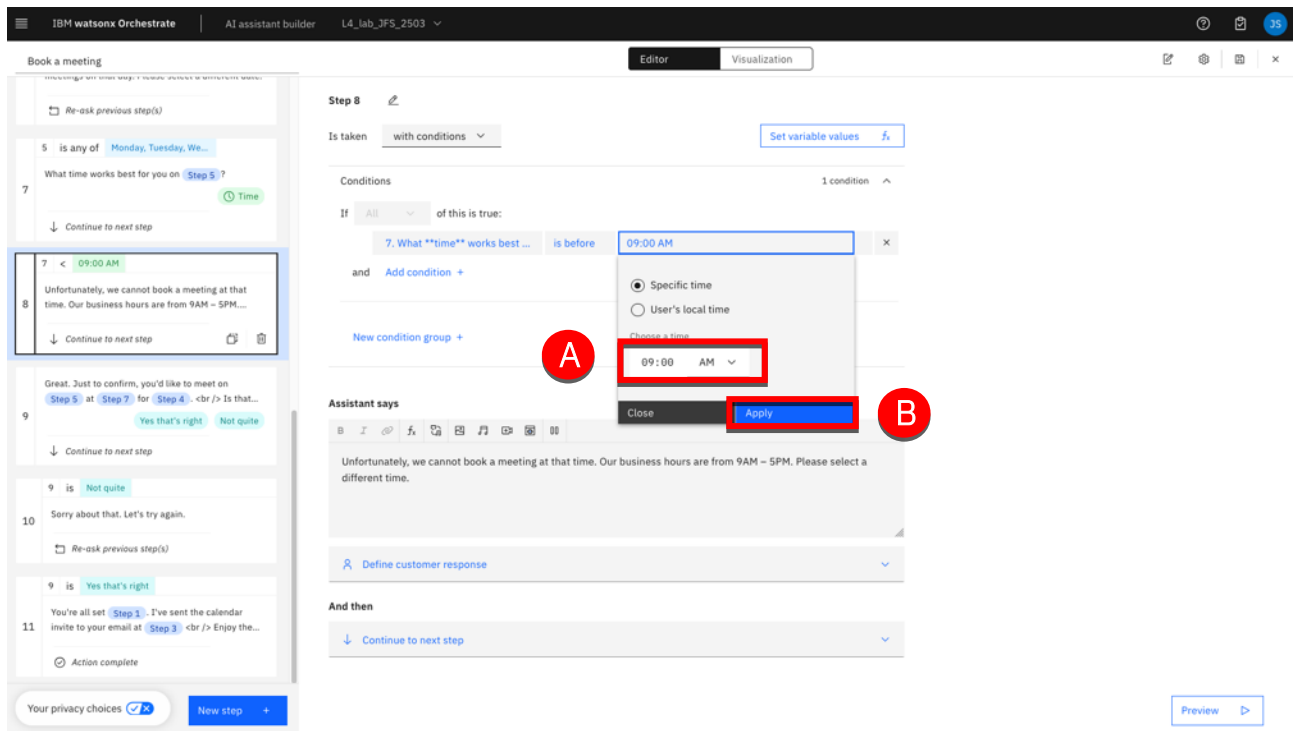
6. Select is before (A).



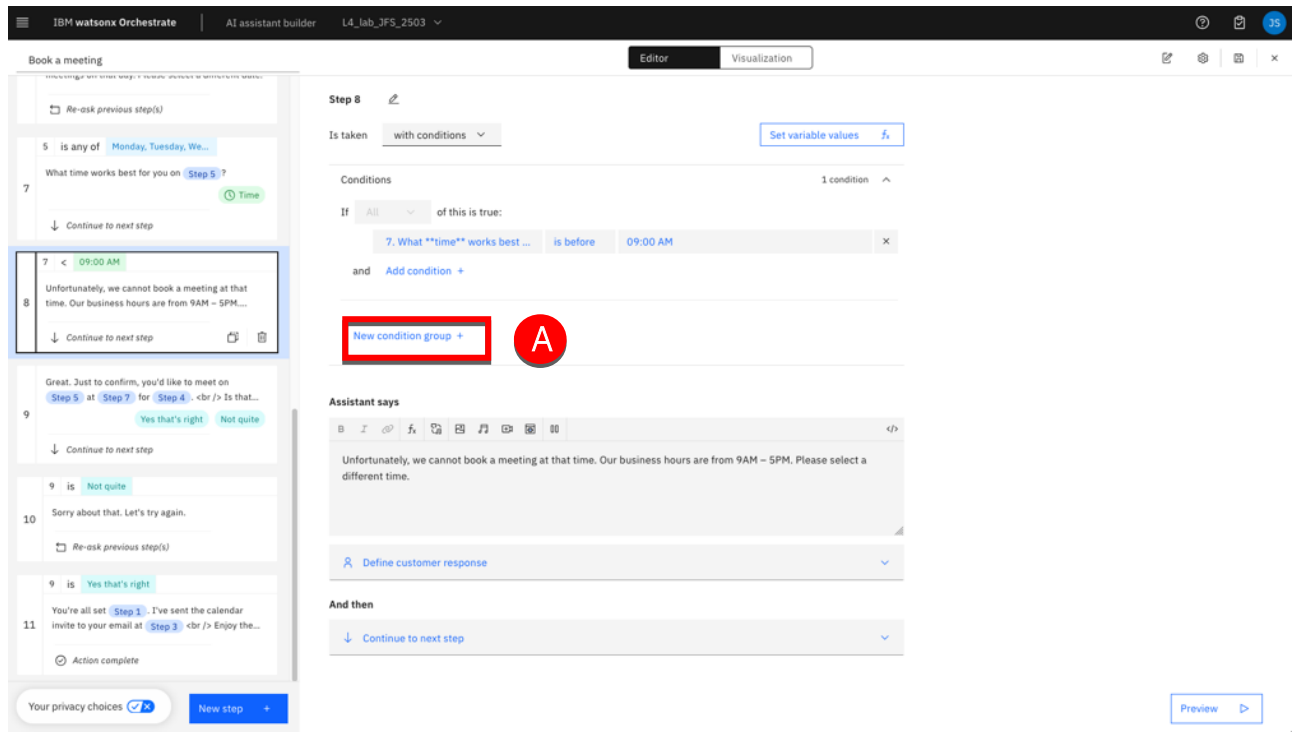
7. Click the value of the condition (A).



8. Change the value of the condition to 9:00am, (A) then click **Apply** (B).

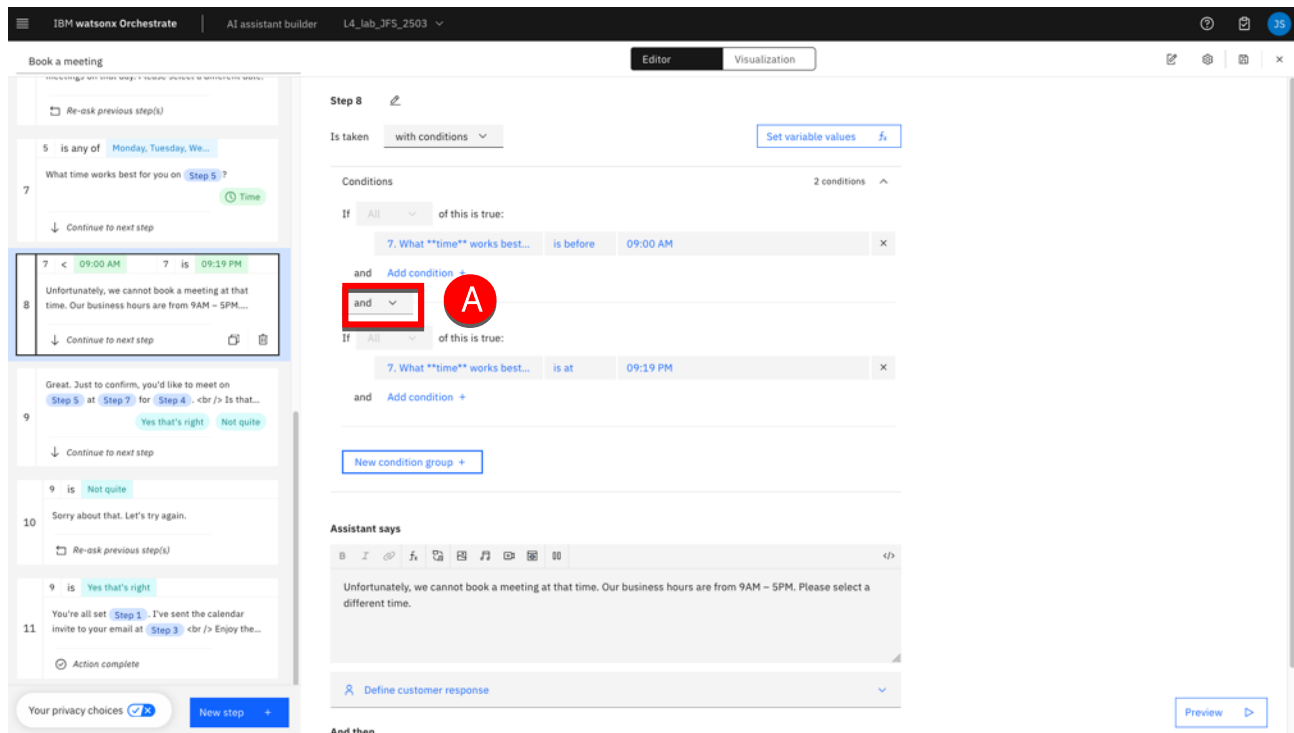


9. Click New condition group (A).



Here, you are adding a second condition group to the step.

10. Click and (A)



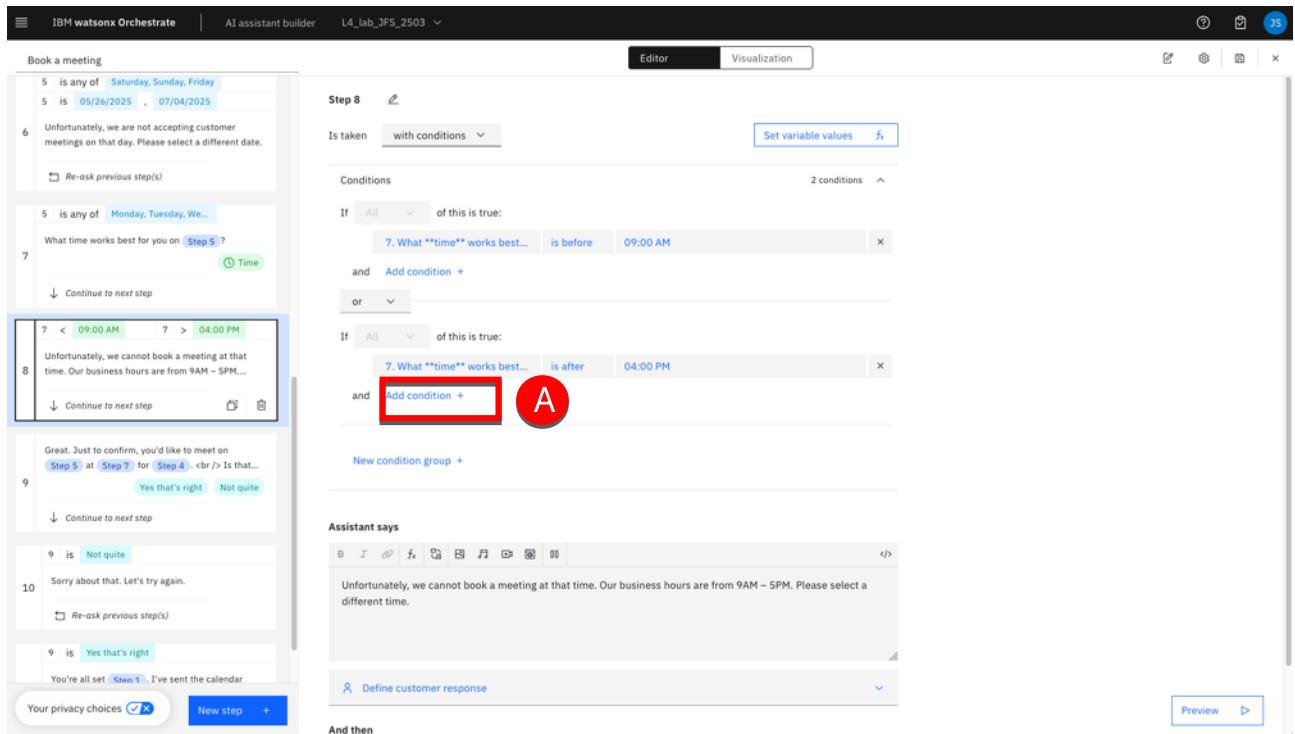
11. In the dropdown menu, select **or** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, a chat preview shows a conversation about booking a meeting. The main area is the 'Editor' tab for 'Step 8'. The 'Is taken' dropdown is set to 'with conditions'. Below this, the 'Conditions' section shows two conditions. The first condition is 'If All of this is true:' followed by '7. What \*\*time\*\* works best for yo... is before 09:00 AM'. The second condition is 'and Add condition +' followed by 'or of this is true:' followed by '7. What \*\*time\*\* works best for yo... is at 10:03 AM'. A red circle with the letter 'A' highlights the 'or' dropdown menu. The 'Assistant says' section contains a message: 'Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM. Please select a different time.' The 'And then' section has a dropdown set to 'Continue to next step'.

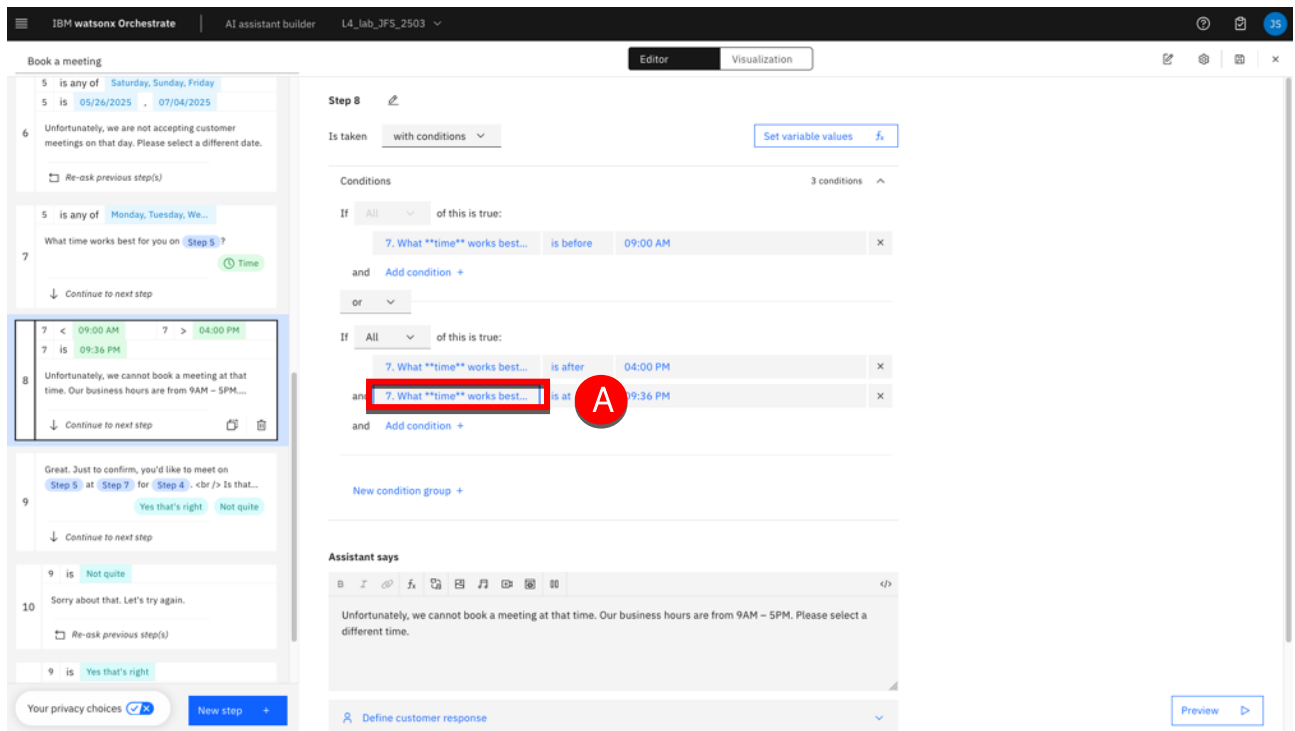
12. For the condition in newly added condition group, repeat steps 5 – 8, this time setting the operator to **is after** and the value to **4:00 PM** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, the chat preview is updated to show a confirmation message: 'Great. Just to confirm, you'd like to meet on Step 5 at Step 7 for Step 4. <br /> Is that... Yes that's right Not quite'. The main area is the 'Editor' tab for 'Step 8'. The 'Is taken' dropdown is set to 'with conditions'. Below this, the 'Conditions' section shows two conditions. The first condition is 'If All of this is true:' followed by '7. What \*\*time\*\* works best... is before 09:00 AM'. The second condition is 'and Add condition +' followed by 'or of this is true:' followed by '7. What \*\*time\*\* works best... is after 04:00 PM'. A red circle with the letter 'A' highlights the 'is after' dropdown menu. The 'Assistant says' section contains a message: 'Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM. Please select a different time.' The 'And then' section has a dropdown set to 'Continue to next step'.

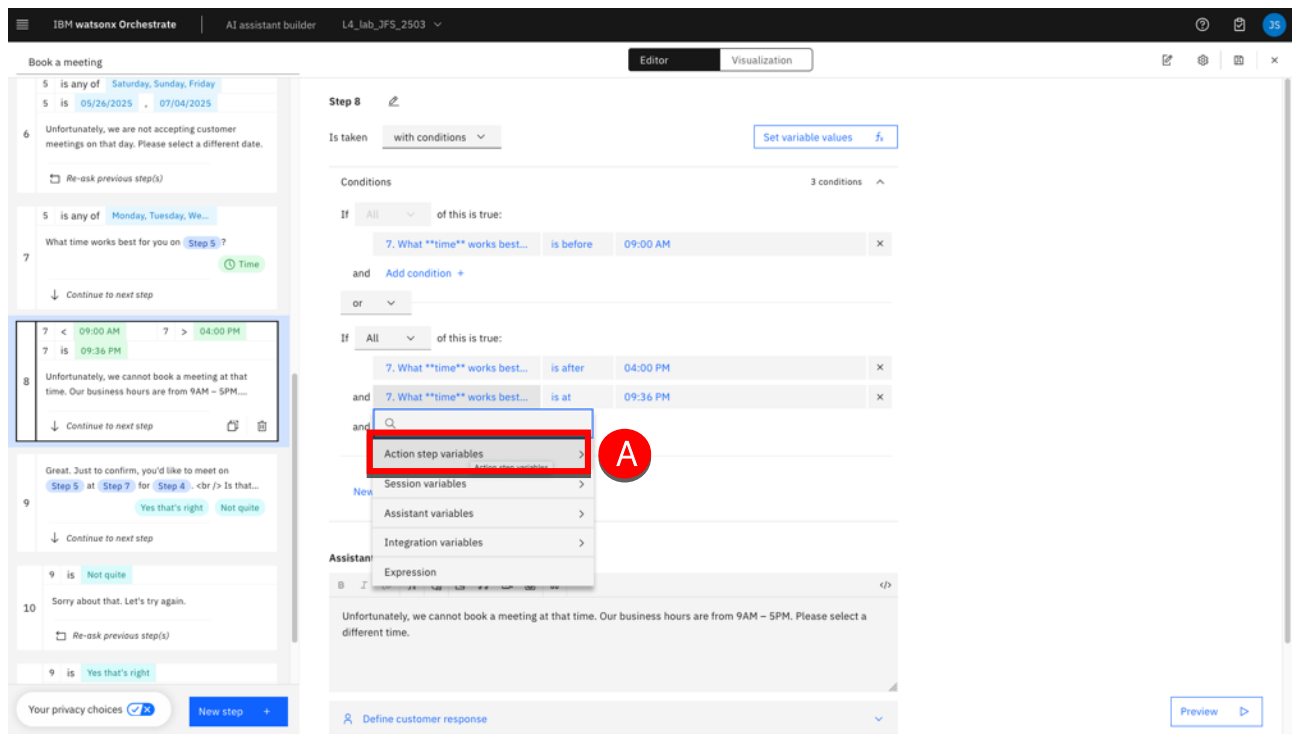
13. In the second condition group, click **Add condition** (A).



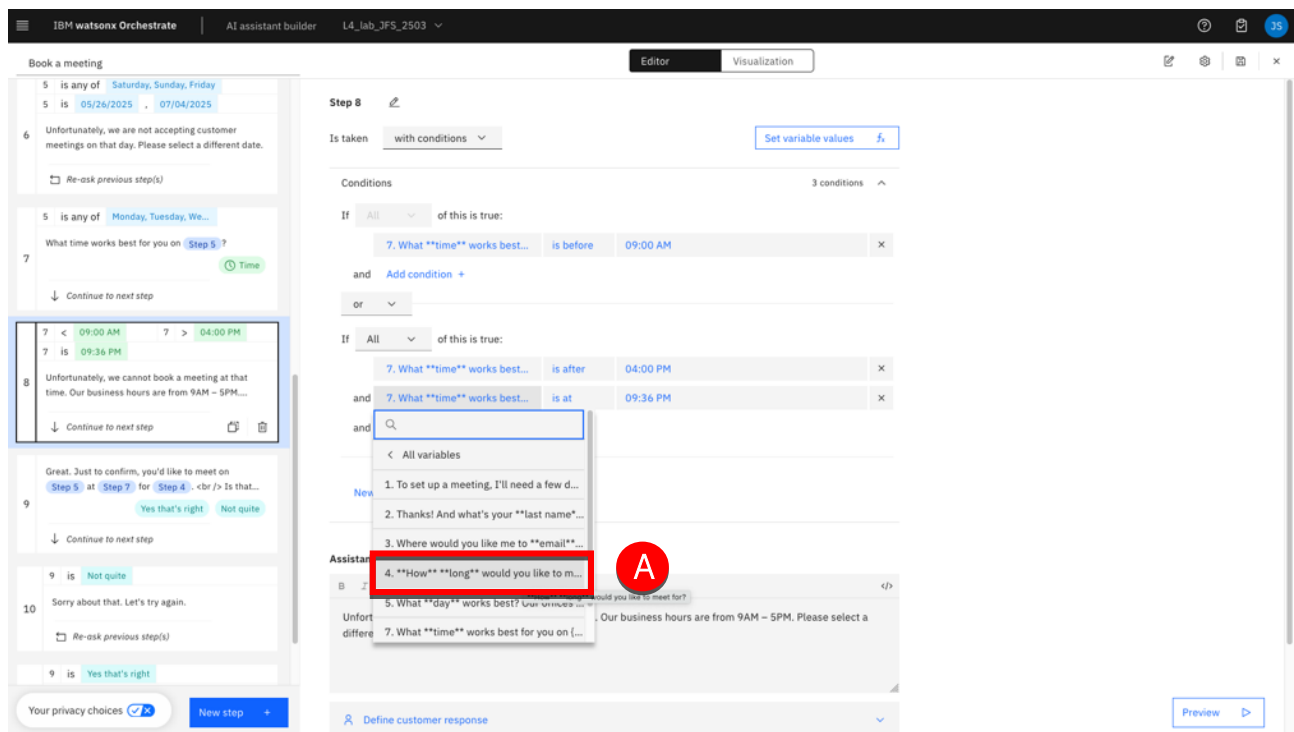
14. Click the variable of the newly added condition (A).



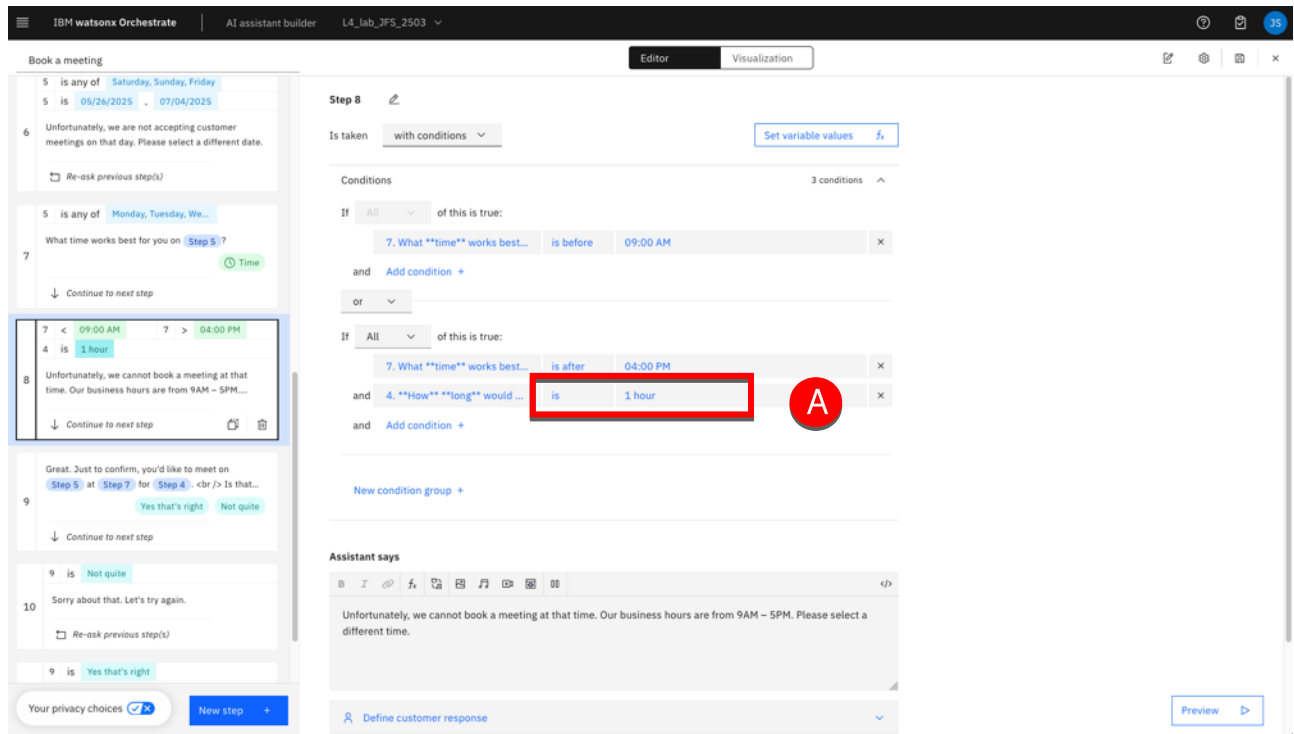
15. In the dropdown menu, click **Action step variables** (A).



16. Click Step 4. How long would you like to meet for? (A).



17. Verify that the operator is set to is and the value is set to 1 hour (A).



This set of conditions will tell the customer to reschedule their meeting if they select a time that is:

*Condition Group 1*

Before 9:00am

OR

*Condition Group 2*

**After 4:00pm** AND they are scheduling a **1 hour** meeting

Next, you will add two more condition groups to build the logic for 30 minute and 15 minute meetings.

18. Repeat steps 9 – 17, this time setting a condition group that specifies **After 4:30pm** AND they are scheduling a **30 minutes** meeting (A).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar shows a flowchart with steps 5 through 10. Step 8 is highlighted with a blue box. The main area shows the configuration for Step 8, which is titled "Book a meeting". The "Is taken" dropdown is set to "with conditions". The "Conditions" section shows a list of 5 conditions. The first condition is "If All of this is true:", followed by two conditions: "7. What \*\*time\*\* works best... is before 09:00 AM" and "4. \*\*How\*\* \*\*long\*\* would ... is 1 hour". The second condition group is "If All of this is true:", followed by two conditions: "7. What \*\*time\*\* works best... is after 04:00 PM" and "4. \*\*How\*\* \*\*long\*\* would ... is 30 minutes". This second condition group is highlighted with a red box and labeled with a red circle containing the letter 'A'. The "Assistant says" section at the bottom shows the text "Great. Just to confirm, you'd like to meet on Step 5 at Step 7 for Step 4. <br /> Is that...".

Verify that your condition groups match the condition groups' configurations in the image.

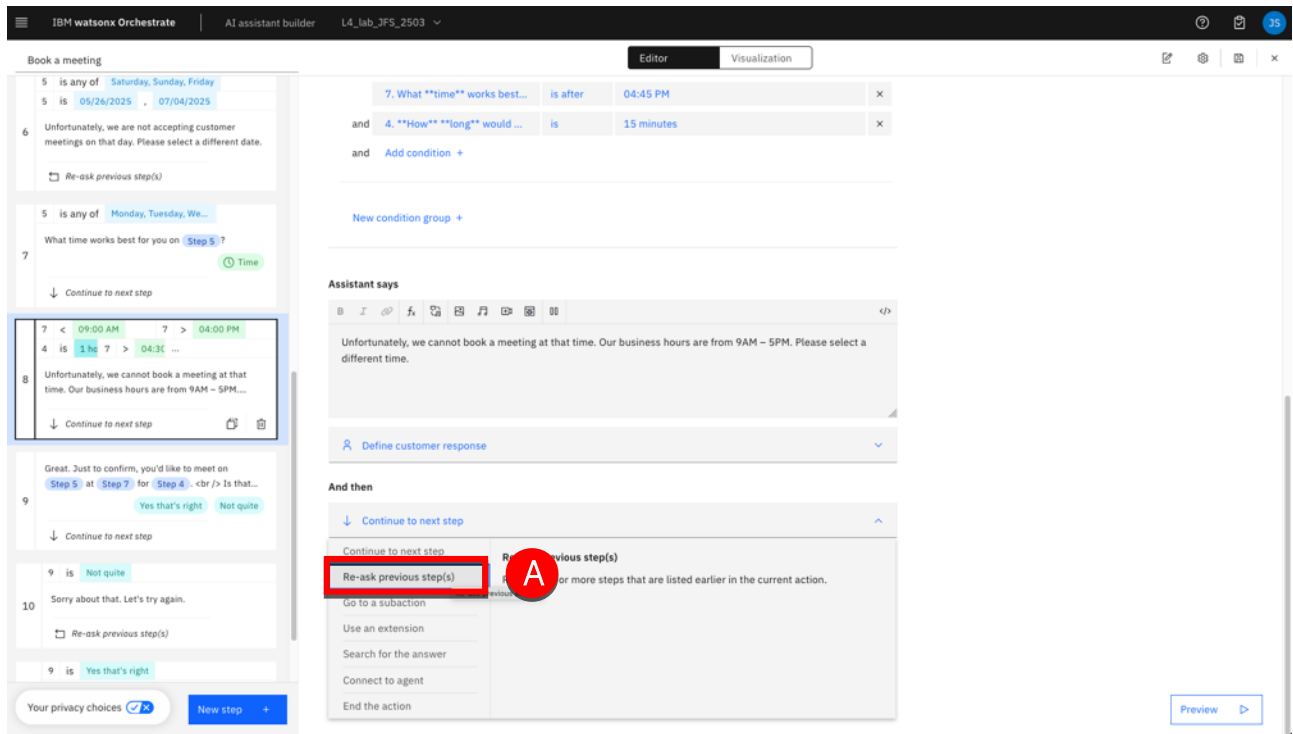
19. Repeat step 18, this time setting a condition group that specifies **After 4:45pm** AND they are scheduling a **15 minutes** meeting (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar displays a flowchart with steps 5 through 10. Step 8 is highlighted, showing a message: "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM...". The main editor area shows a configuration for a condition group. The condition is set to "All" and "of this is true:". The conditions are: "7. What \*\*time\*\* works best..." is after 04:00 PM, and "4. \*\*How\*\* \*\*long\*\* would ..." is 1 hour. A red box labeled 'A' highlights a new condition group being added, with the condition: "7. What \*\*time\*\* works best..." is after 04:45 PM, and "4. \*\*How\*\* \*\*long\*\* would ..." is 15 minutes. The "Assistant says" section shows the message: "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM. Please select a different time." The "And then" section shows a dropdown menu with "Continue to next step" selected.

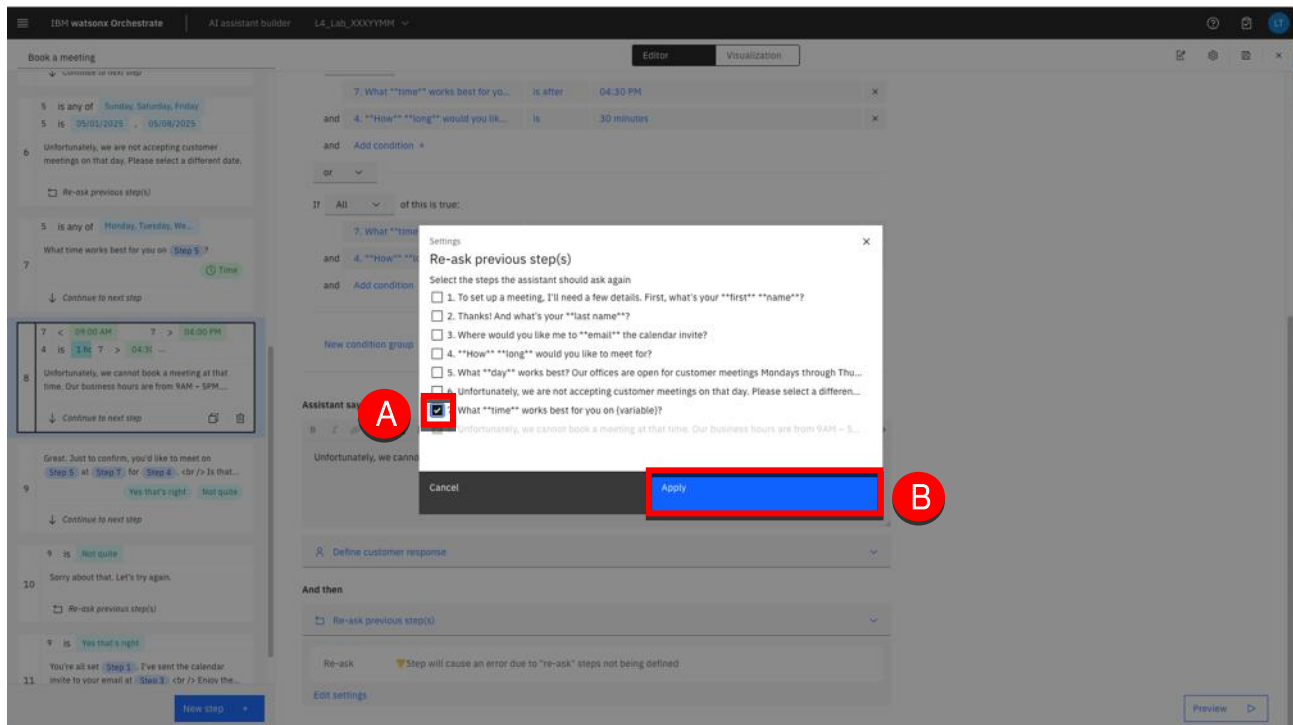
20. Scroll down (A), then click **Continue to the next step** (B).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface, similar to the previous one. The left sidebar shows the flowchart with step 8 highlighted. The main editor area shows the condition group configuration. A red box labeled 'A' is positioned on the right side of the editor, indicating a scroll action. The "Assistant says" section shows the message: "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM. Please select a different time." The "And then" section shows a dropdown menu with "Continue to next step" selected, highlighted by a red box labeled 'B'. The "Preview" button is visible in the bottom right corner.

21. In the dropdown menu, select **Re-ask previous step (A)**.



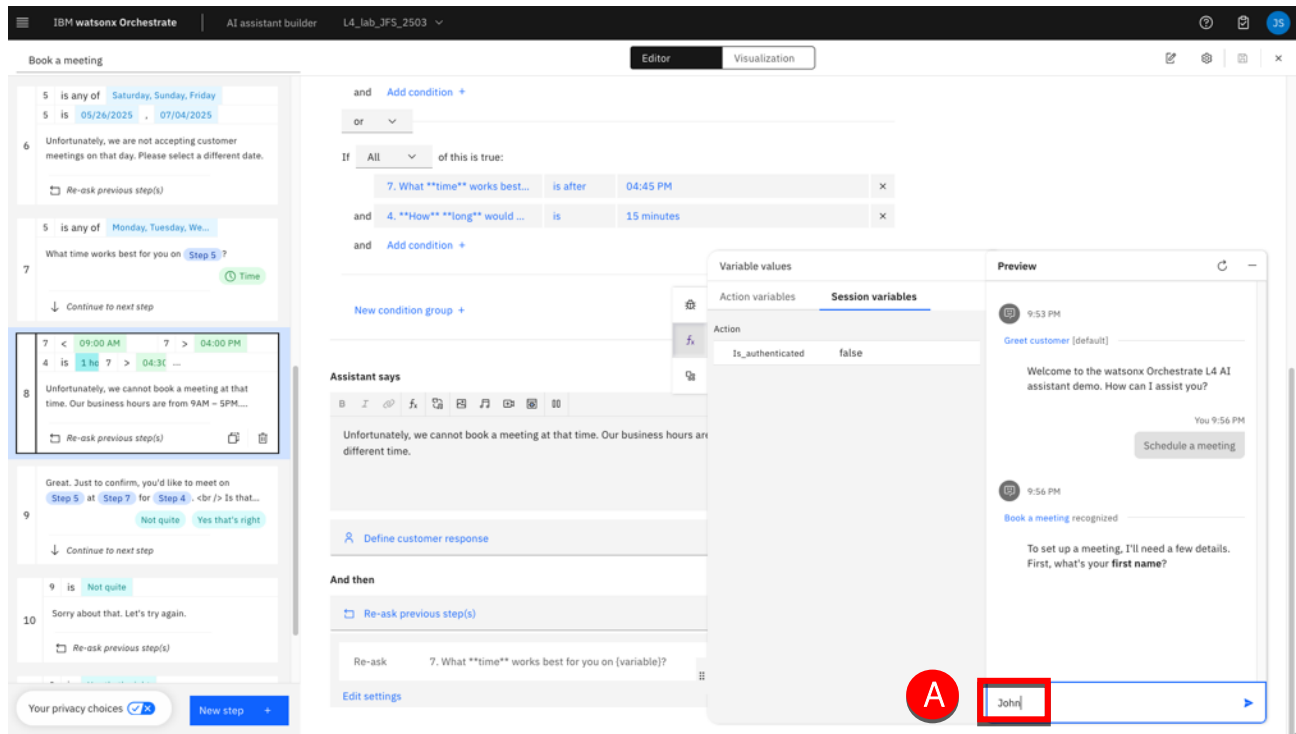
22. In the pop-up, select step **7 (A)** then click **Apply (B)**.



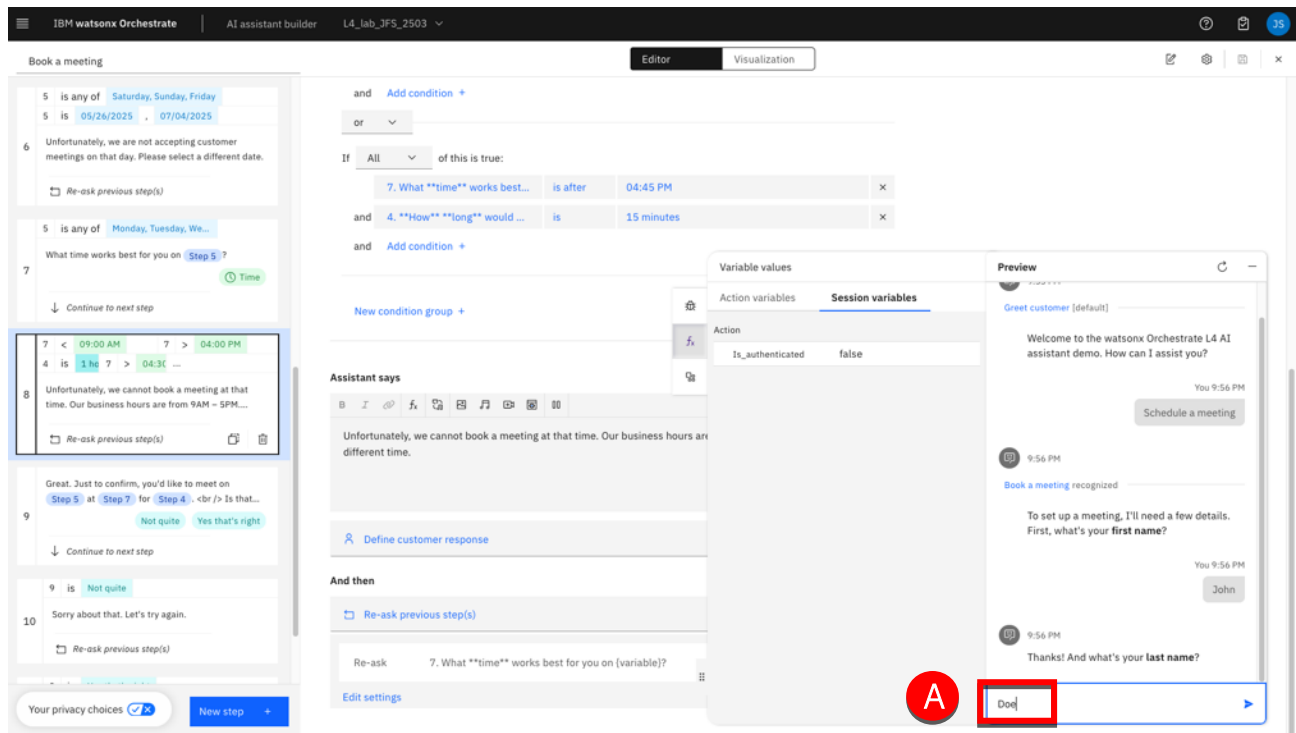
23. Save the action (A), then open and refresh the Preview window (B). Click the **Variable values** icon (C).

24. Type `Schedule a meeting` (A).

25. Type a valid first name (A).



26. Type a valid last name (A).



27. Type a valid email (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Book a meeting

Step 8

Is taken with conditions

Conditions

If All of this is true:

7. What \*\*time\*\* works best... is before 09:00 AM

and Add condition +

If All of this is true:

7. What \*\*time\*\* works best... is after 04:00 PM

and 4. \*\*How\*\* \*\*long\*\* would ... is 1 hour

and Add condition +

If All of this is true:

7. What \*\*time\*\* works best... is after 04:30 PM

and 4. \*\*How\*\* \*\*long\*\* would ... is 30 minutes

and Add condition +

If All of this is true:

7. What \*\*time\*\* works best... is after 04:45 PM

and 4. \*\*How\*\* \*\*long\*\* would ... is 15 minutes

and Add condition +

Variable values

Action variables

Greet customer

Action name Book a meeting

Action variables

To set up a ... John

Thanks! An... Doe

Where woul... jdoe@acm ...

\*\*How\*\* \*\*L... 30 minutes

What \*\*day... 1 hour

Unfortunate... 15 minutes

Great. Just ...

Unfortunate...

Sorry about ...

You're all se...

Preview

10:47 PM

Book a meeting recognized

To set up a meeting, I'll need a few details. First, what's your first name?

You 10:47 PM

John

10:47 PM

Thanks! And what's your last name?

You 10:48 PM

Doe

10:48 PM

Where would you like me to email the calendar invite?

You 10:49 PM

jdoe@acme.com

10:49 PM

How long would you like to meet for?

15 minutes

30 minutes

1 hour

Sorry about ...

Type something...

28. Click 30 minutes (A).

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS\_2503

Book a meeting

Step 8

Is taken with conditions

Conditions

If All of this is true:

7. What \*\*time\*\* works best... is before 09:00 AM

and Add condition +

If All of this is true:

7. What \*\*time\*\* works best... is after 04:00 PM

and 4. \*\*How\*\* \*\*long\*\* would ... is 1 hour

and Add condition +

If All of this is true:

7. What \*\*time\*\* works best... is after 04:30 PM

and 4. \*\*How\*\* \*\*long\*\* would ... is 30 minutes

and Add condition +

If All of this is true:

7. What \*\*time\*\* works best... is after 04:45 PM

and 4. \*\*How\*\* \*\*long\*\* would ... is 15 minutes

and Add condition +

Variable values

Action variables

Greet customer

Action name Book a meeting

Action variables

To set up a ... John

Thanks! An... Doe

Where woul... jdoe@acm ...

\*\*How\*\* \*\*L... 30 minutes

What \*\*day... 1 hour

Unfortunate... 15 minutes

Great. Just ...

Unfortunate...

Sorry about ...

You're all se...

Preview

10:47 PM

Thanks! And what's your last name?

You 10:48 PM

Doe

10:48 PM

Where would you like me to email the calendar invite?

You 10:49 PM

jdoe@acme.com

10:49 PM

How long would you like to meet for?

15 minutes

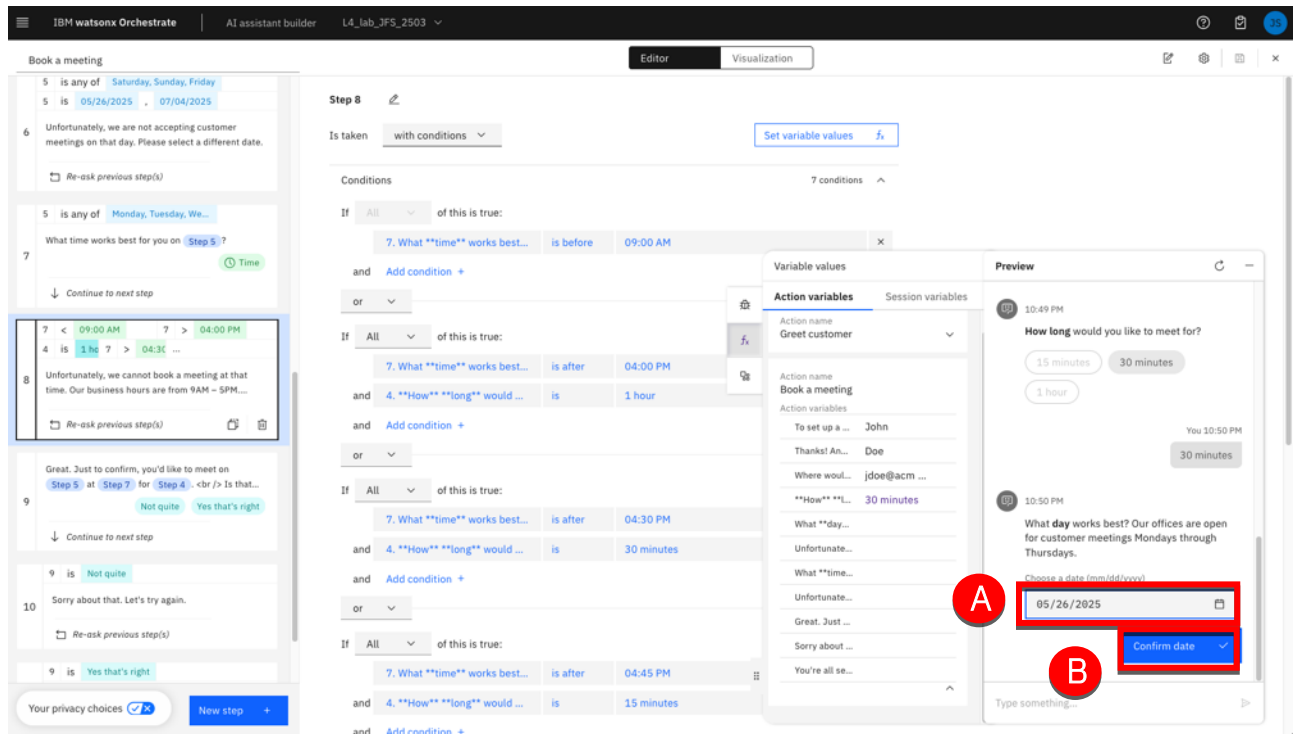
30 minutes

1 hour

Sorry about ...

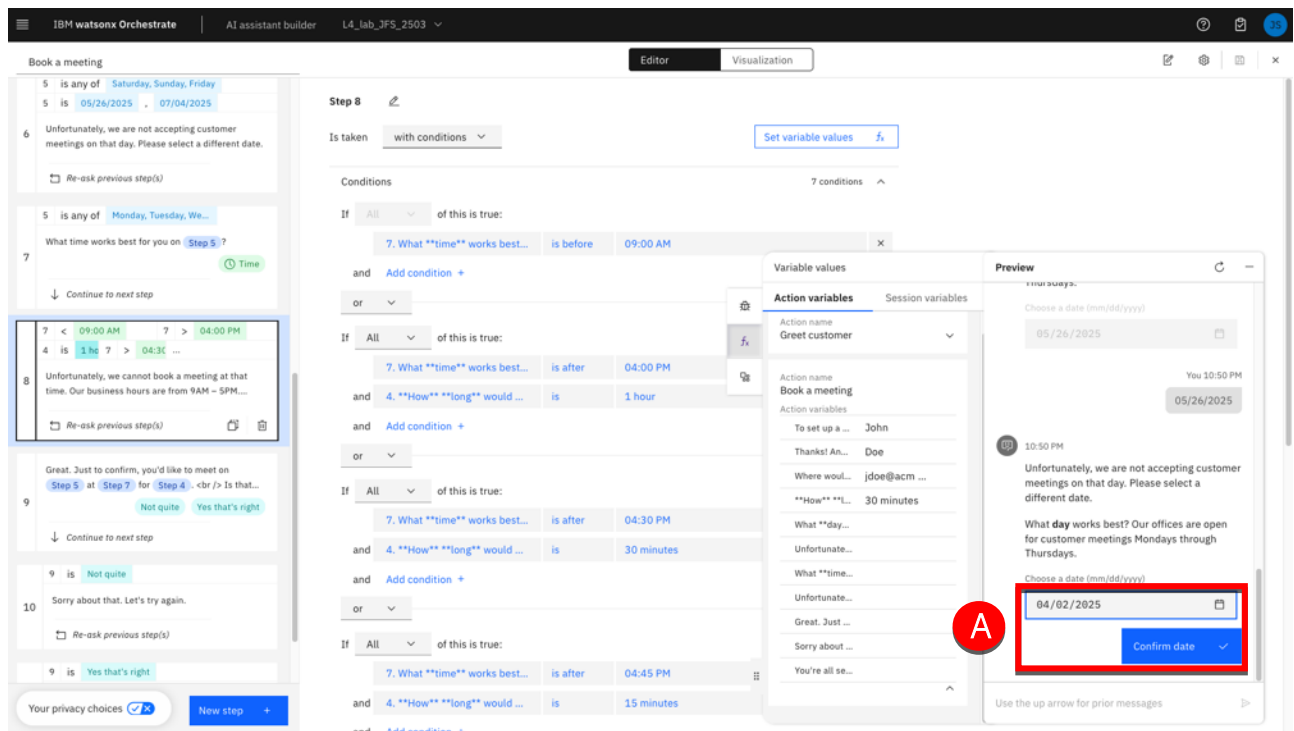
Type something...

29. Enter the date of one of the business holidays you specified earlier (A). Click **Confirm date** (B).

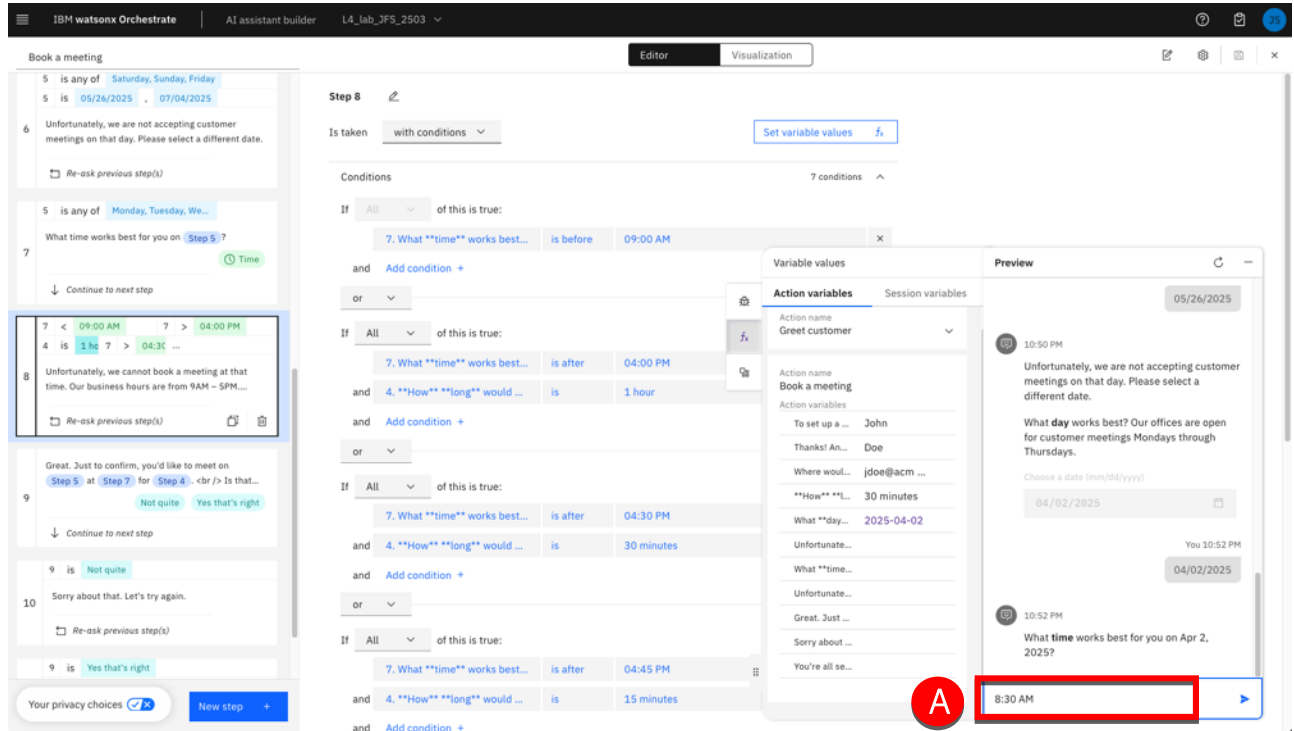


Note that the AI assistant did not accept the inputted date.

30. Submit another valid date (Monday – Thursday) (A).

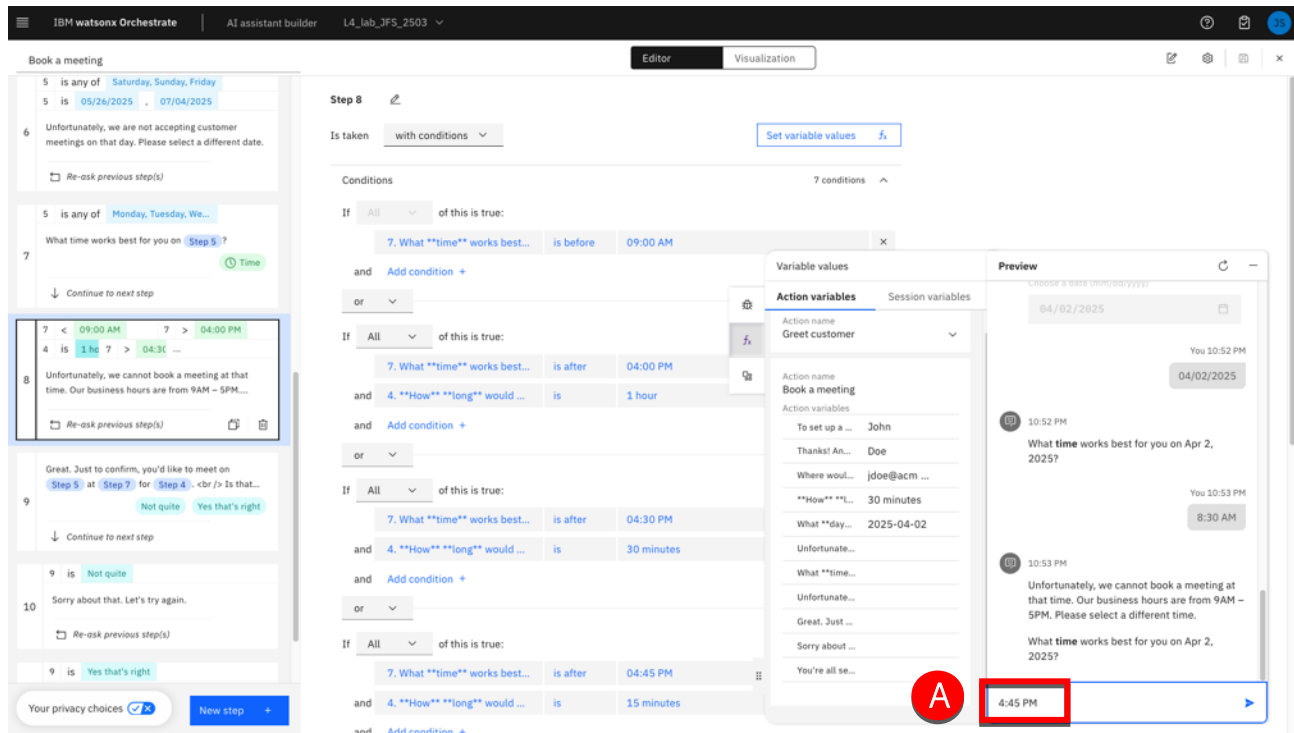


31. Type 8:30 AM (A).



Note that the Assistant did not accept this time, because it is before business hours.

32. Type 4:45 PM (A).



Note that the Assistant did not accept this time, because a 30 minute meeting at 4:45 would end after business hours.

33. Type 4 : 30 PM (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The workflow is titled "Book a meeting". The steps are as follows:

- Step 5: "Is any of Sunday, Saturday, Friday" (Date selection)
- Step 6: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." (Re-ask previous step(s))
- Step 7: "What time works best for you on Step 5?" (Time selection)
- Step 8: "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM..." (Re-ask previous step(s))
- Step 9: "Great. Just to confirm, you'd like to meet on Step 5 at Step 7 for Step 8. Is that..." (Confirmation)
- Step 10: "Sorry about that. Let's try again." (Re-ask previous step(s))
- Step 11: "You're all set Step 1. I've sent the calendar invite to your email at Step 3. <br /> Enjoy the..." (End of workflow)

The "Preview" pane on the right shows the conversation flow. A red box highlights the "4:30 PM" time selection in the preview pane, with a red circle and letter "A" next to it.

34. Notice that the AI assistant accepted the valid time, click **Yes, that's right** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The workflow is titled "Book a meeting". The steps are as follows:

- Step 5: "Is any of Saturday, Sunday, Friday" (Date selection)
- Step 6: "Unfortunately, we are not accepting customer meetings on that day. Please select a different date." (Re-ask previous step(s))
- Step 7: "What time works best for you on Step 5?" (Time selection)
- Step 8: "Unfortunately, we cannot book a meeting at that time. Our business hours are from 9AM - 5PM..." (Re-ask previous step(s))
- Step 9: "Great. Just to confirm, you'd like to meet on Step 5 at Step 7 for Step 8. Is that..." (Confirmation)
- Step 10: "Sorry about that. Let's try again." (Re-ask previous step(s))
- Step 11: "You're all set Step 1. I've sent the calendar invite to your email at Step 3. <br /> Enjoy the..." (End of workflow)

The "Preview" pane on the right shows the conversation flow. A red box highlights the "Yes that's right" button in the preview pane, with a red circle and letter "A" next to it.

35. You have completed this action.

The screenshot displays the IBM Watsonx Orchestrate AI assistant builder interface. The main workspace is divided into three panels: a left sidebar with a step list, a central editor, and a right preview panel.

- Left Sidebar:** A vertical list of steps (1-10) for a 'Book a meeting' workflow. Step 8 is highlighted with a blue border. The steps include date selection, time selection, and confirmation logic.
- Central Editor:** Shows the configuration for 'Step 8'. It is set to 'Is taken with conditions'. The conditions are defined as a series of 'If All' groups:
  - Condition 1: '7. What \*\*time\*\* works best...' is before '09:00 AM'.
  - Condition 2: '7. What \*\*time\*\* works best...' is after '04:00 PM'.
  - Condition 3: '7. What \*\*time\*\* works best...' is after '04:30 PM'.
  - Condition 4: '4. \*\*How\*\* \*\*long\*\* would ...' is '1 hour'.
  - Condition 5: '4. \*\*How\*\* \*\*long\*\* would ...' is '30 minutes'.
  - Condition 6: '7. What \*\*time\*\* works best...' is after '04:45 PM'.
  - Condition 7: '4. \*\*How\*\* \*\*long\*\* would ...' is '15 minutes'.
- Right Preview Panel:** Shows a chat conversation with the assistant. The assistant's response for Step 8 is: 'Great. Just to confirm, you'd like to meet on Apr 2, 2025 at 4:30 PM for 30 minutes. Is that correct?'. The user responds 'Yes that's right'. The assistant then confirms the booking: 'You're all set John. I've sent the calendar invite to your email at jdoe@acme.com'. The preview panel also shows a 'Variable values' table with the following data:

| Variable         | Value           |
|------------------|-----------------|
| Action name      | Greet customer  |
| Action name      | Book a meeting  |
| Action variables |                 |
| To set up a ...  | John            |
| Thanks! An...    | Doe             |
| Where woul...    | jdoe@acm ...    |
| **How** **L...   | 30 minutes      |
| What **day...    | 2025-04-02      |
| Unfortunate...   |                 |
| What **time...   | 16:30:00        |
| Unfortunate...   |                 |
| Great. Just ...  | Yes that's r... |
| Sorry about ...  |                 |
| You're all se... | true            |

Note: Optionally, you can test other meeting configurations, such as the 1 hour and 15 minute meeting times to verify the business logic of the assistant.

By using condition groups, you programmed complex logic into an action step.

## Embed the Web Chat

Now that your assistant is complete, let's explore how you can use it from an existing web page. To learn how to embed your assistant, you will need to have access to a website from where you can edit its HTML code to insert your assistant's embedding script.

Let's start by creating a demo webpage on GitHub using GitPages.

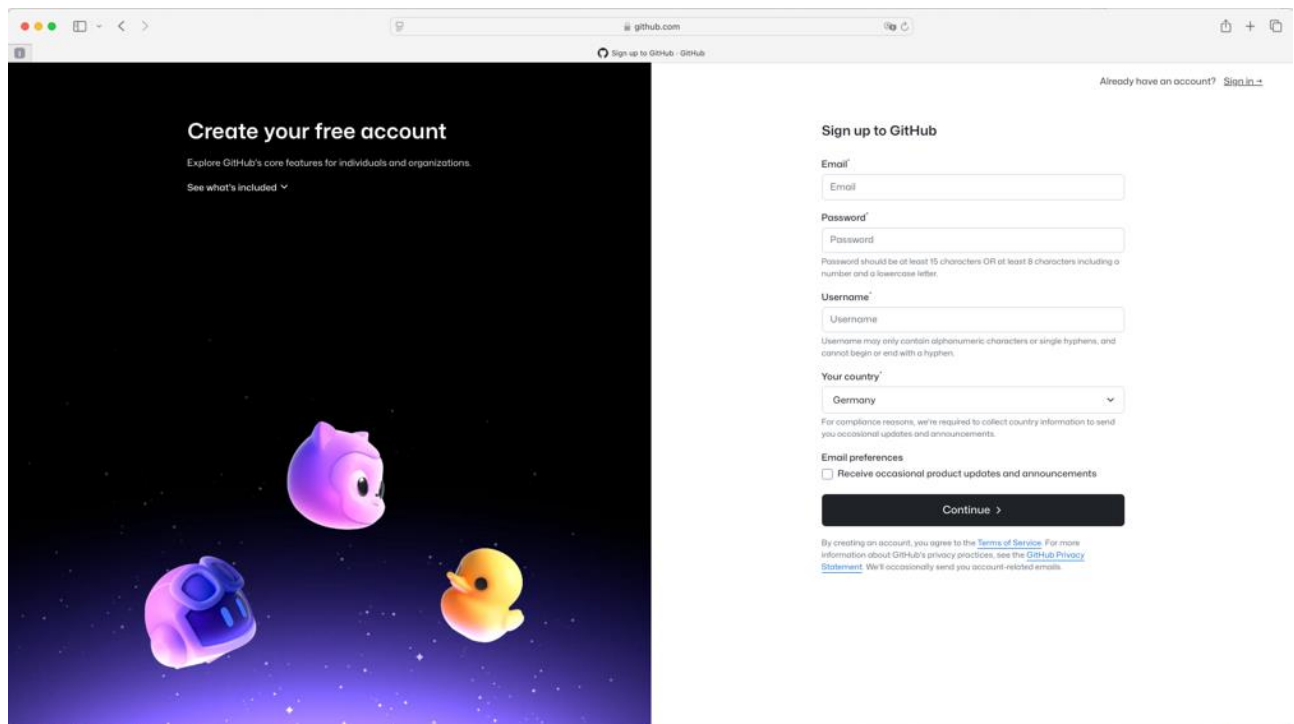
### Create a demo website

To learn how to embed your assistant in an existing website, you will simulate a bank website. In this section, we suggest you create this site using GitHub pages.

**Note:** You can embed your assistant in any HTML page if you already have one ready to receive it.

### Create or log in to your GitHub account

If you already have a GitHub account, log in your personal account. Otherwise create an account [here](https://www.github.com). ([www.github.com](https://www.github.com) Signup section)

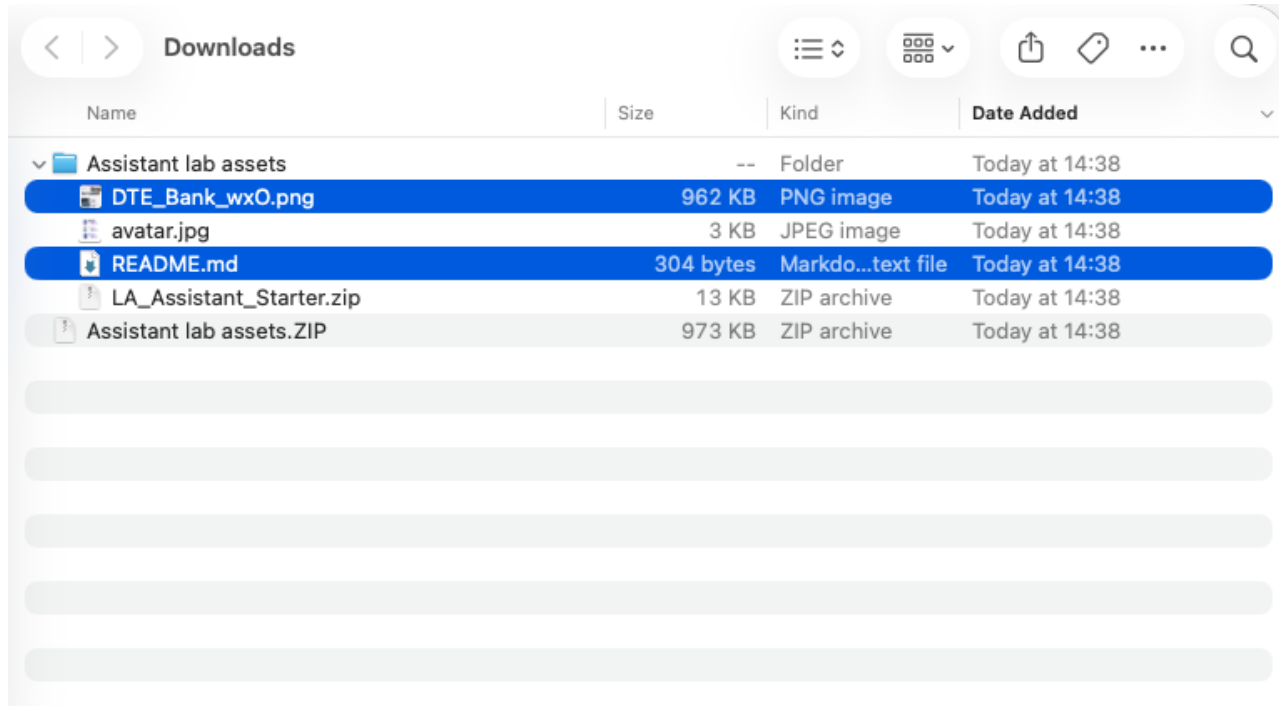


### Get the required files

You will need 2 files to simulate a bank web site:

- The site image
- The Readme file of the GitHub repository

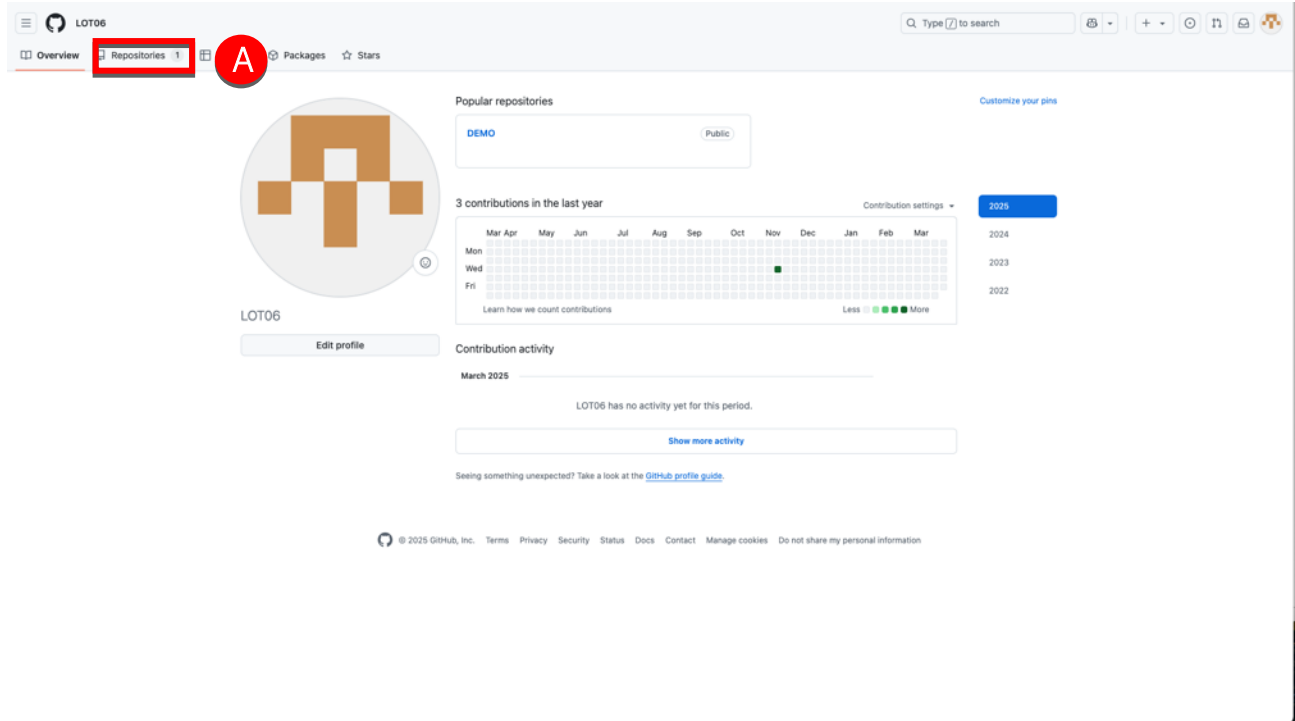
You will find these files in the archive file you have extracted in the pre-requisite section of this guide.



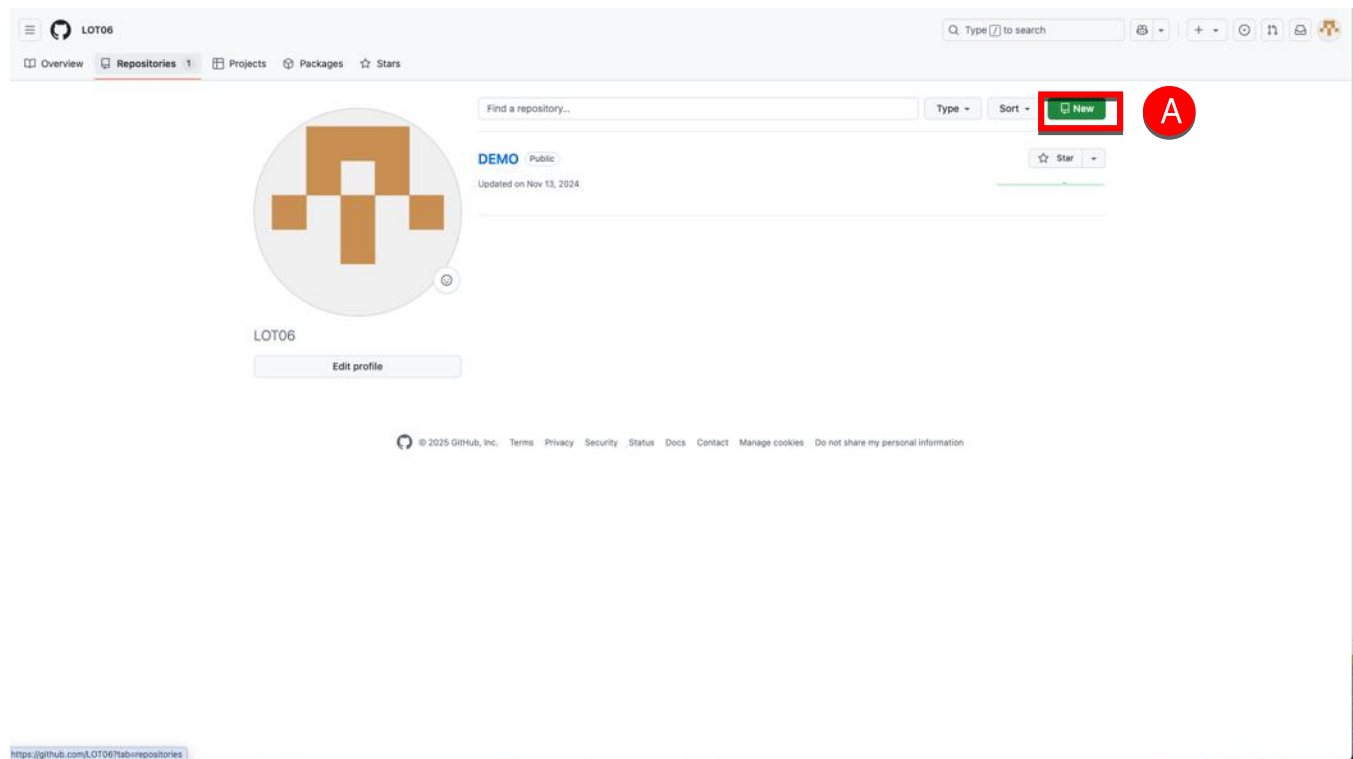
## Create the repository

You will now create a repository for the GitHub page of your simulated website.

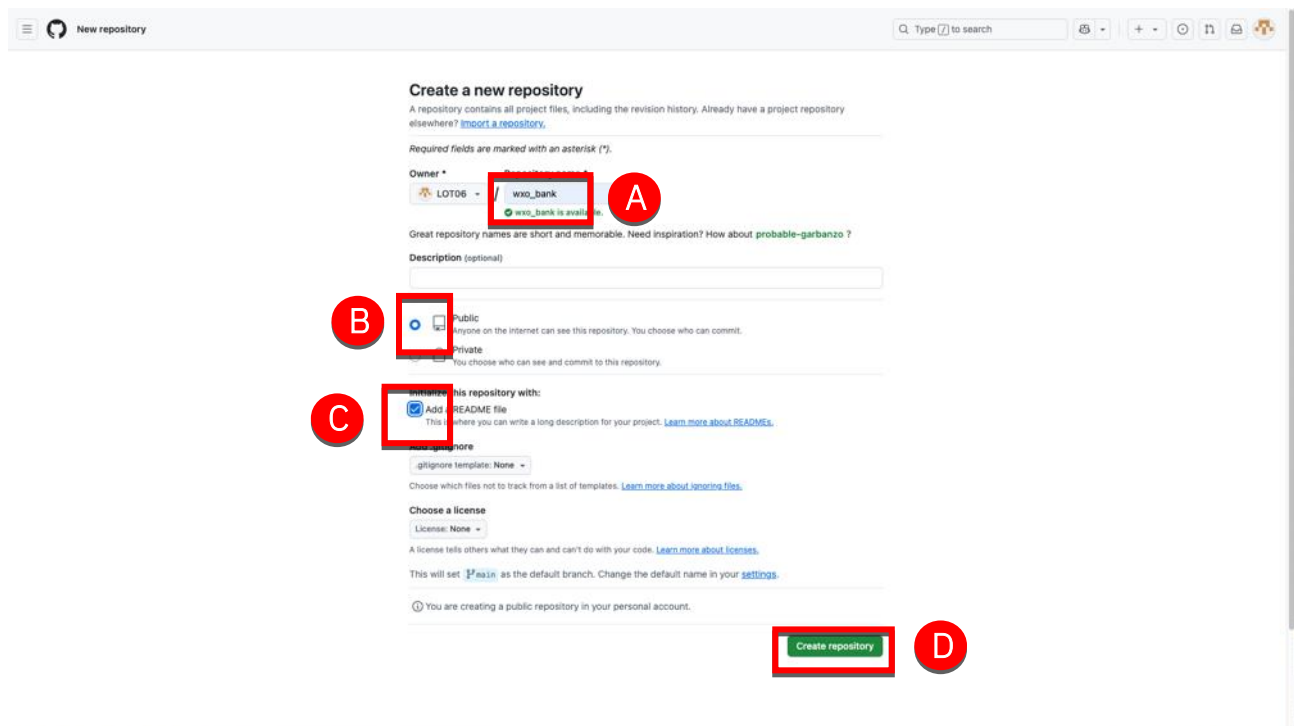
1. Login to your personal GitHub account.
2. Click your profile picture, then click **Your profile**.
3. Click **Repositories** (A).



4. Click **New** (A).

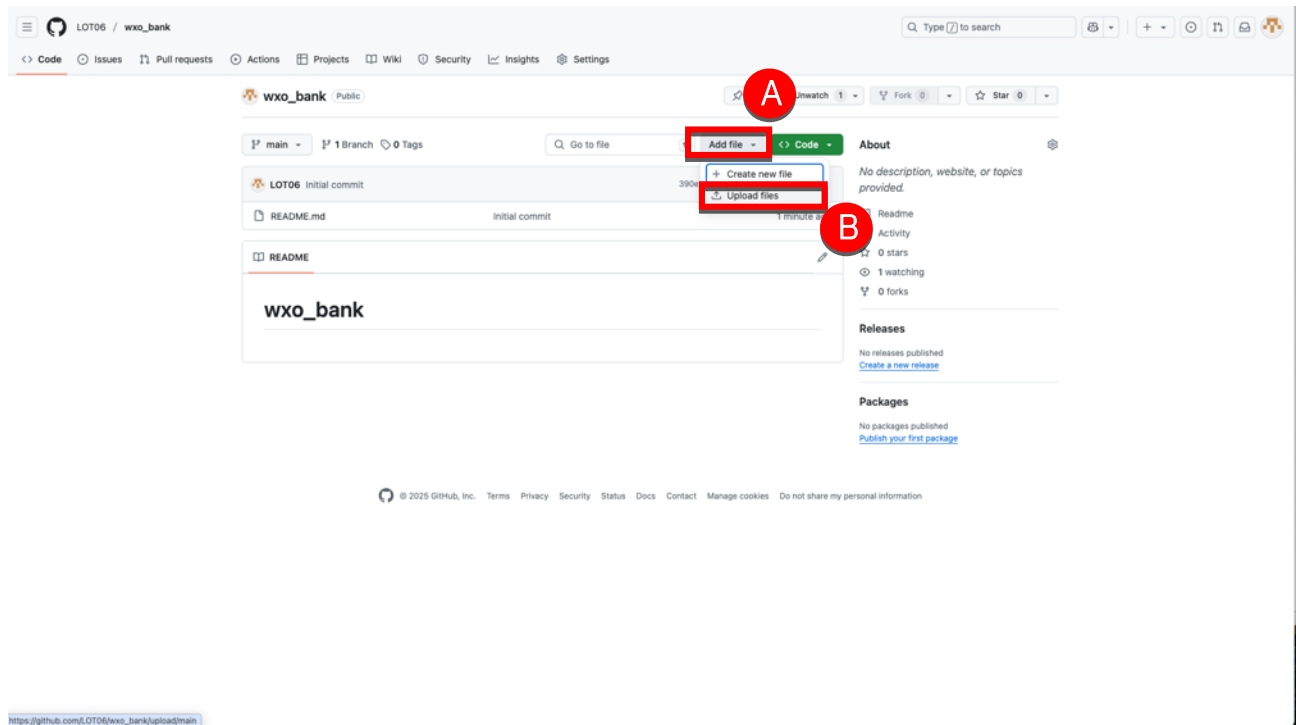


5. Type **wxo\_bank** in the **Repository name** (A), select **Public** (B), select **Add a readme file** (C) then click **Create repository** (D).

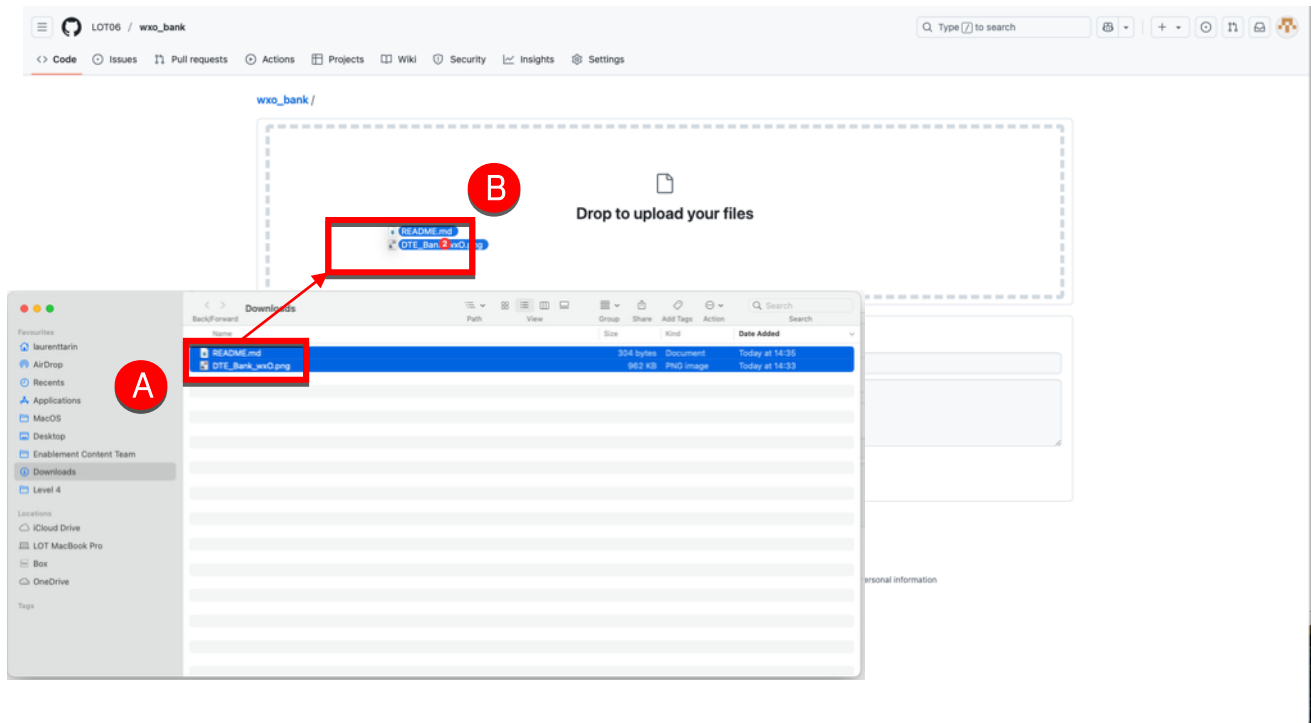


## Add the files to the repository

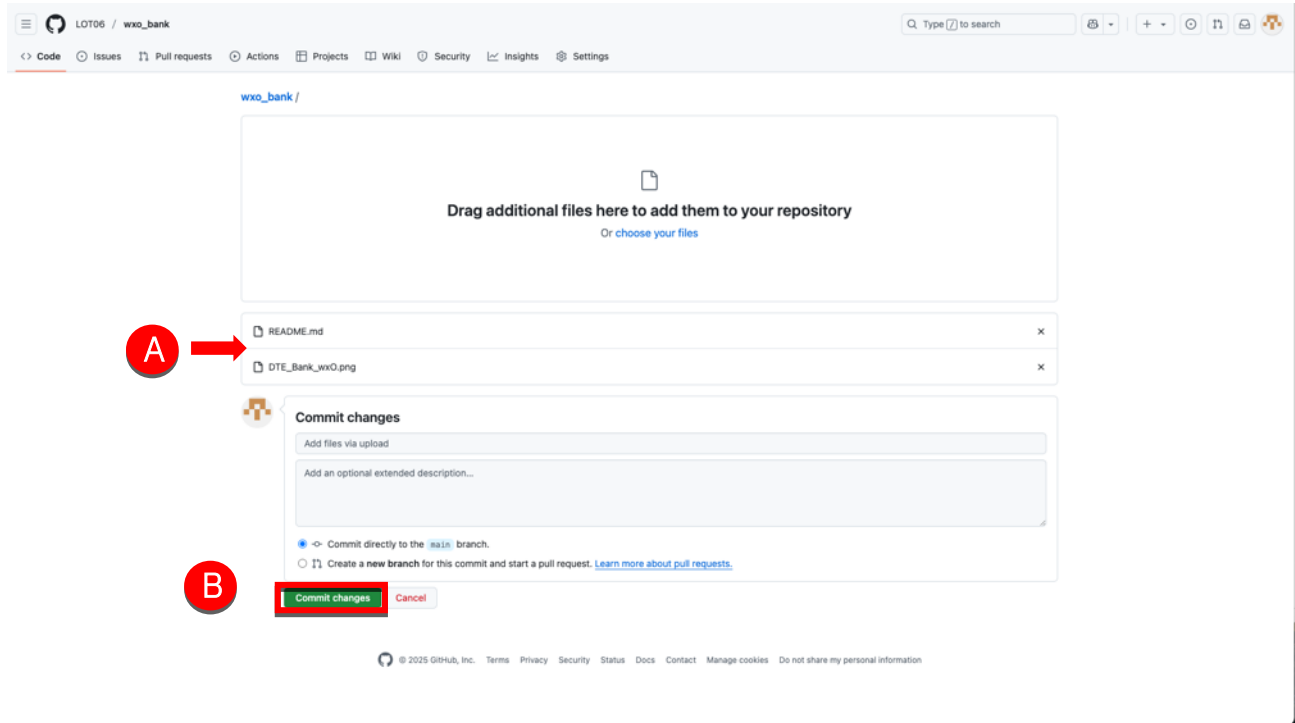
1. Click **Add file** (A) then select **Upload files** (B).



2. Select the 2 files you downloaded in the previous step (A) and drop them in the upload area (B).

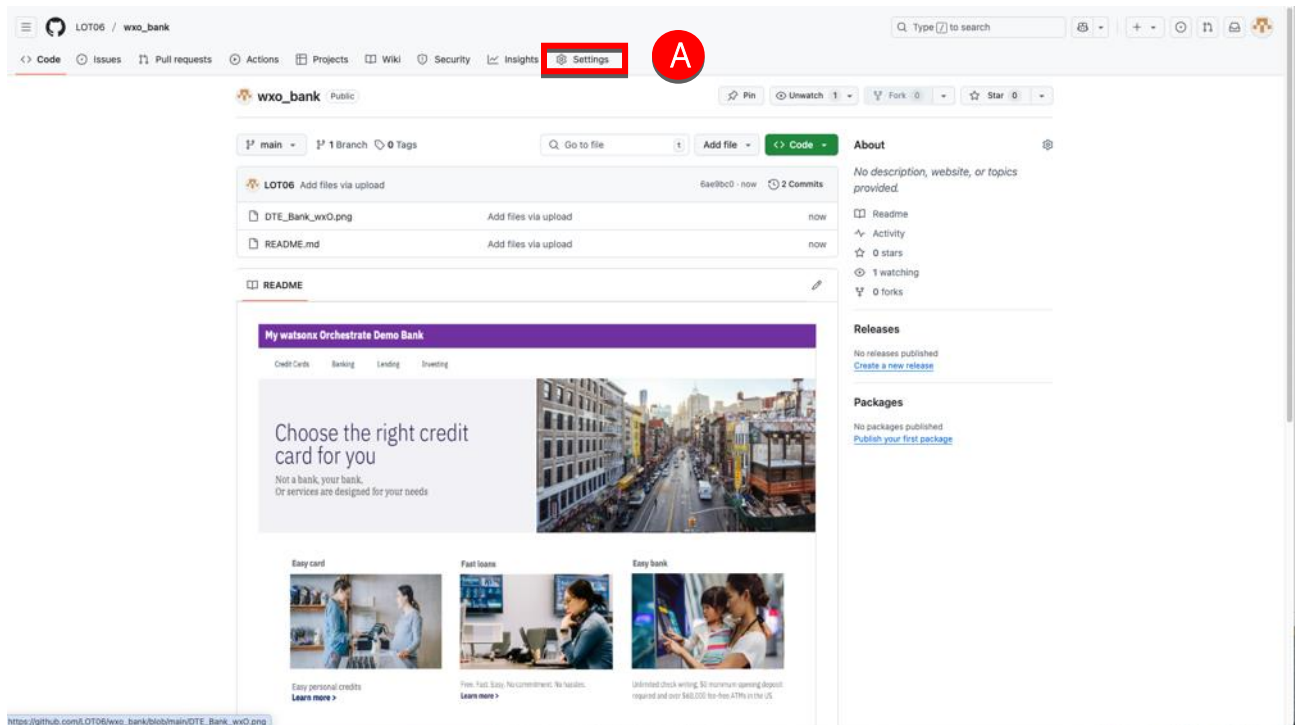


- Wait for the upload to be completed (A) then click **Commit changes** (B).

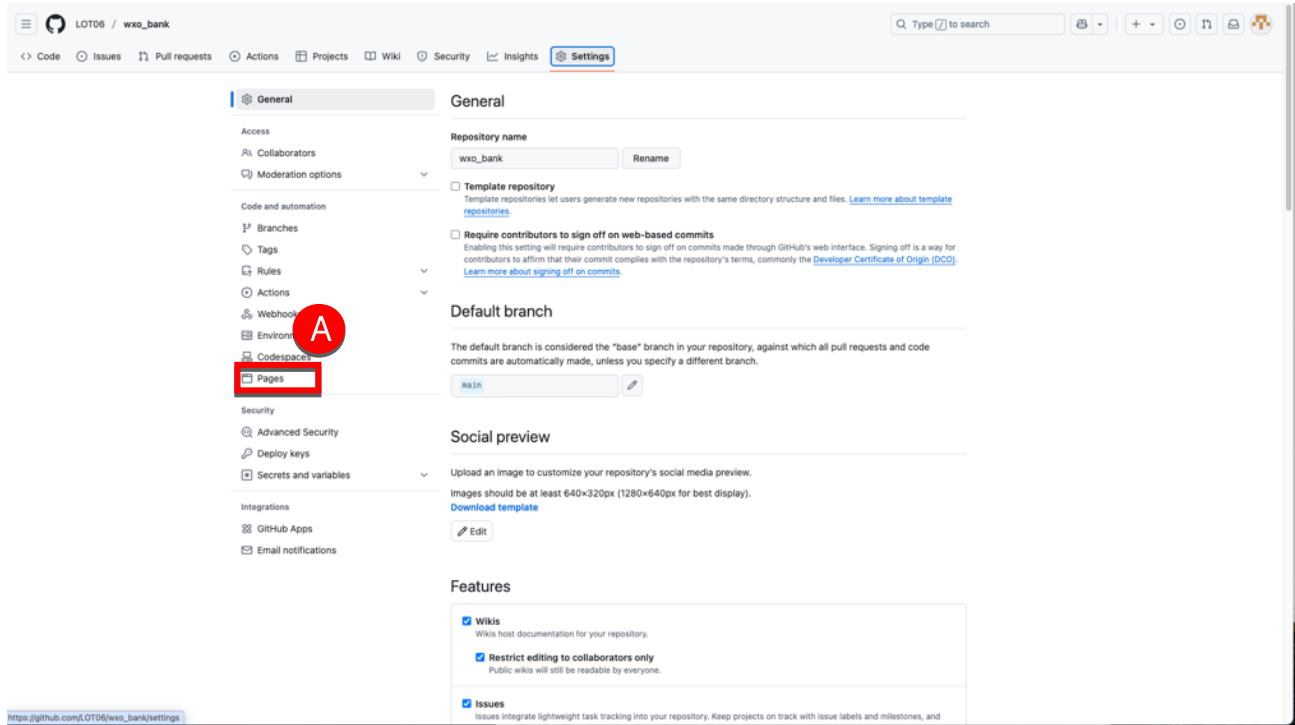


## Configure your GitHub page

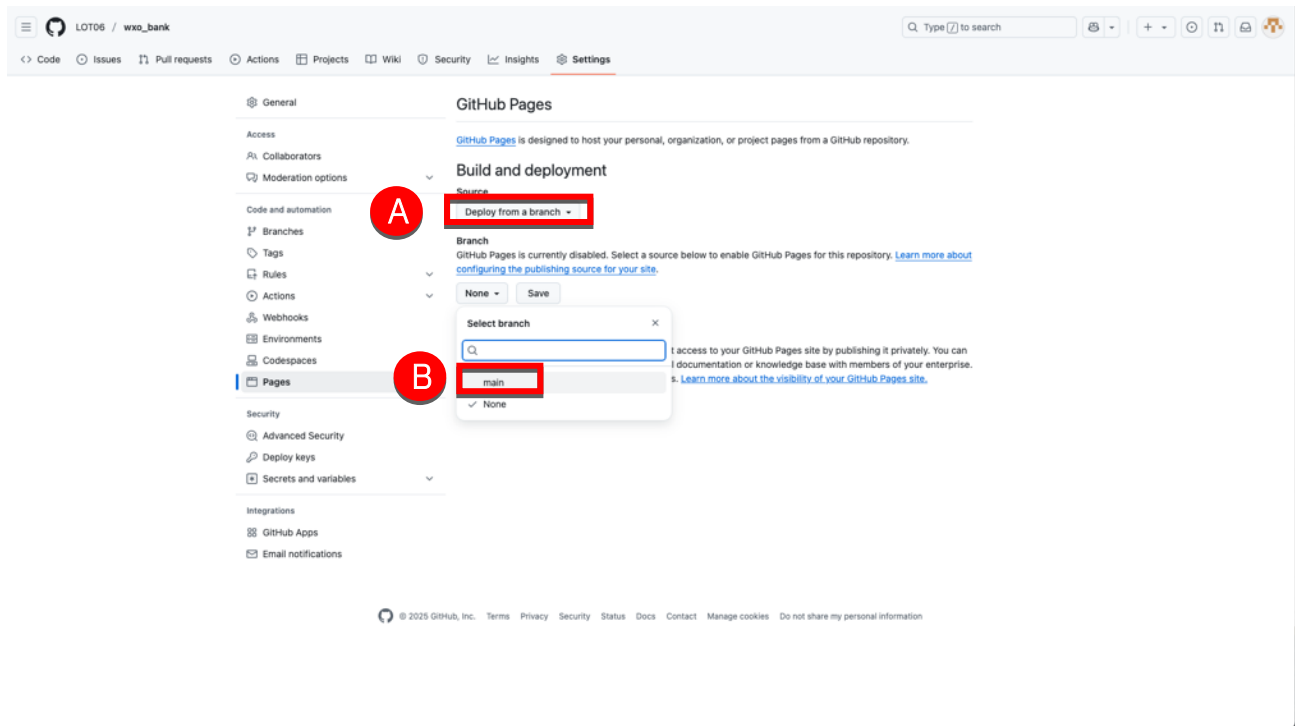
- Click **Settings** (A).



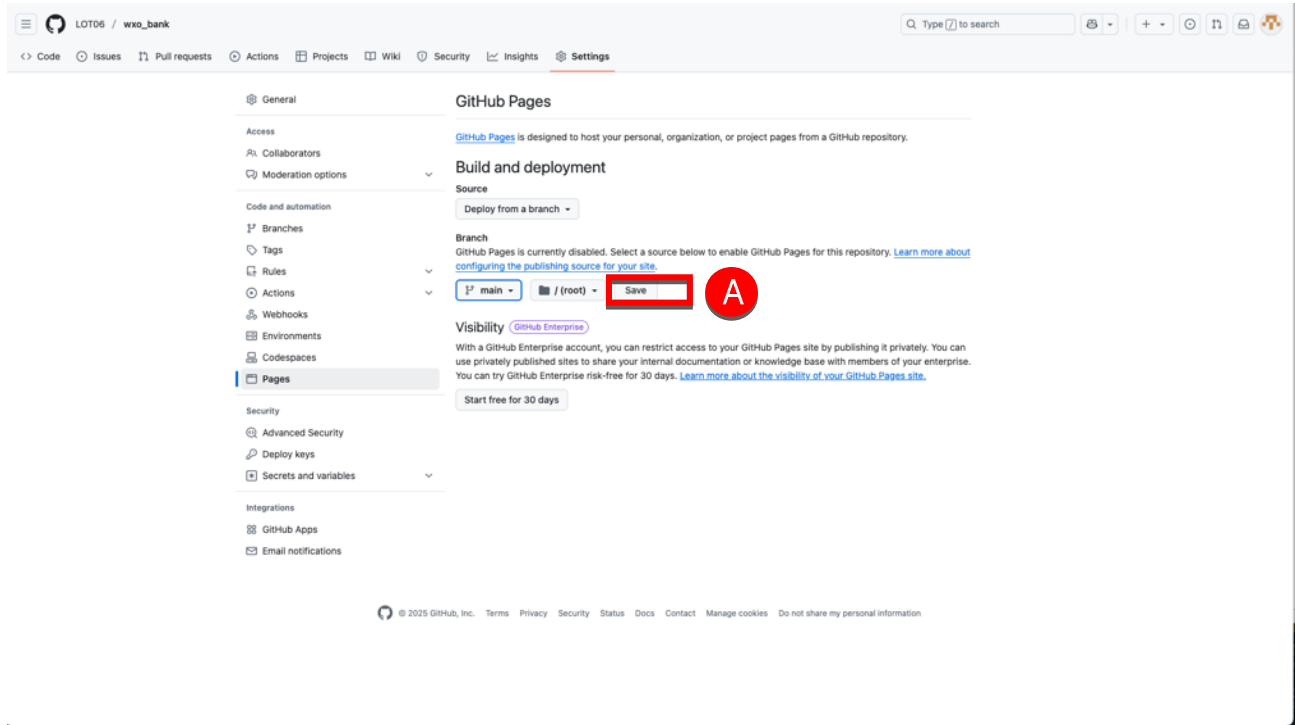
2. Click **Pages** (A).



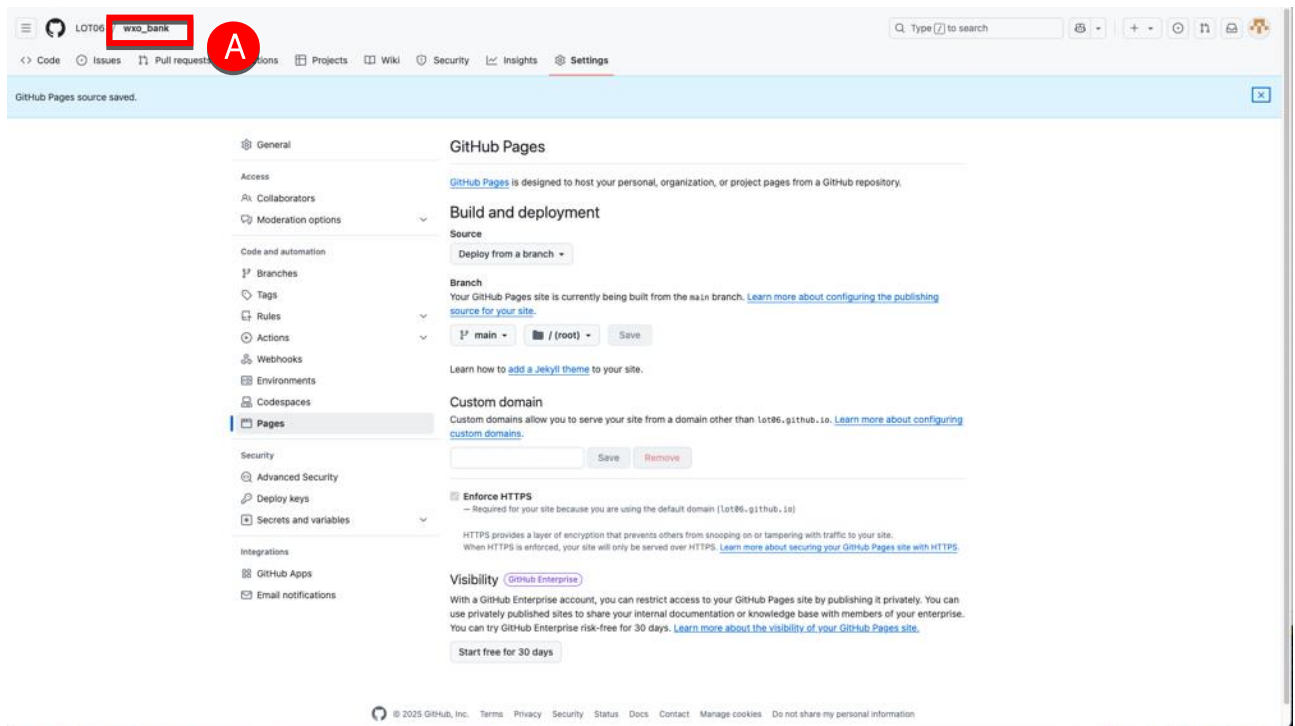
3. Make sure **Deploy from a branch** is selected (A) then expand **None** (B) to select **main** (C).



4. Click **Save** (A).

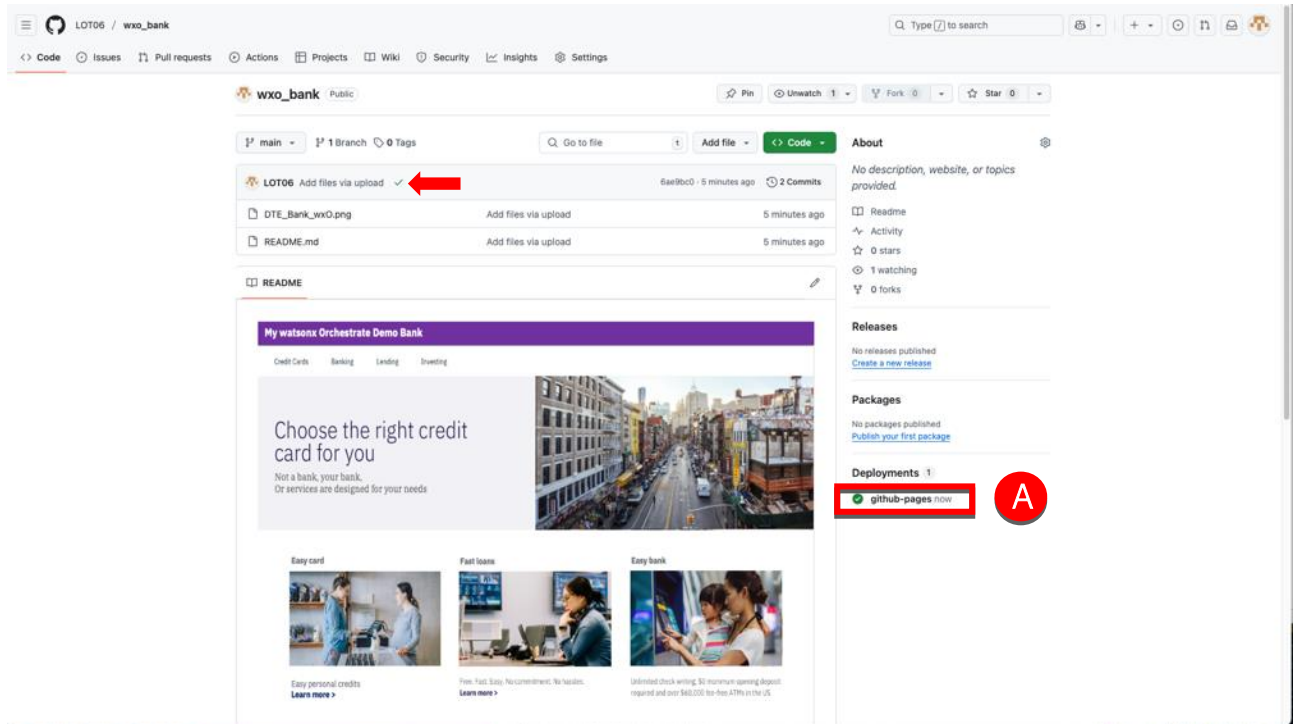


5. Click your **wxx\_bank** repository link (A).

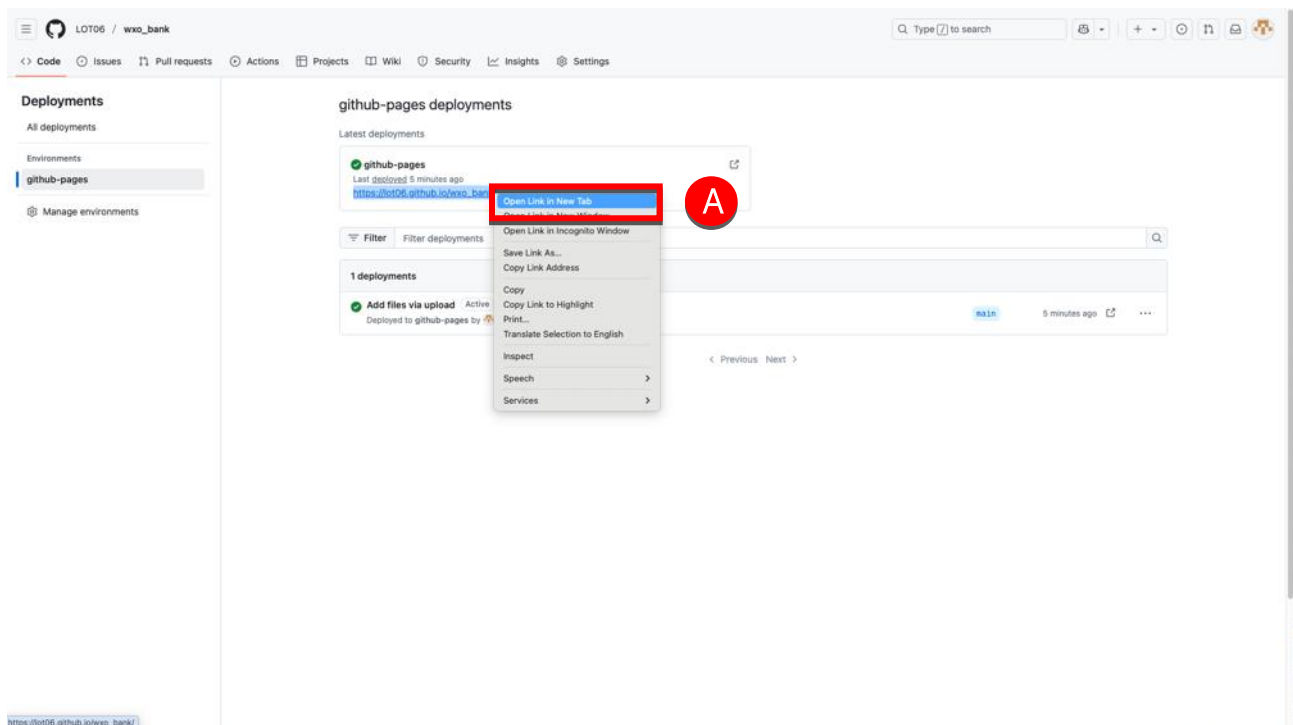


Wait a minute or two for the GitHub-page deployment to be performed.

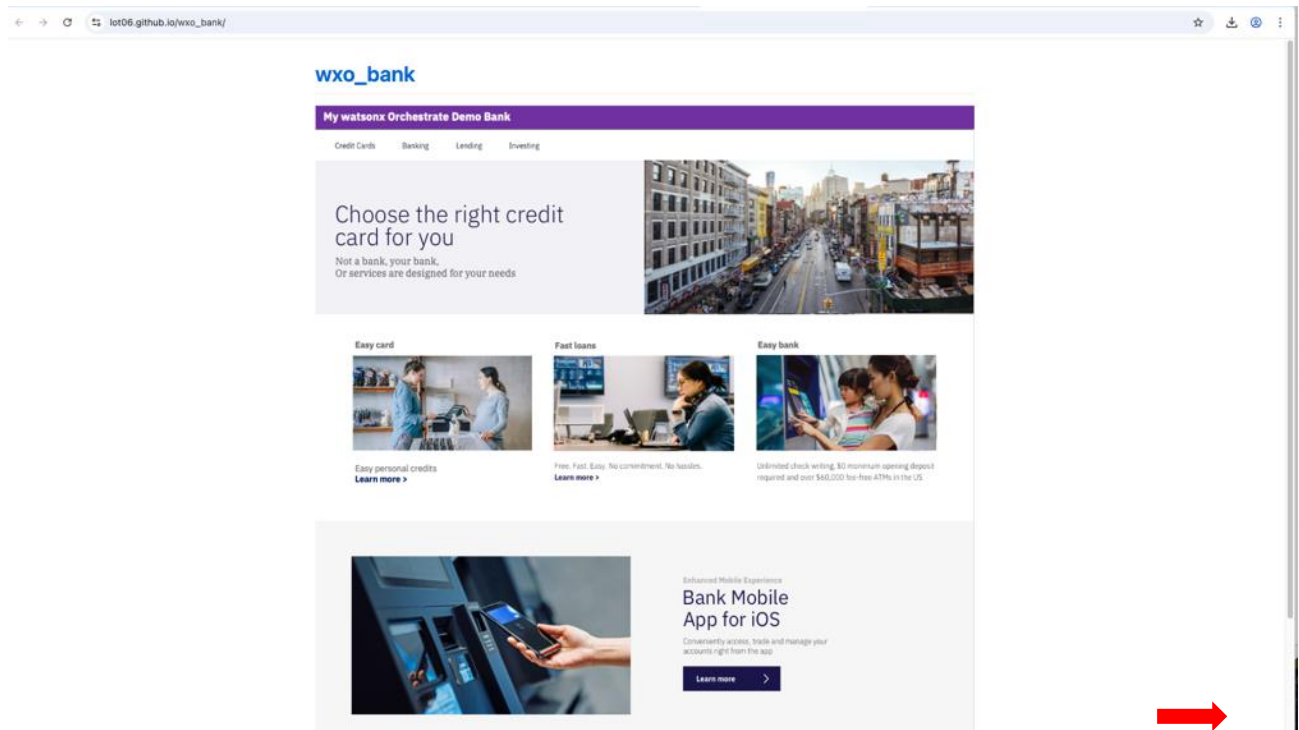
6. Refresh your page if needed until you see the GitHub-page deployment in green. Click the **GitHub-pages** link (A).



7. Your simulated website is ready. Right-click the public website link and select **Open Link in New Tab** (A).



8. Go to the new tab.

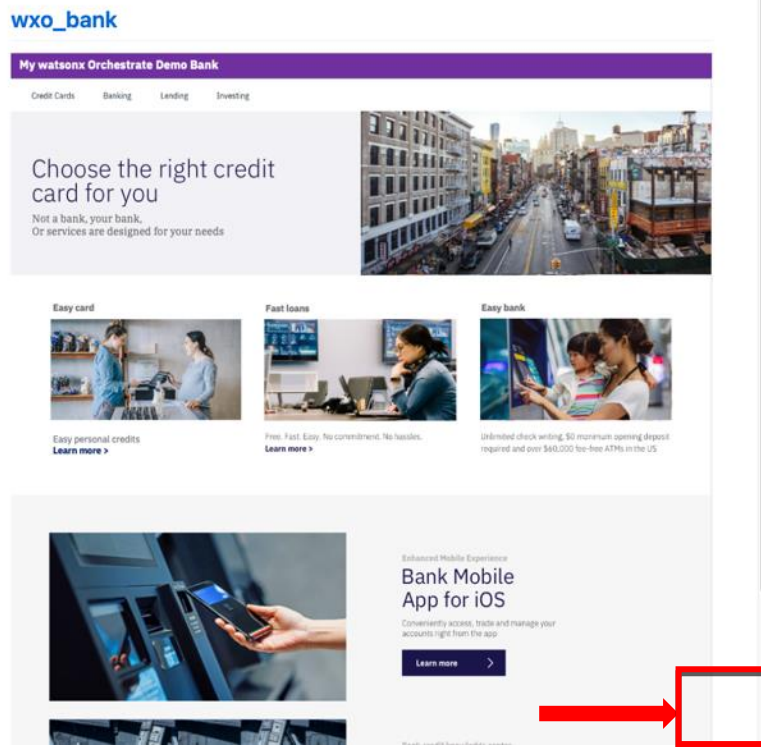


Your bank website is ready. Note that there is no assistant yet.

## Embed the assistant

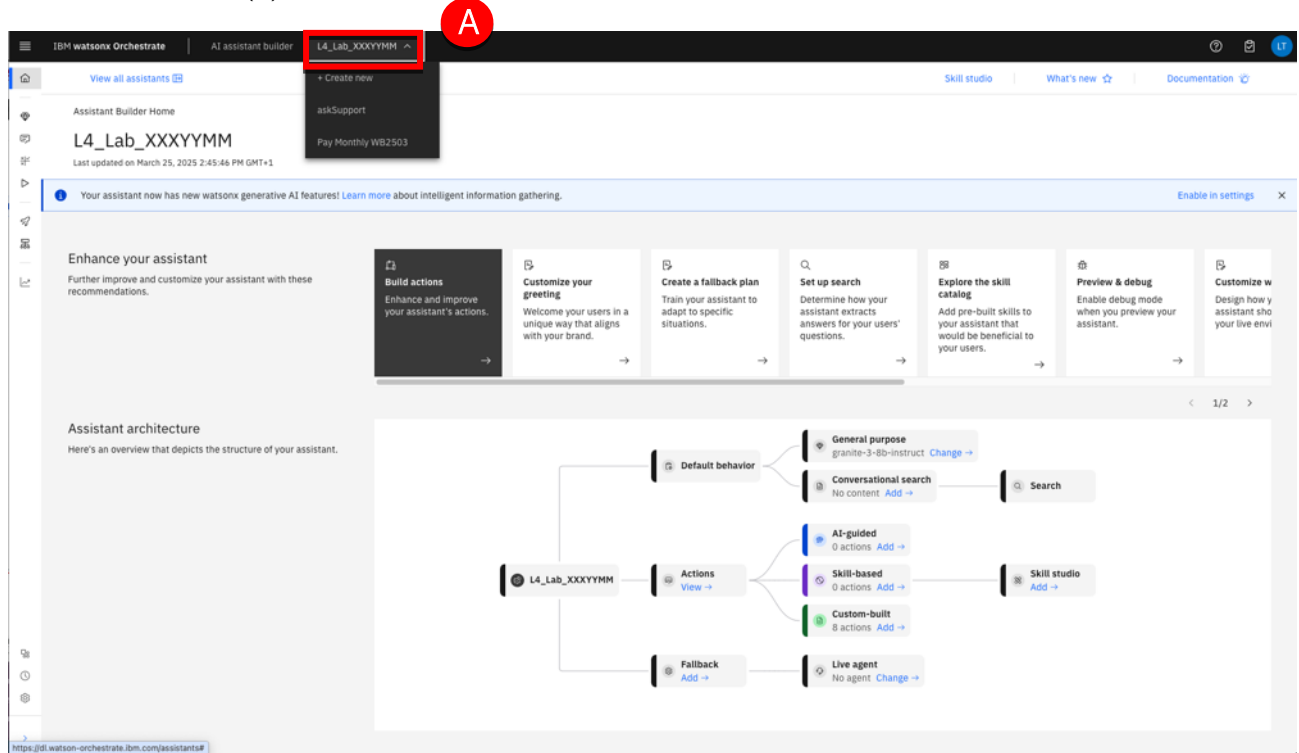
At this step, you created a static demo webpage hosted by GitHub pages. In this section, you edit this webpage.

1. Open the web page you will add the assistant to.



**Note:** For now, there is no assistant at all.

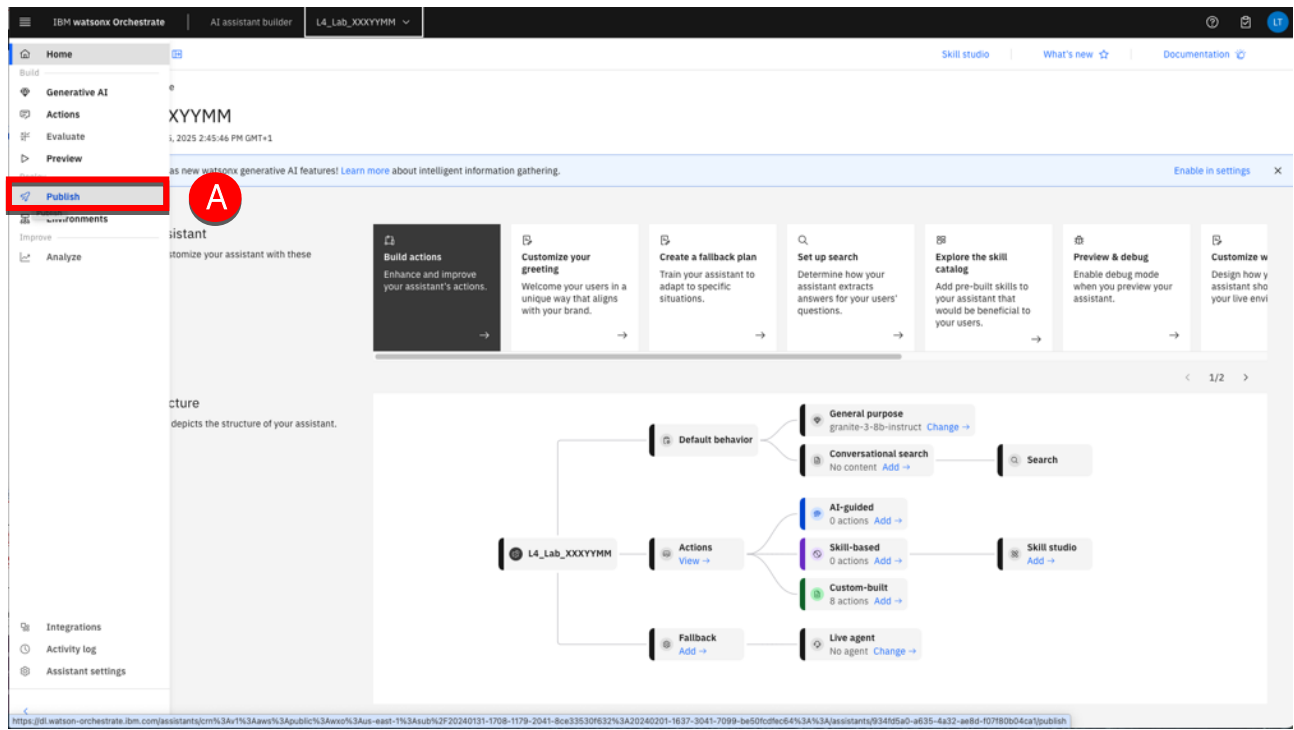
2. Navigate back to your watsonx Orchestrate Assistant Builder. Then, select your assistant from the assistant selector (A).



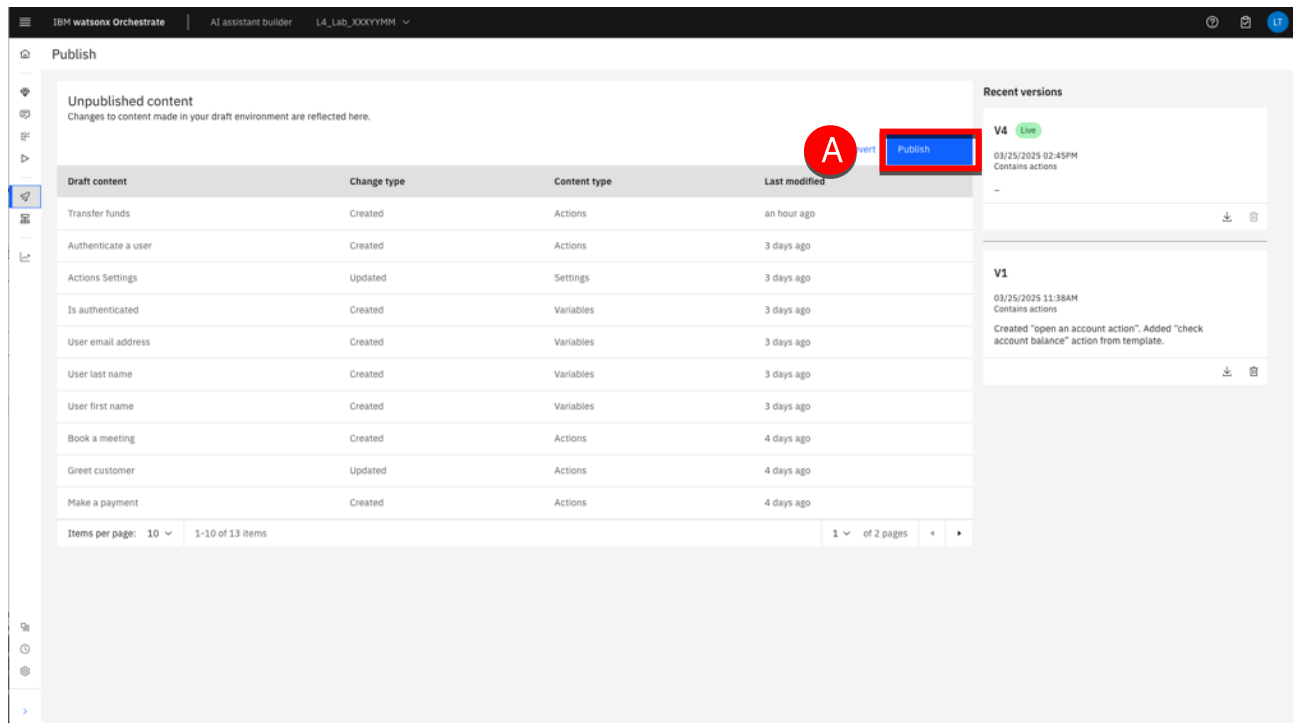
First you must publish the assistant to the Live environment. Then you will be able to customize your assistant's look and feel before getting the embeddable script.

## Publish to the Live environment

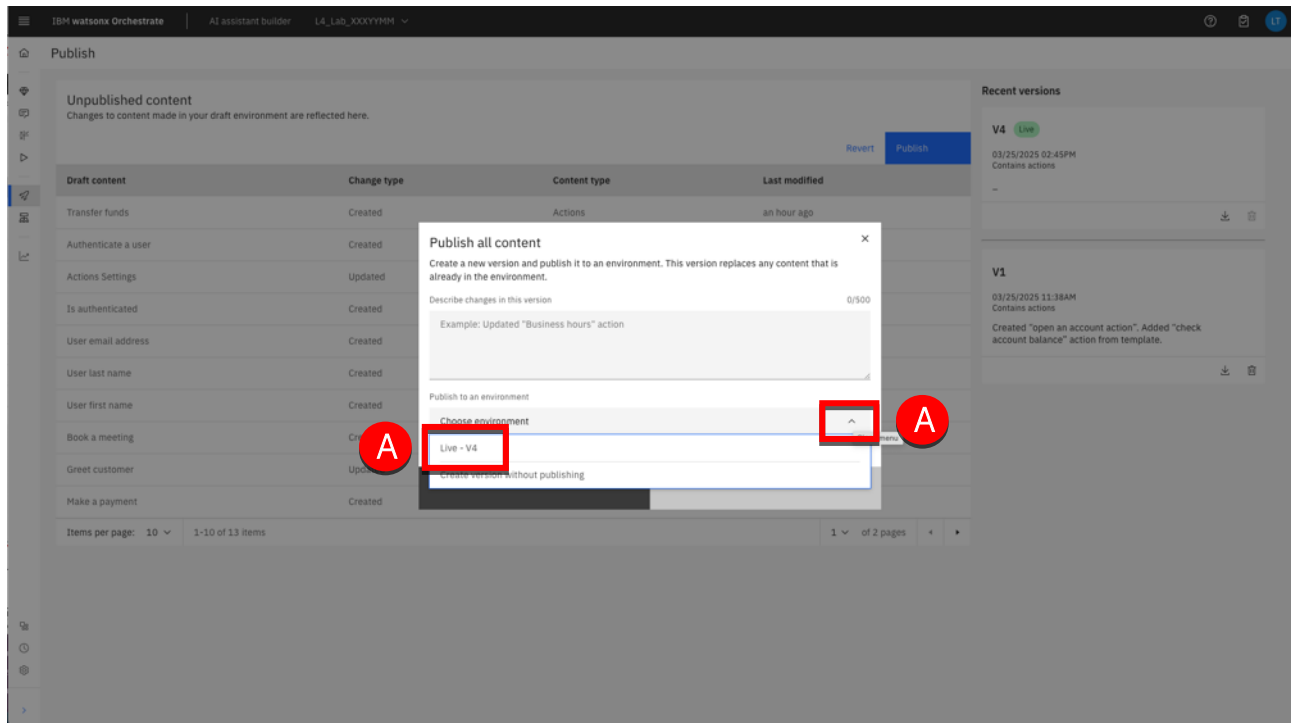
1. Click **Publish** (A).



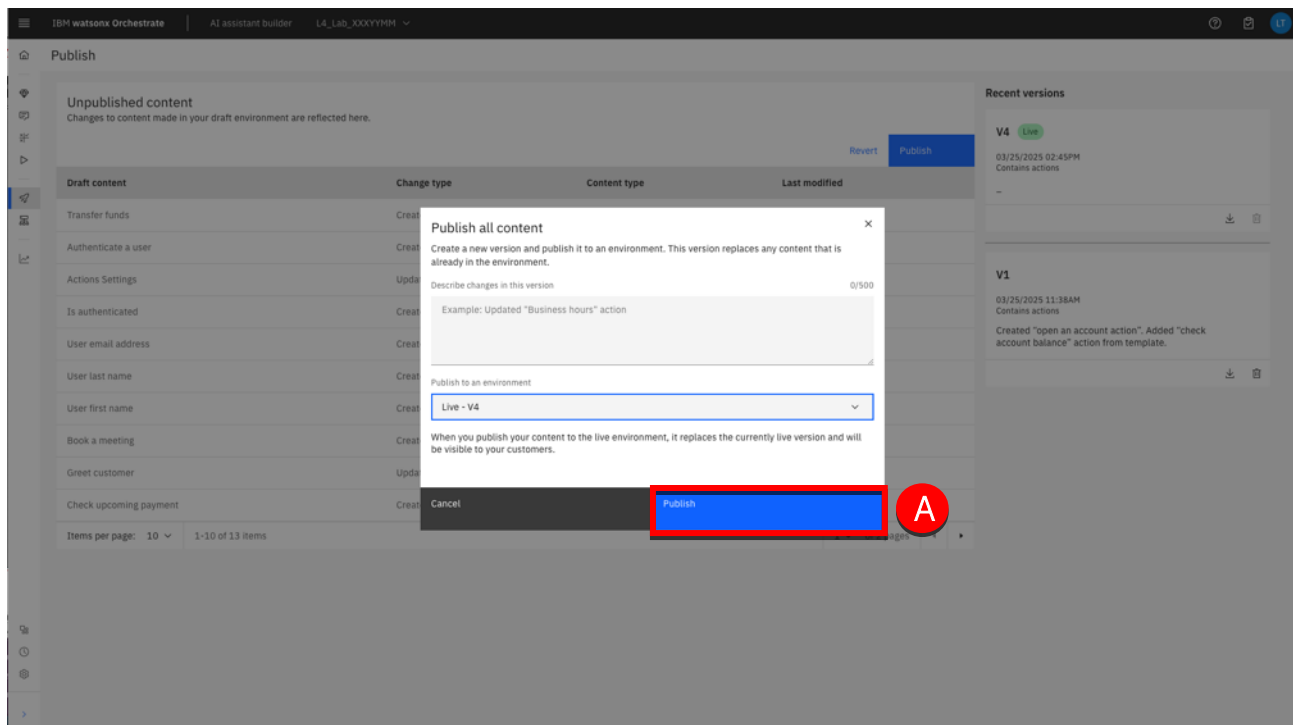
2. Click **Publish** (A).



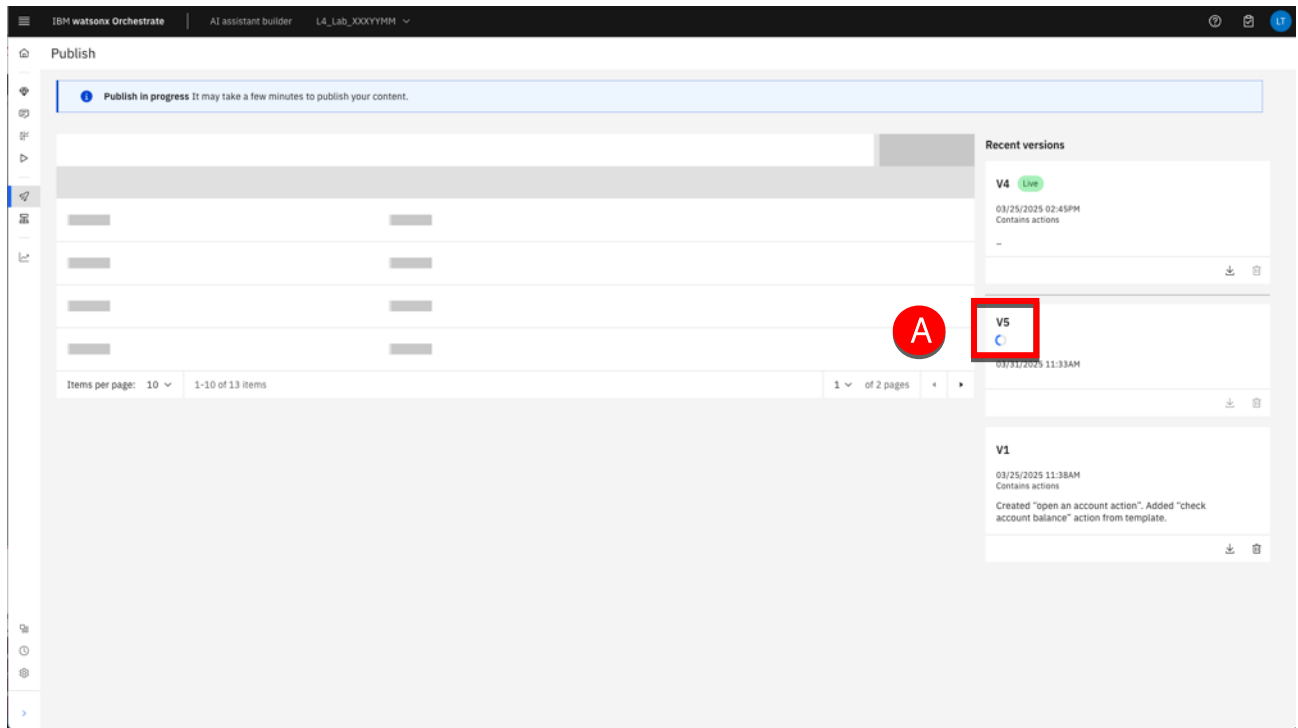
- Expand the **Publish to an environment** (A) then select **Live -V#** (B) (where # would be the number of the latest version your live environment has deployed).



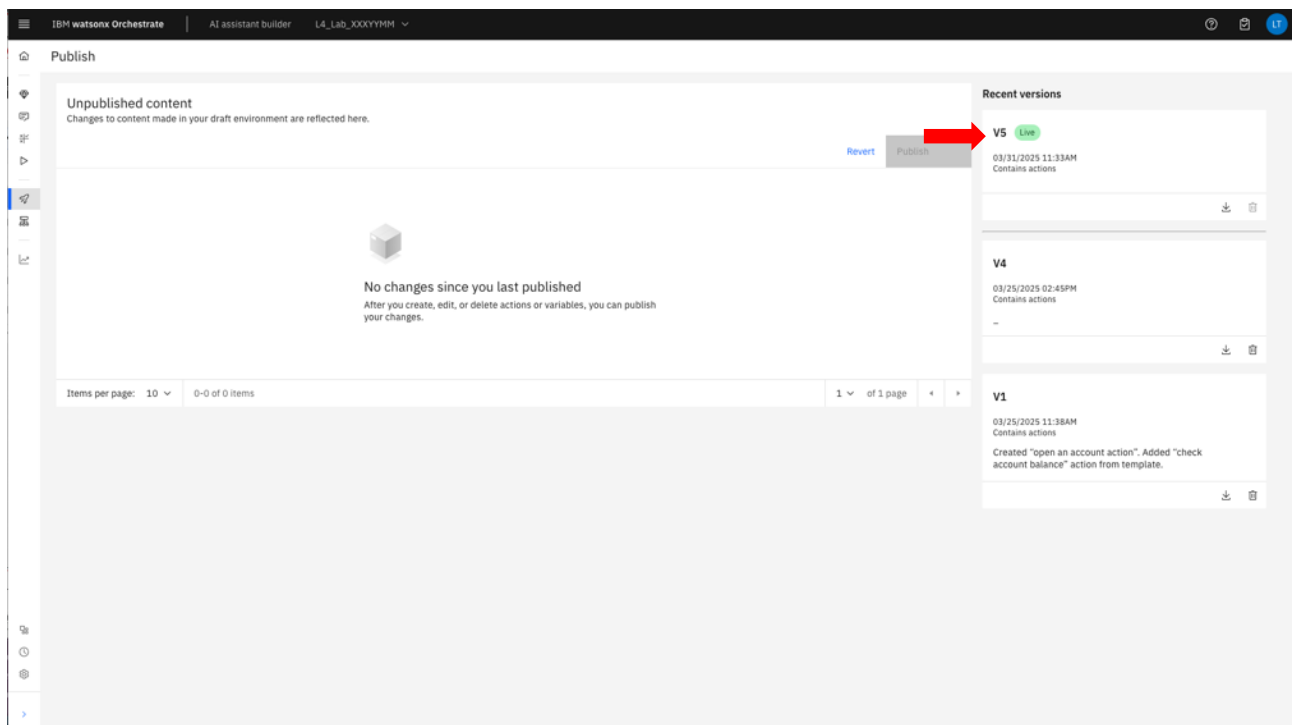
- Click **Publish** (A).



5. Wait a few seconds for the publish to be performed.

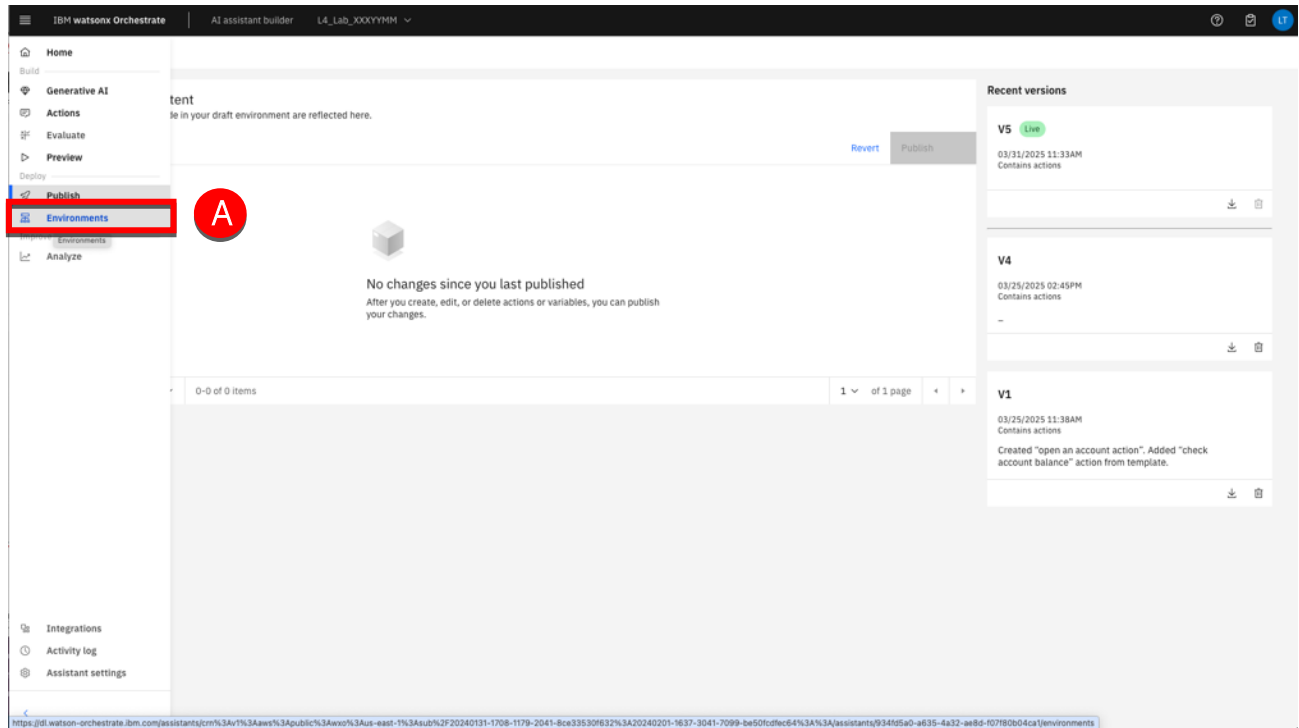


Note: After the publish is completed, a version **#+1** is automatically created on your Live environment.

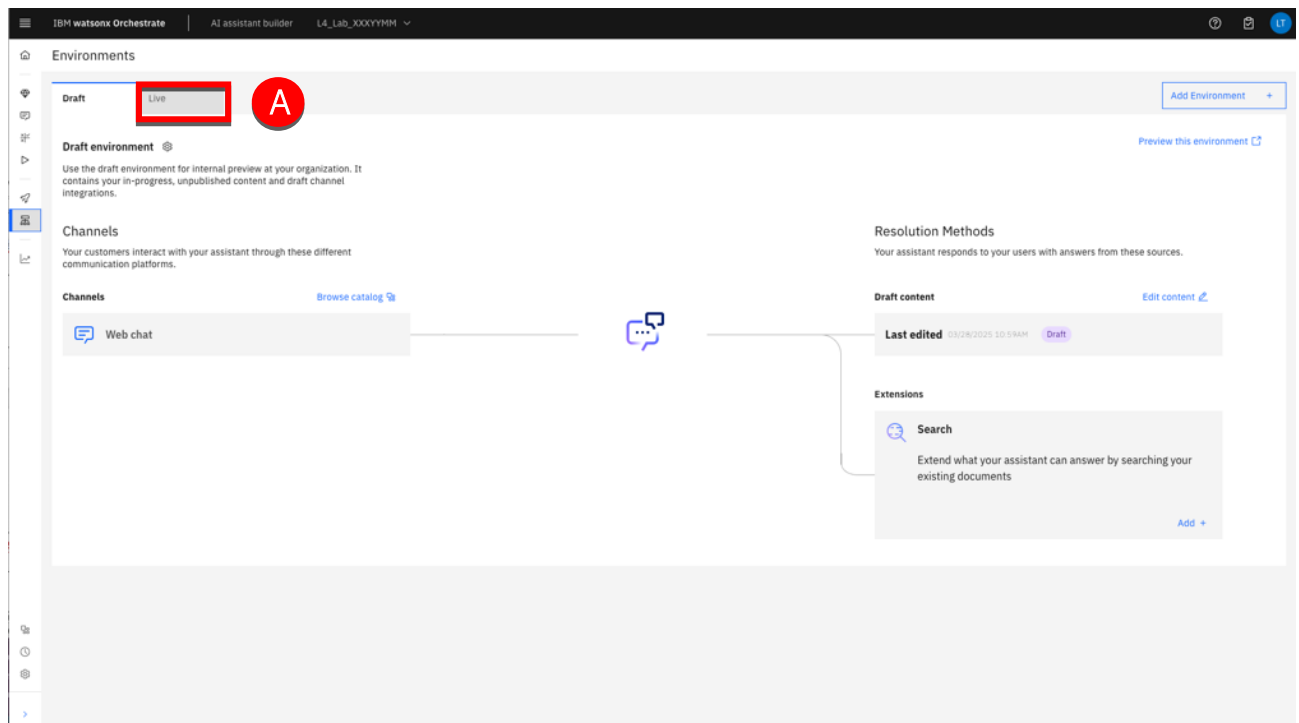


The assistant is now published in the Live environment. You can now customize it and get the script to embed it on the web page.

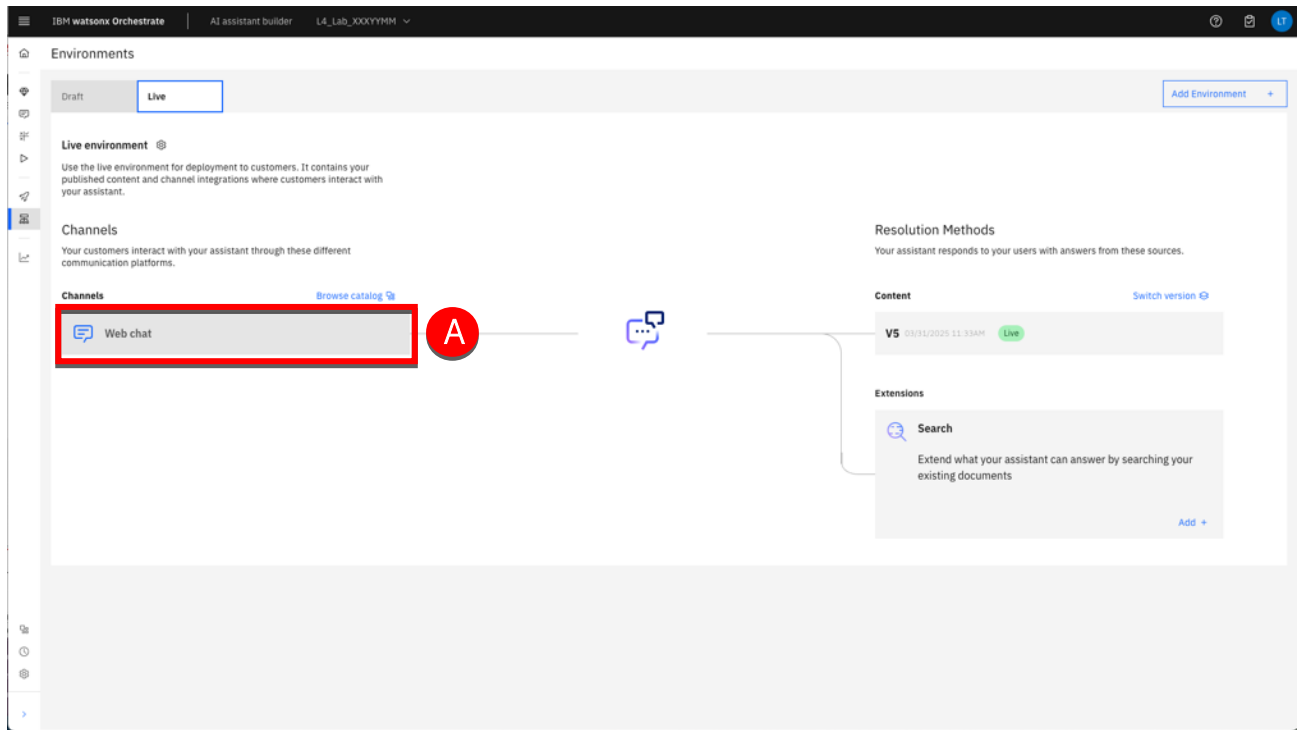
6. Select **Environments (A)**.



7. Select **Live (A)**.



8. Click **Web chat** (A).

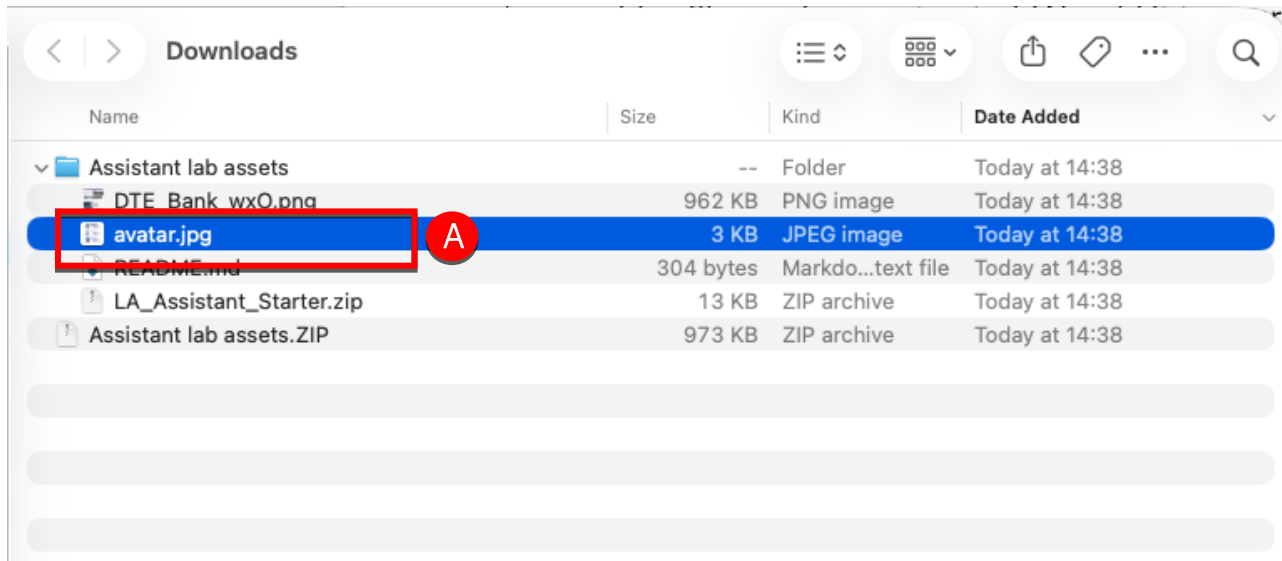


We can now Customize the Assistant look and feel.

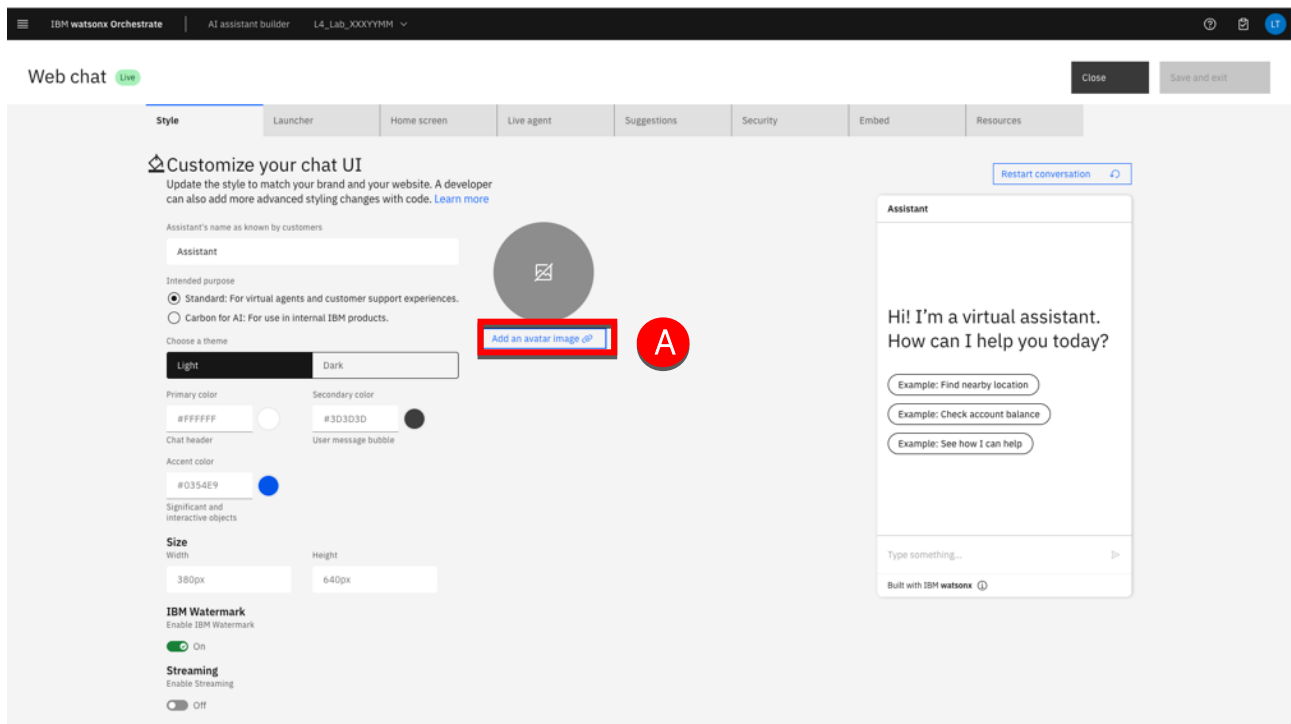
### Customize chat UI (Optional)

From this view, you can customize your assistant colors. Feel free to select the colors of your choice. Let's add an avatar to the assistant.

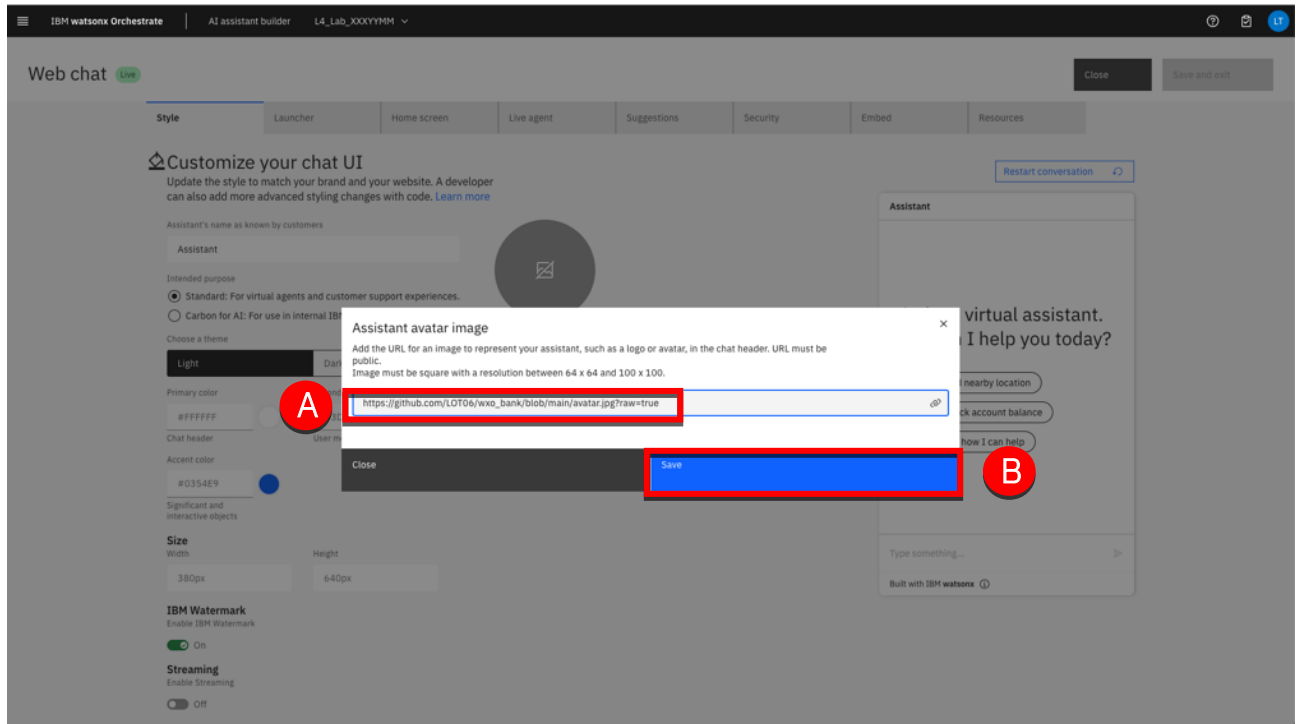
As we must provide the URL of the avatar image, you can add the avatar image contained in the archive file you have extracted (A), add it to your github repository, and get the image URL requested in the following step.



1. Click Add an avatar image (A).



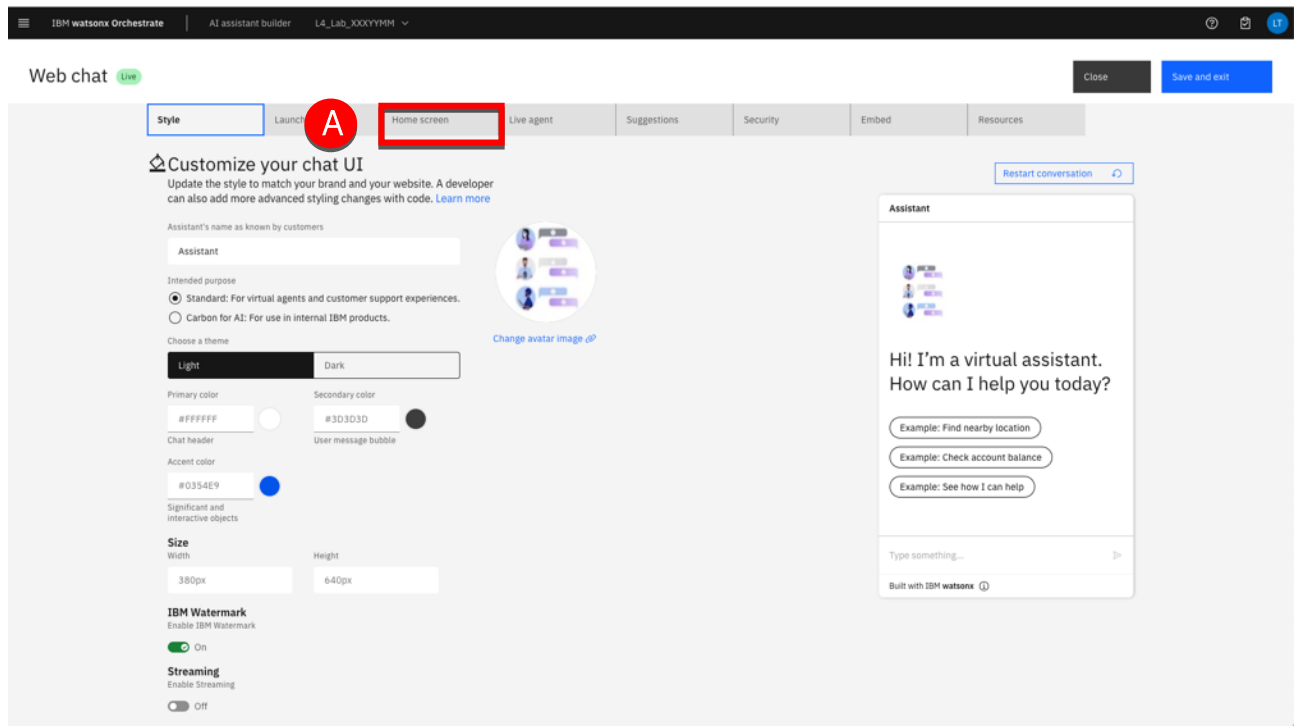
2. Paste the image URL (A) you have collected after adding the image in your own github repository, then click **Save** (B).



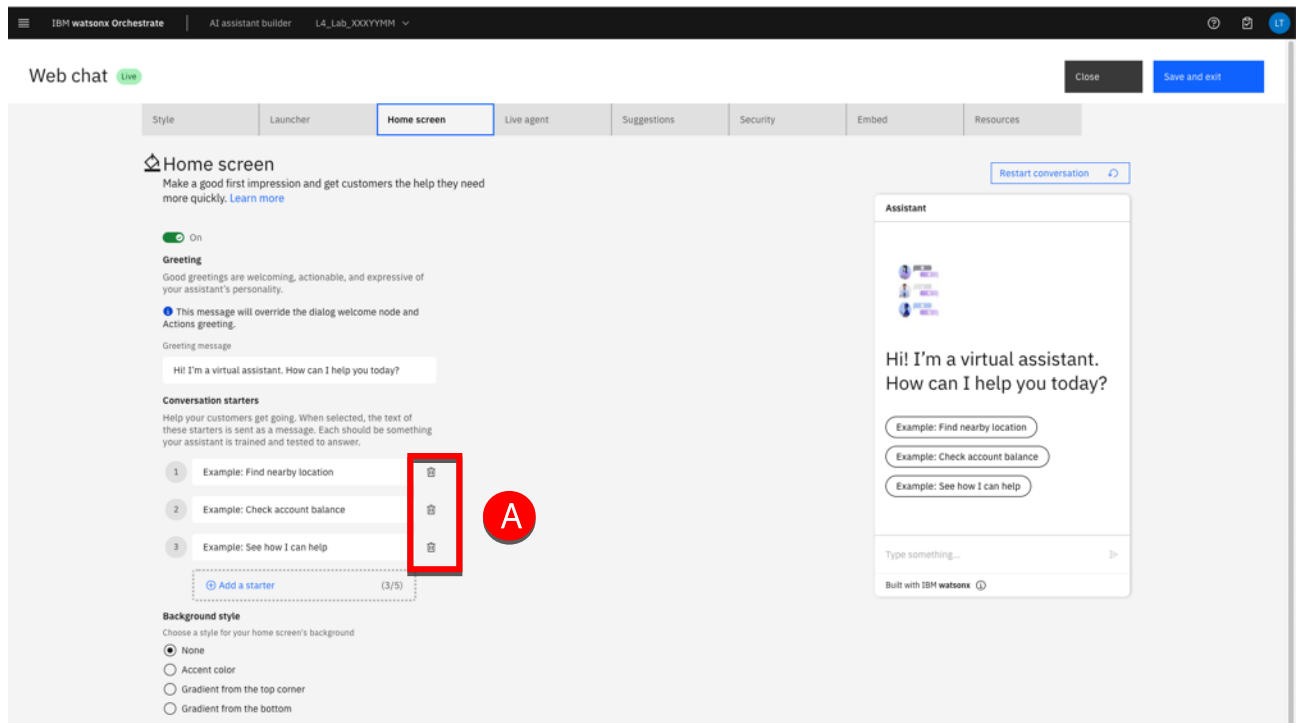
## Customize the Home screen

You can now customize the conversation starters with some actions the assistant is able to perform.

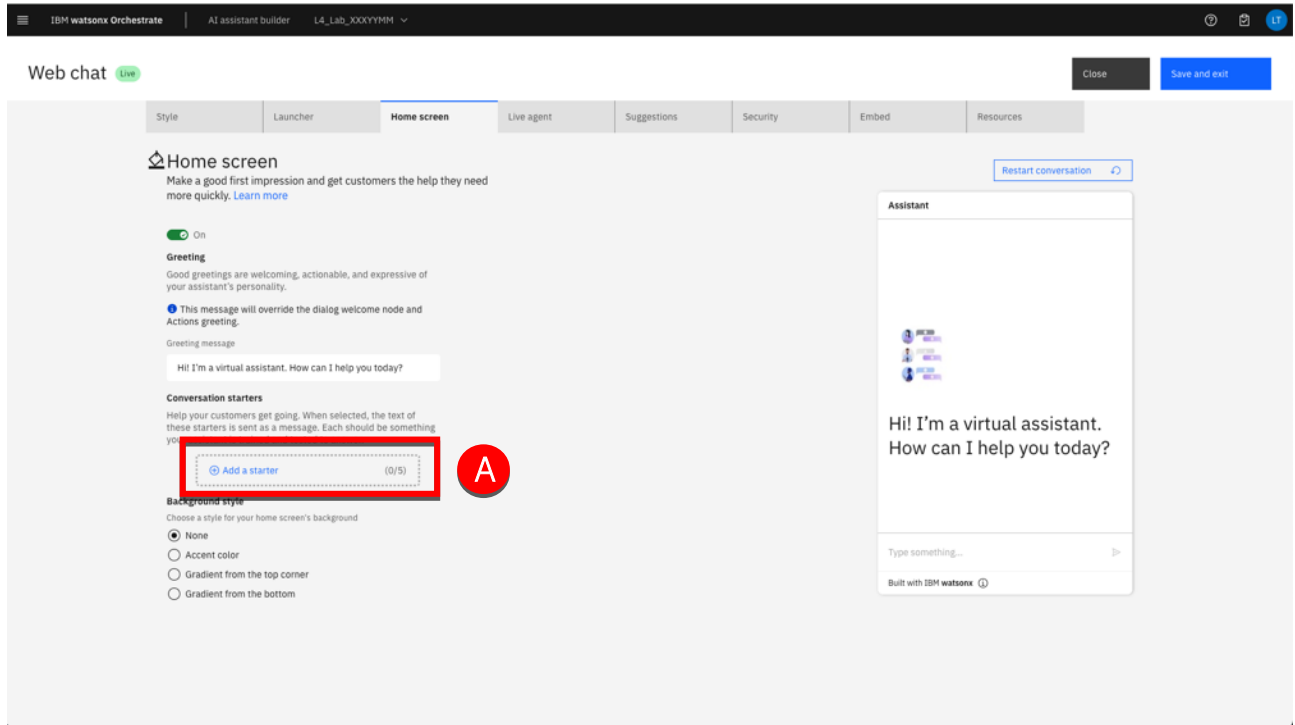
1. Select Home screen (A).



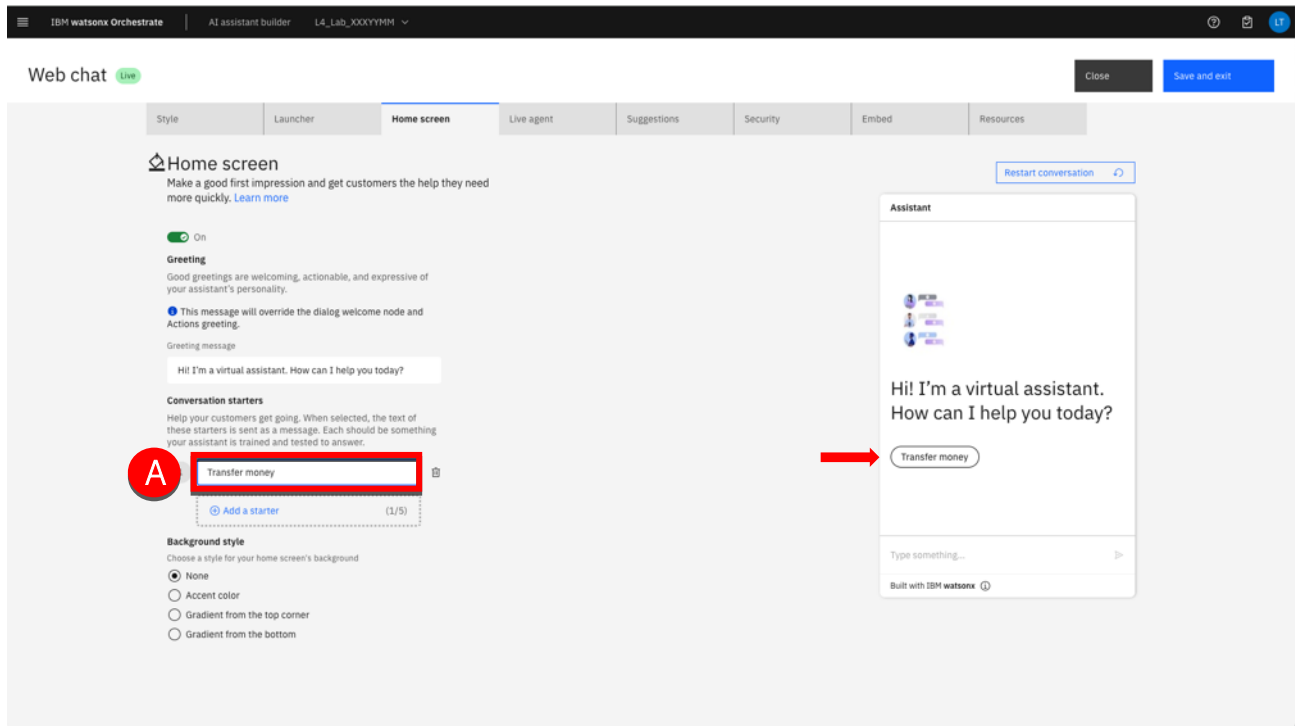
2. Click the trash icons to remove the three default Example phrases (A).



3. Click **Add a starter** (A).



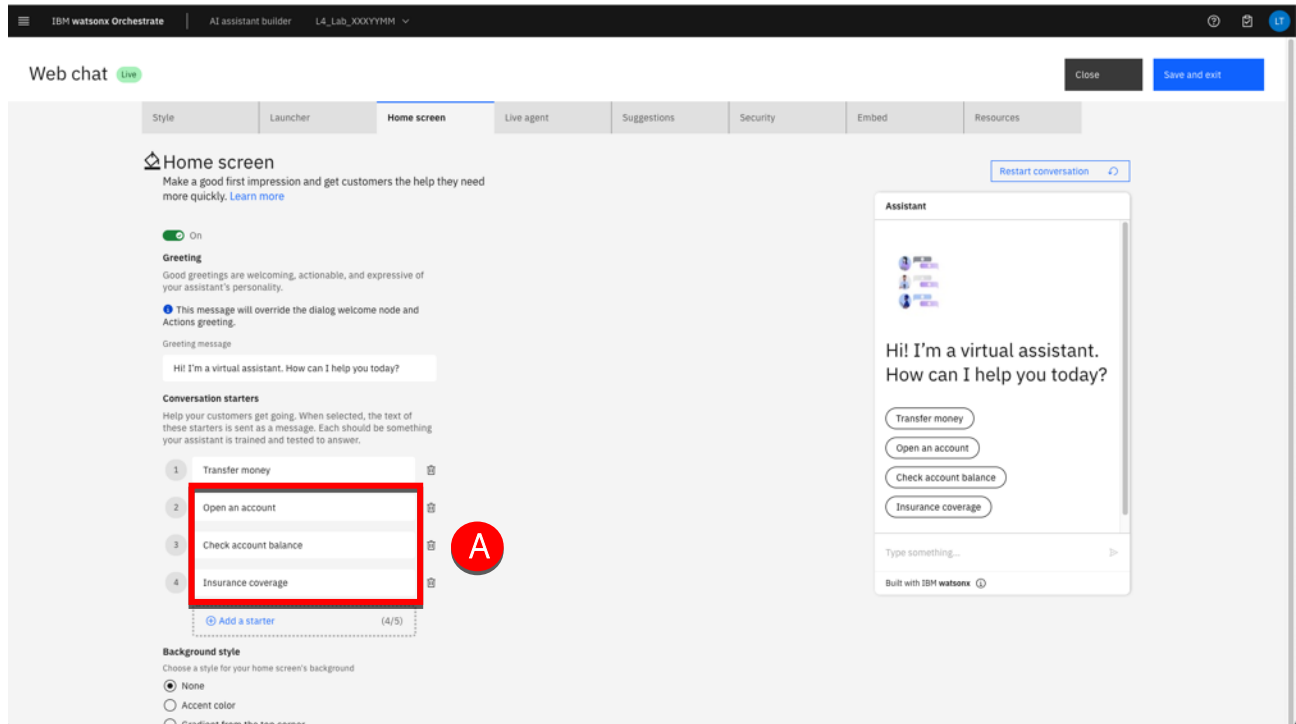
4. Type **Transfer money** (A).



**Note:** The corresponding invocation button is automatically added to the assistant Home screen.

5. Repeat step 3 to add the following invocation phrases (A):

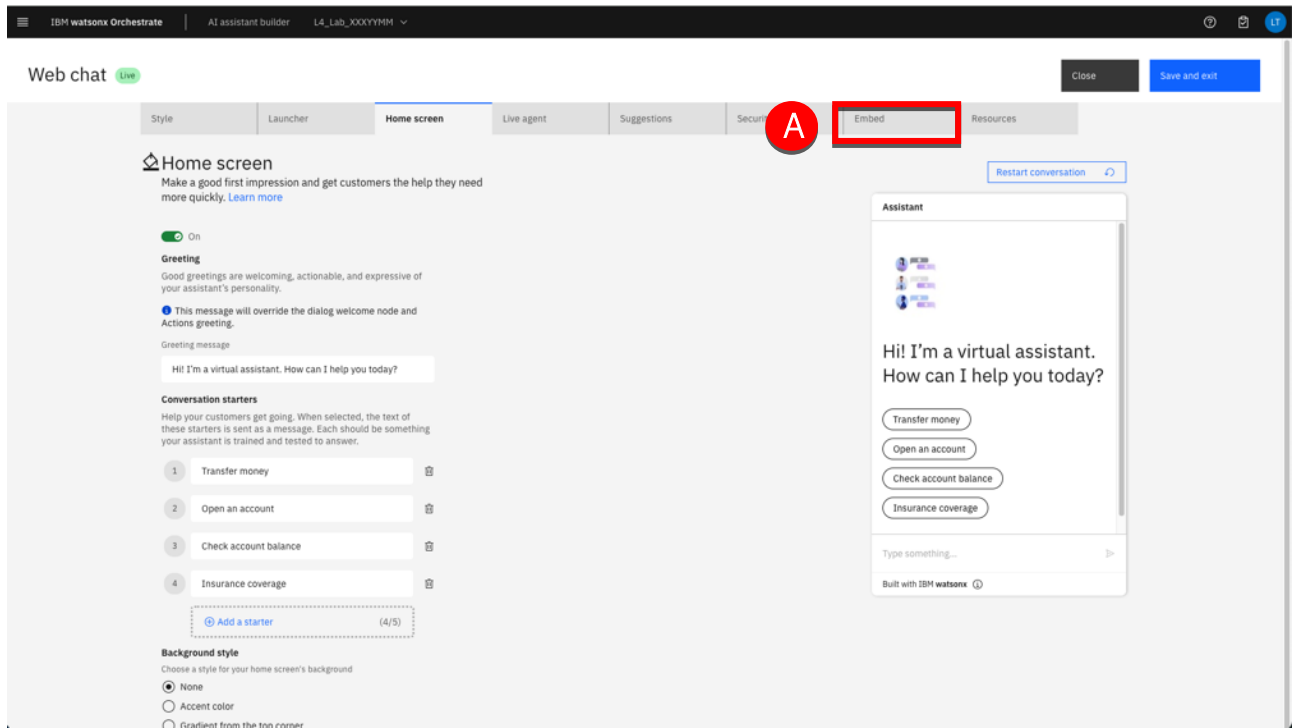
- Open an account
- Check account balance
- Insurance coverage



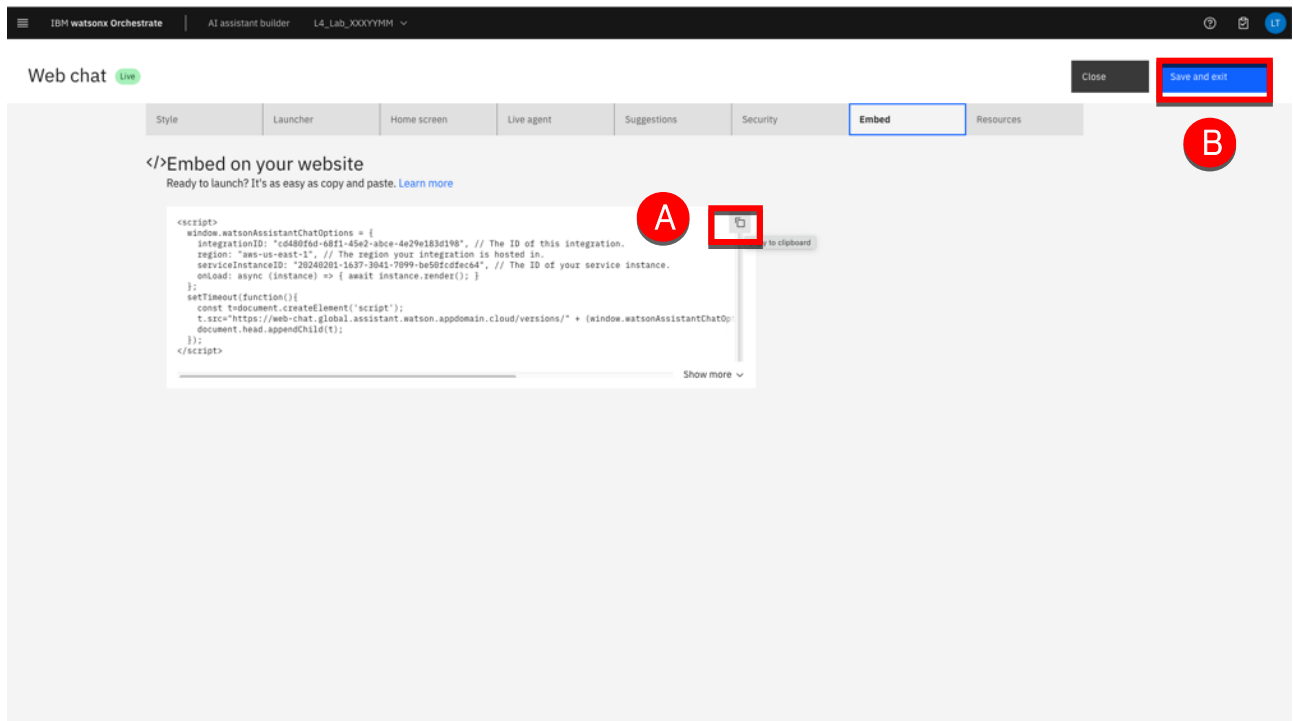
Optionally, you can play with the other color parameters to tune your assistant's look and feel. Let's now get the assistant invocation script.

## Get the Script code

1. Select **Embed** (A).



2. Click the copy to clipboard icon (A) then click Save and exit (B).

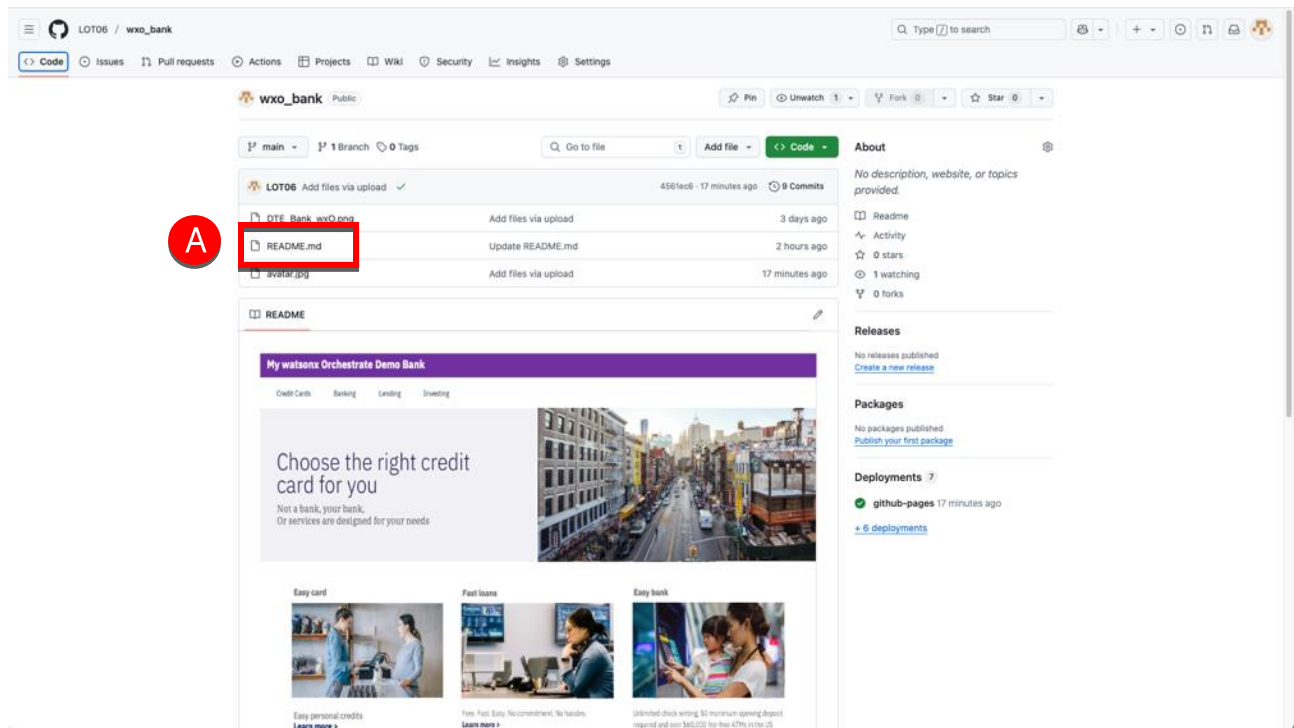


Let's now paste the script in our webpage HTML code.

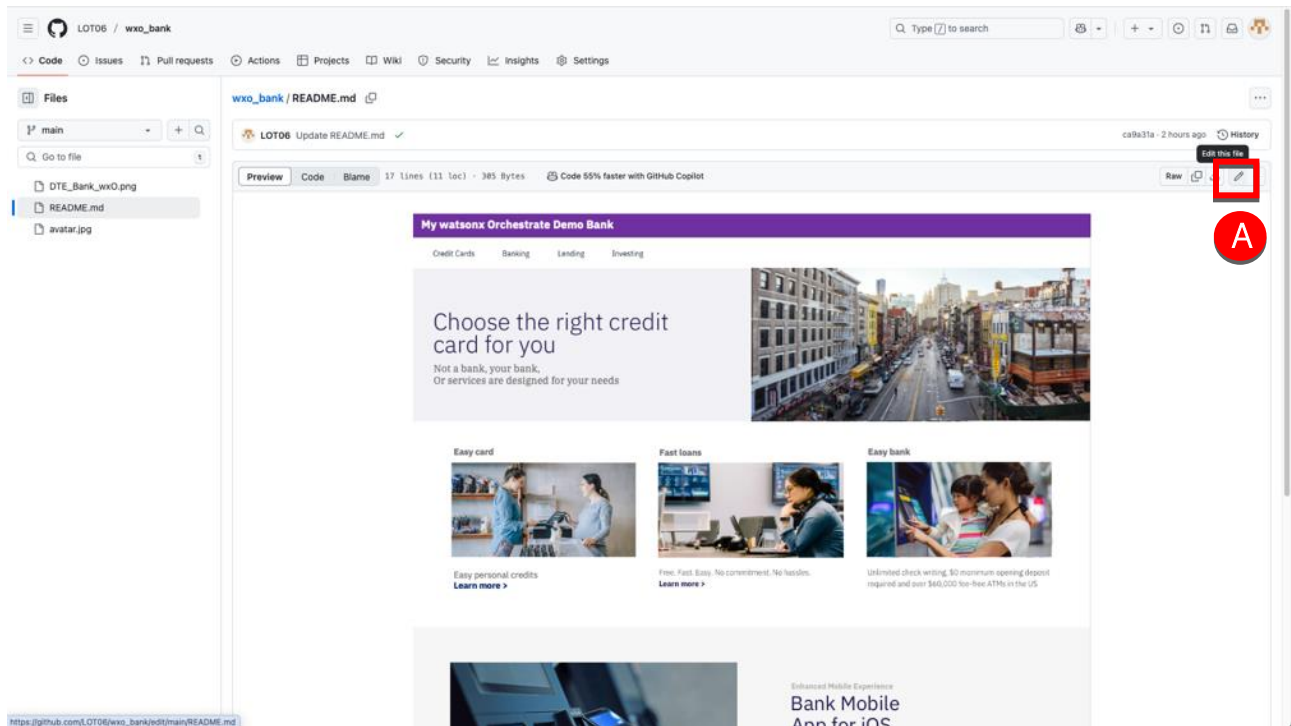
## Embed the script

You will now edit your webpage. We will use the GitHub editor, as our static webpage is hosted by GitHub Pages.

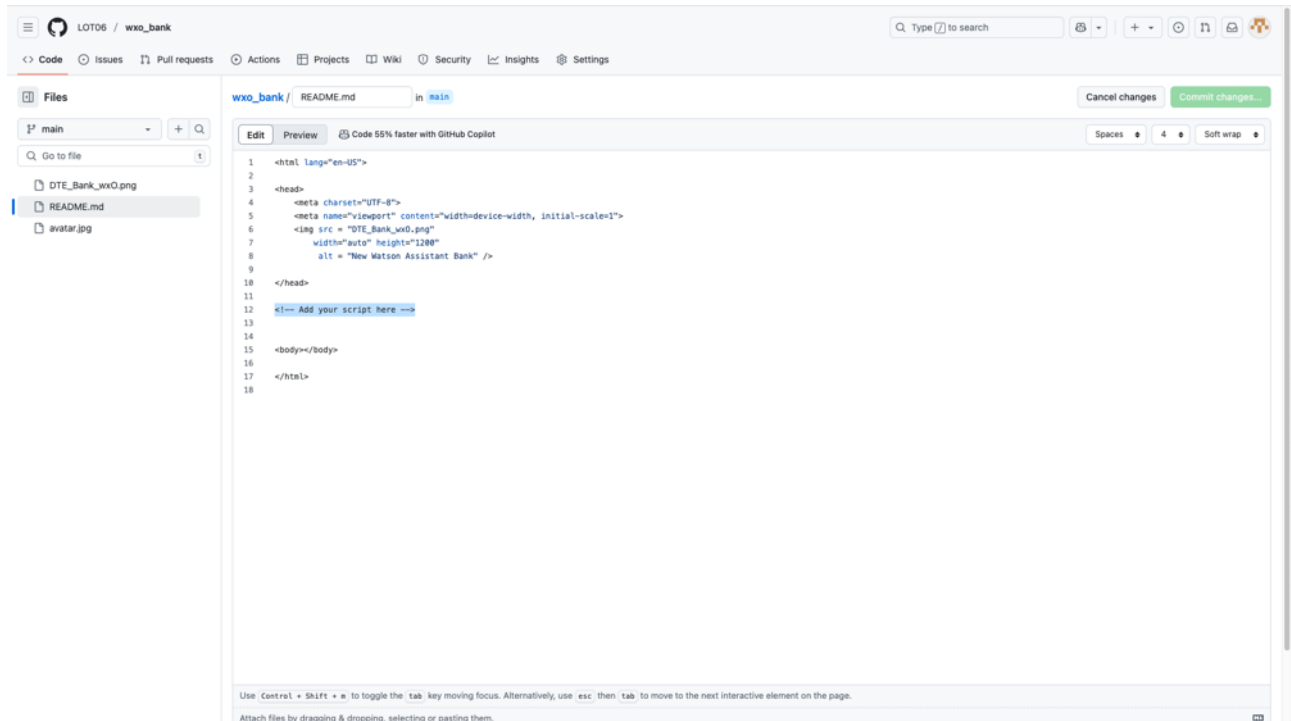
1. Click your **README.md** file from your GitHub repository page (A).



2. Click the **Edit file** icon (A).

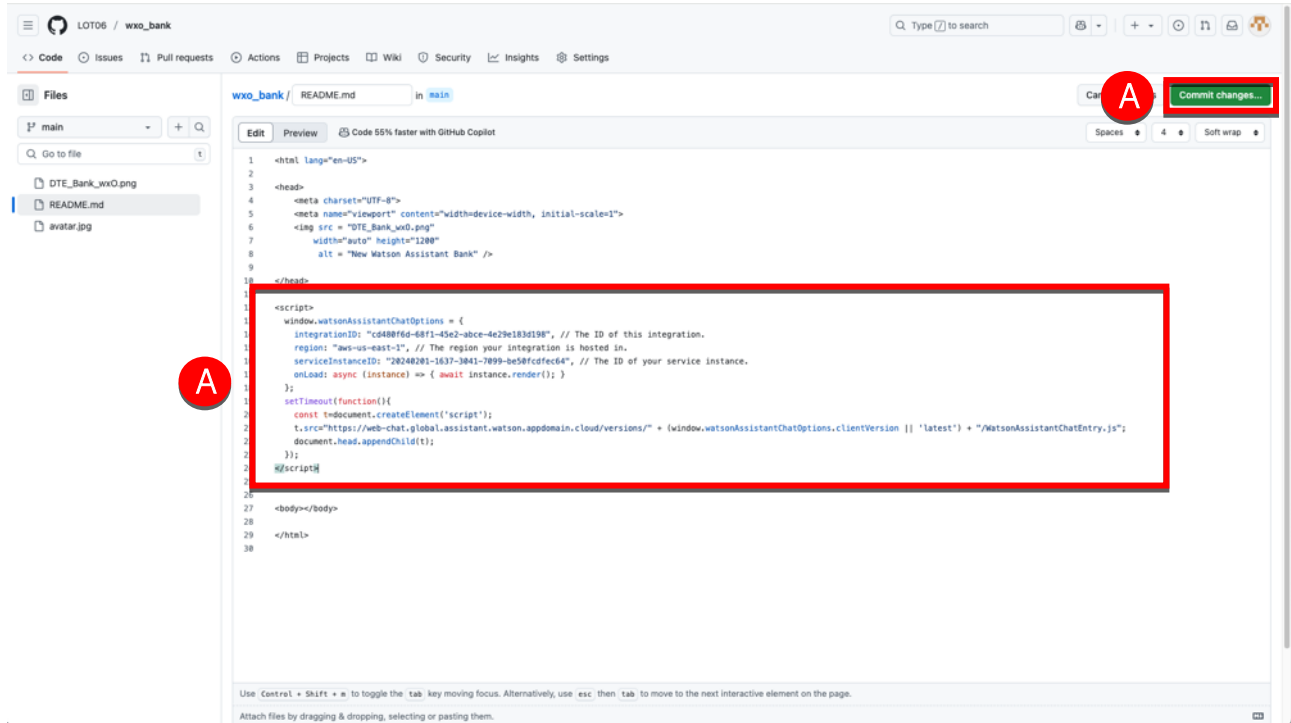


3. Select the `<!--Add your script here -->` line (A).

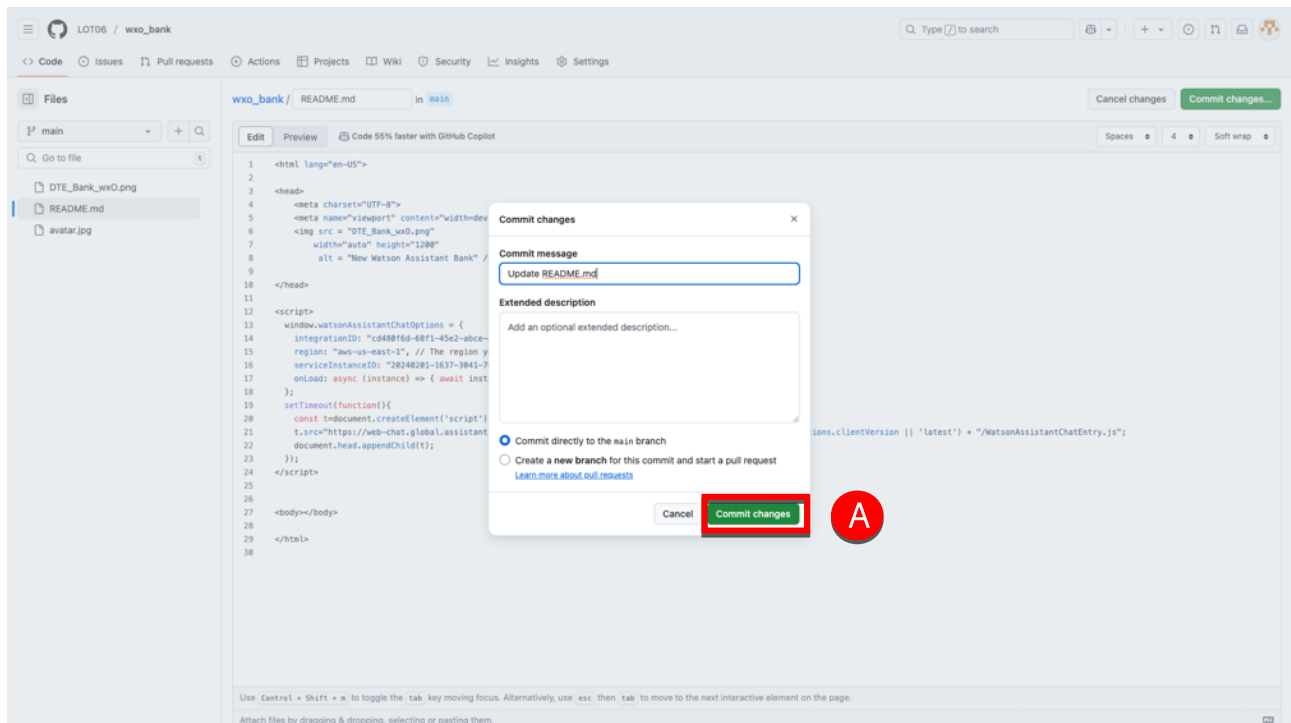


**Note:** The embedded script must be added before the `<body>` of your HTML page.

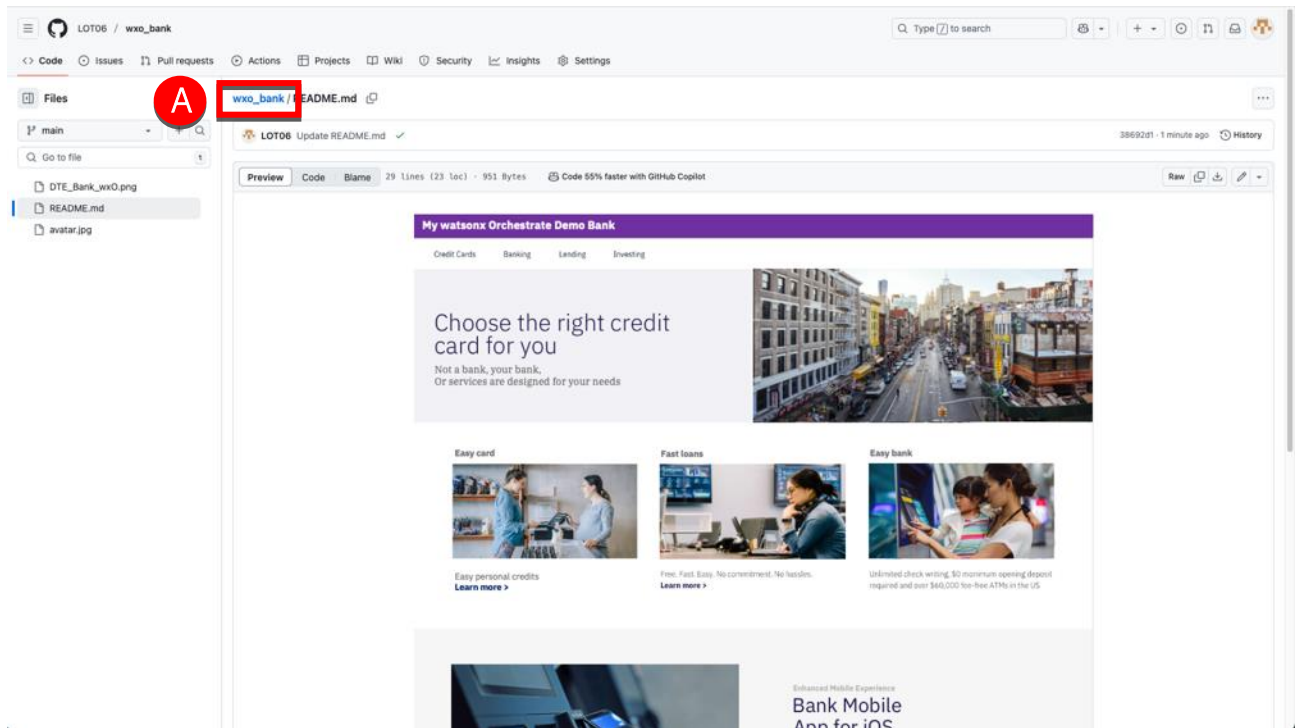
4. Paste the script code copied from the previous chapter (A) then click **Commit changes** (B).



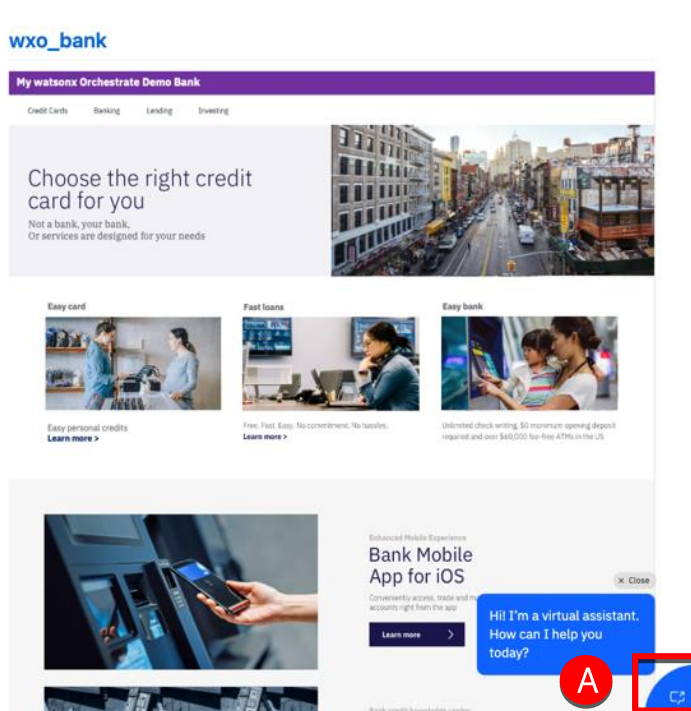
5. Click **Commit changes** (A).



- Click the repository link (A).

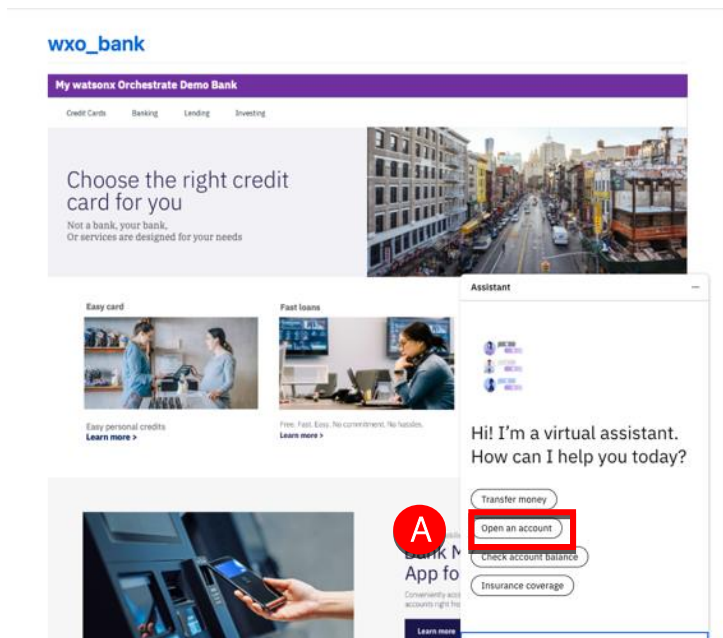


- Wait for Your recent commit to process. If necessary, refresh the page.
- Return to your simulated bank web page and refresh it. Click your assistant icon in the lower right corner (A).

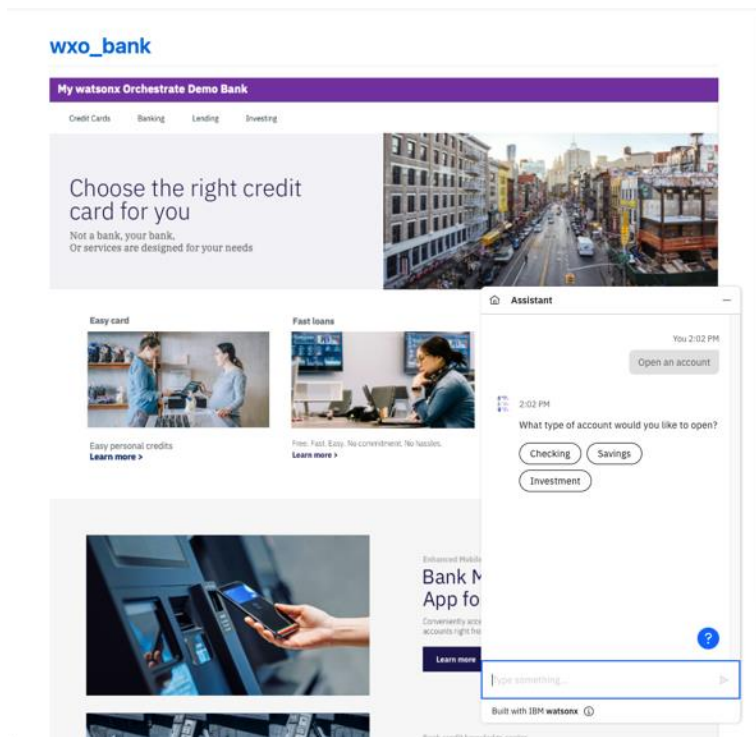


Note that after the refresh, your assistant is now embedded in your webpage.

9. Click **Open an account** (A).



10. Continue with the actions. Your assistant is now embedded in your web page.



Congratulations. You have now completed watsonx Orchestrate Assistant 2 lab.

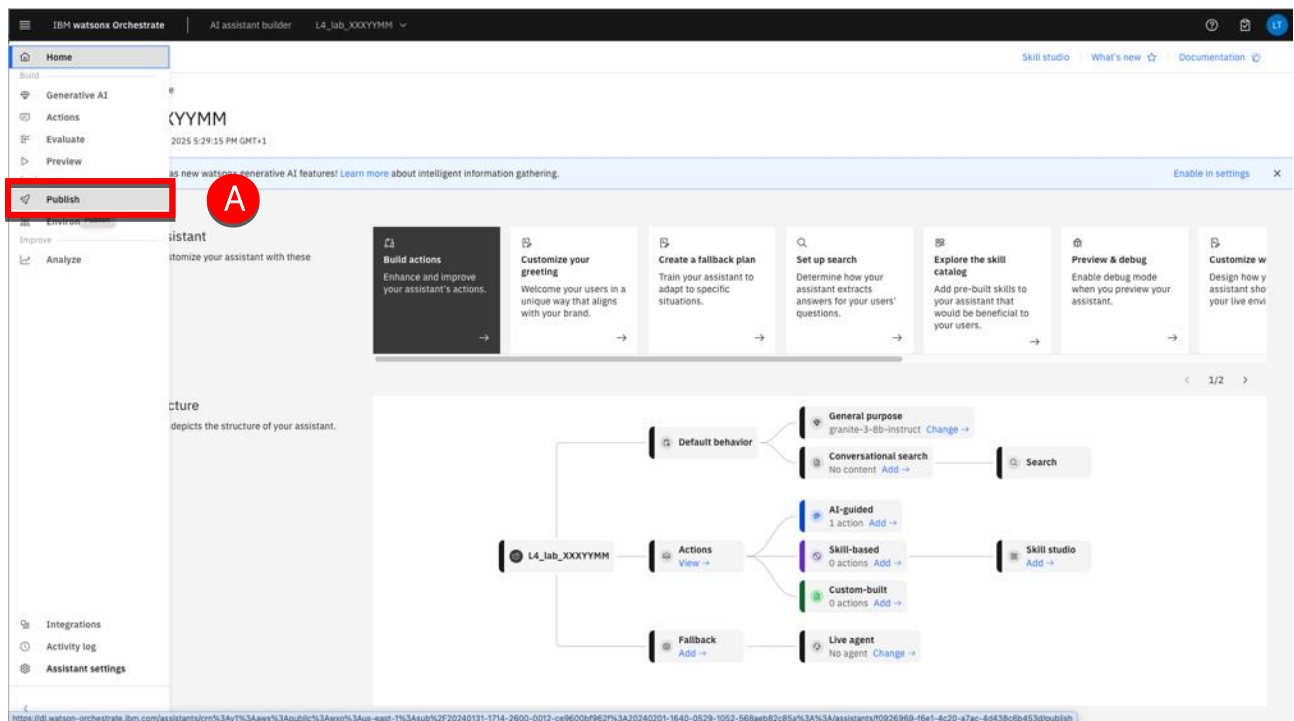
## Annex - Export / re-import/ delete your assistant archive

This section describes how to export your work in an archive that can be re-imported later, should you need to resume your work.

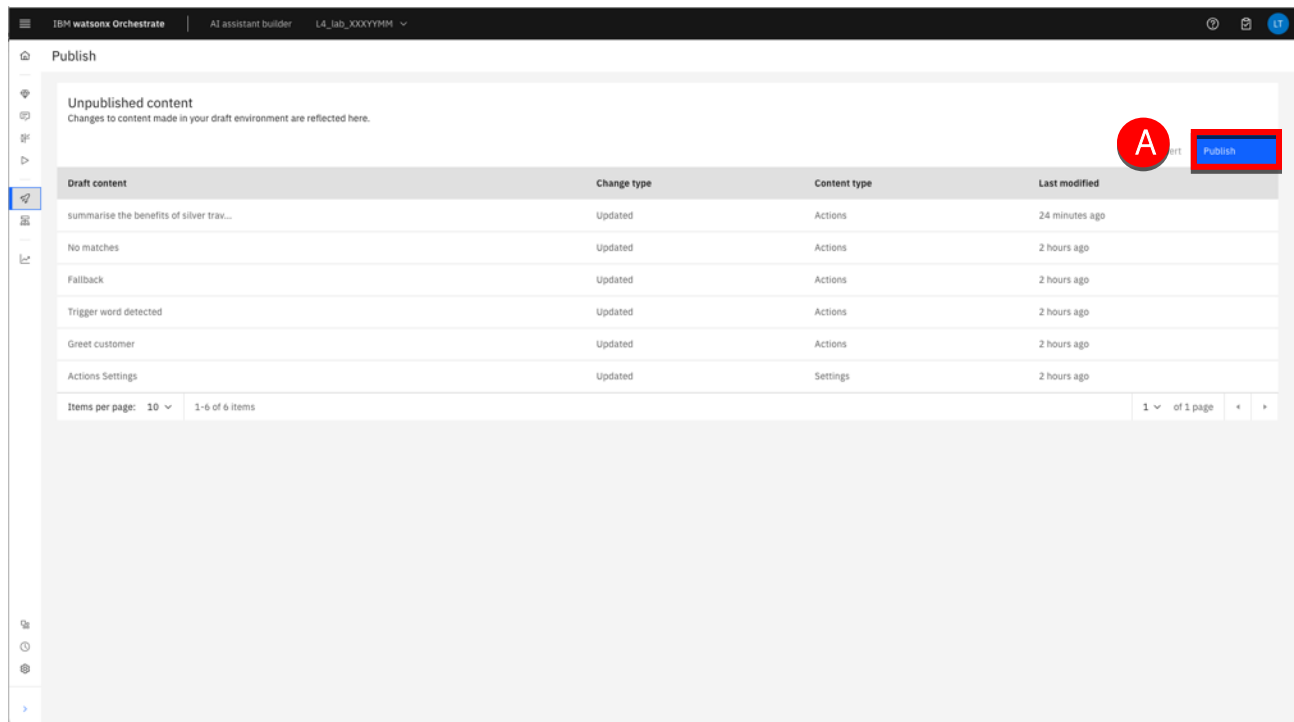
### Export an assistant

#### Create a version

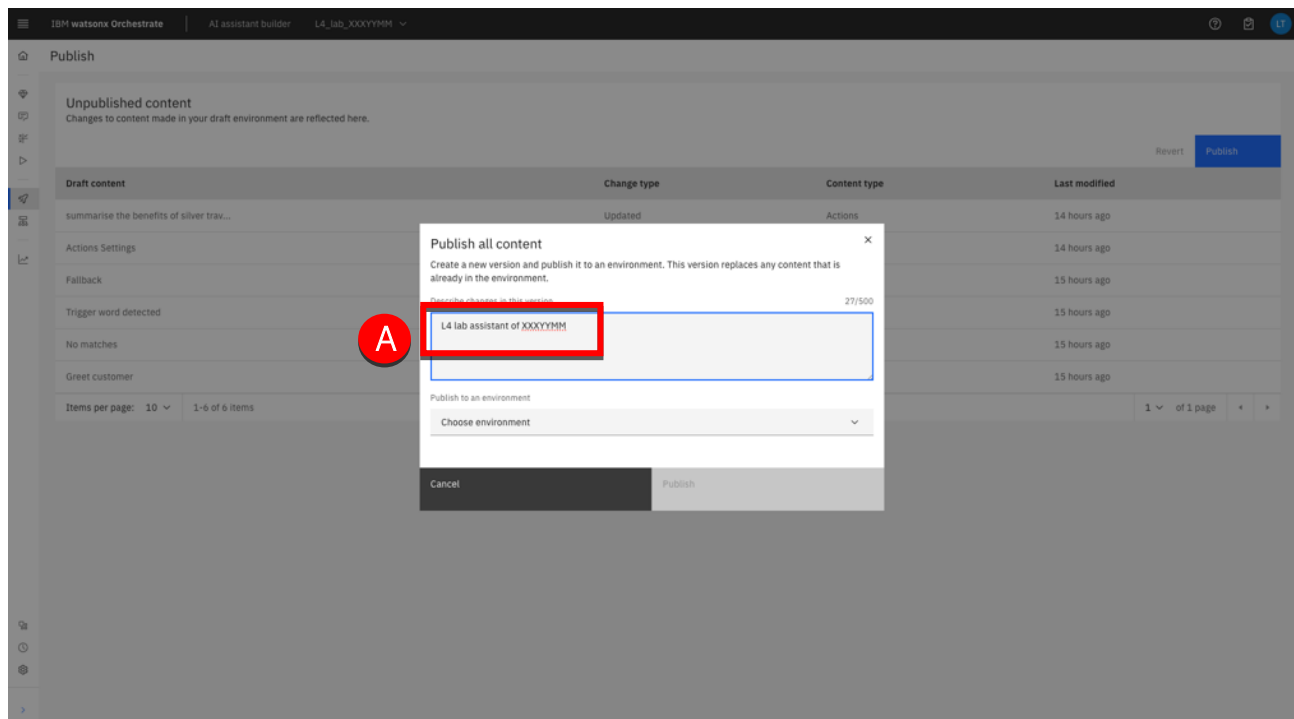
1. Go to the assistant home page.
2. Click **Publish** (A).



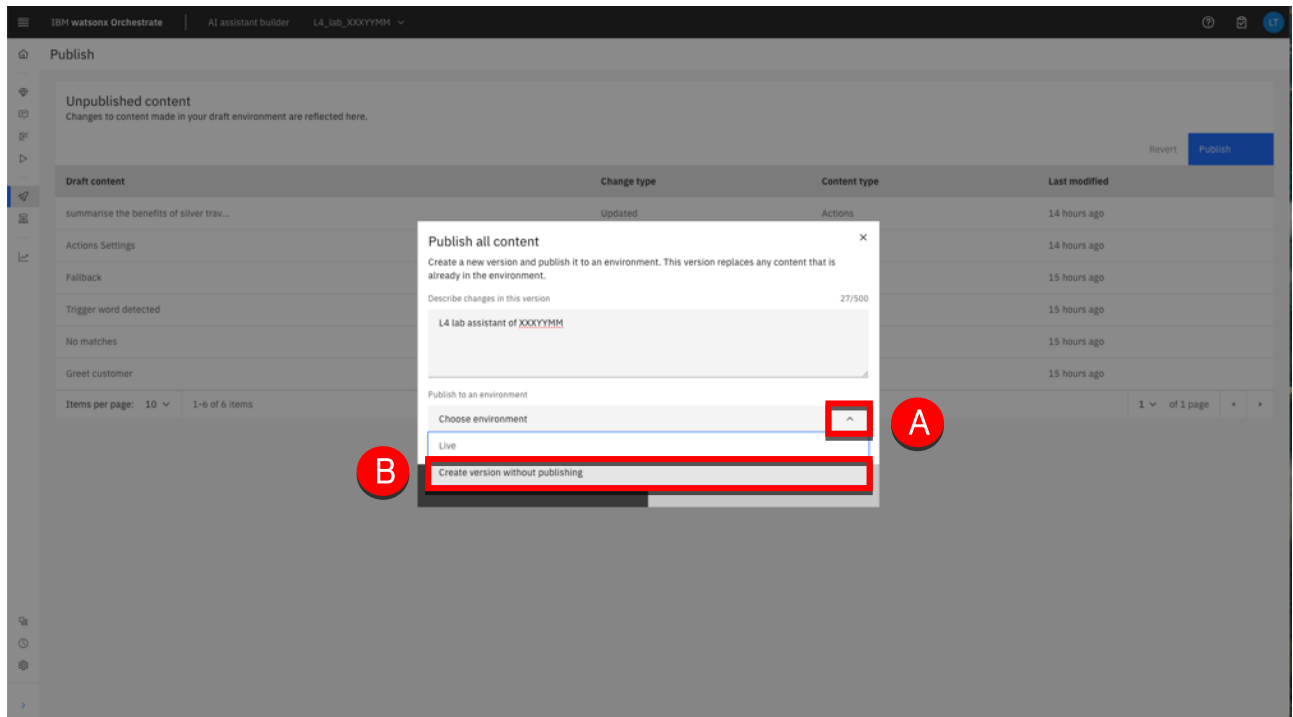
3. Click **Publish** (A). If Publish is greyed out, you may skip ahead to the [Create the archive](#) section.



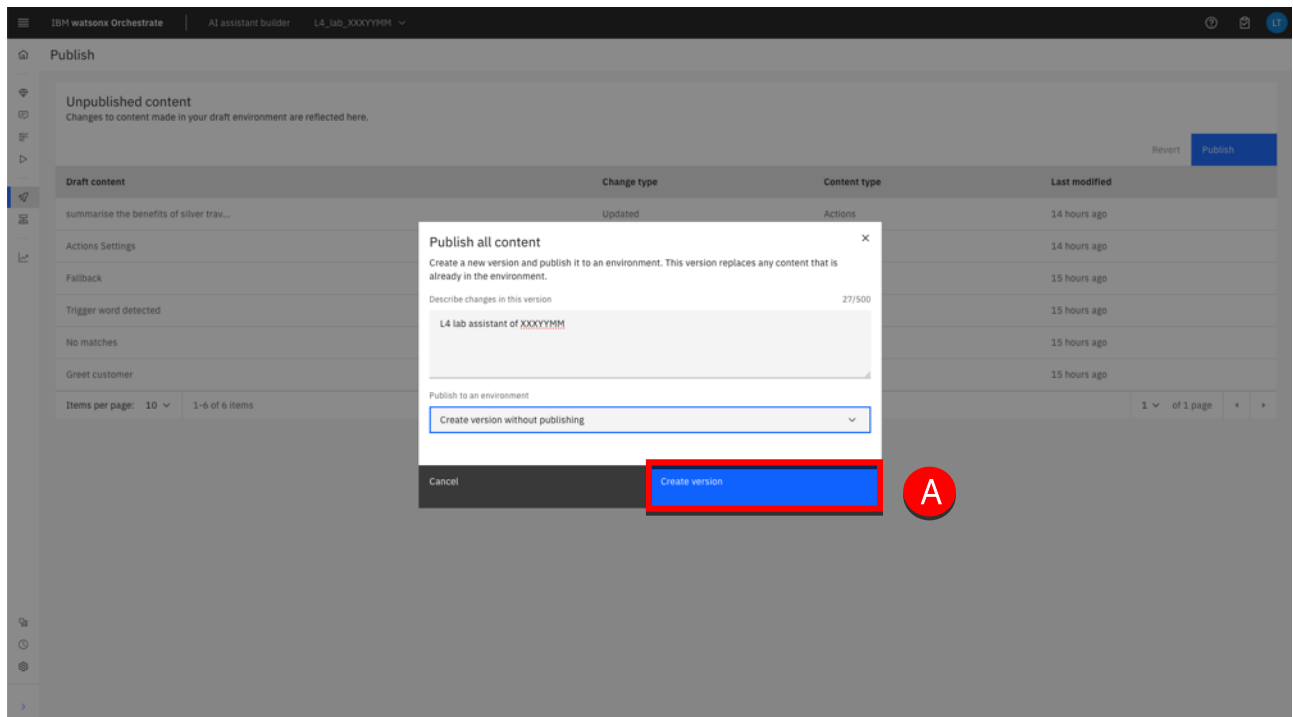
4. Enter L4 lab assistant of XXXYYMM (A) (XXXYYMM being you proper identification code).



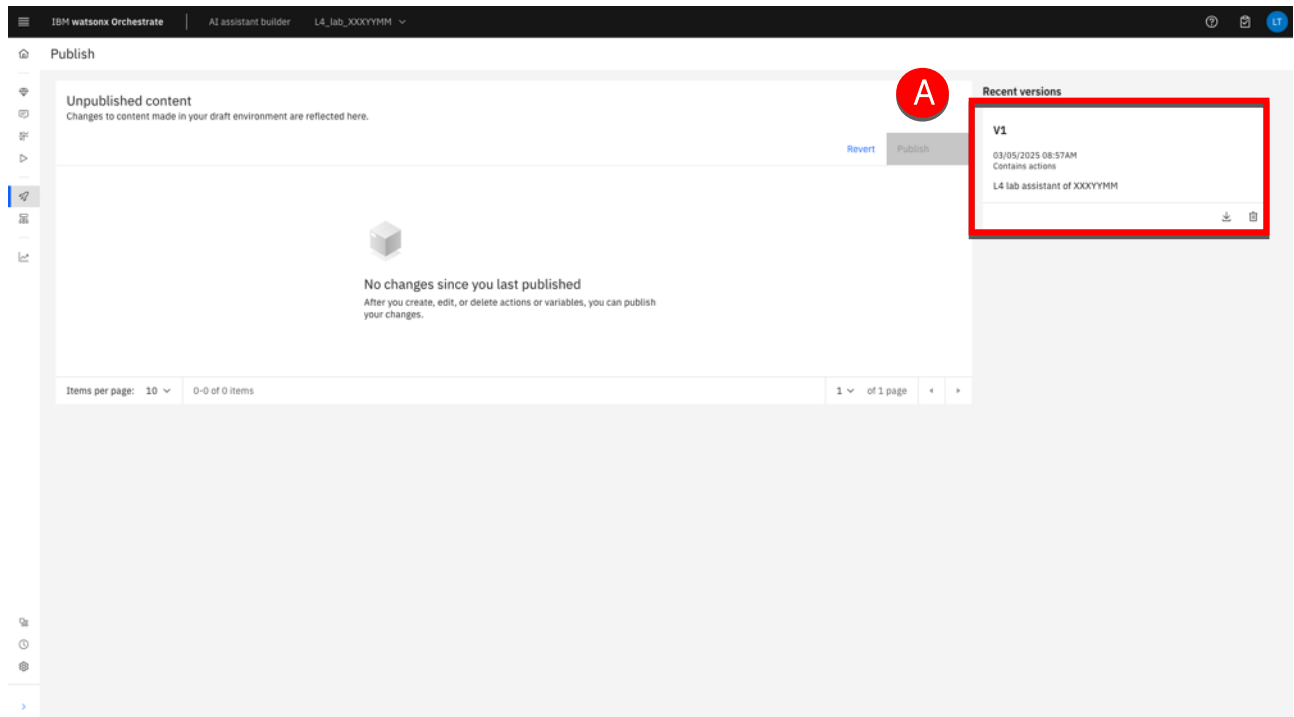
5. Expand the **Publish to an environment** menu (A) then select **Create version without publishing** (B).



6. Click **Create version** (A).

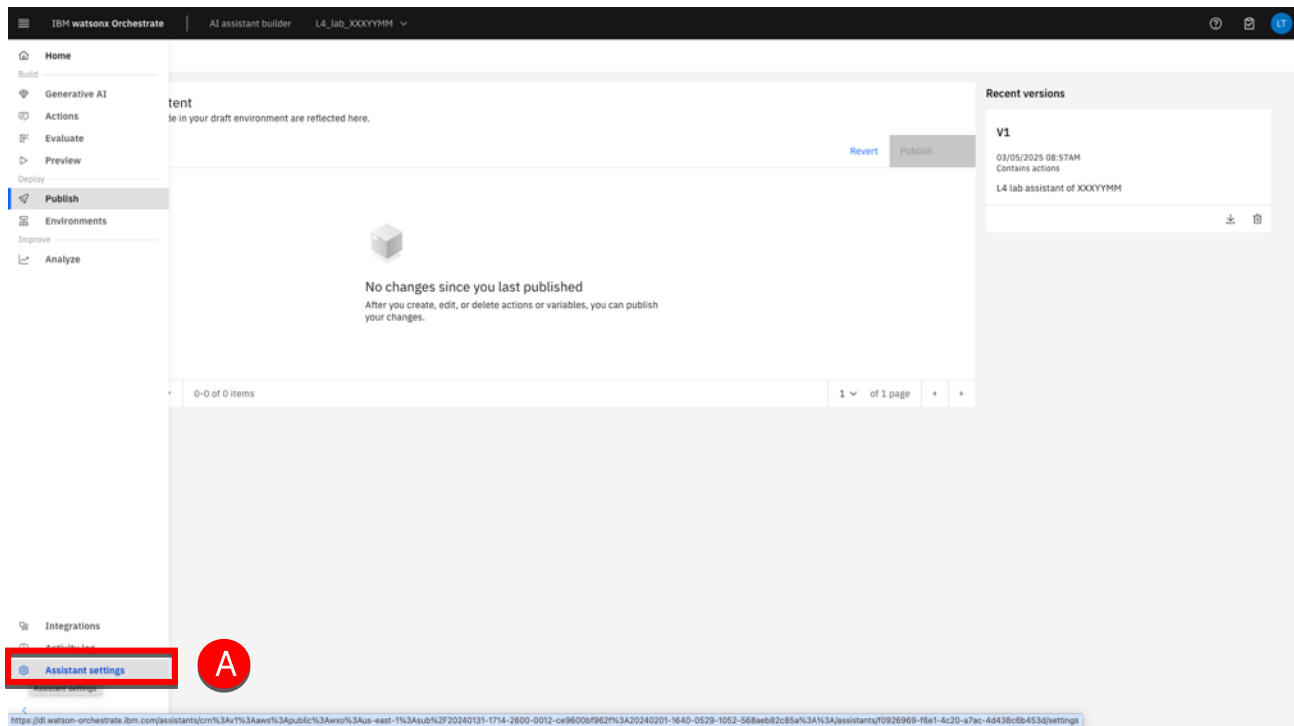


7. Wait for your version to be created (A).

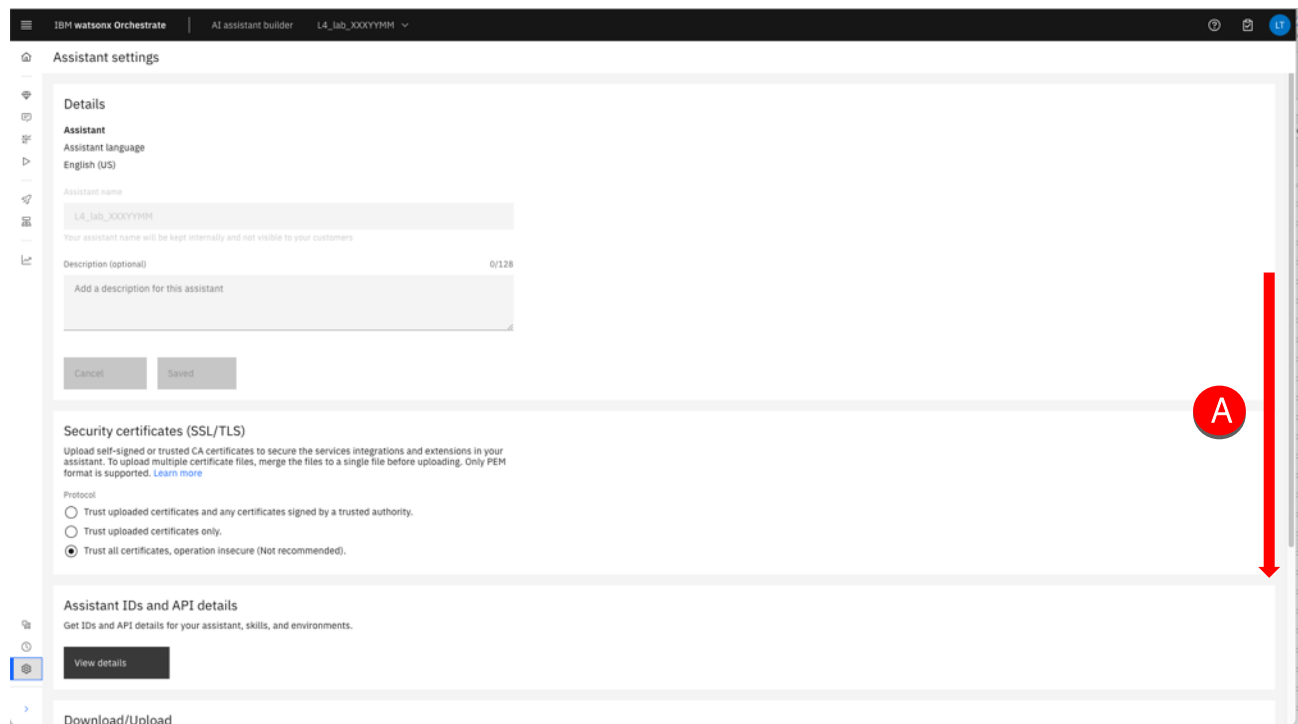


## Create the archive

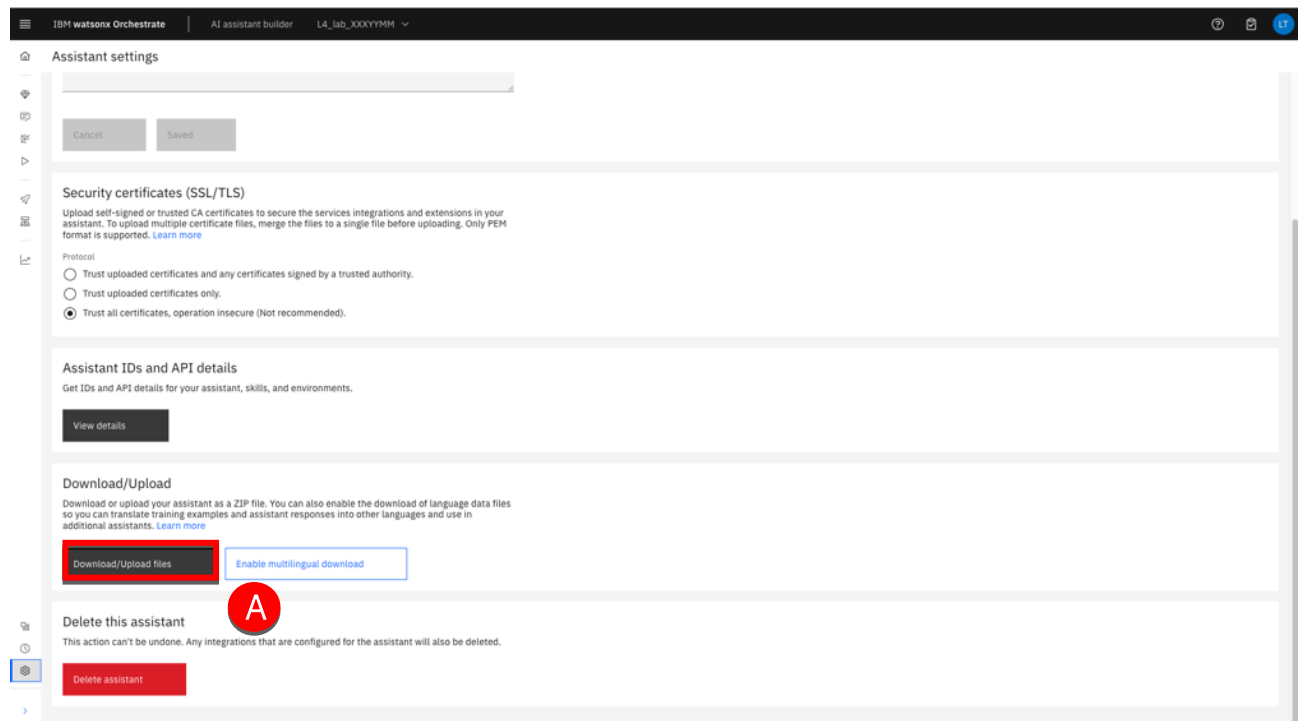
1. Click Assistant settings (A).



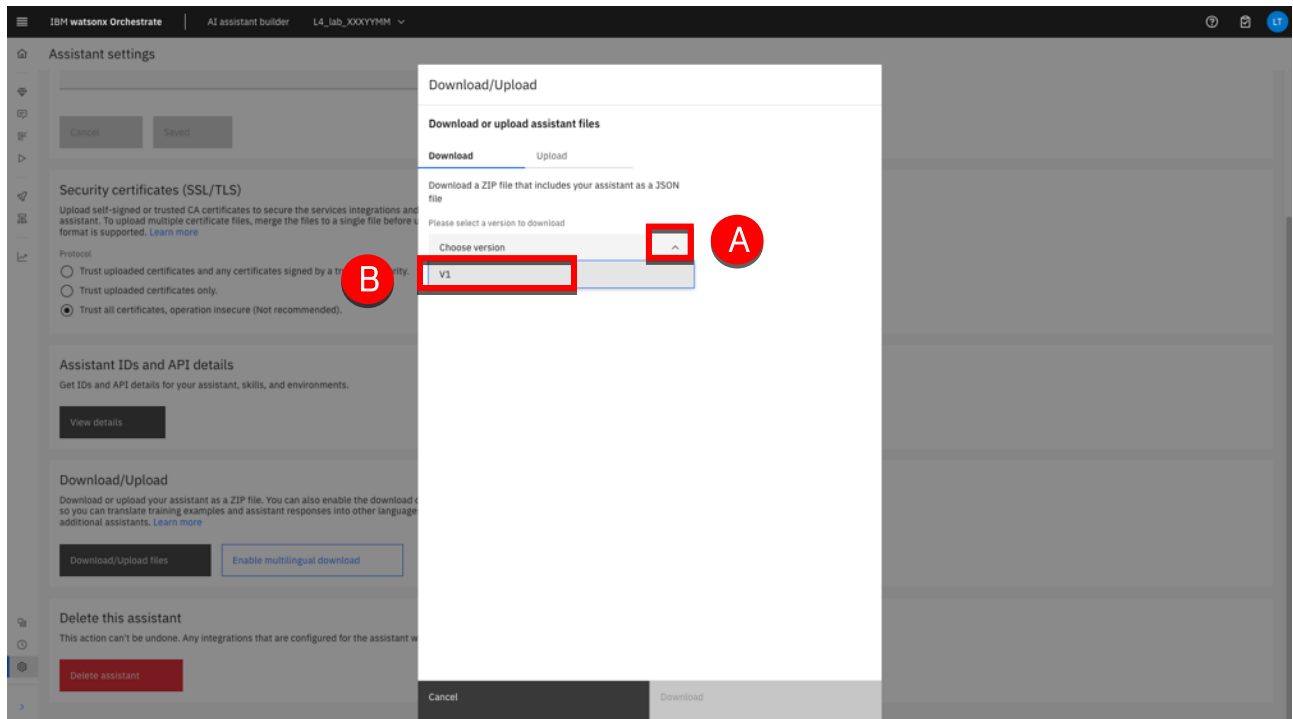
2. Scroll down (A).



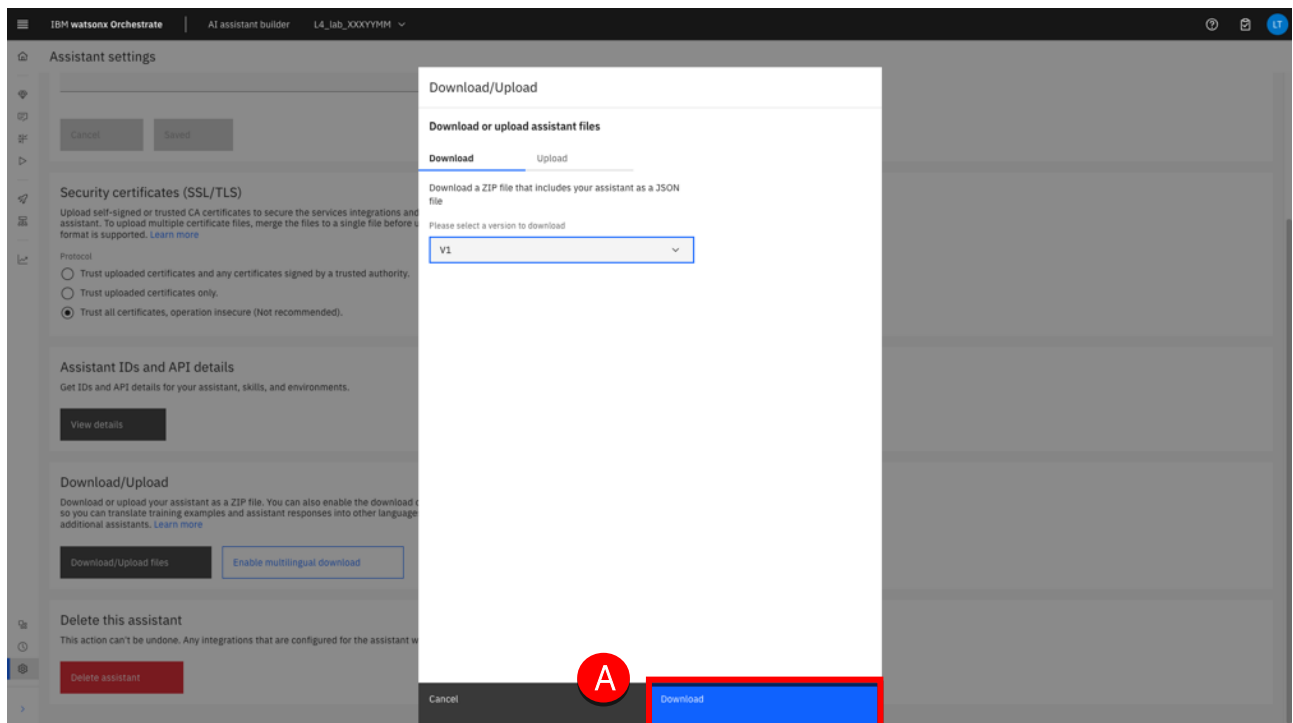
3. Click Download/Upload files (A).



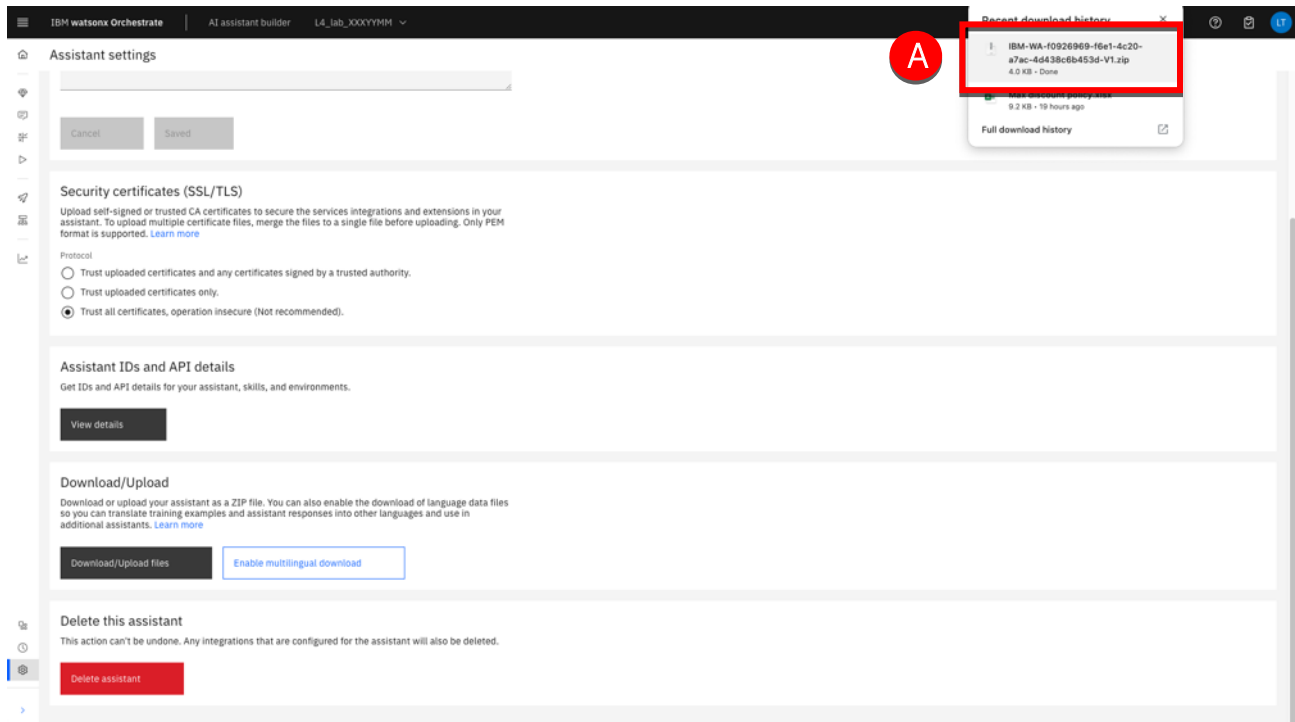
4. Expand the **Choose version** menu (A) then select your last published version (B).



5. Click **Download** (A).



6. Get your archive file from your download folder (A) (May change depending on your browser).

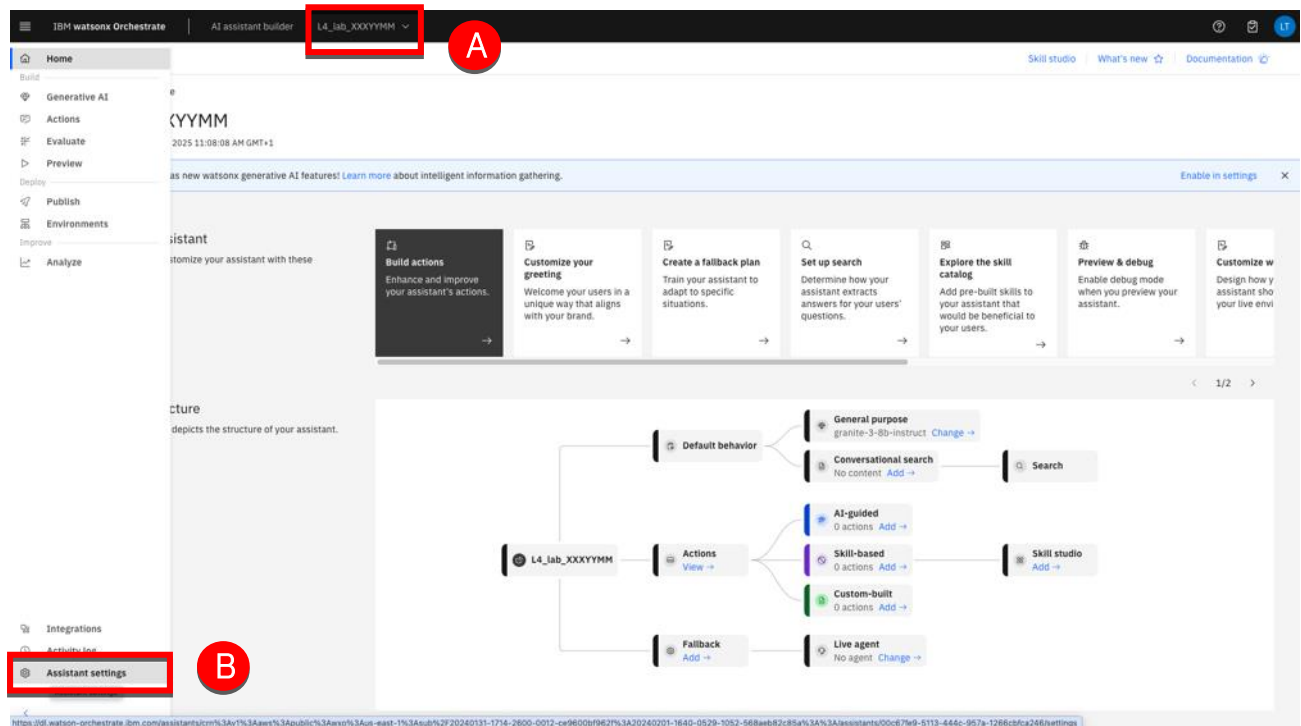


**Note:** the name of the generated archive file will look like IBM-WA-f0926969-f6e1-4c20-a7ac-4d438c6b453d-V1.zip.

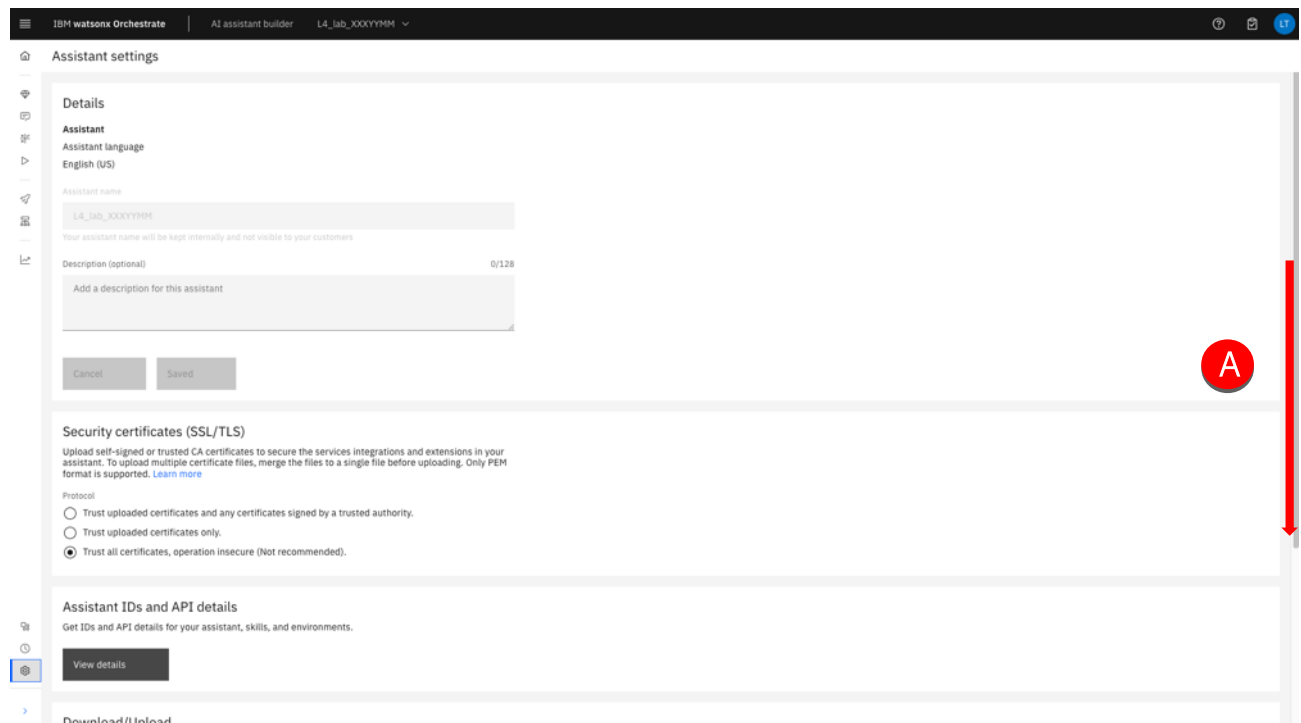
## Import an assistant

When you import actions from an archive, these actions are added to the current Assistant in AI assistant builder. It is important to create a new empty assistant with a proper name before importing any archive file.

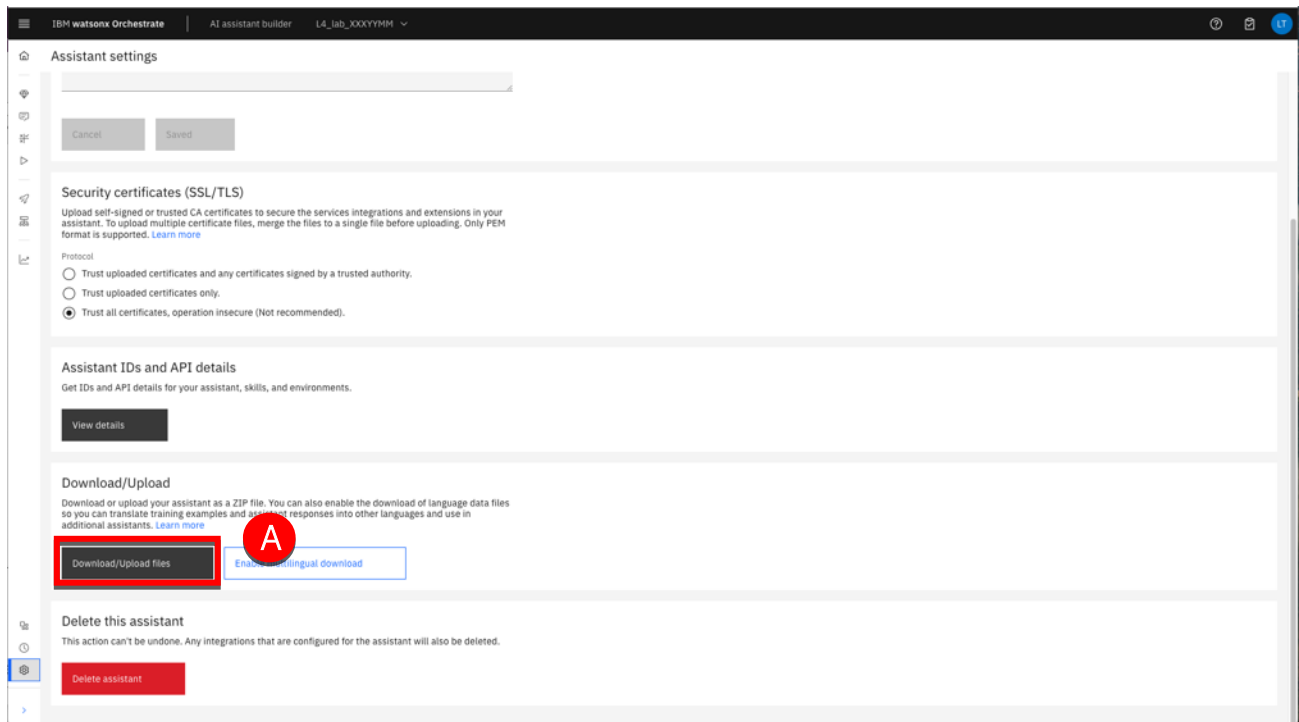
1. Create a new assistant following the [Create your L4 assistant instructions](#) in this guide.
2. Check that you are on your new assistant (A), click Assistant settings (B).



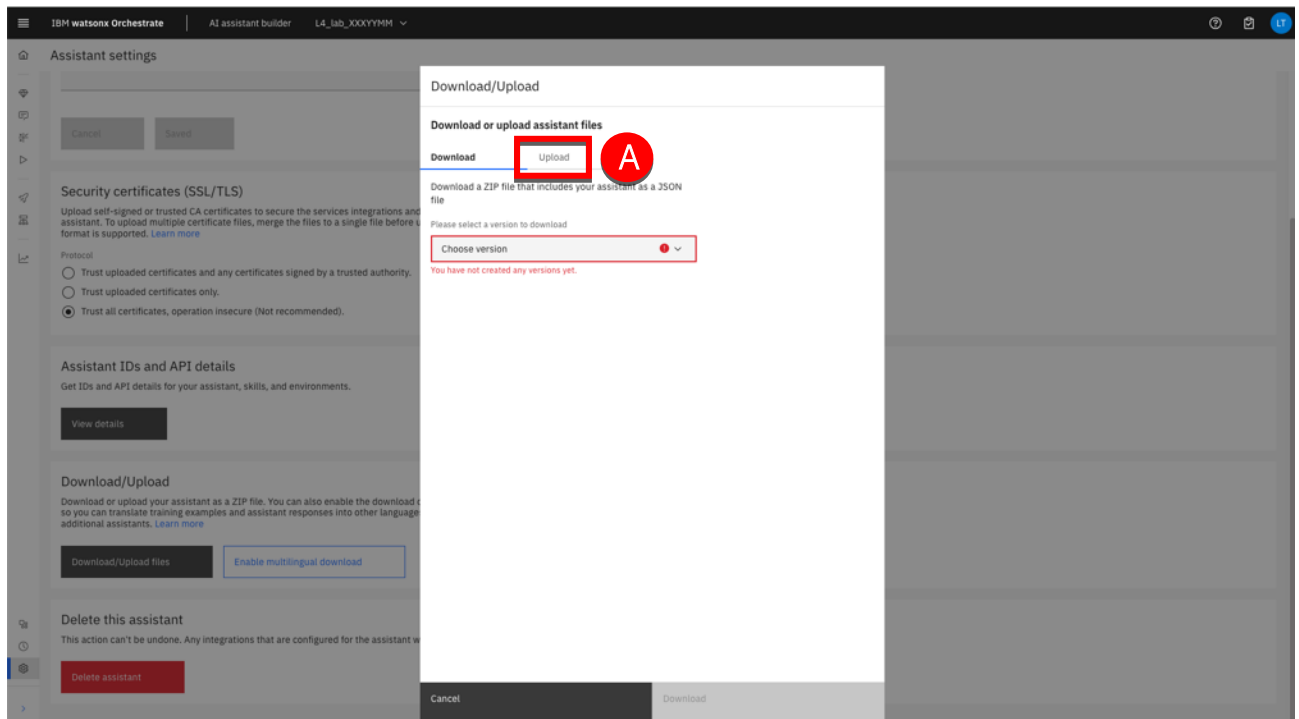
3. Scroll down (A).



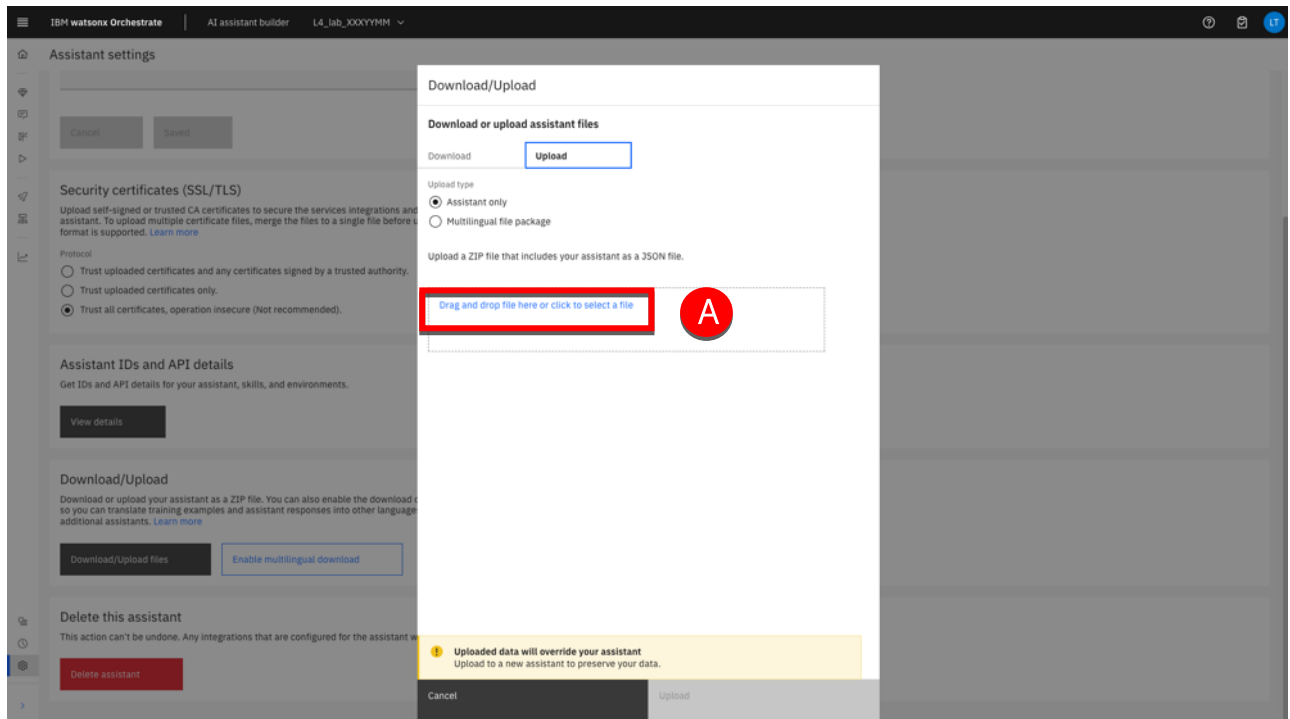
- Click **Download/Upload files** (A).



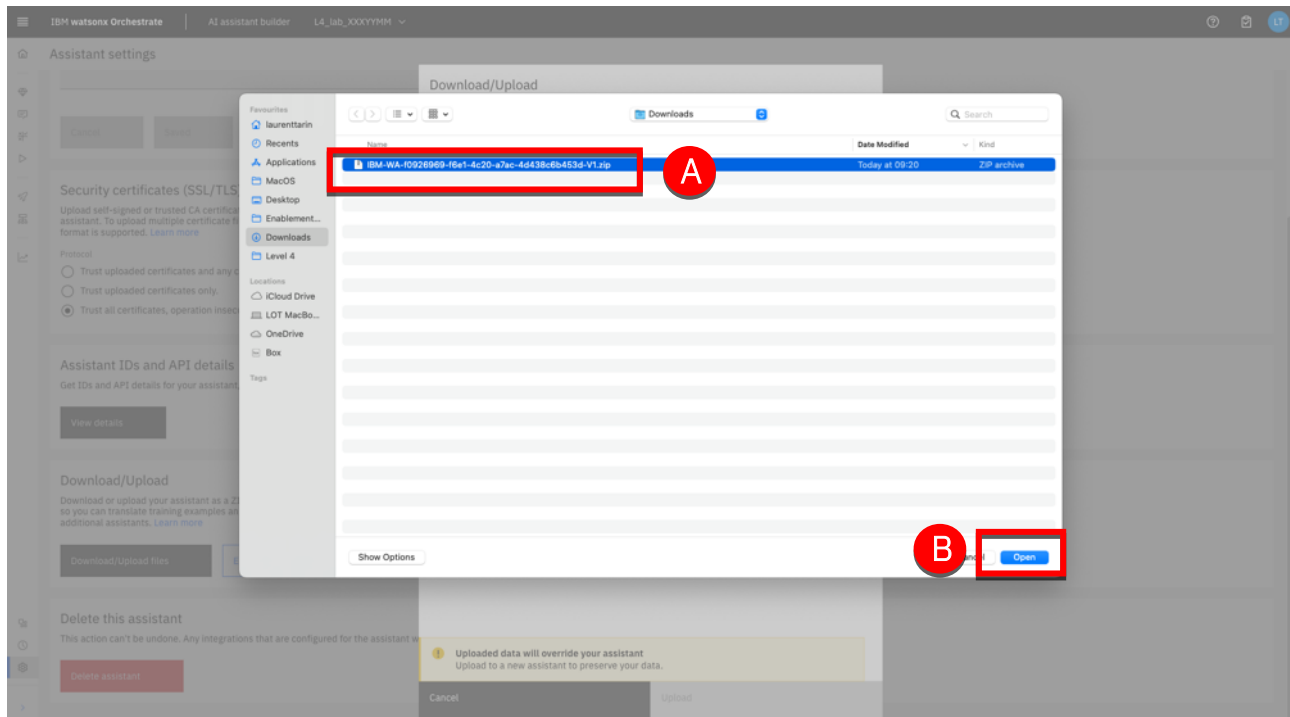
- Select the **Upload** tab (A).



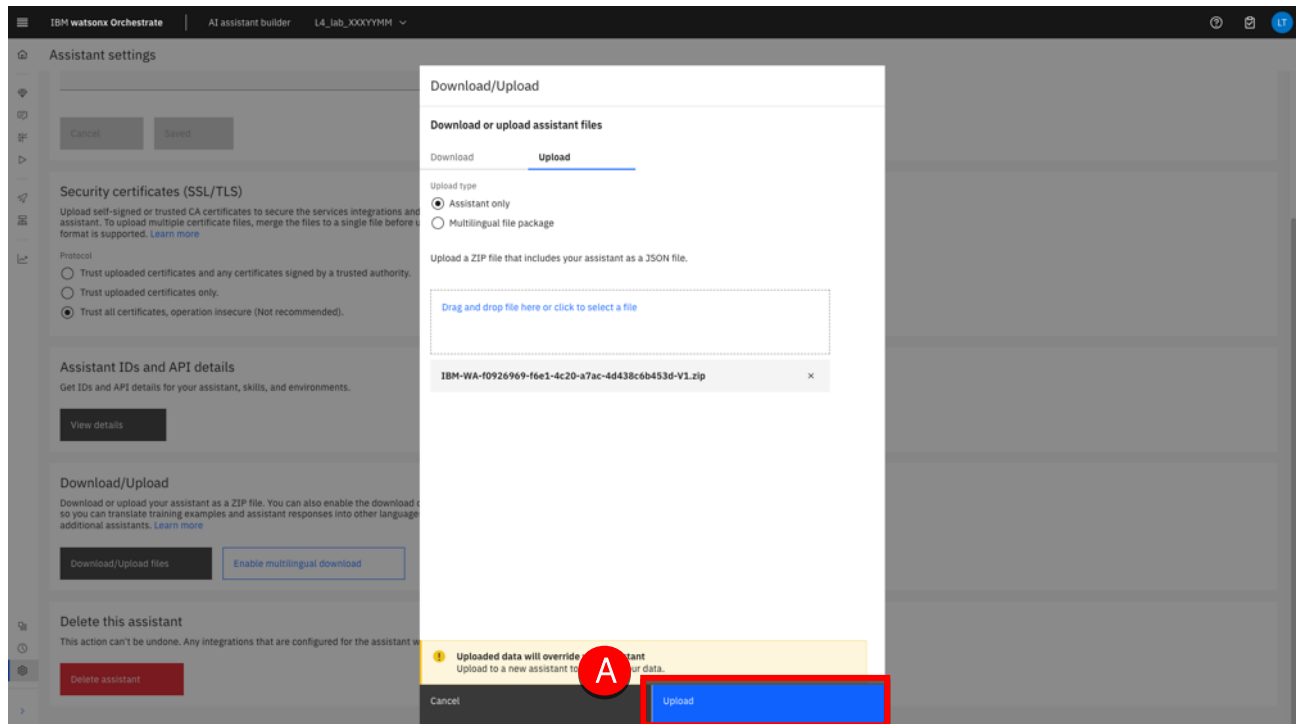
6. Click the **Drag and drop file here or click to select a file** link (A).



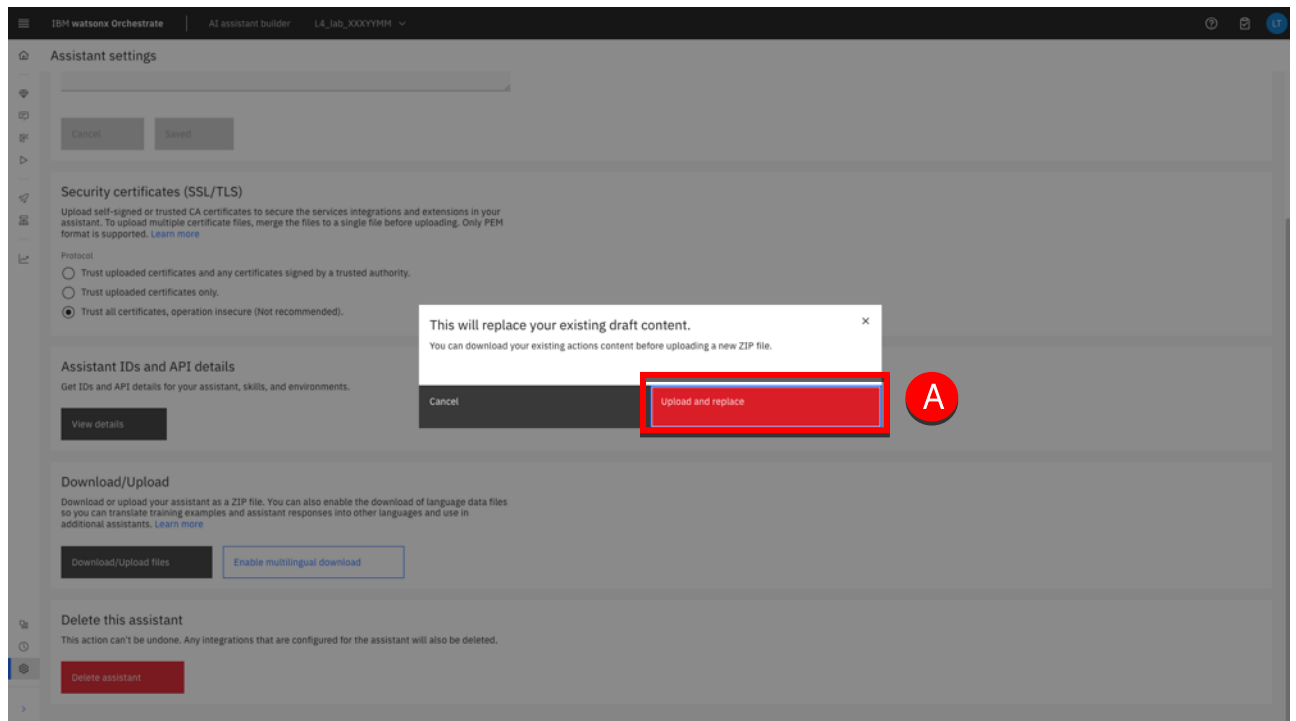
7. Select your archive file (A) then click **Open** (B).



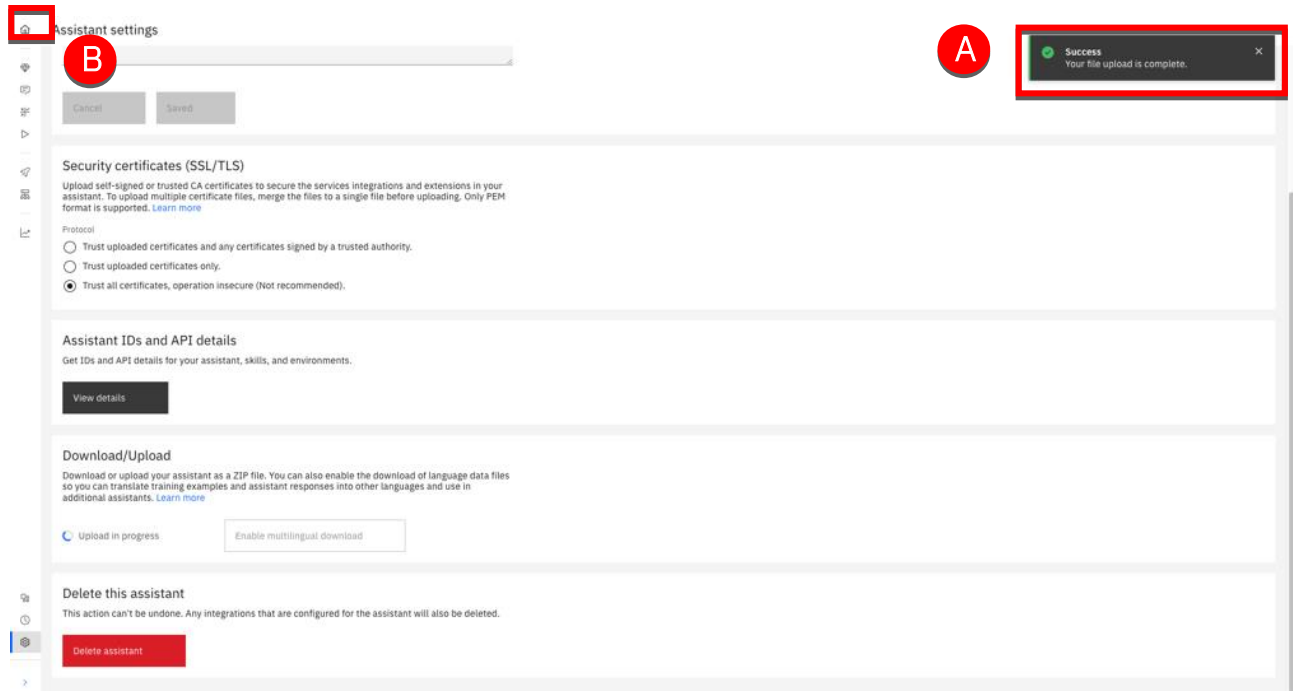
8. Click **Upload** (A).



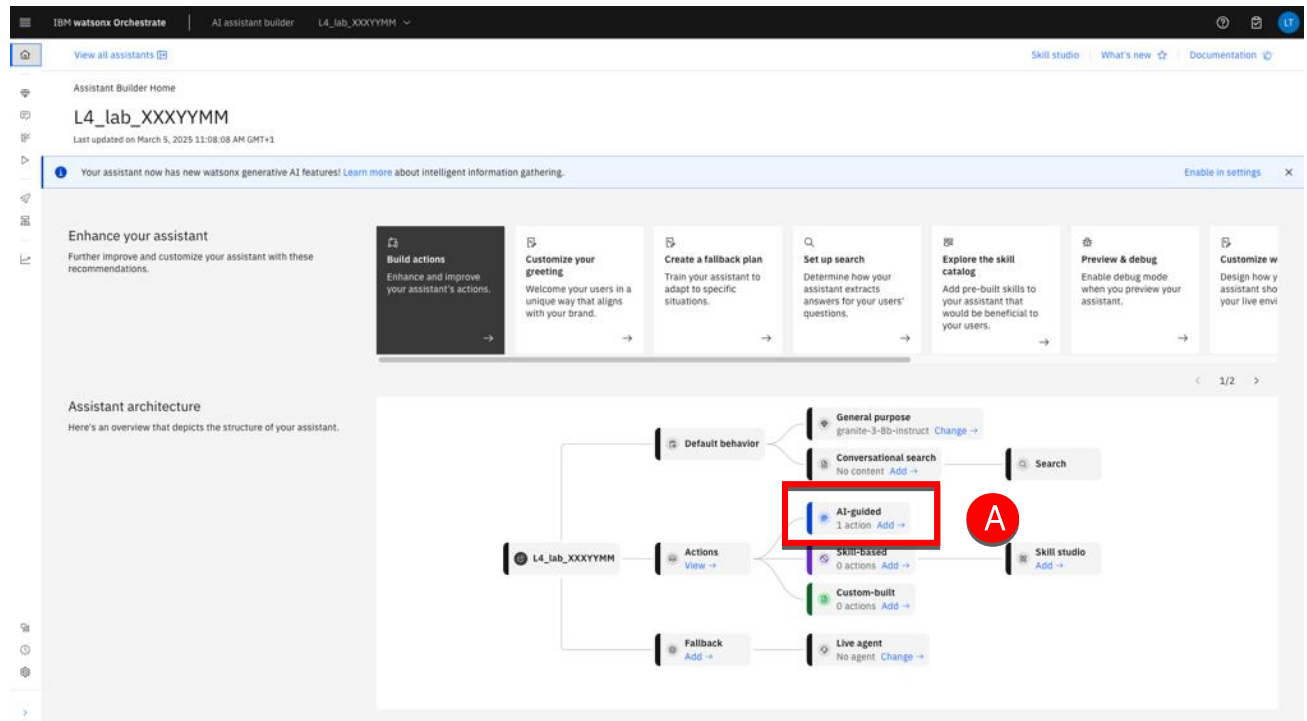
9. Click **Upload and replace** (A).



10. Wait for the **Success** notification that will be shortly displayed on the top right of your screen (A). Click the home icon (B).



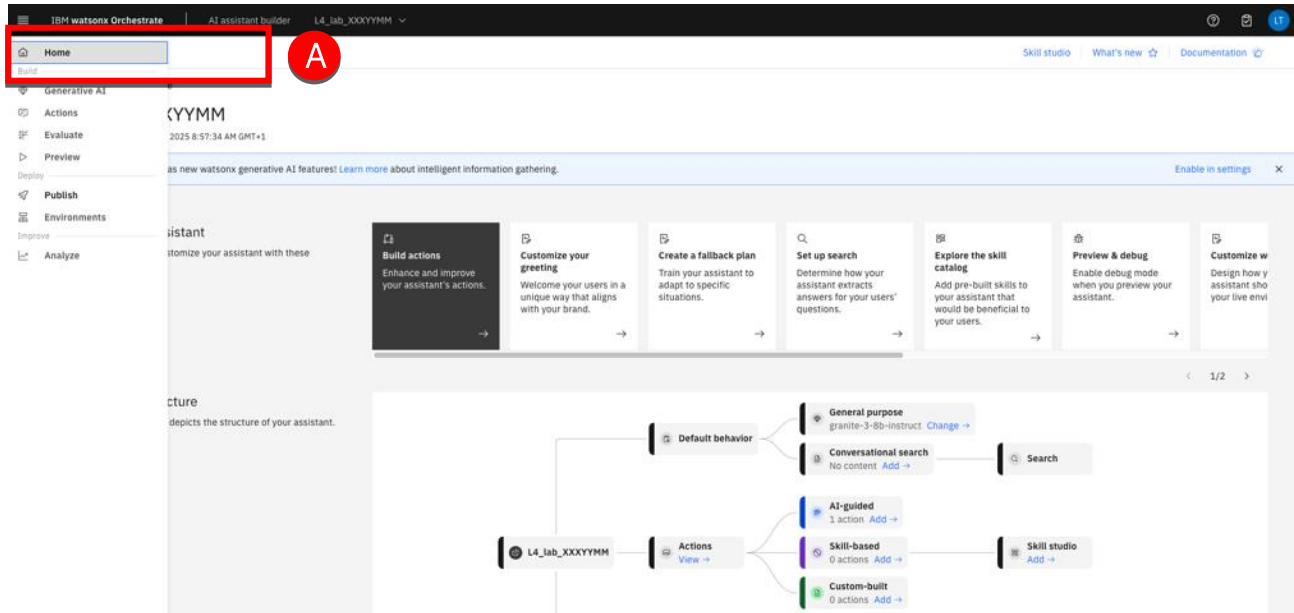
11. Check your previous actions are there (A) (AI-guided actions in the example bellow).



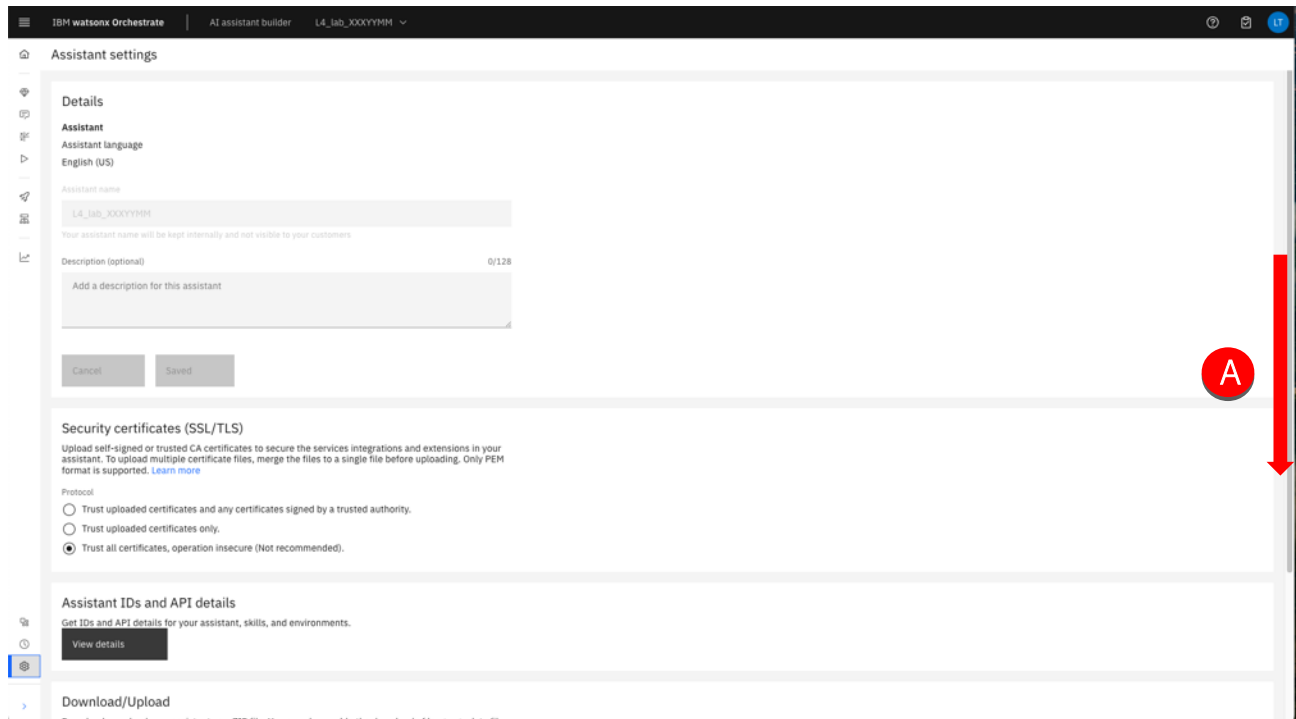
## Delete your assistant

Assistant must be deleted when you have performed all the hands-on labs related to assistants or when you have exported your archive before taking a break in the learning plan. Follow these instructions to delete your assistant.

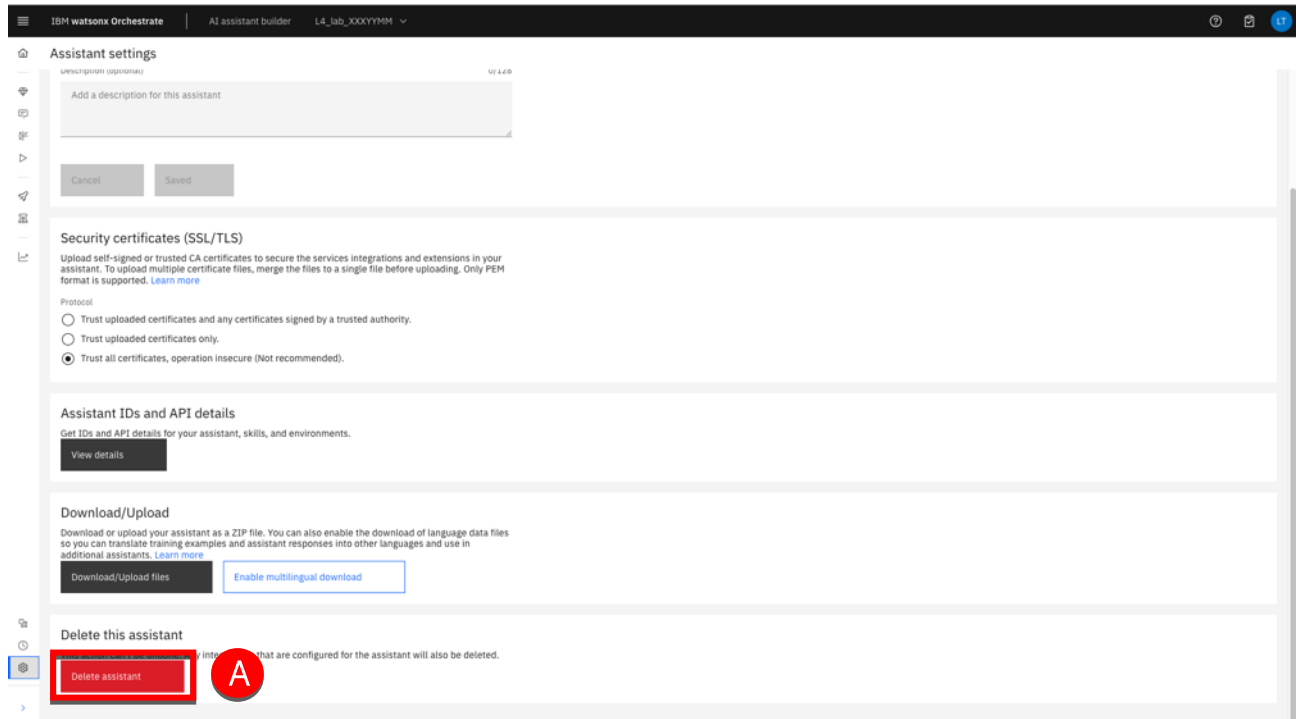
1. Return to the AI assistant builder home page.
2. Select the assistant to delete (A). Click **Assistant settings** (B).



3. Scroll down (A).



4. Click Delete (A).



5. Enter DELETE (A) then click **Delete** (B).



